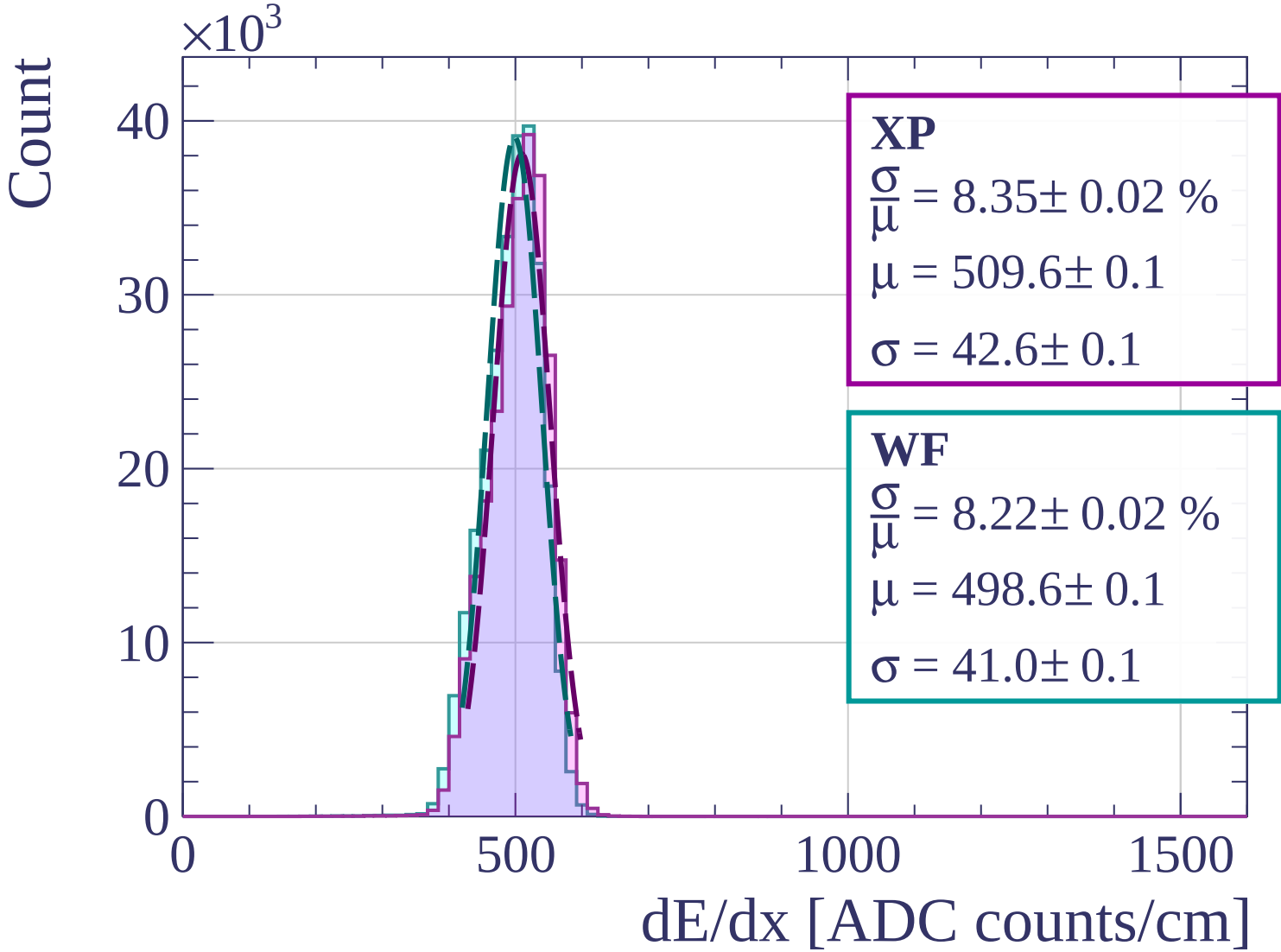
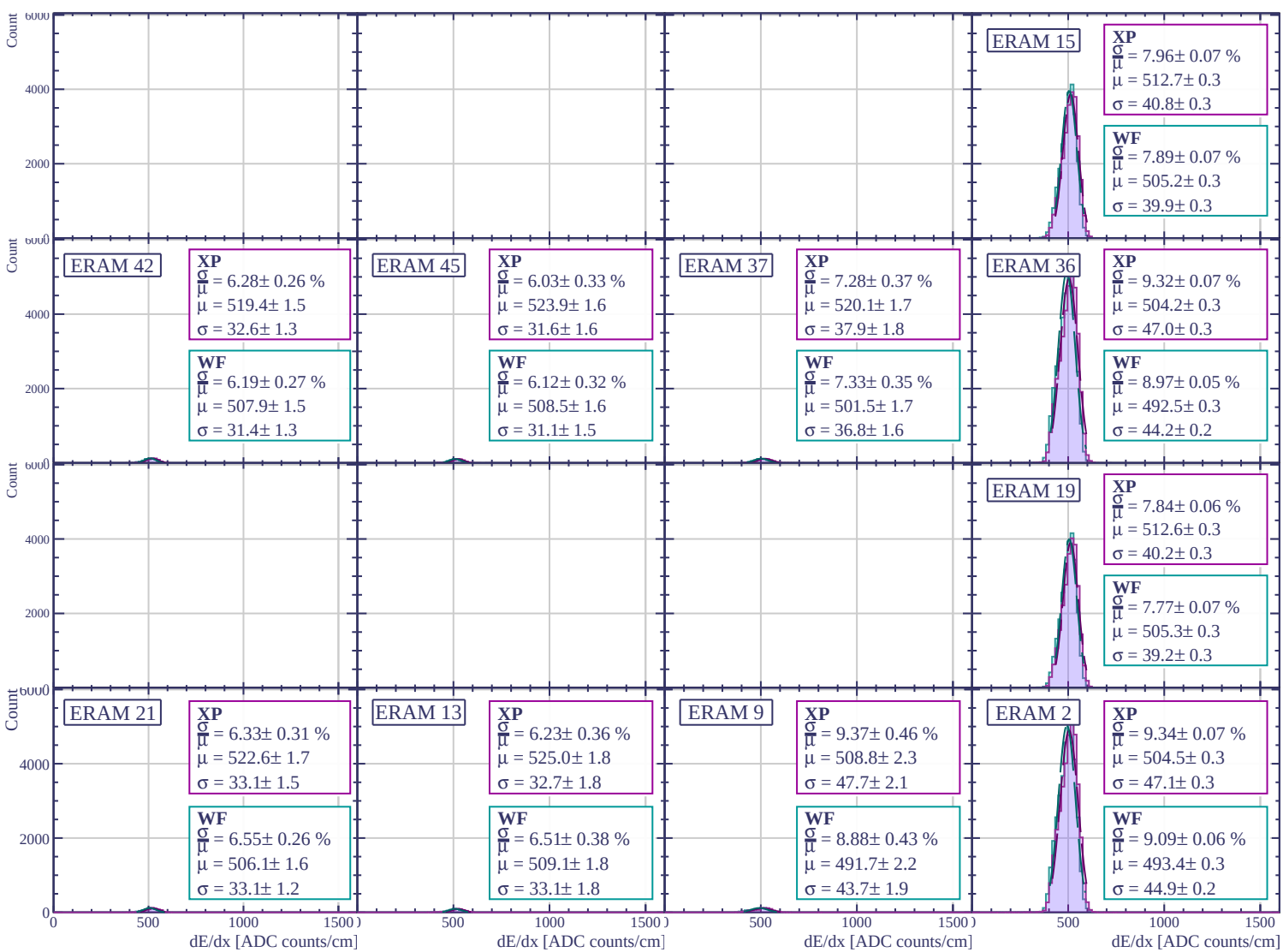
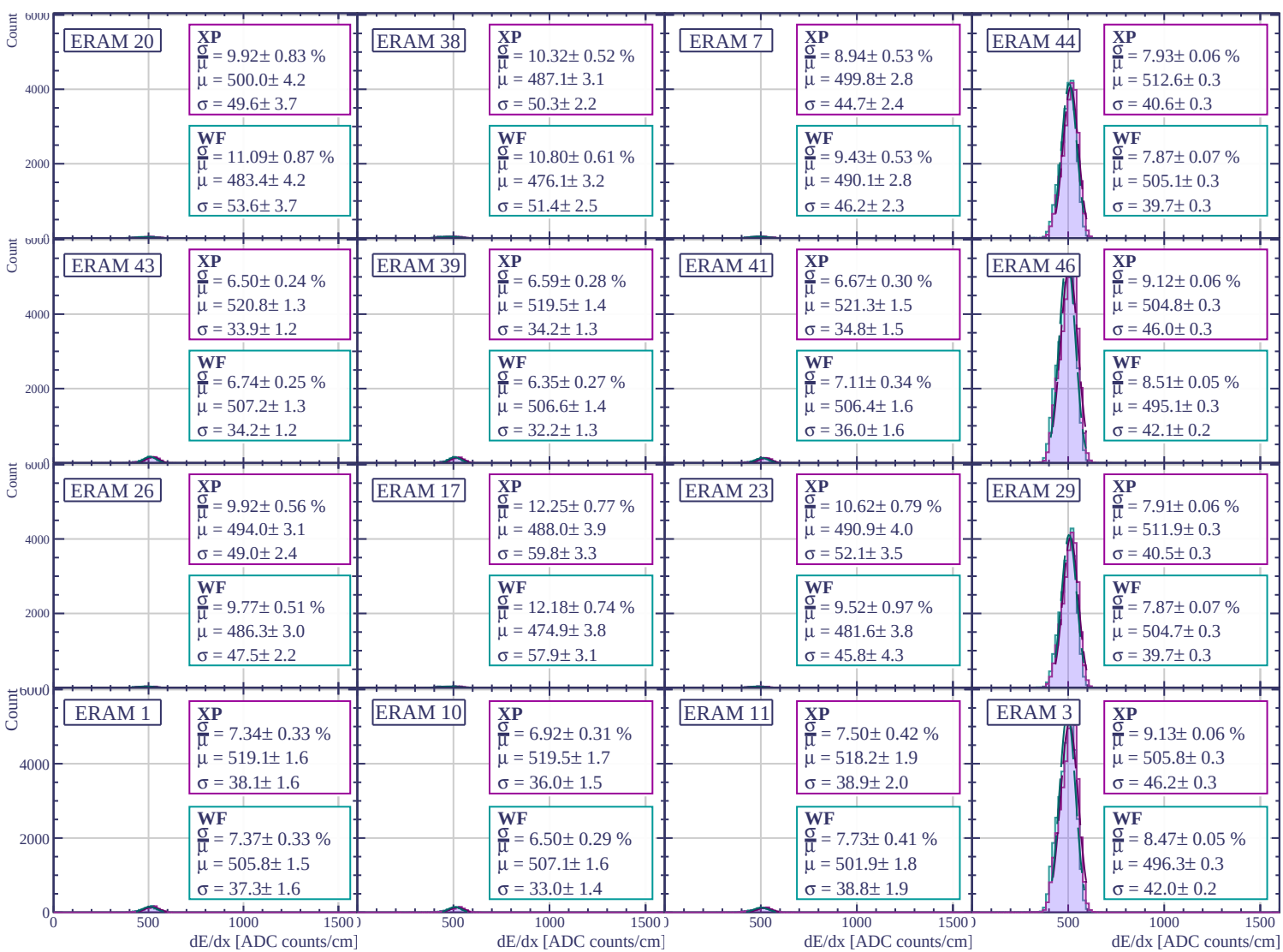
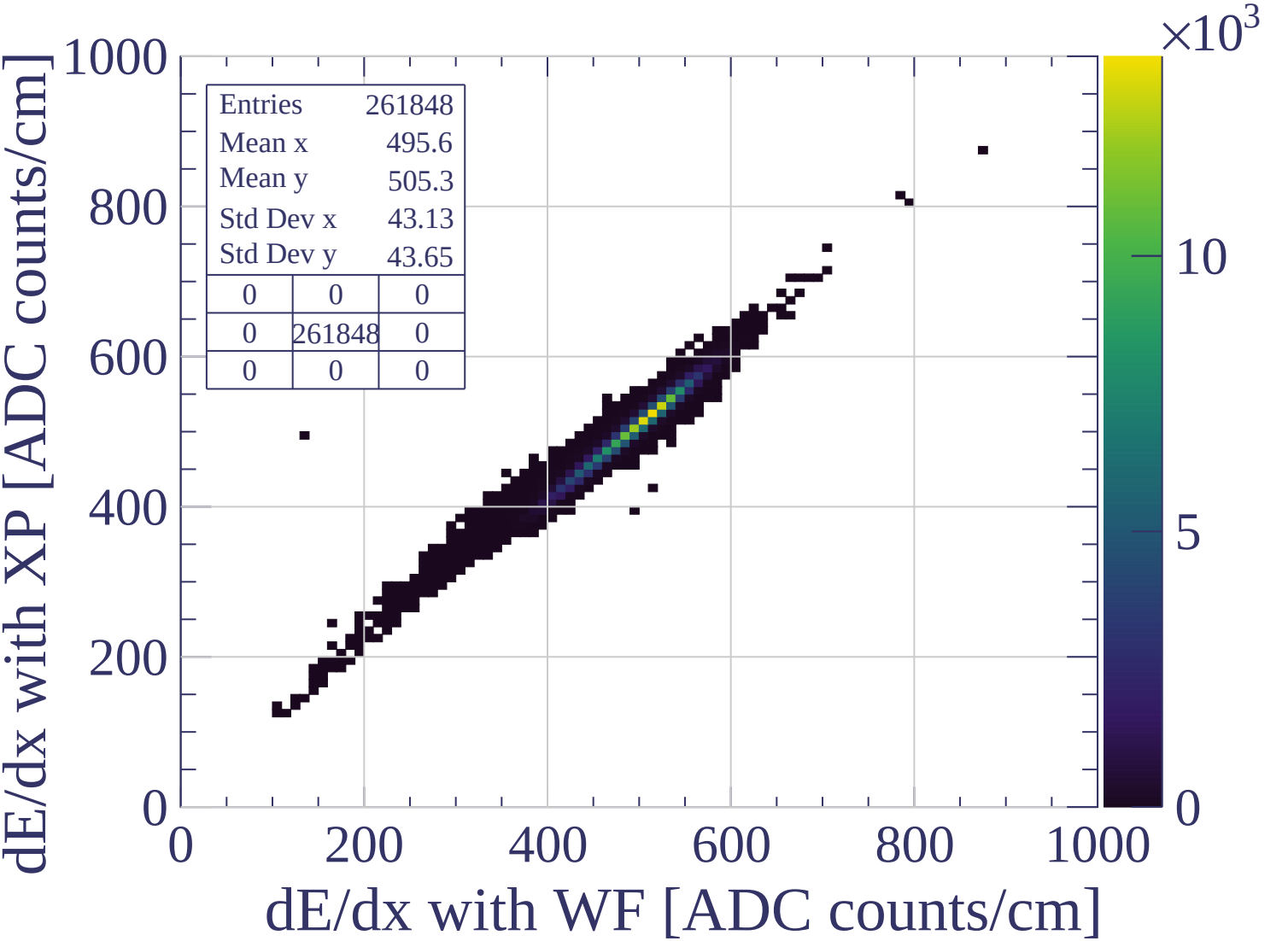
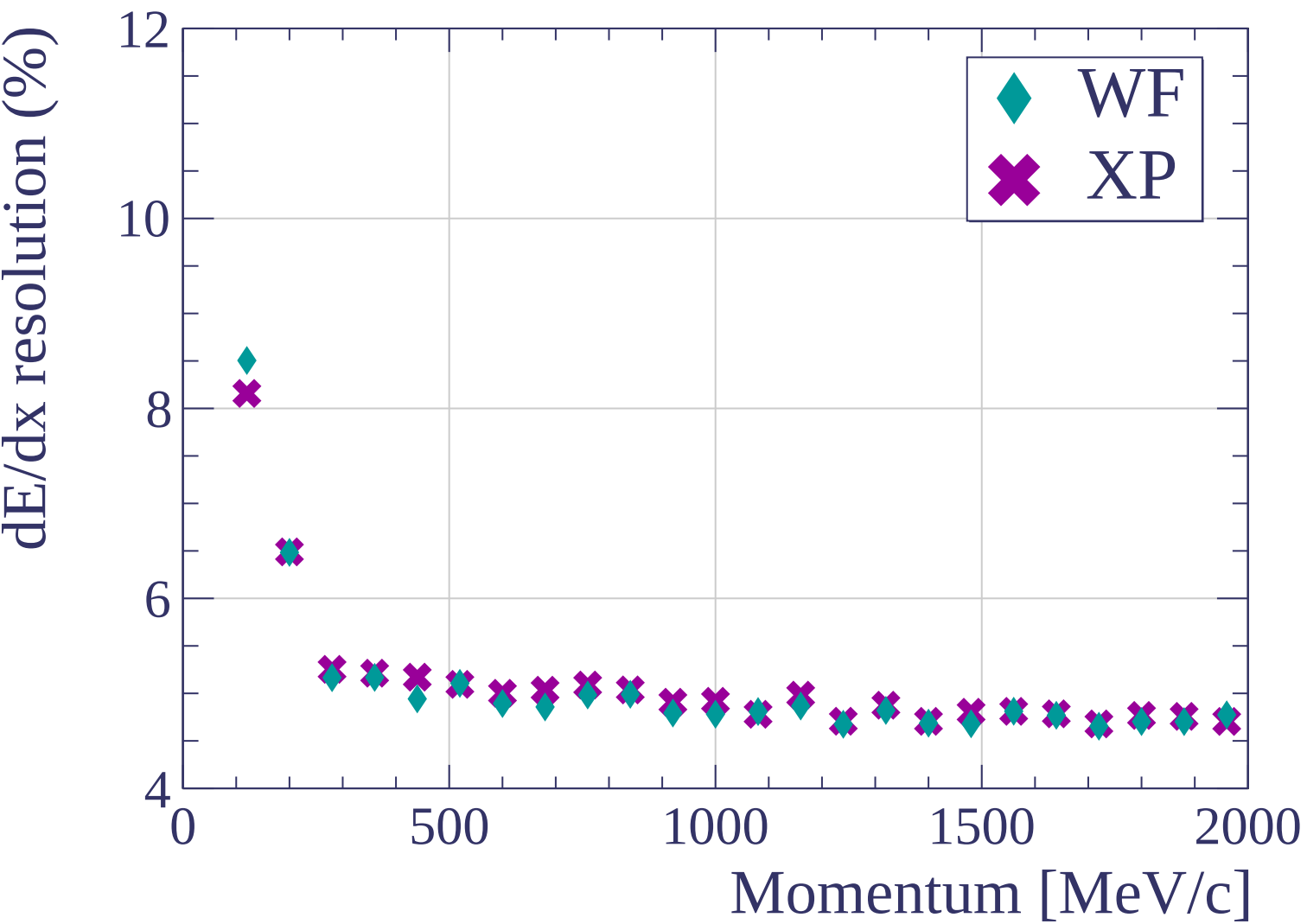


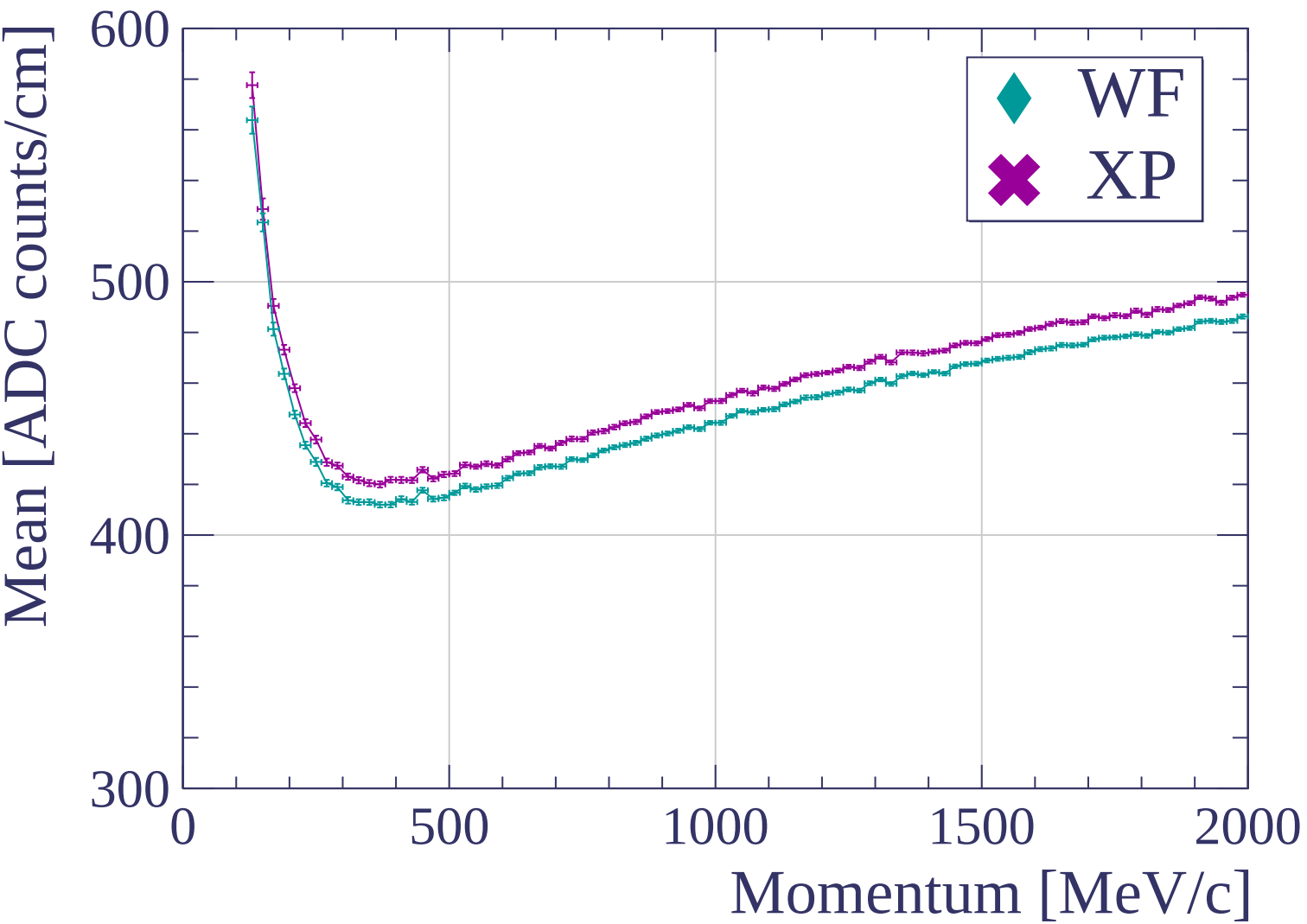


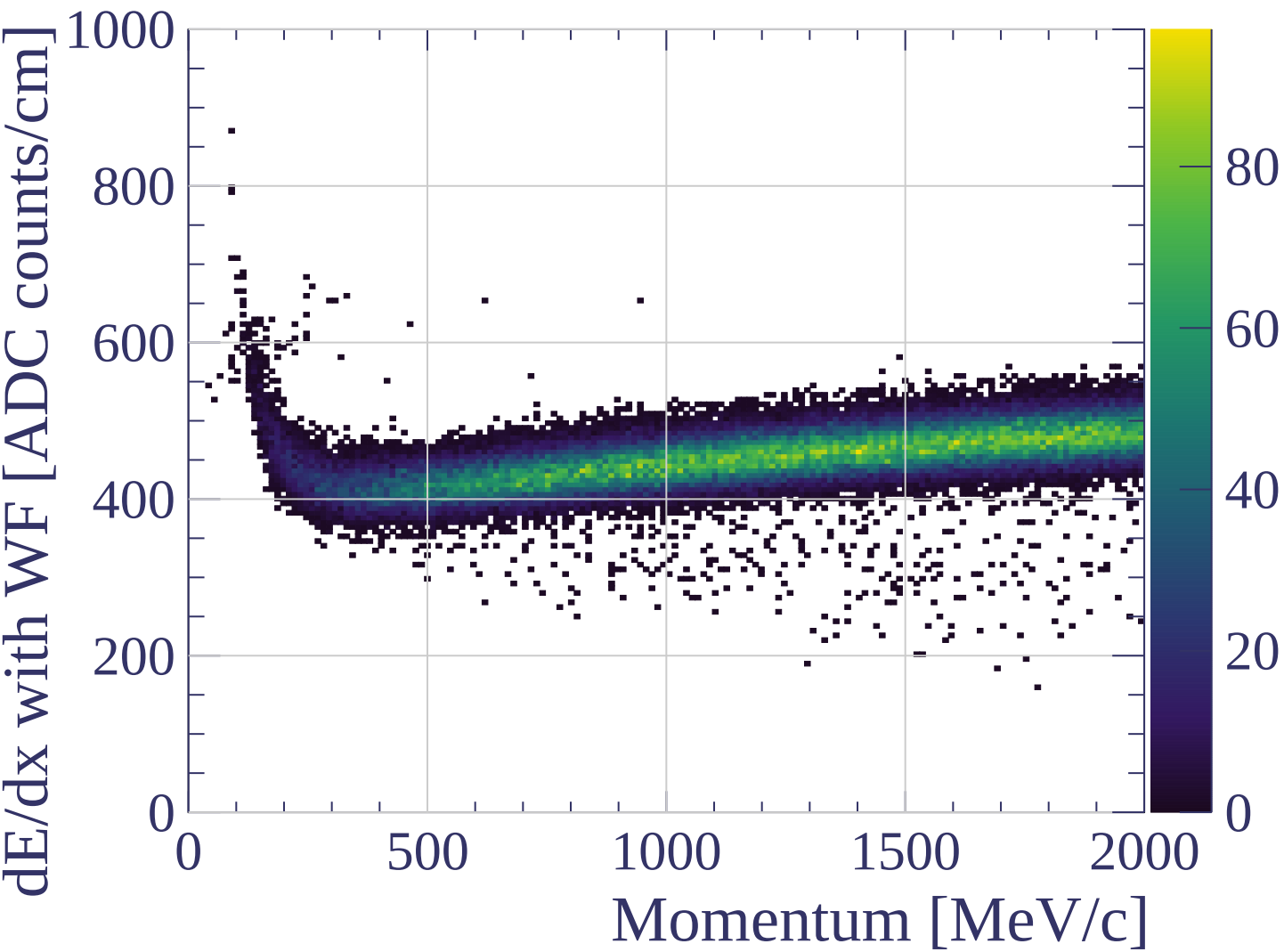


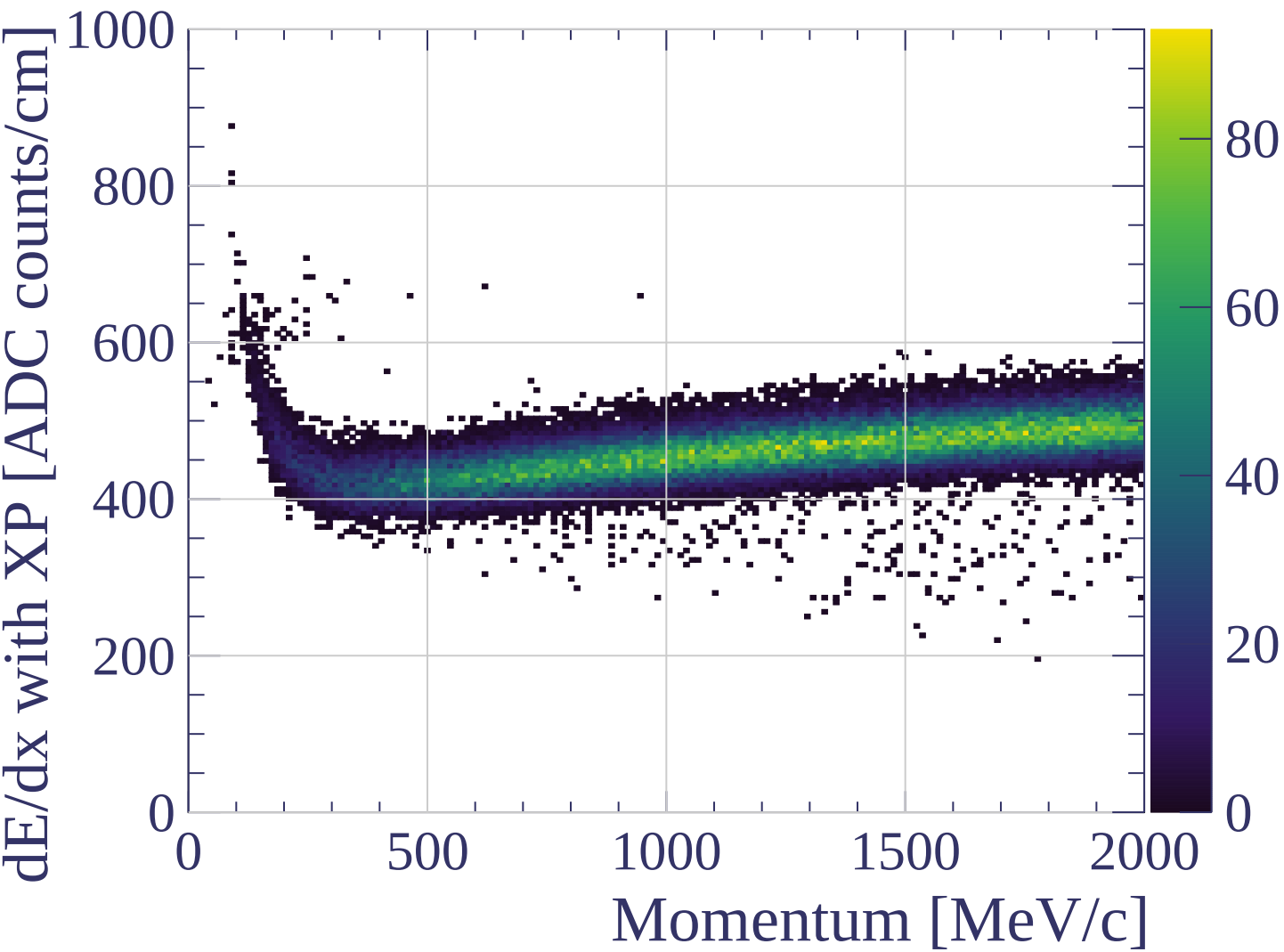


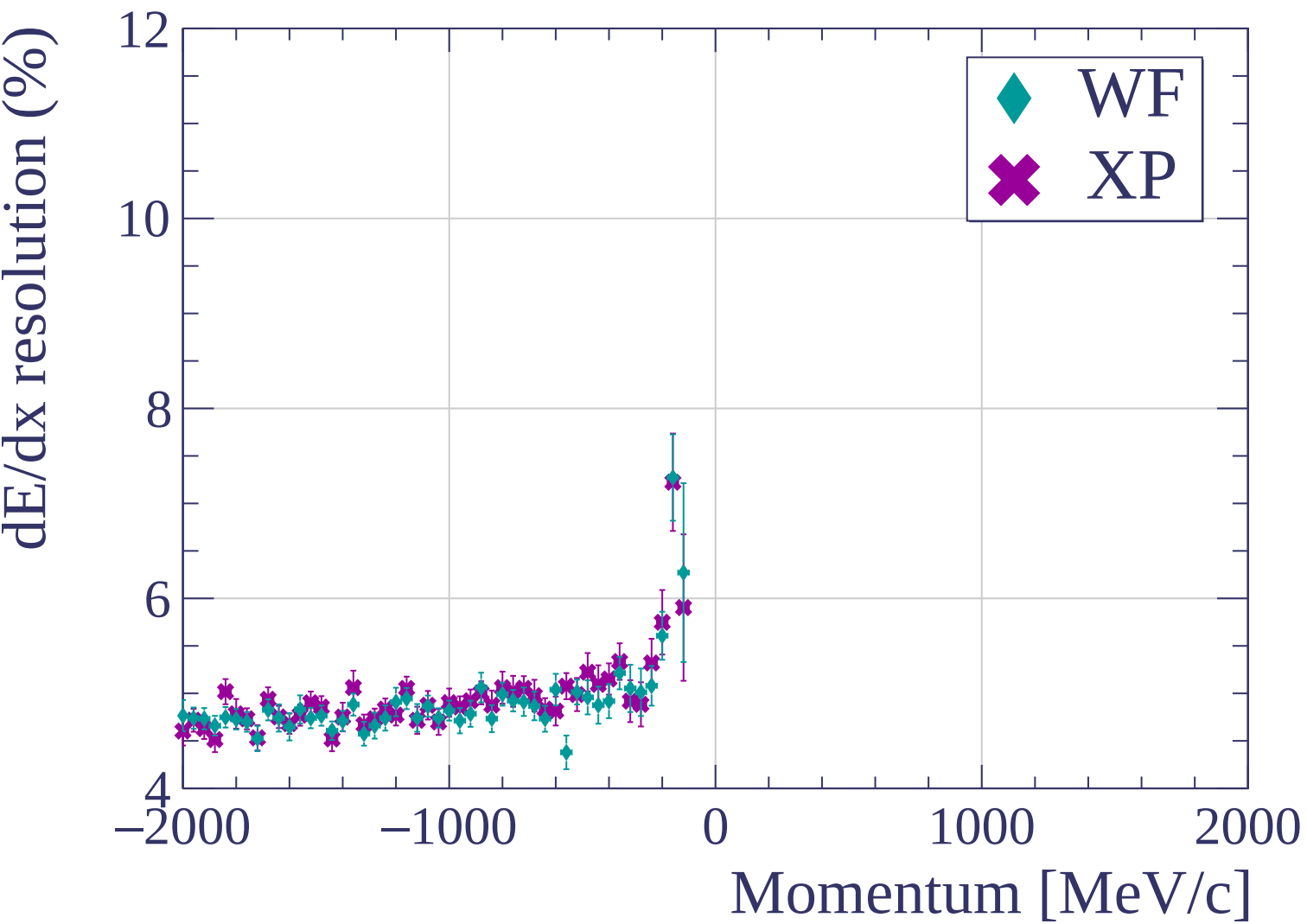


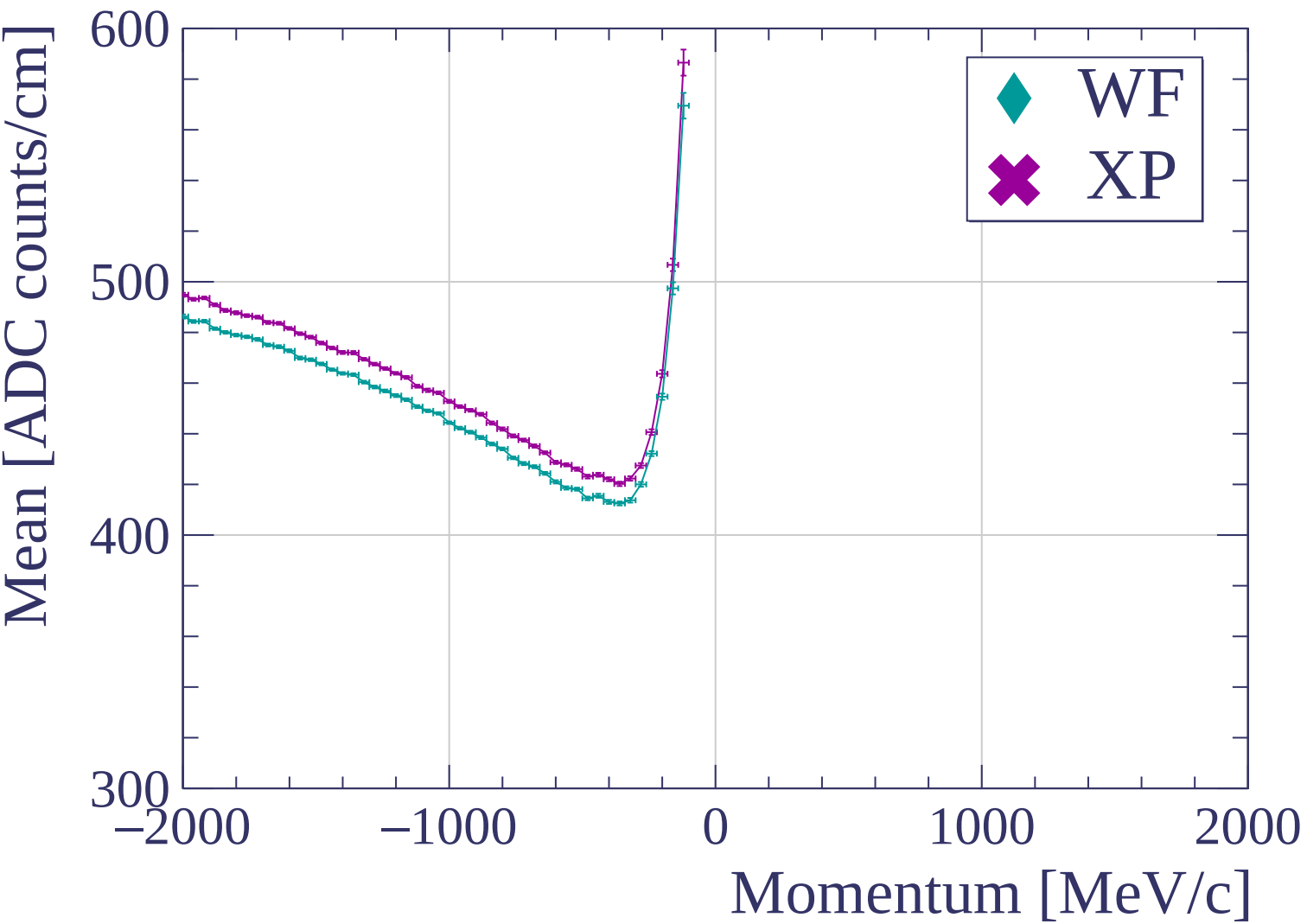


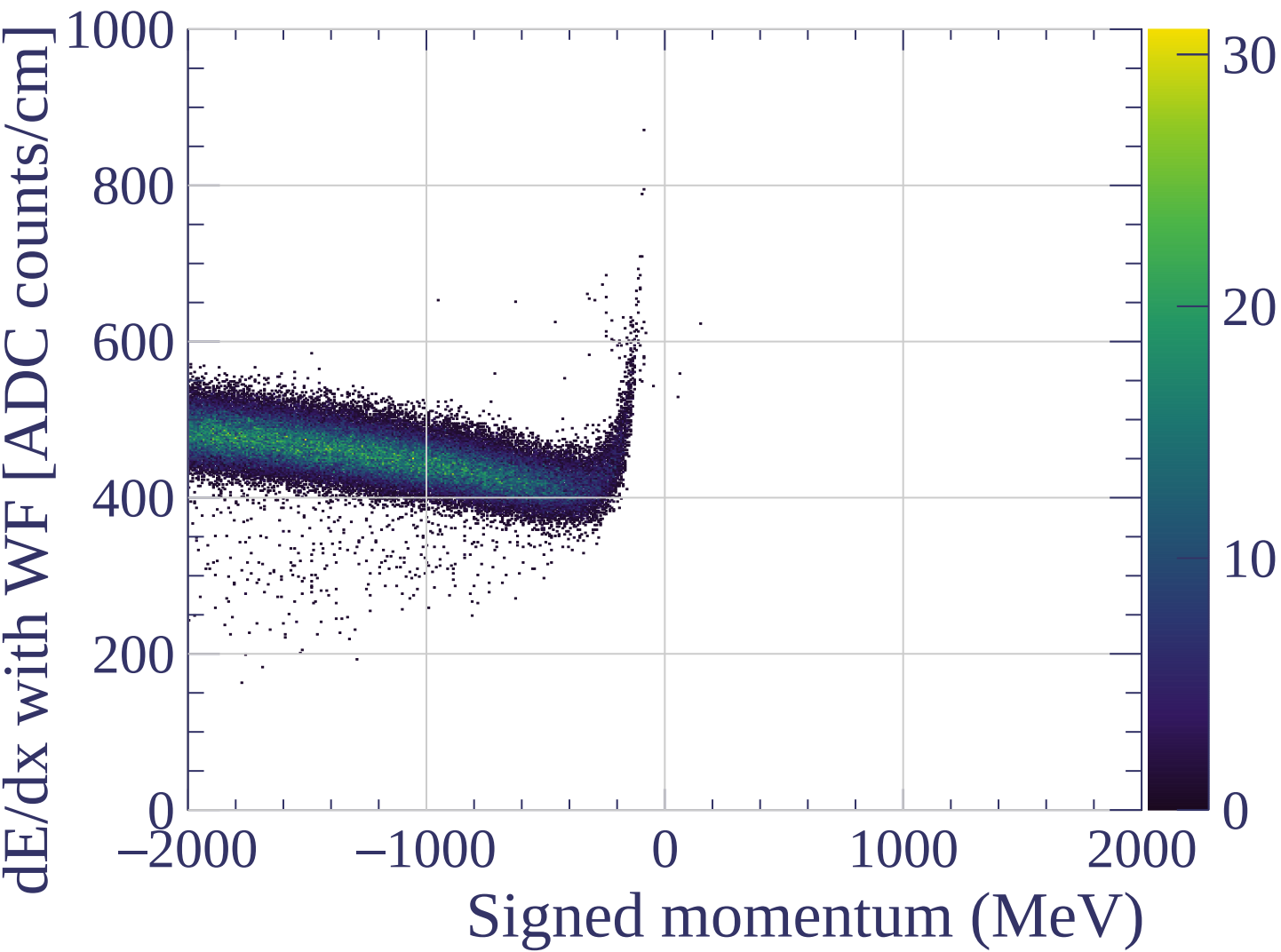


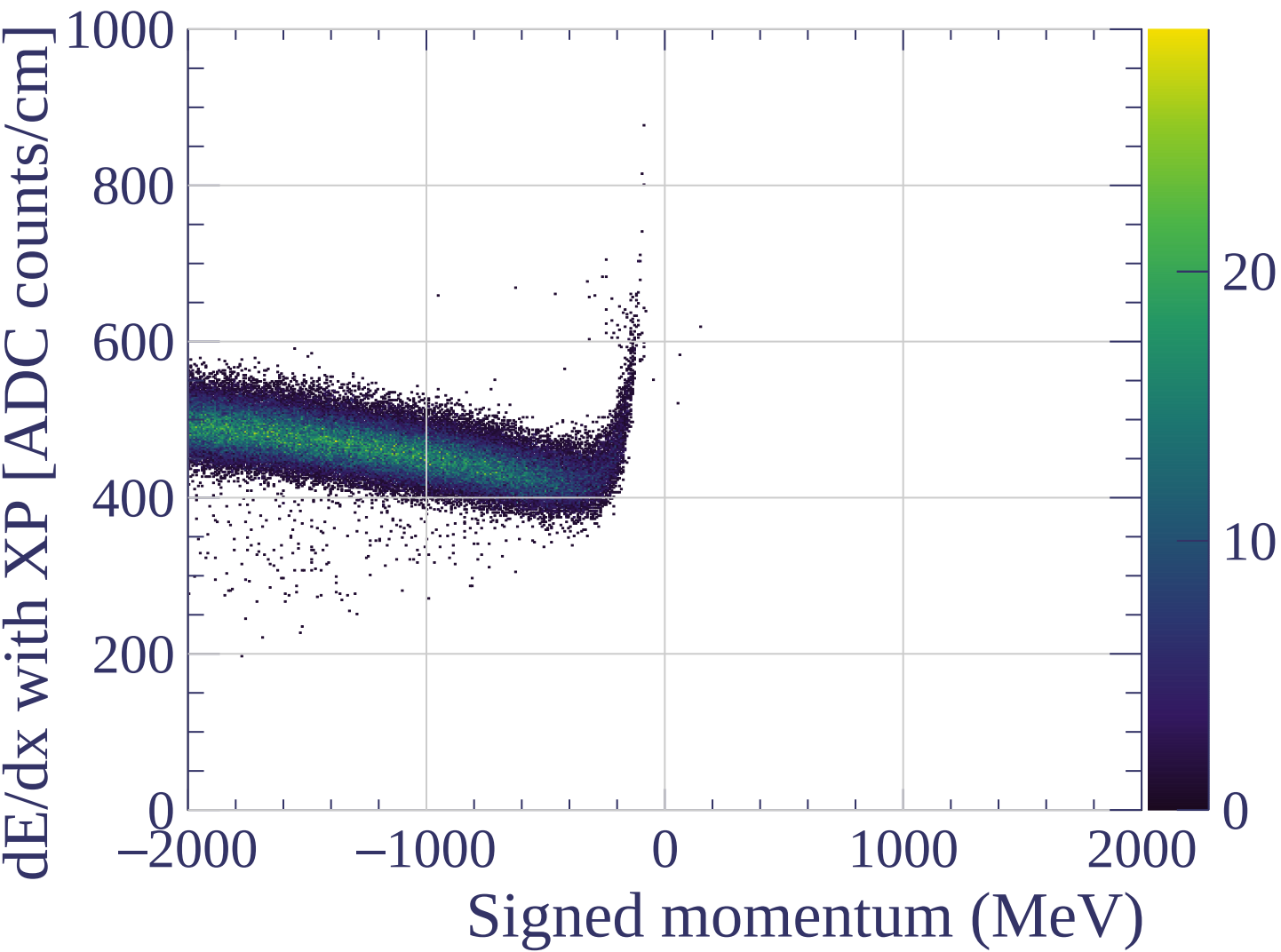




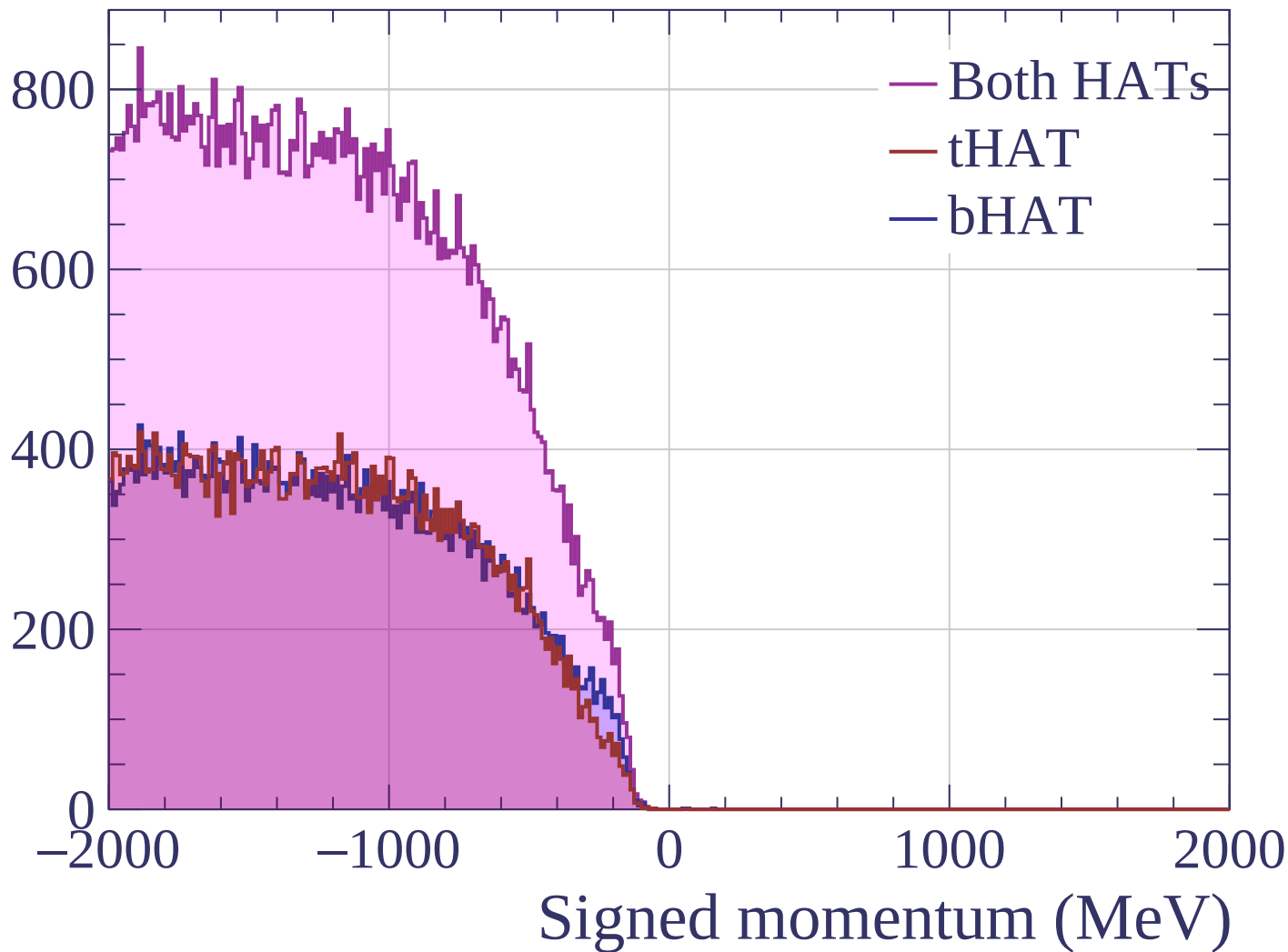


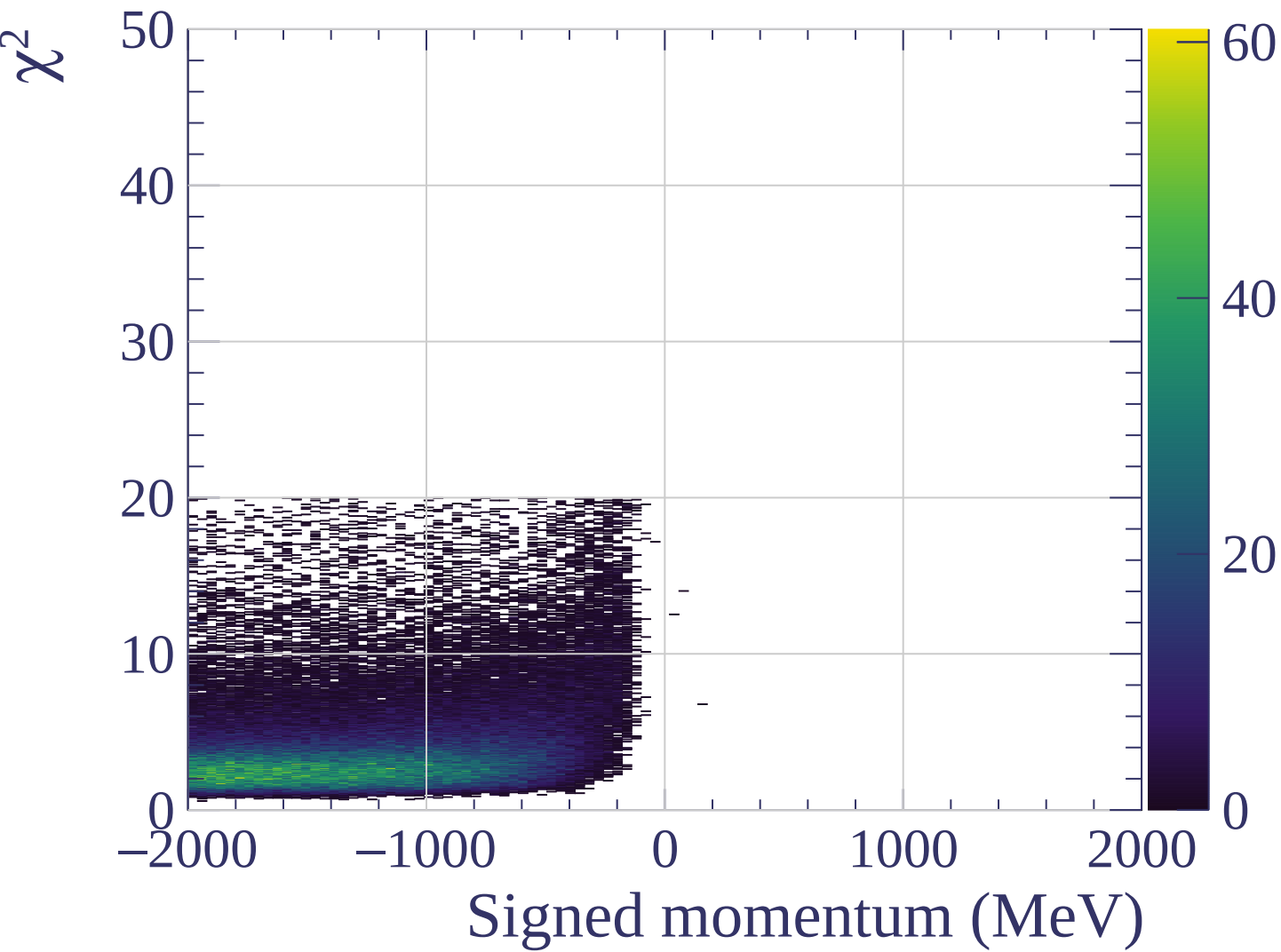


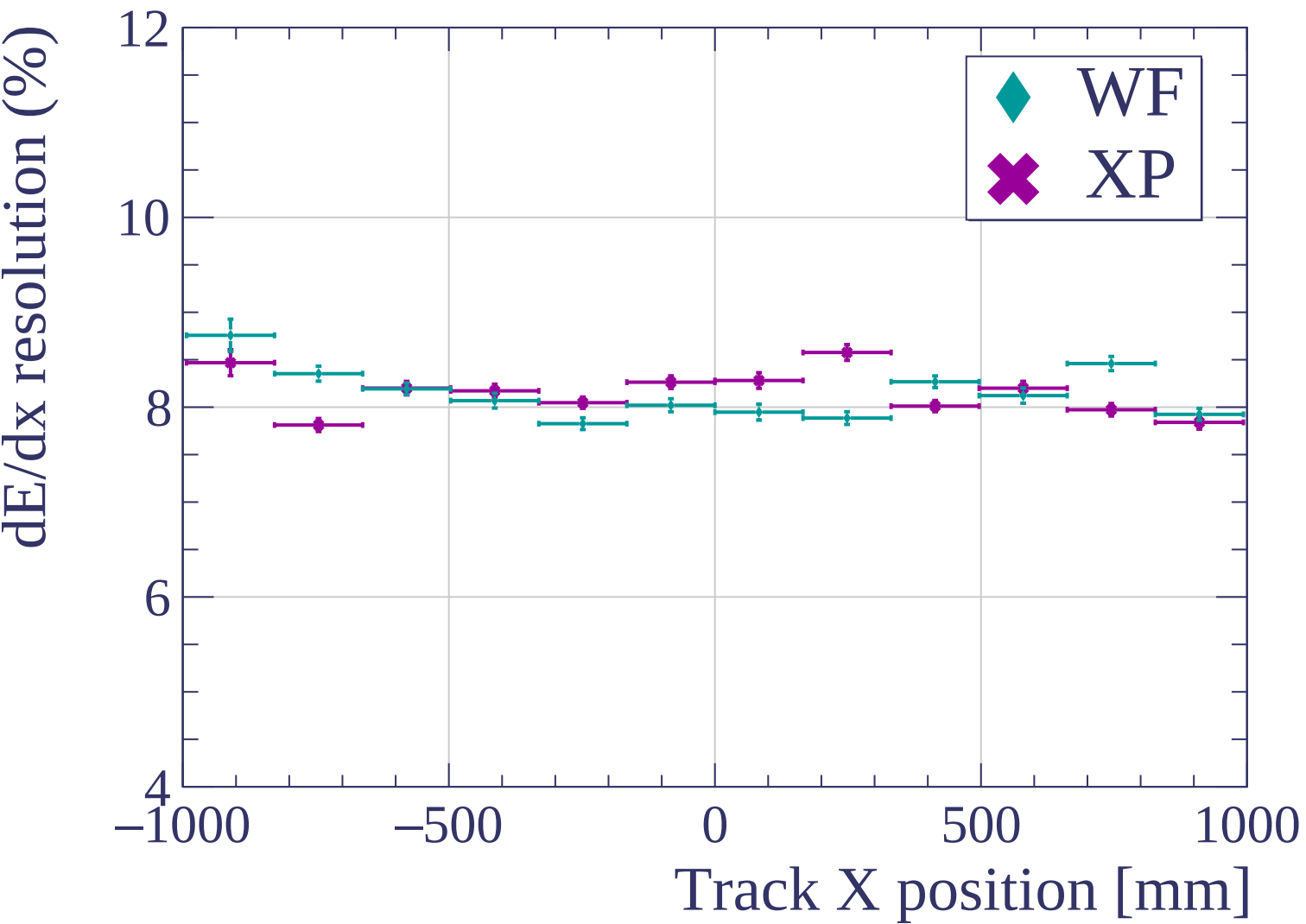


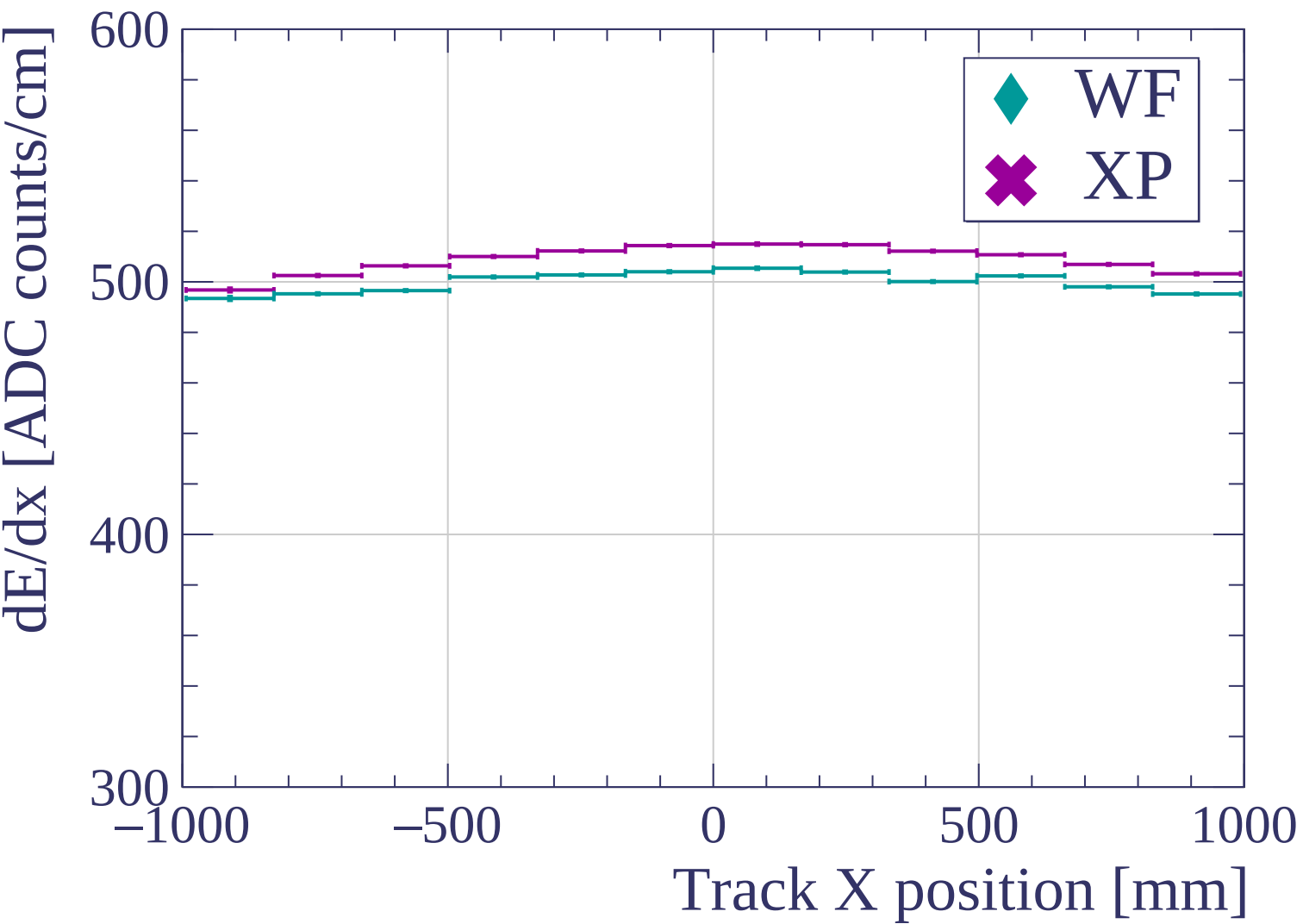


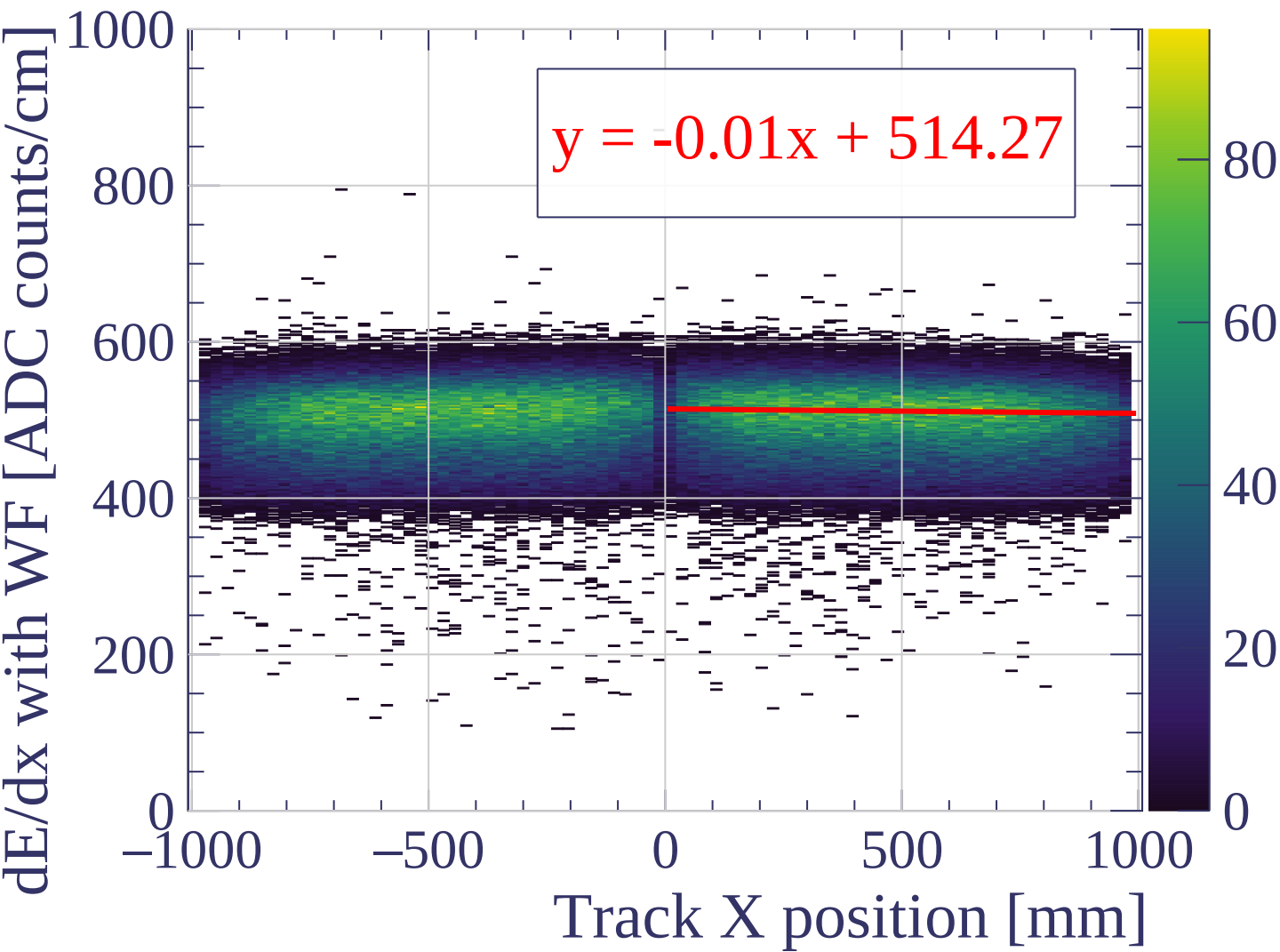
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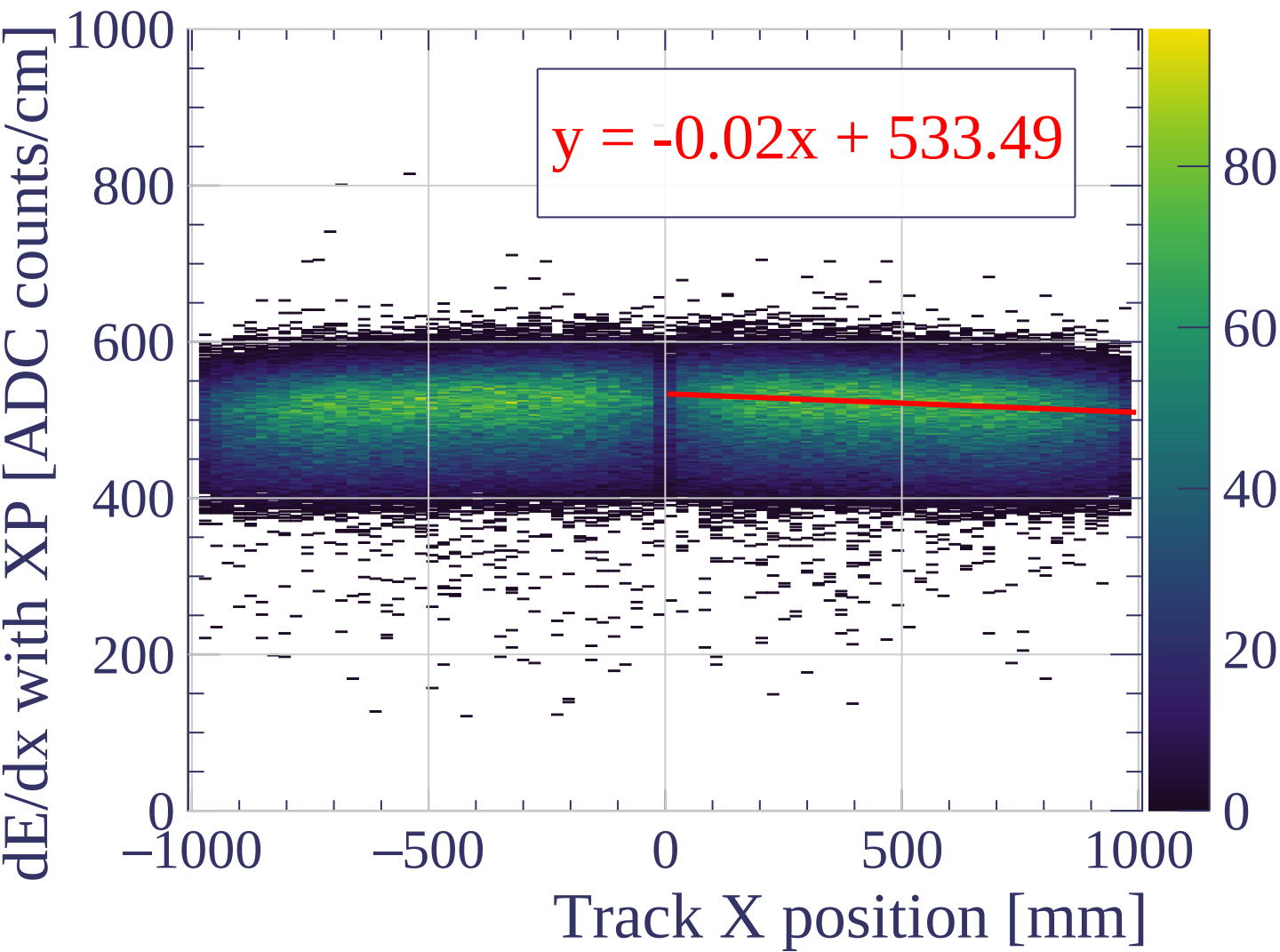


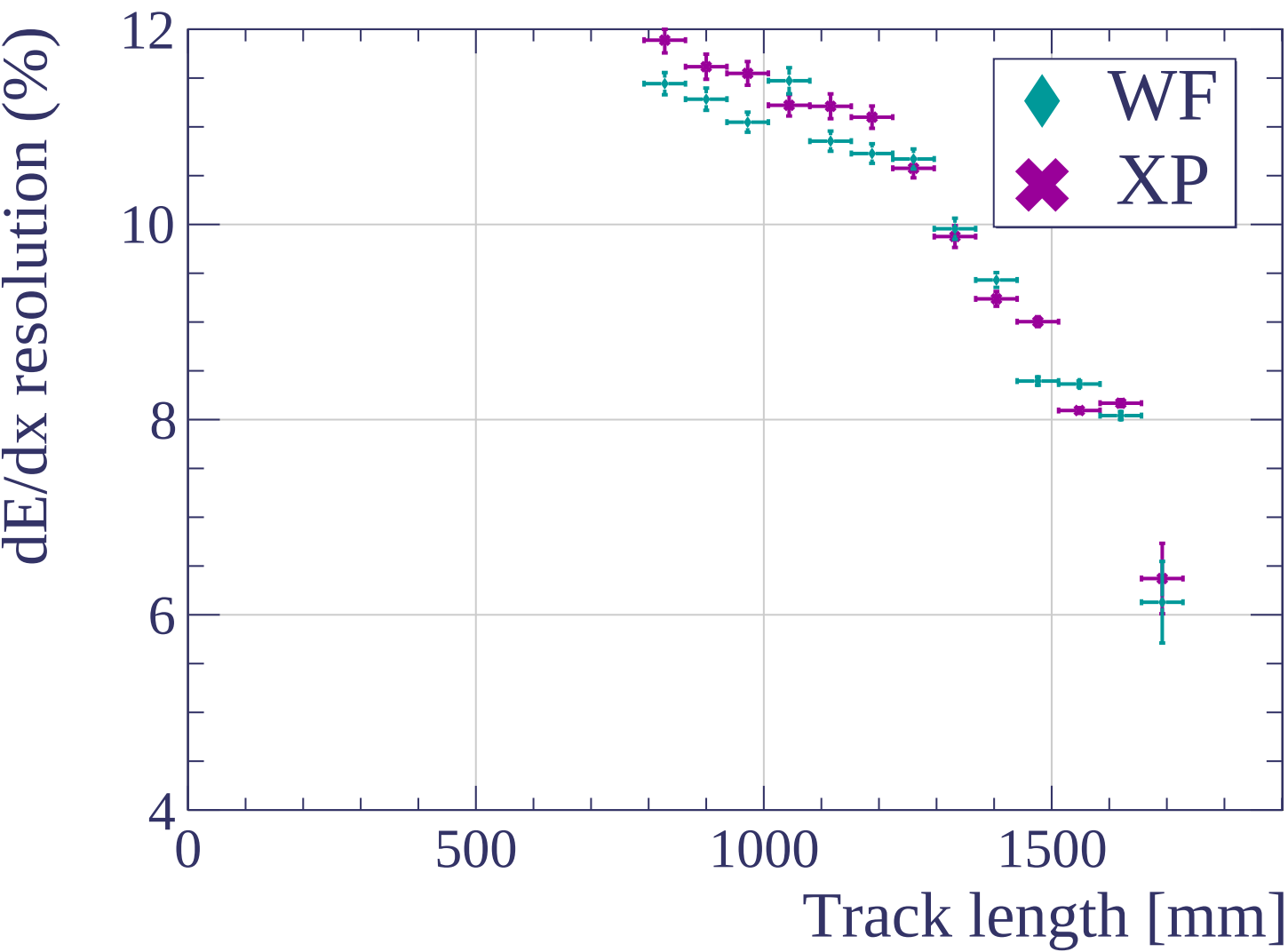


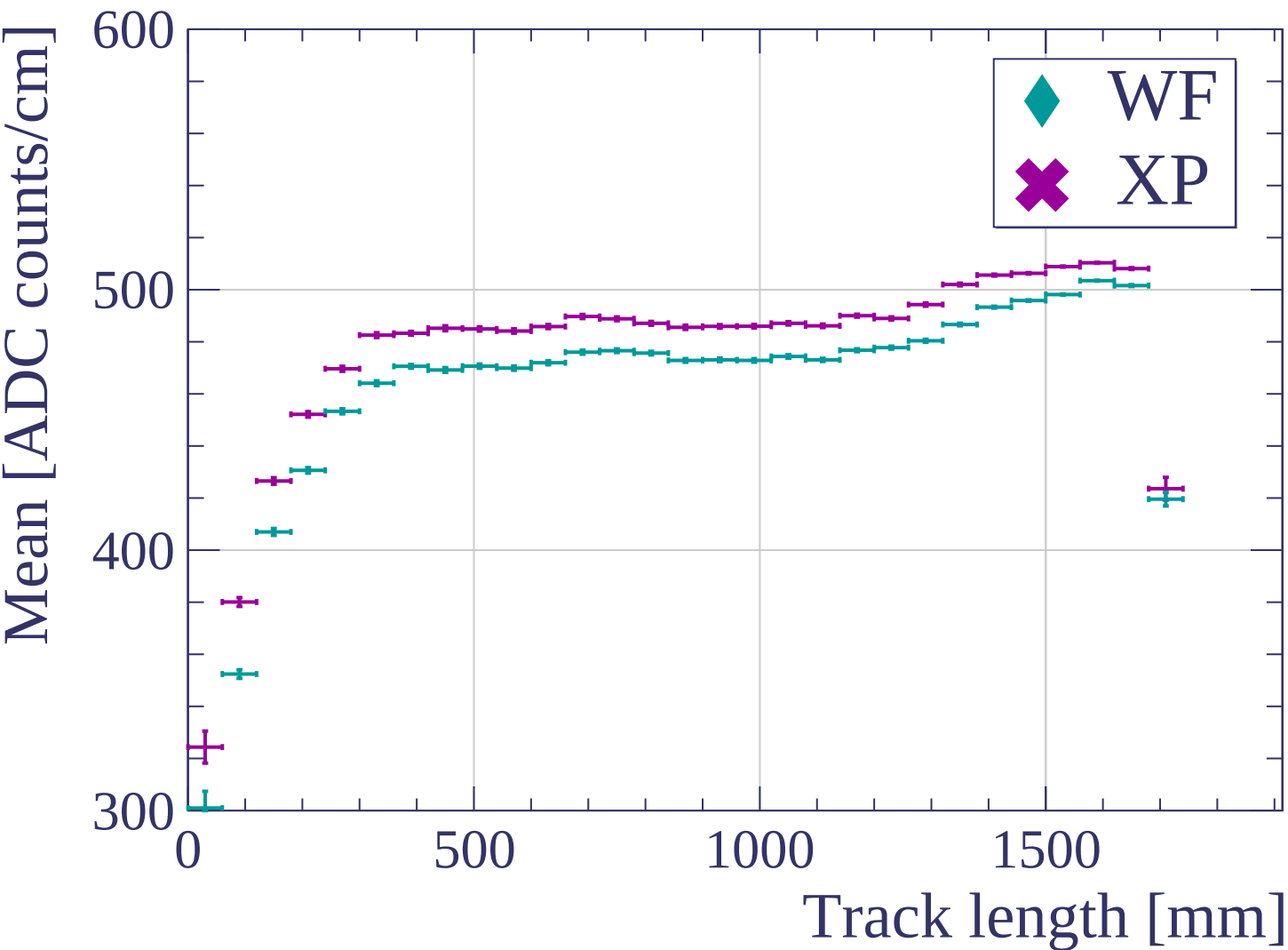


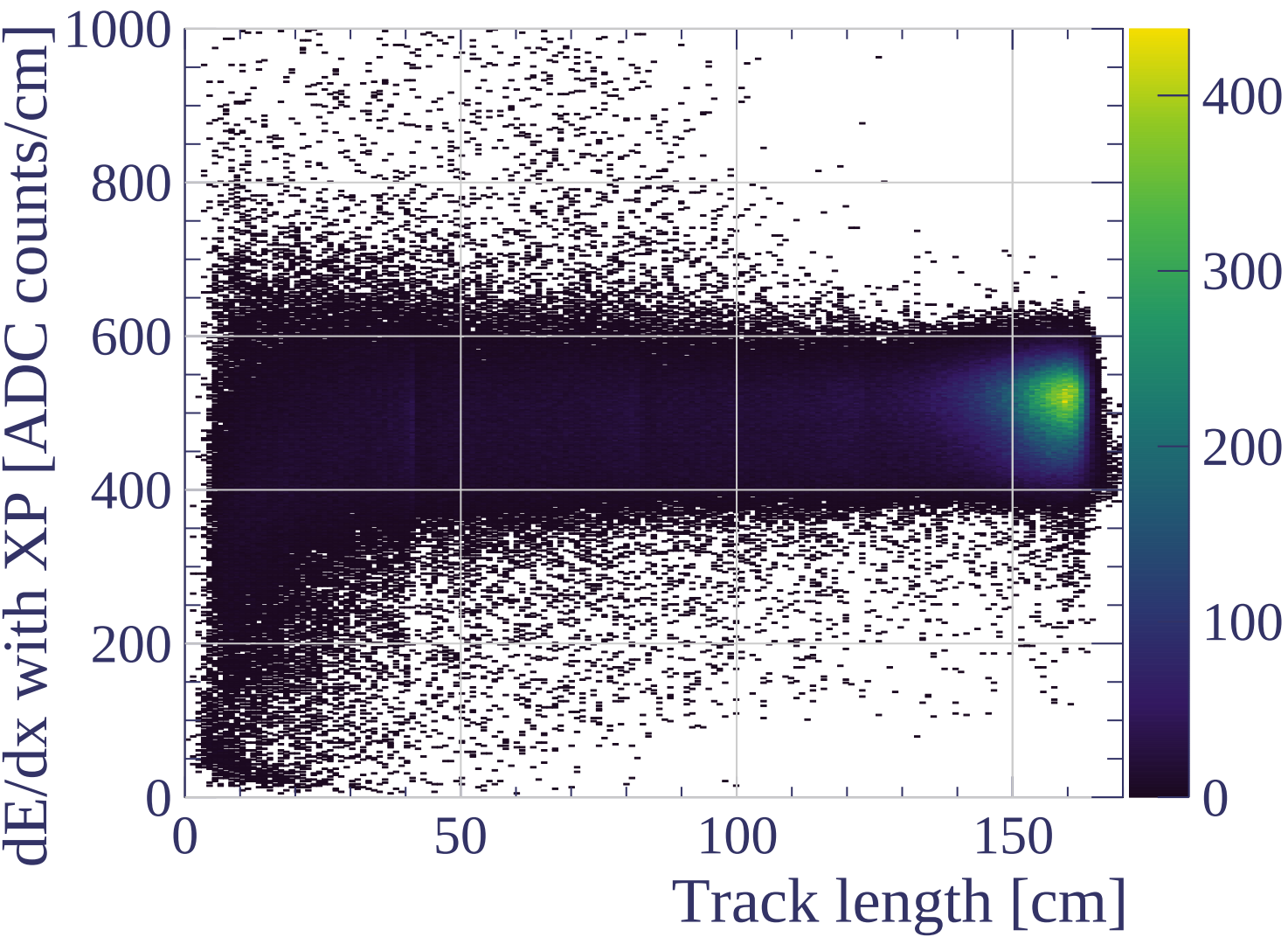


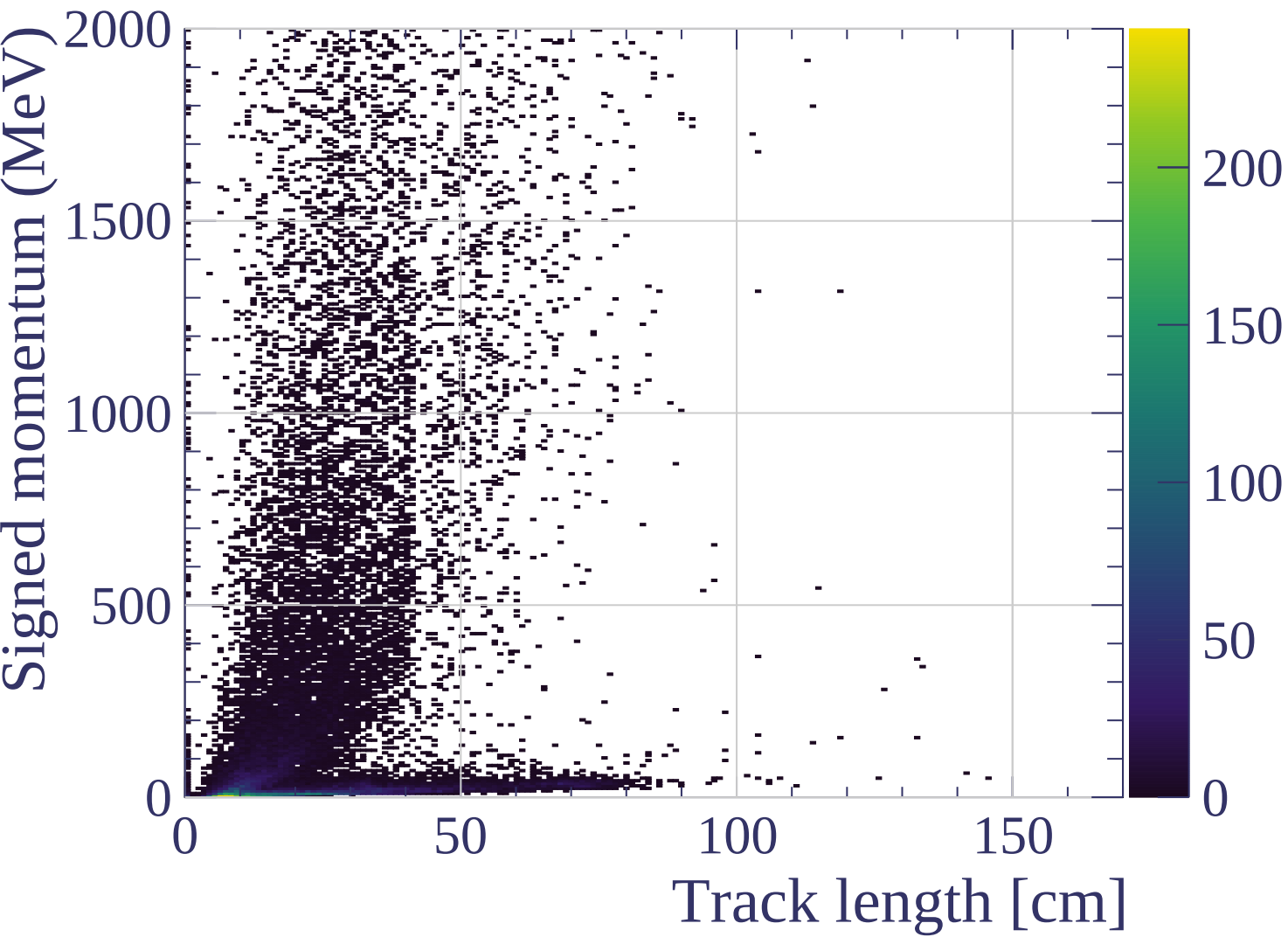


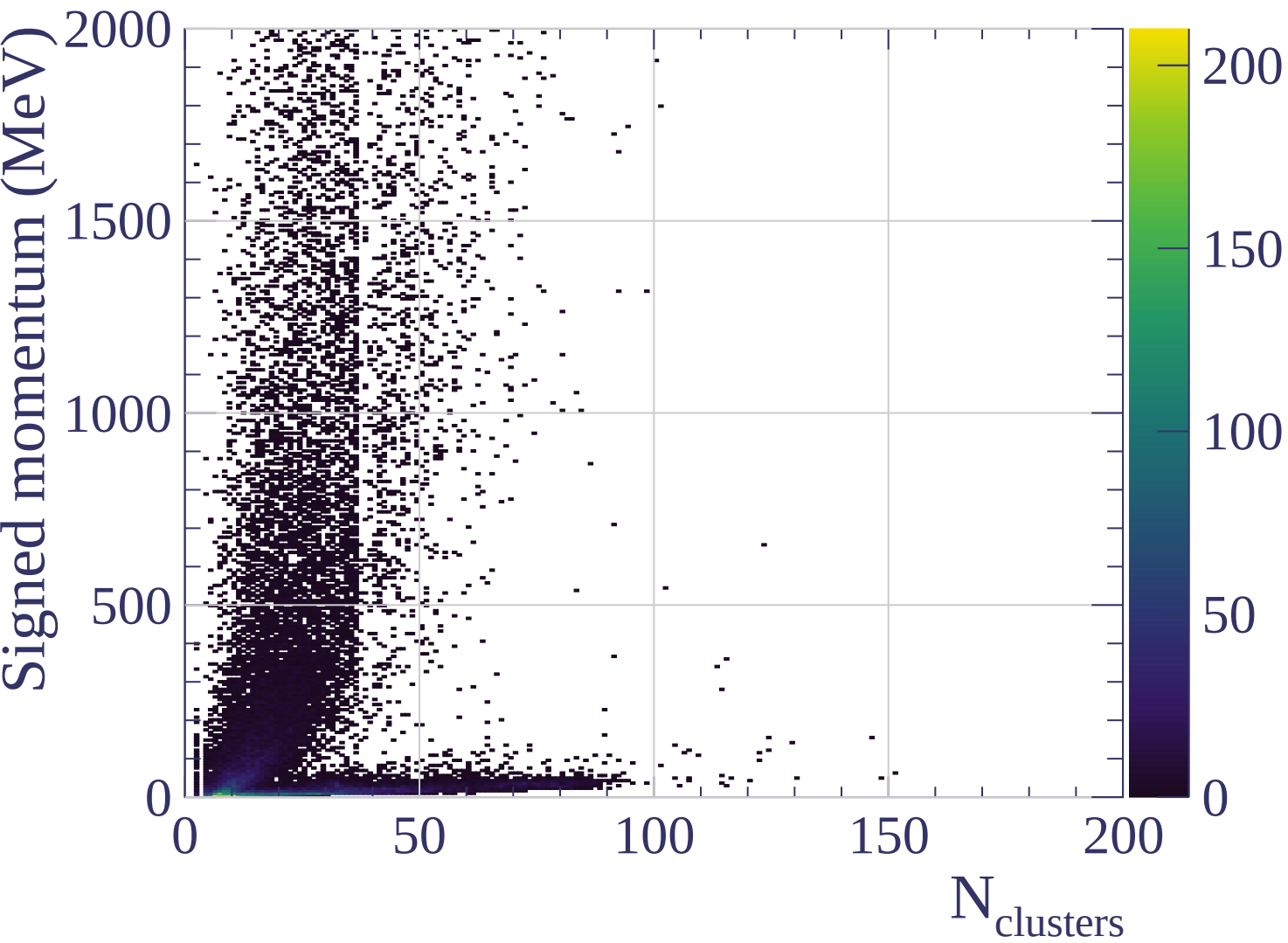


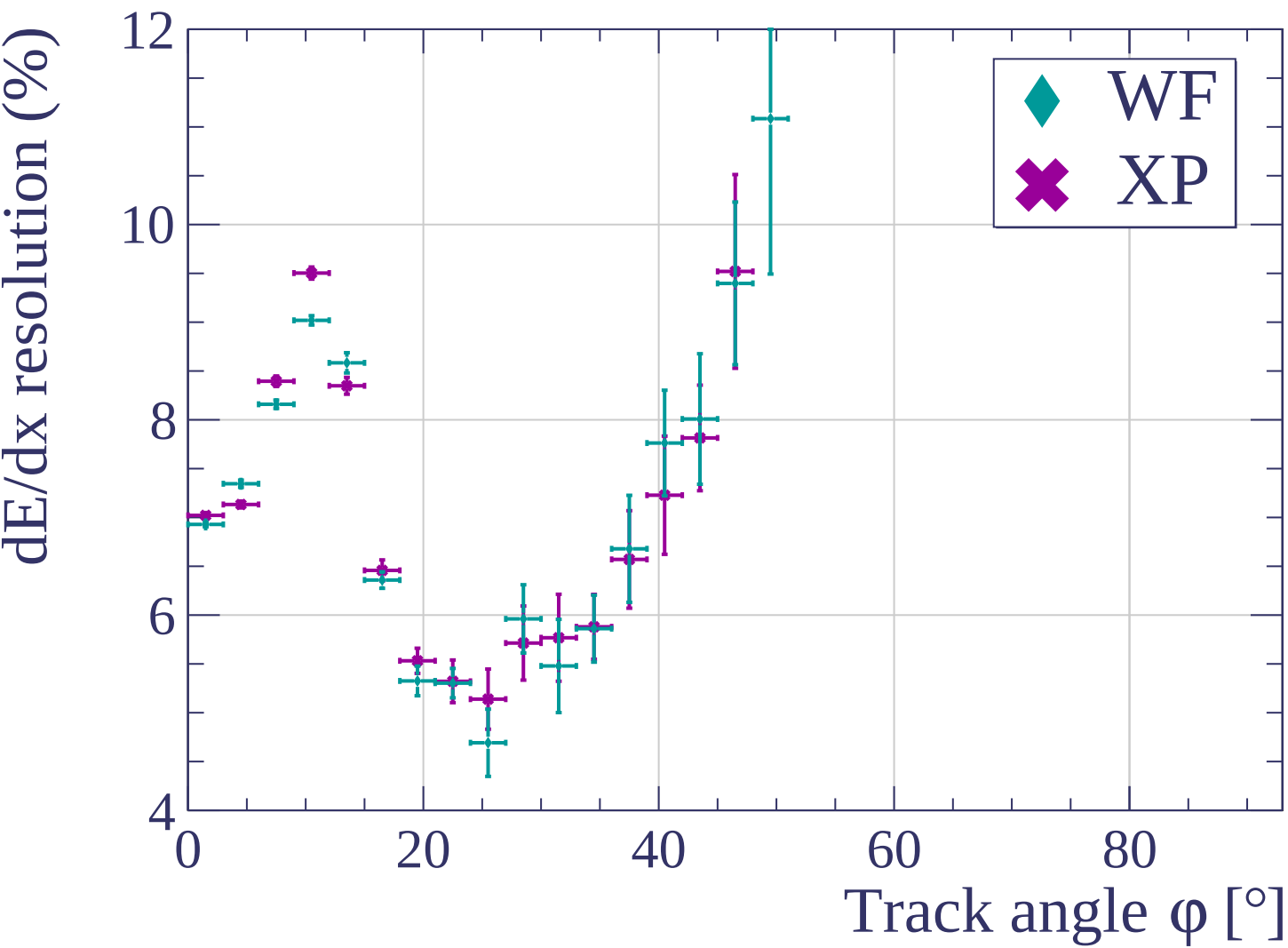


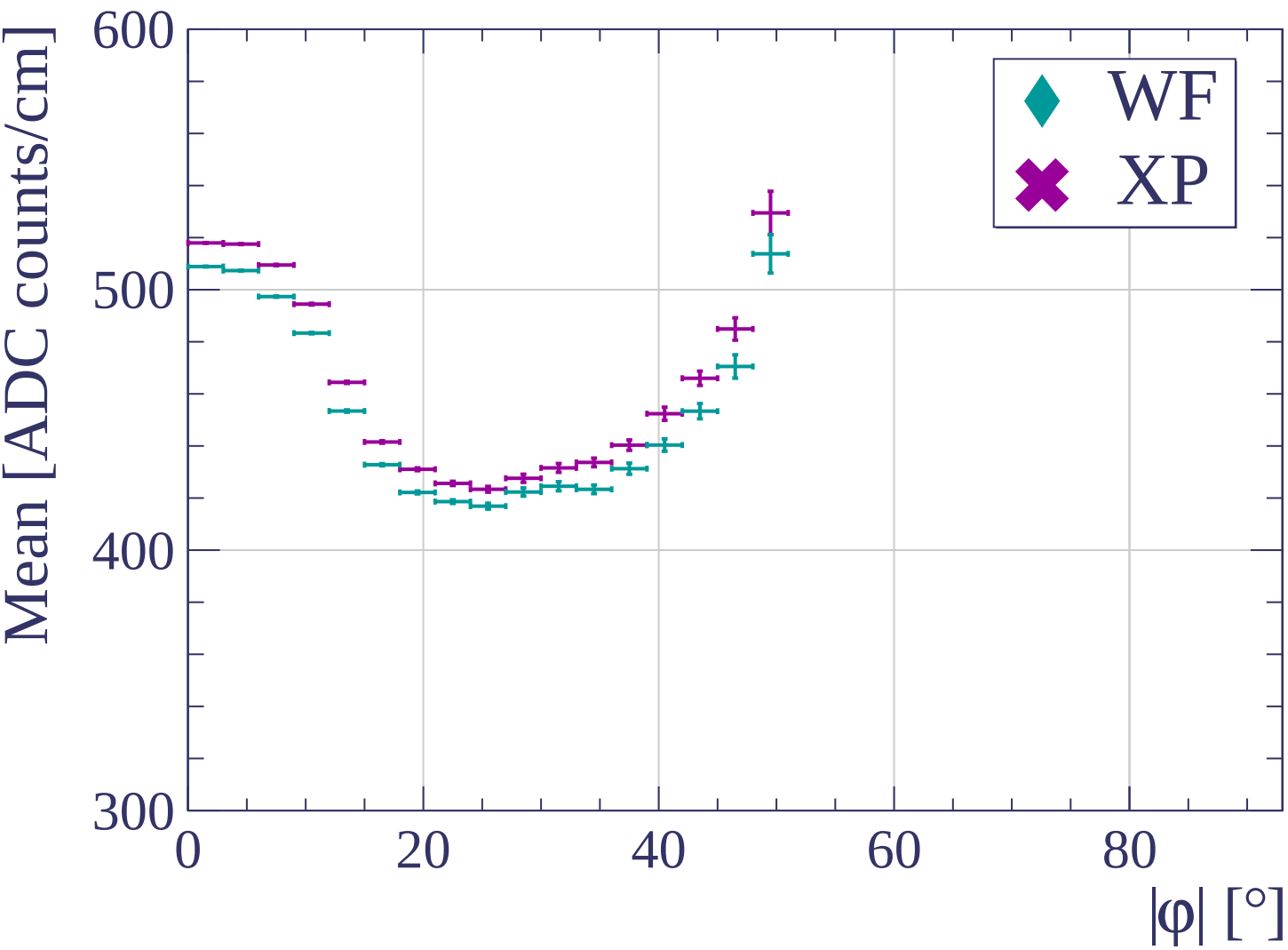


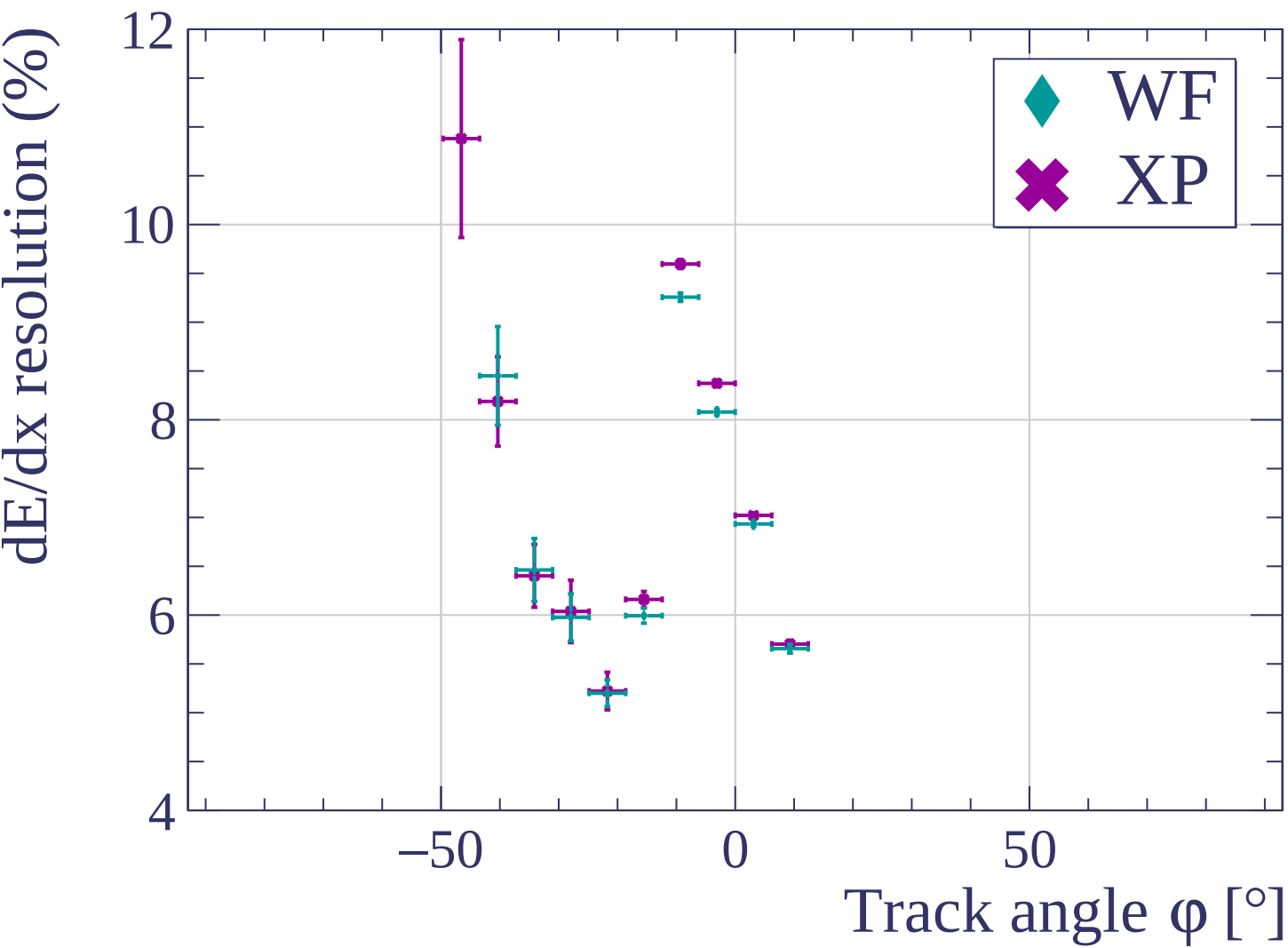


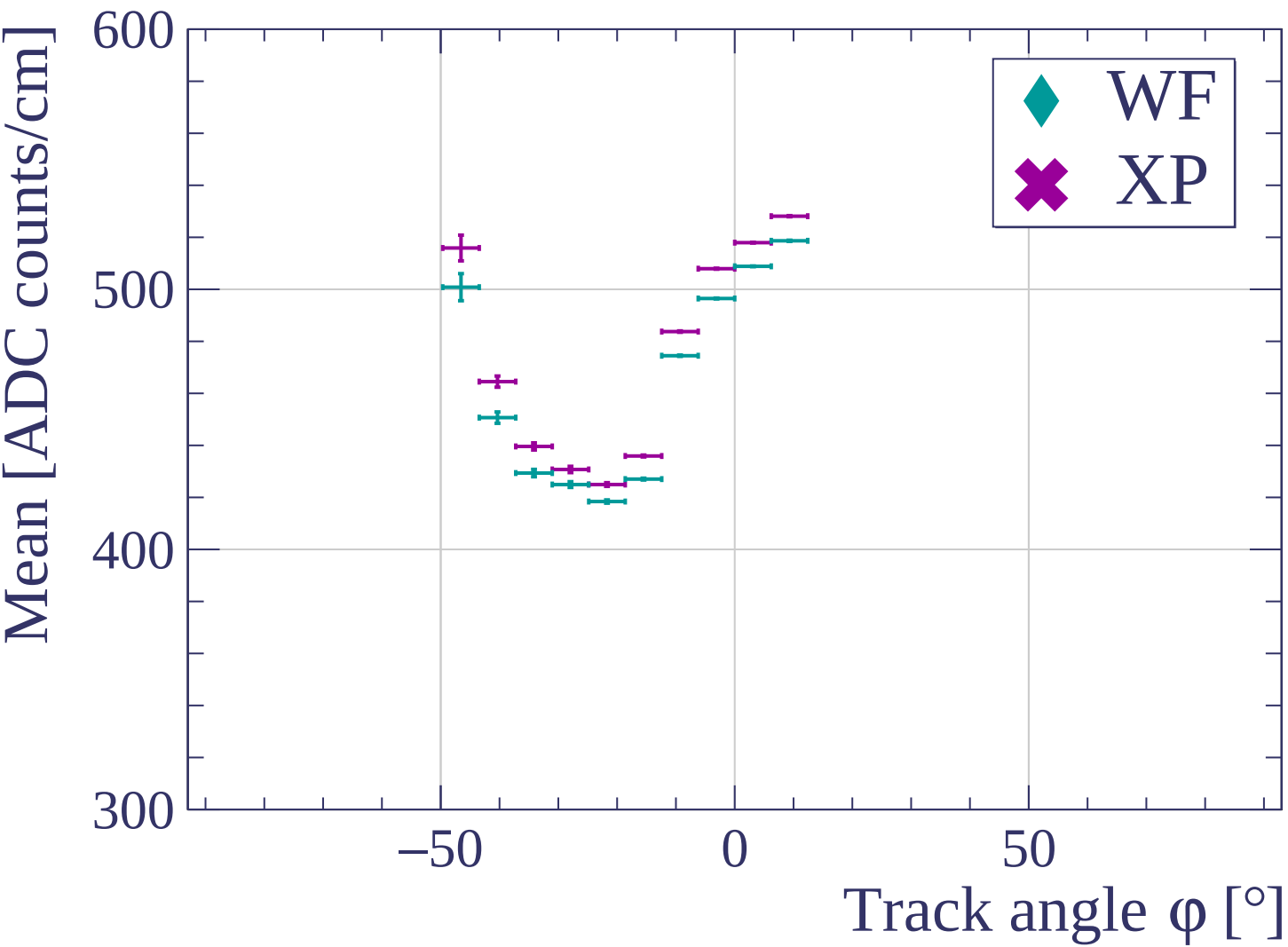


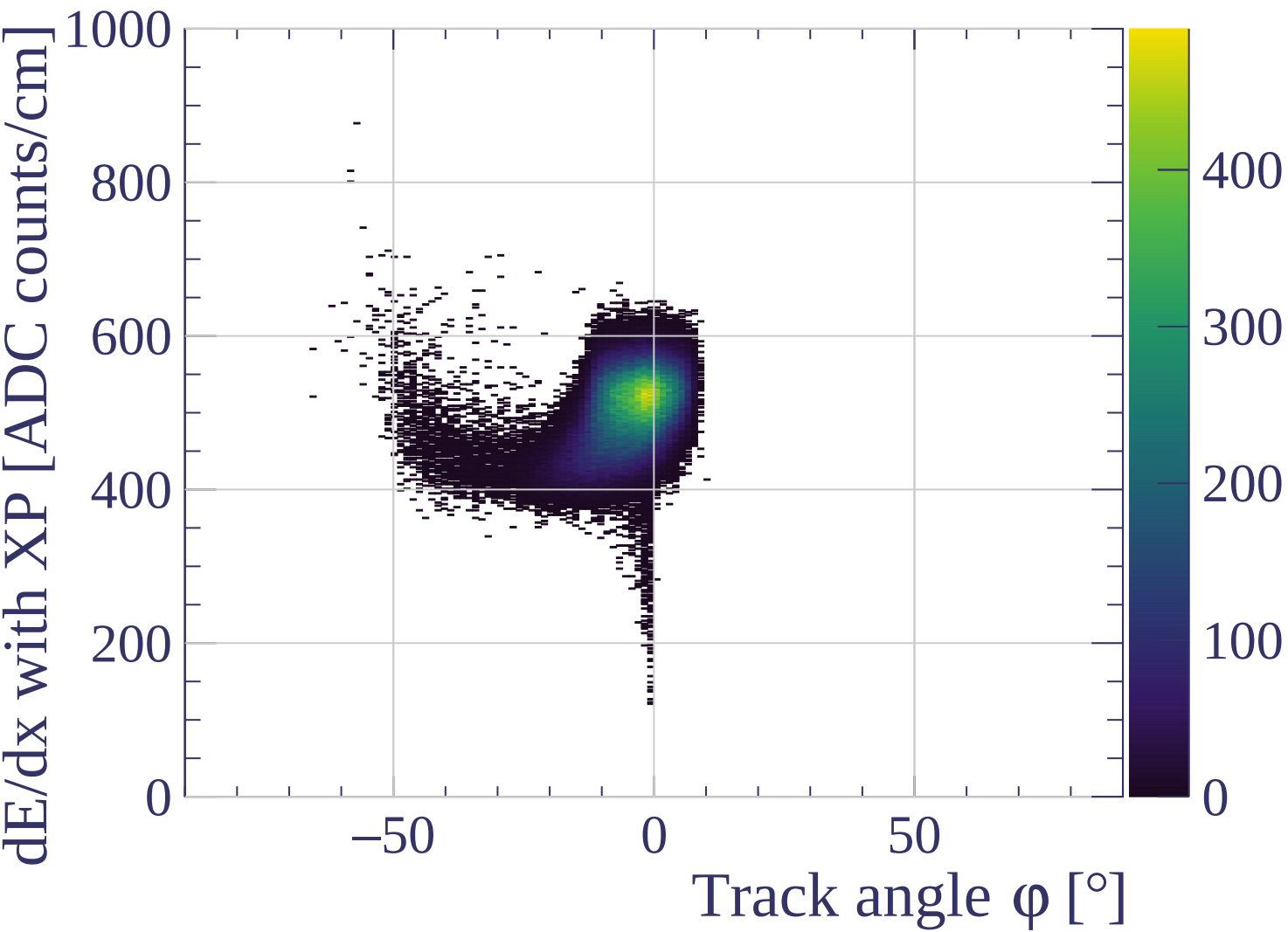




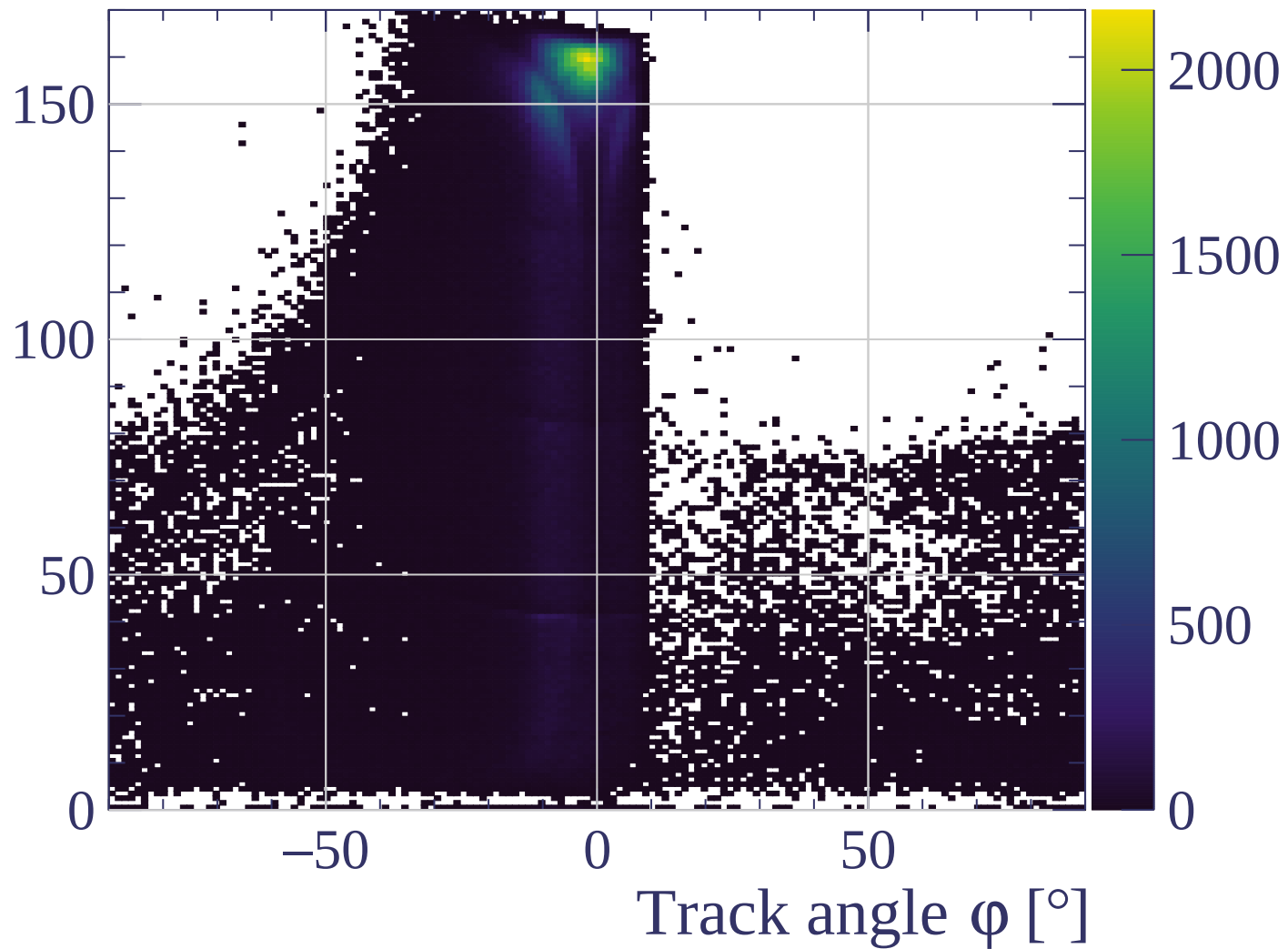


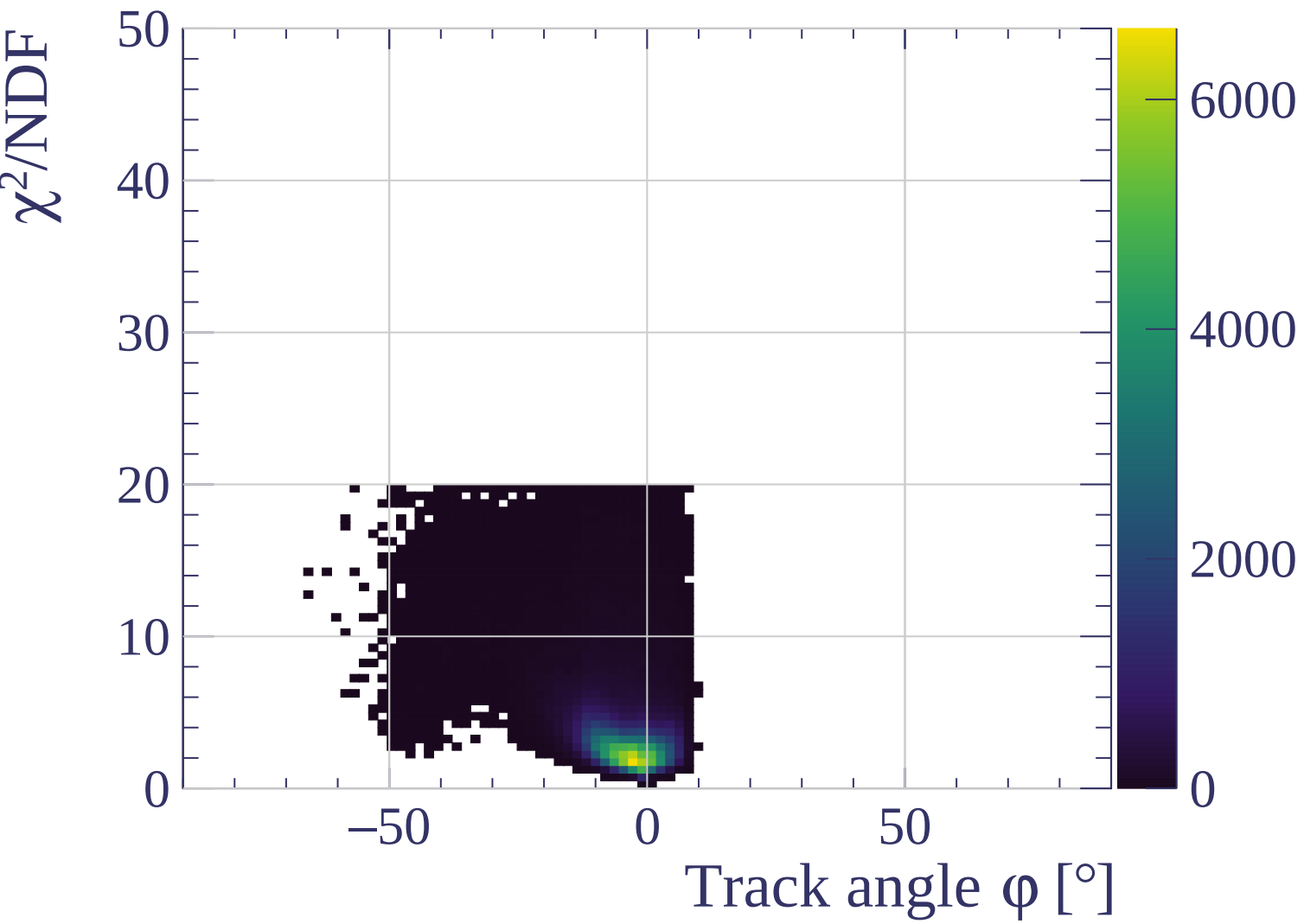


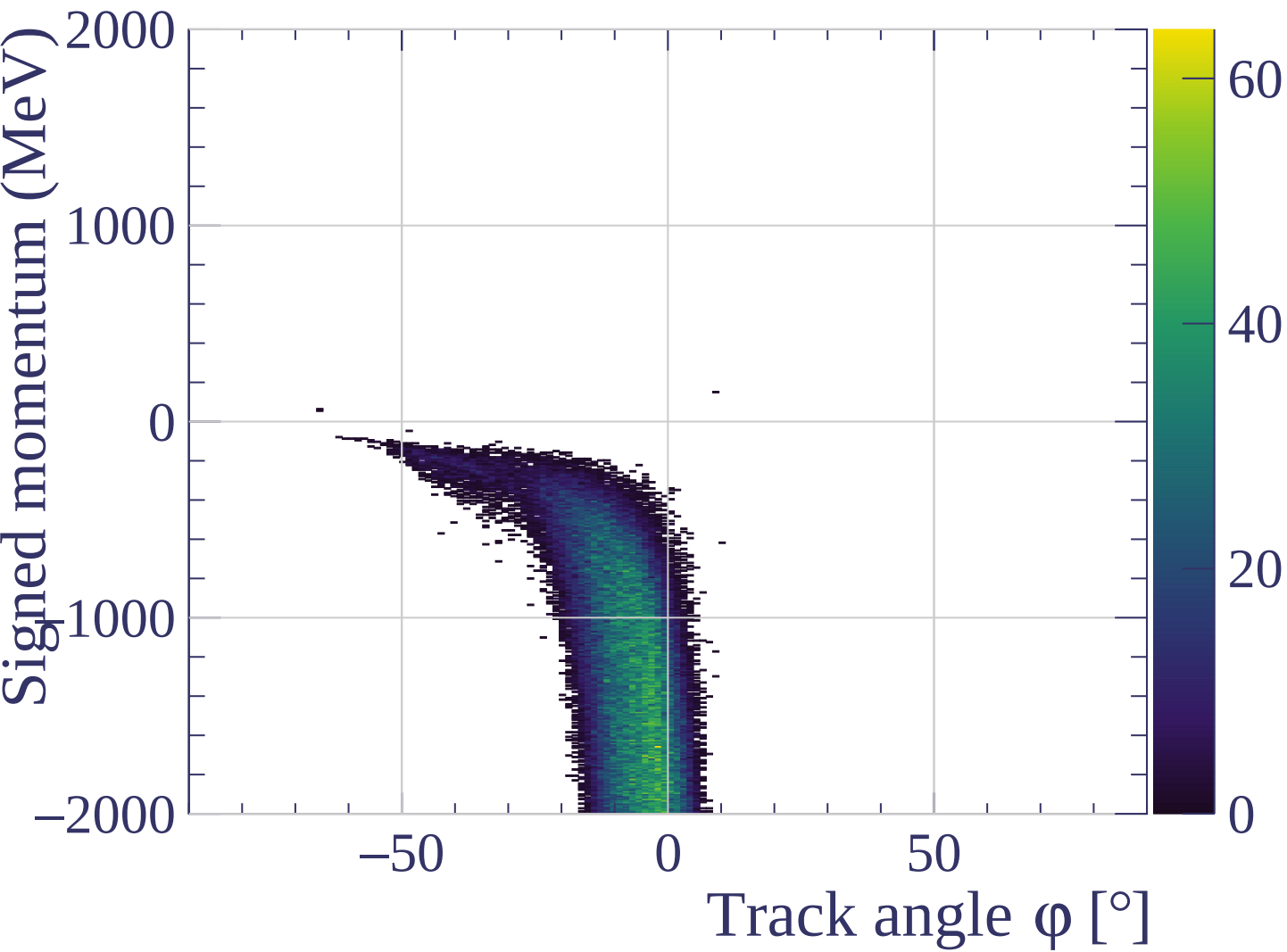


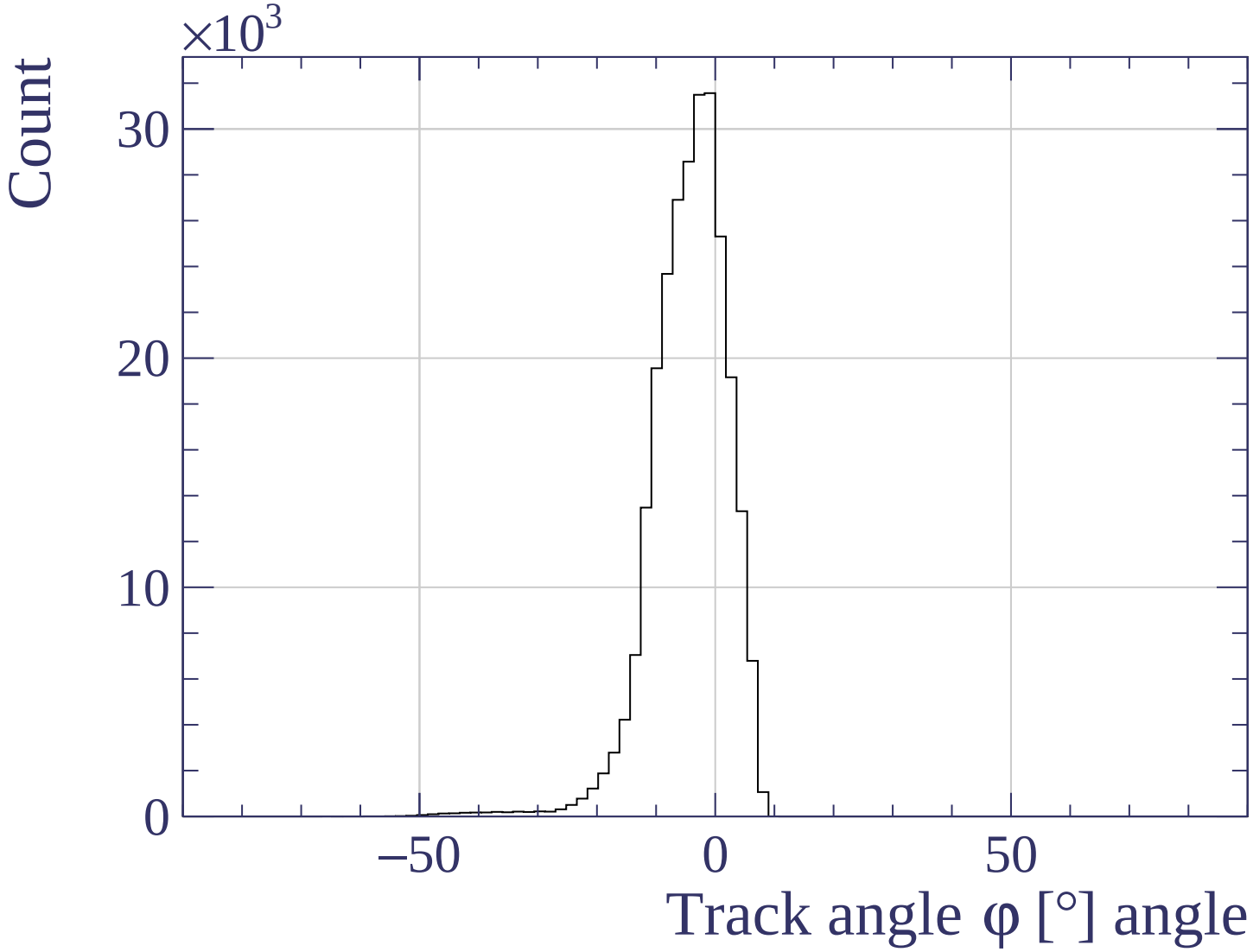


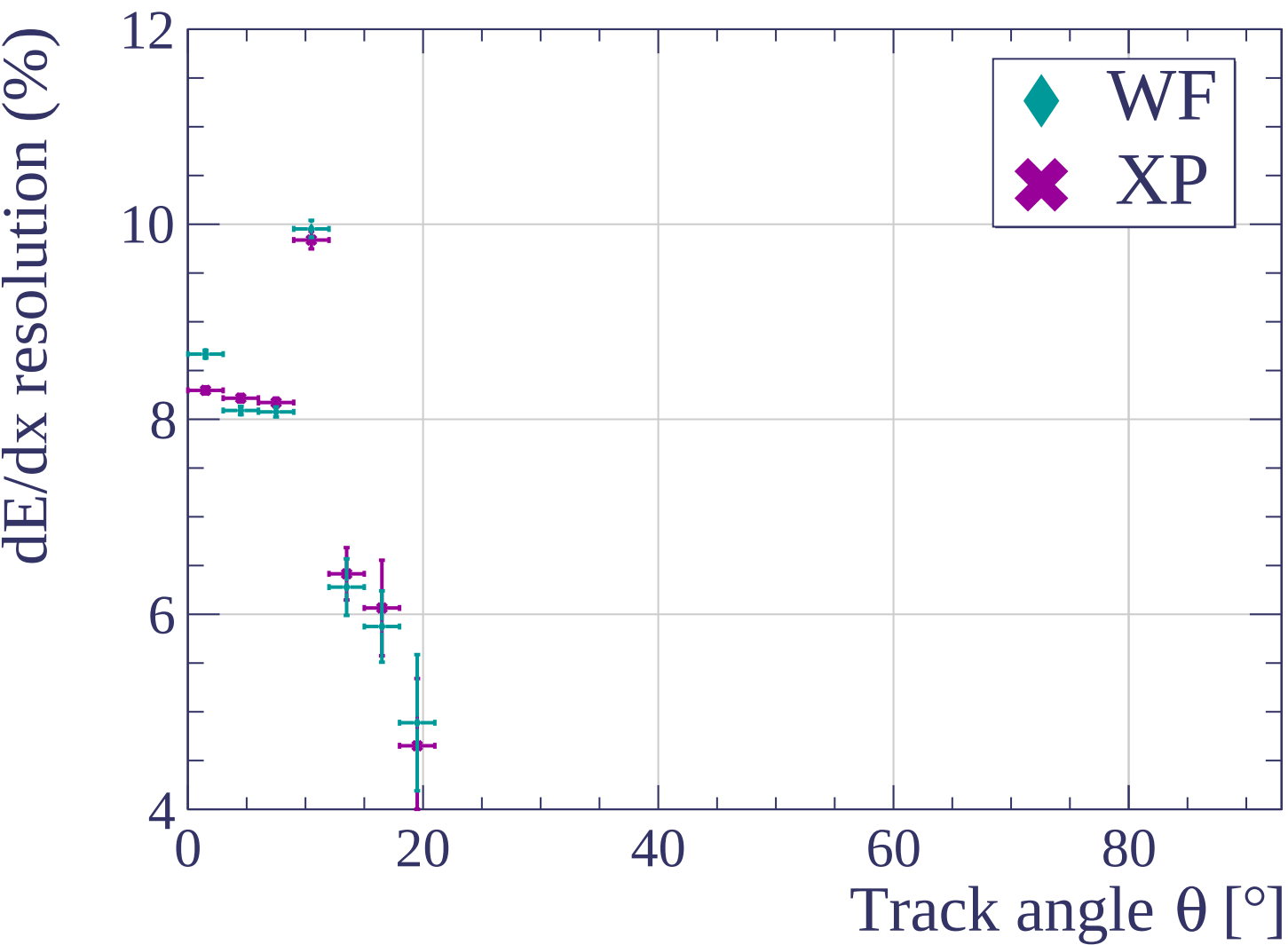
Track length [cm]

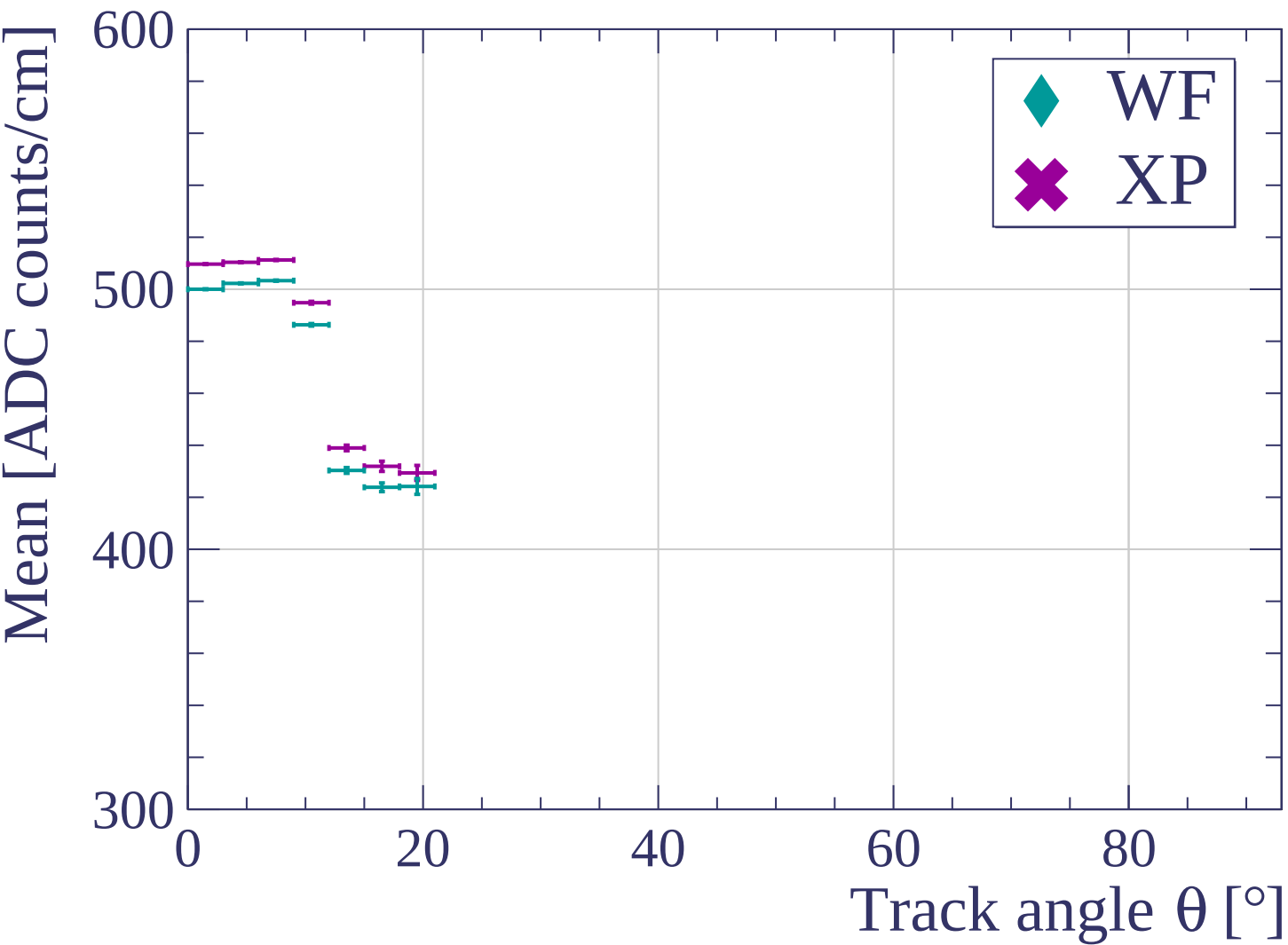


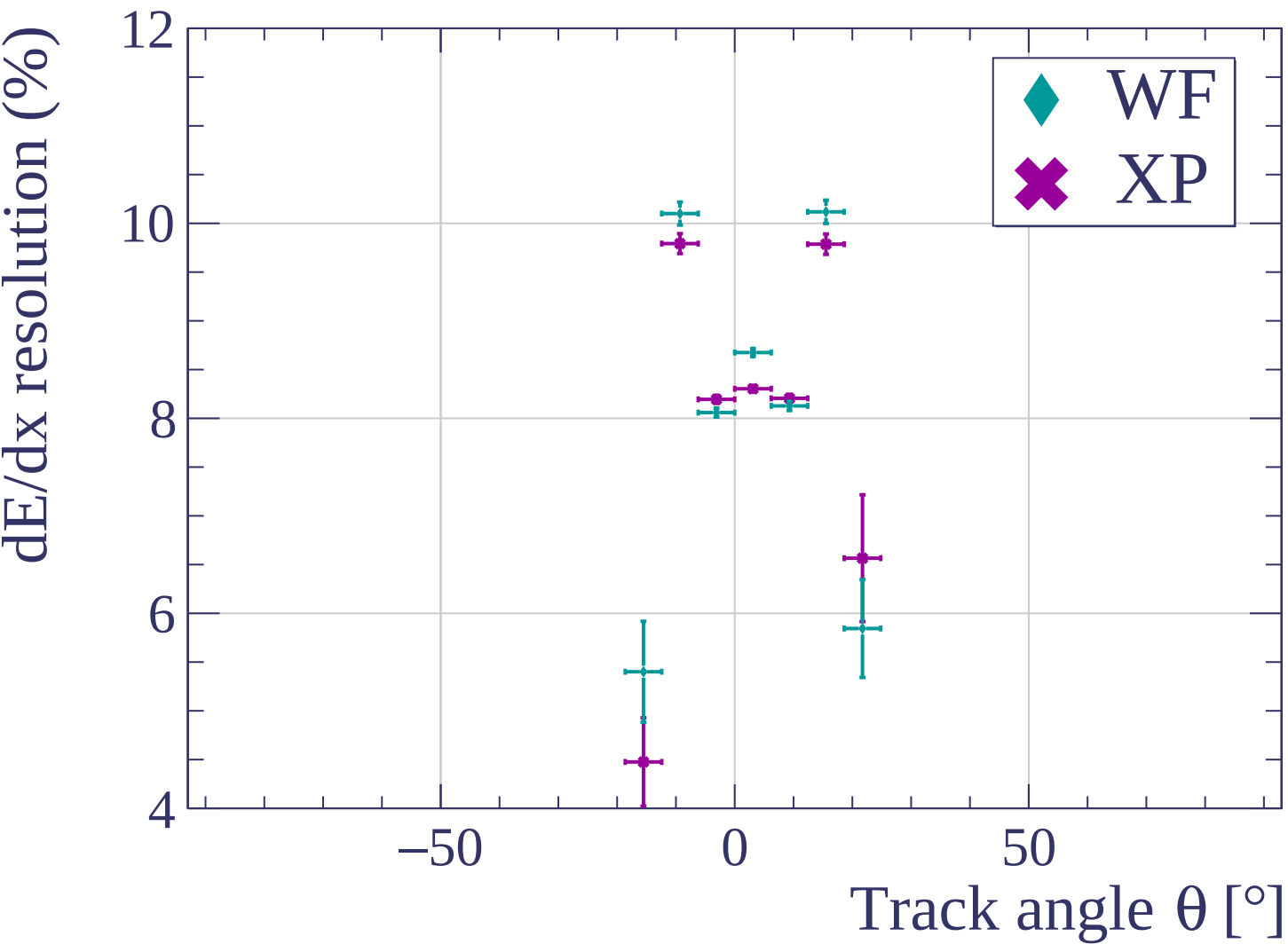


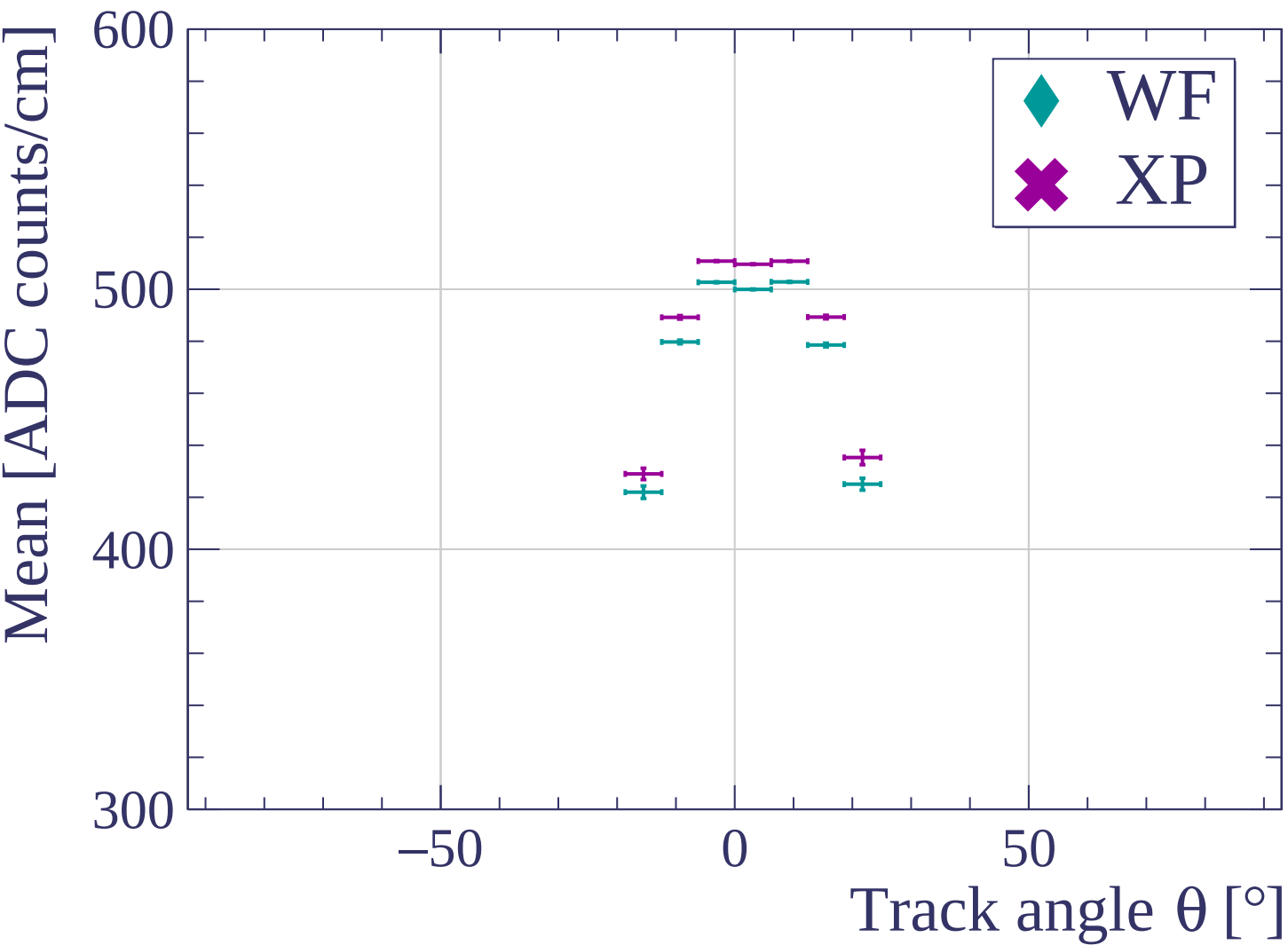


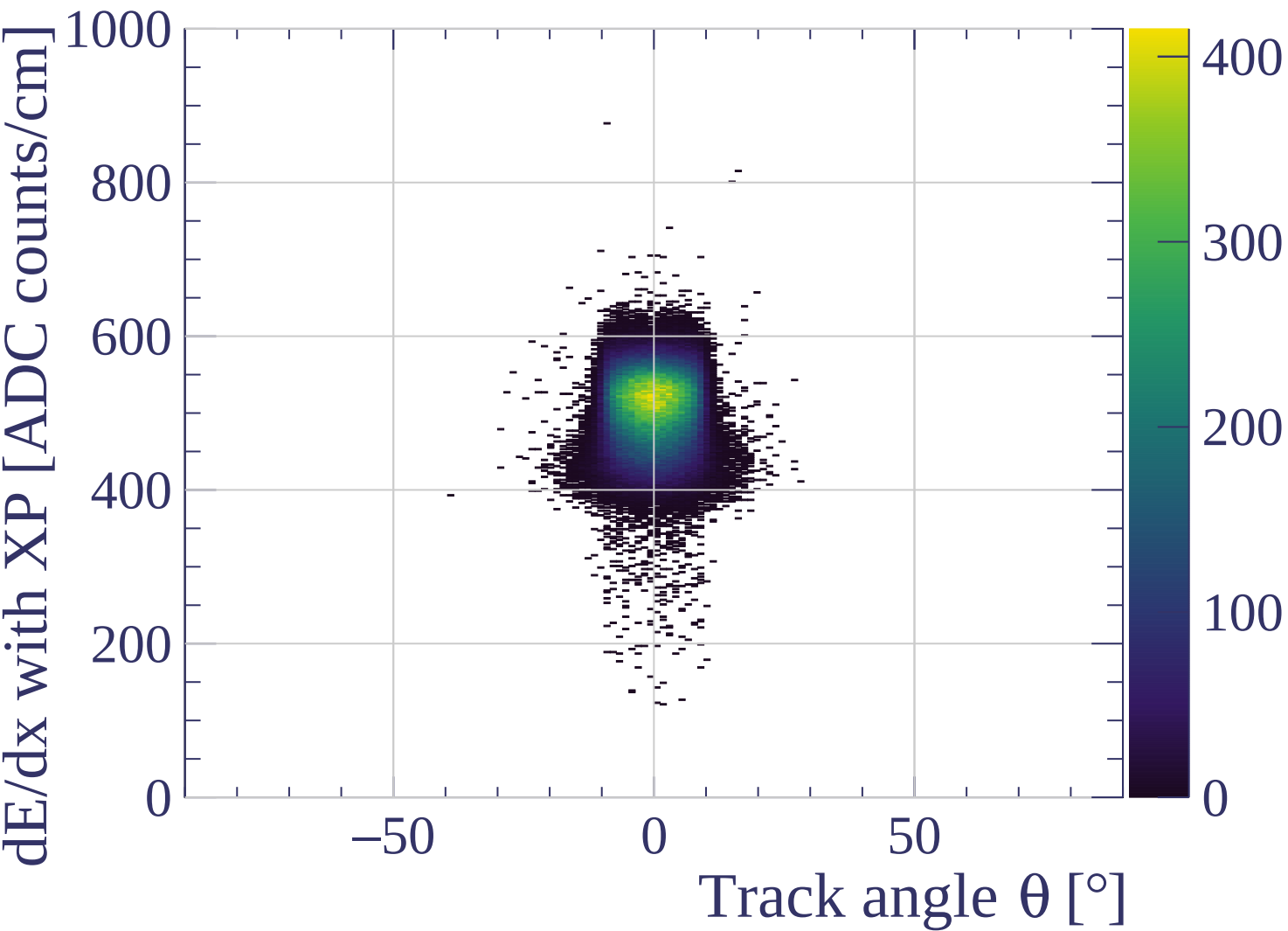




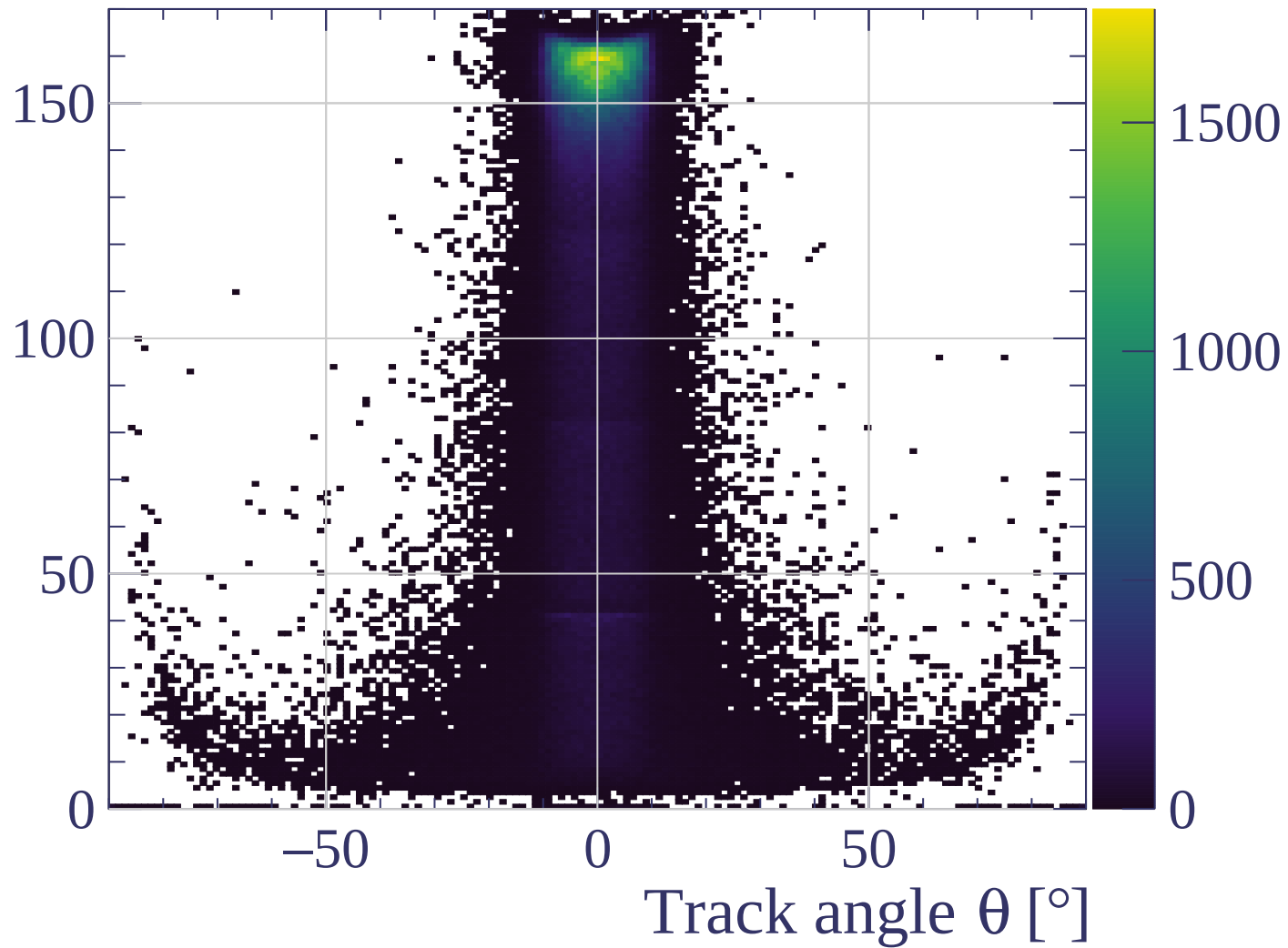


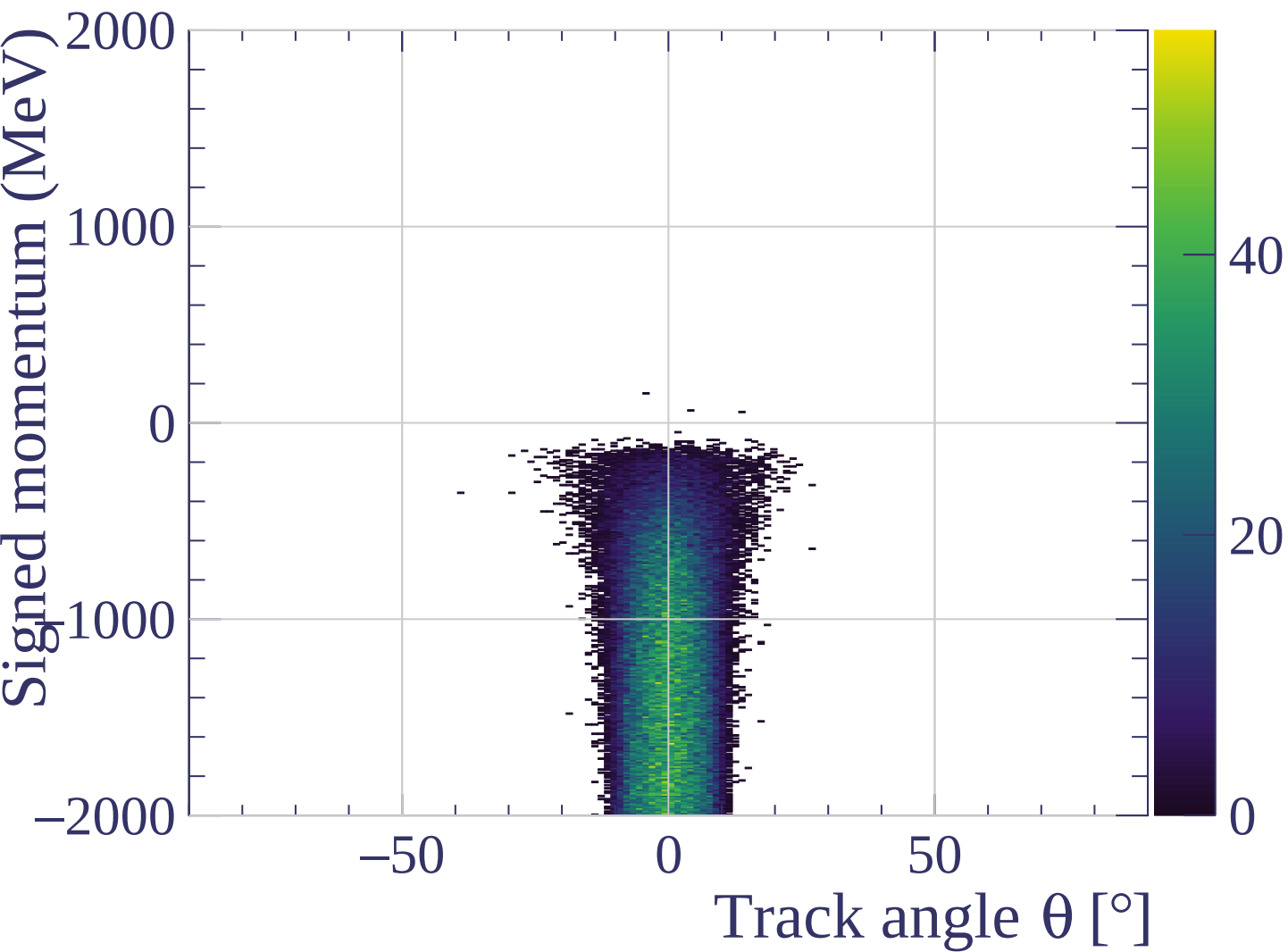


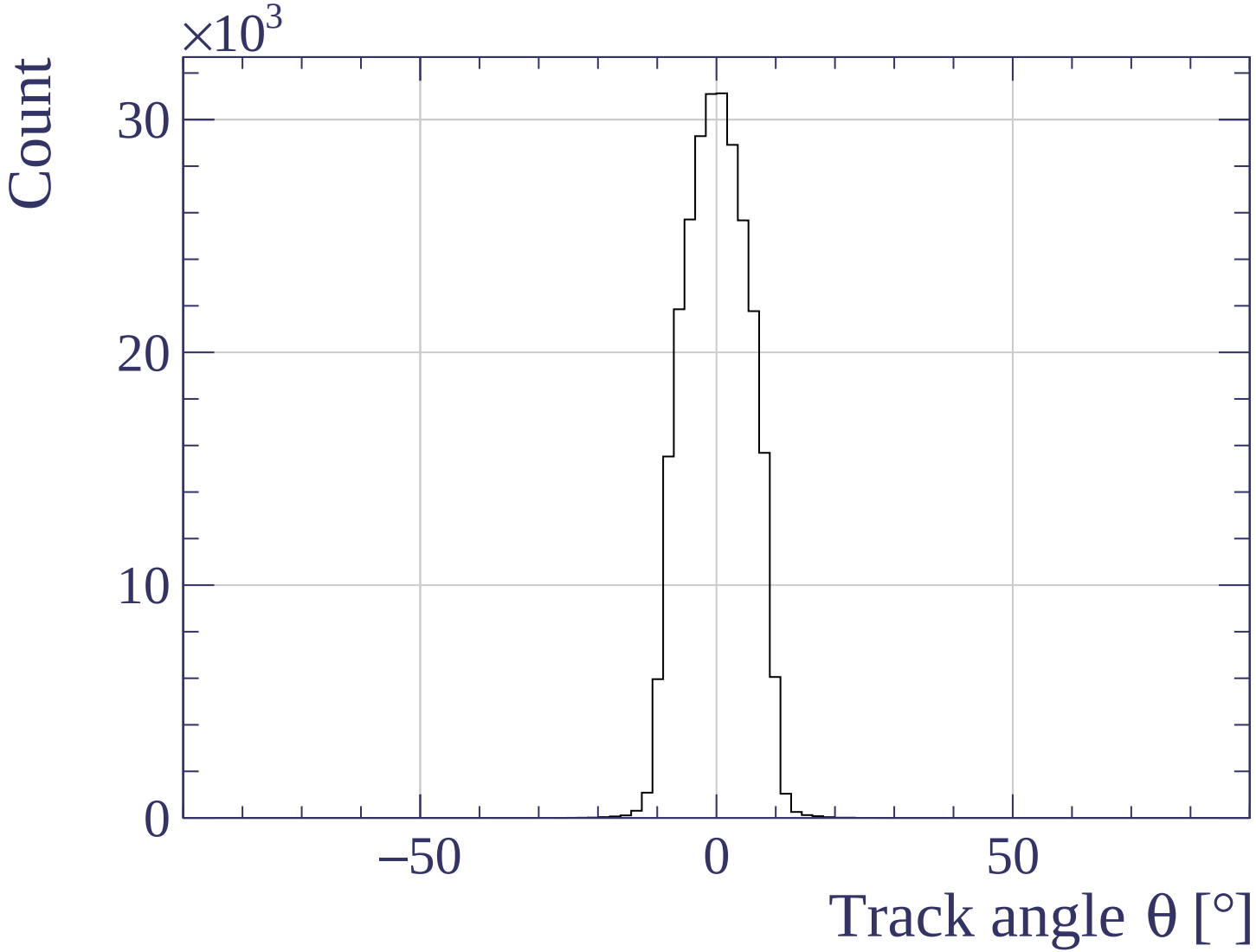




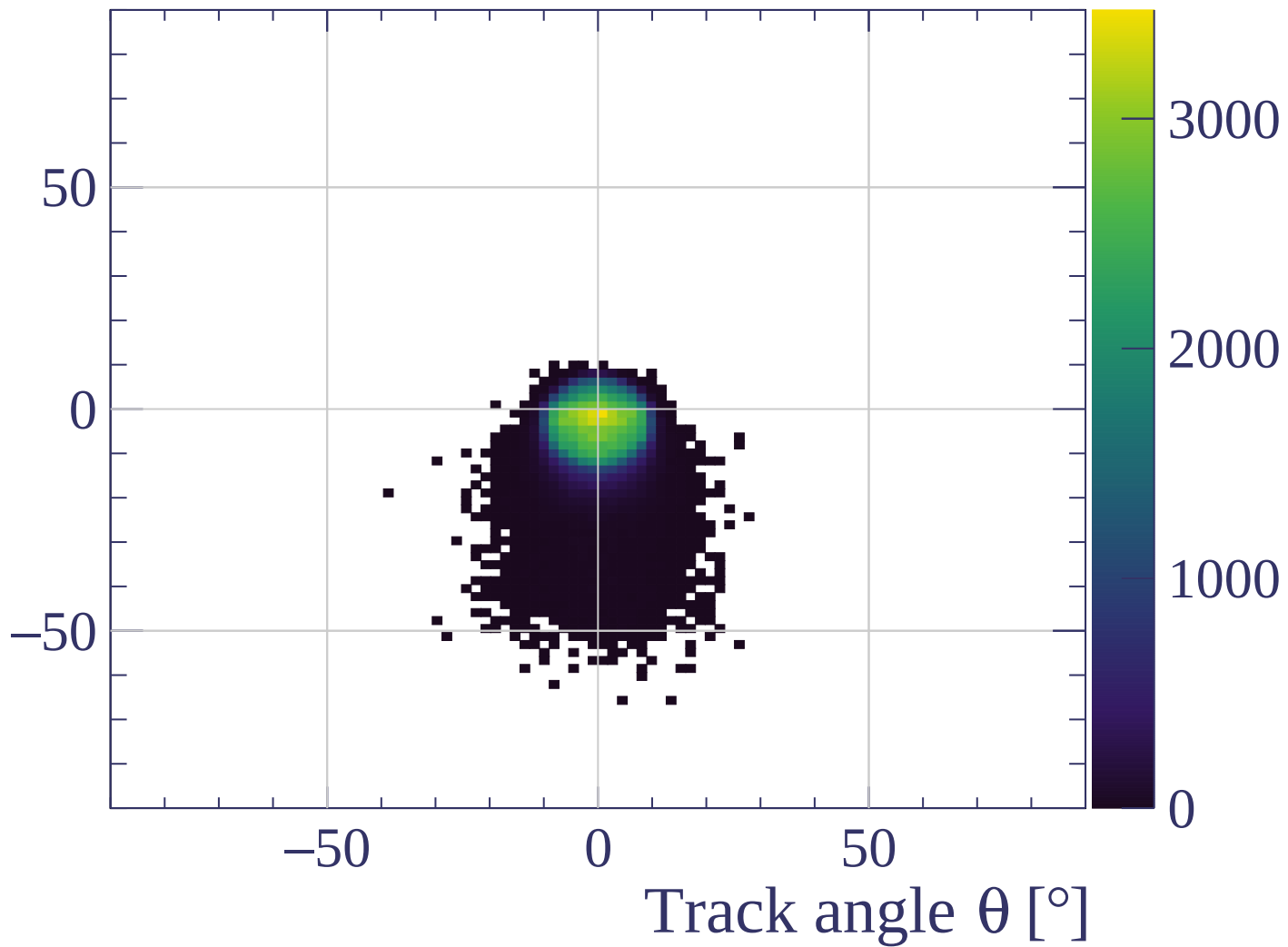
Track length [cm]



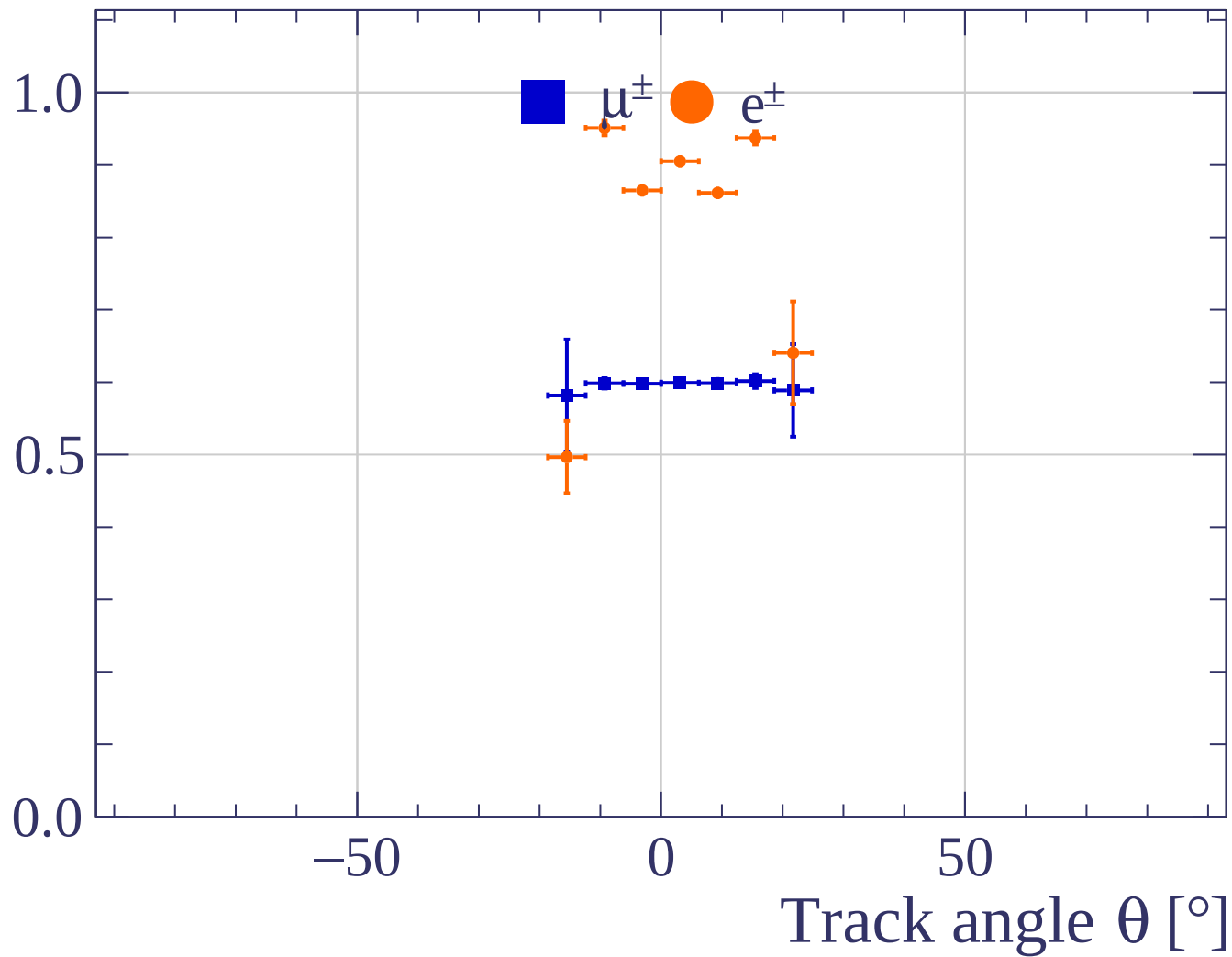




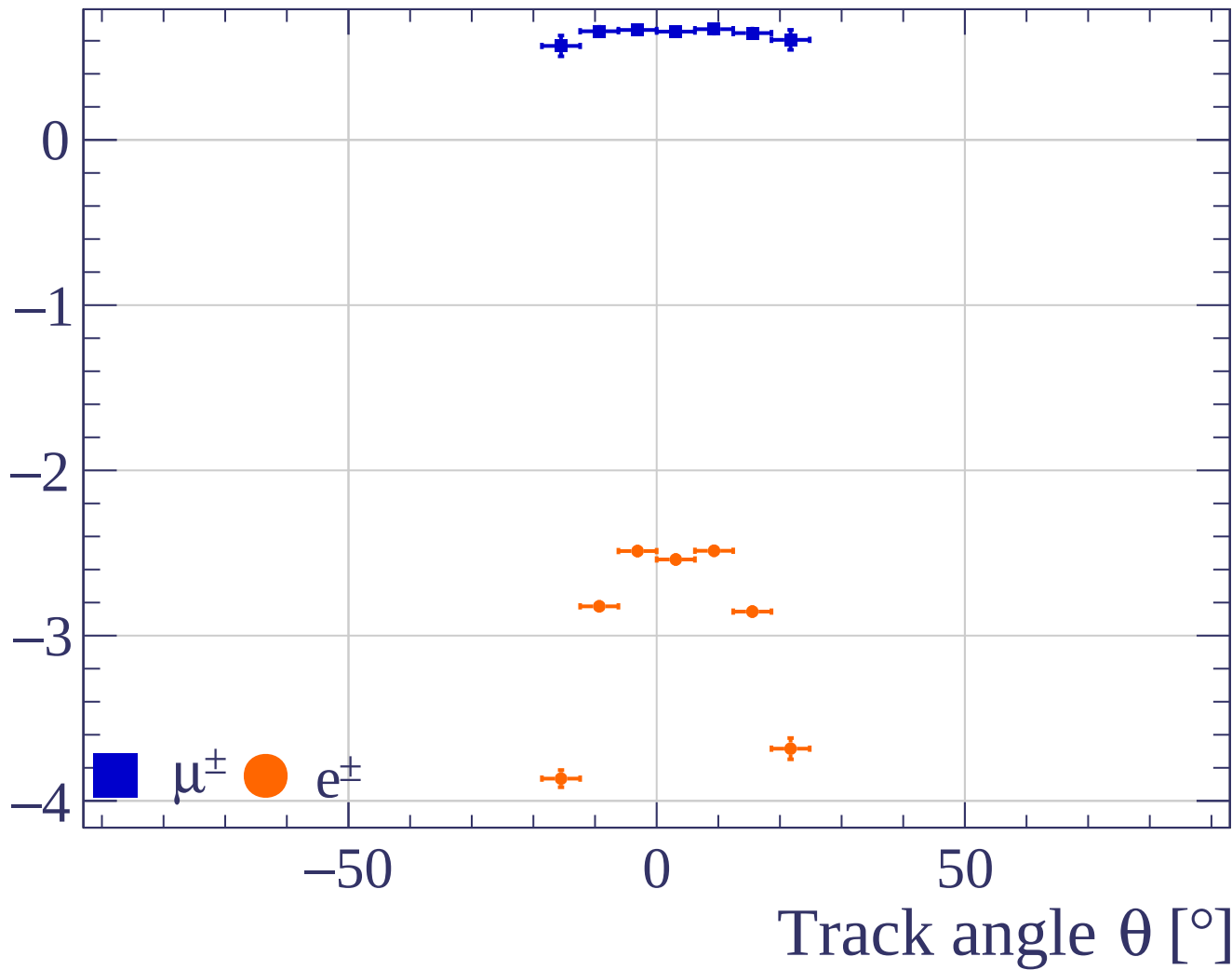
Track angle ϕ [°]



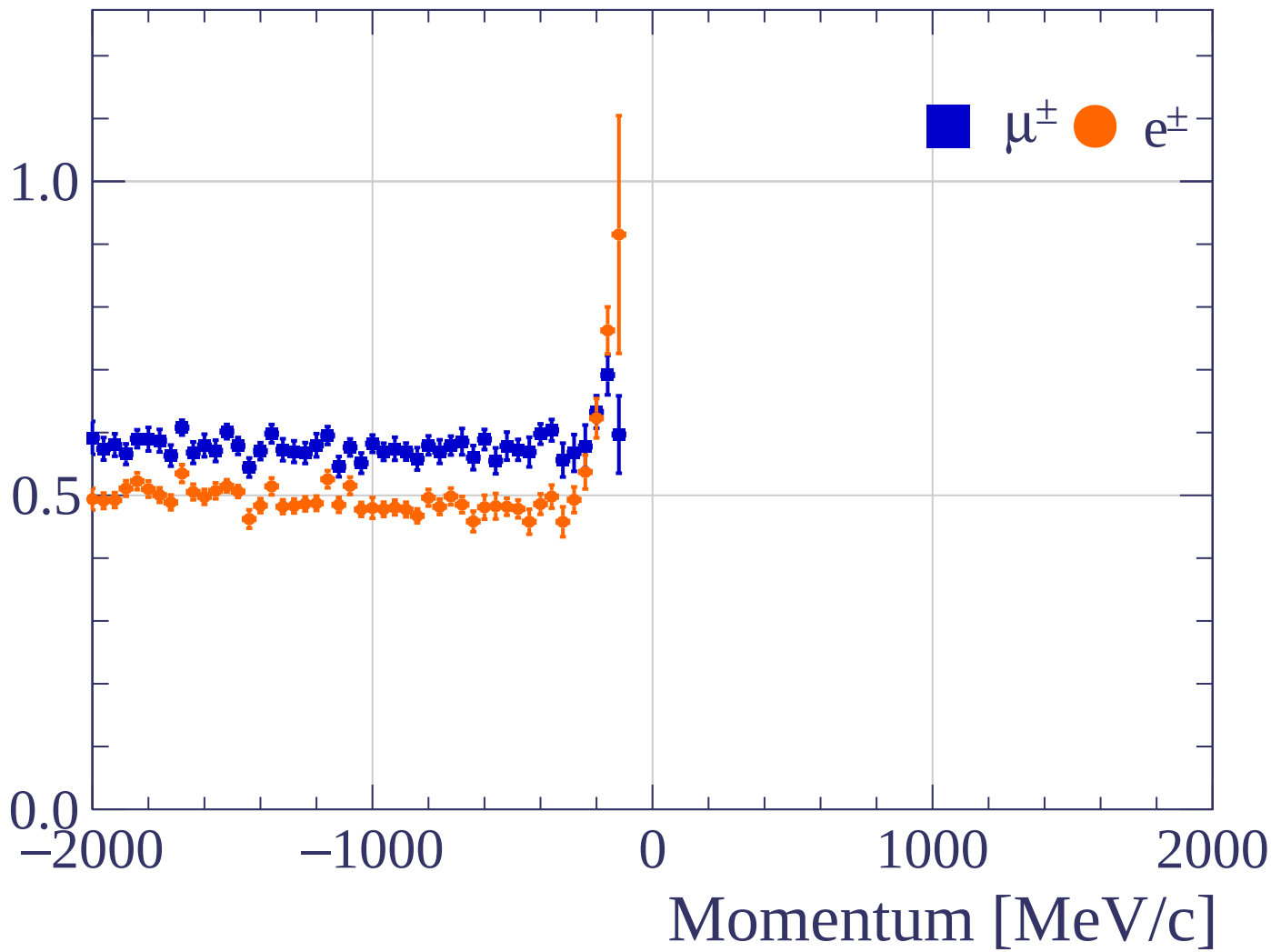
Pulls standard deviation



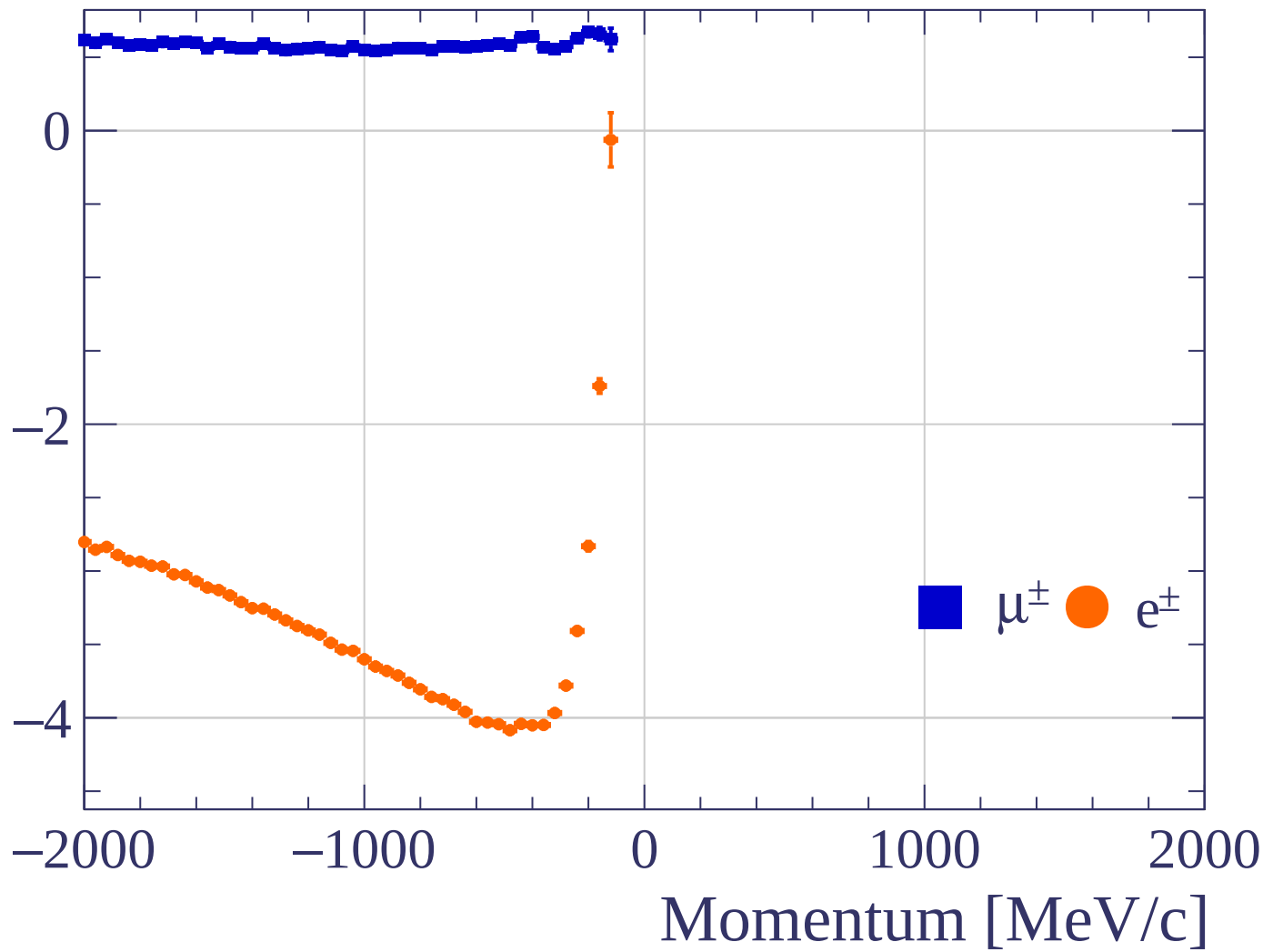
Pulls mean

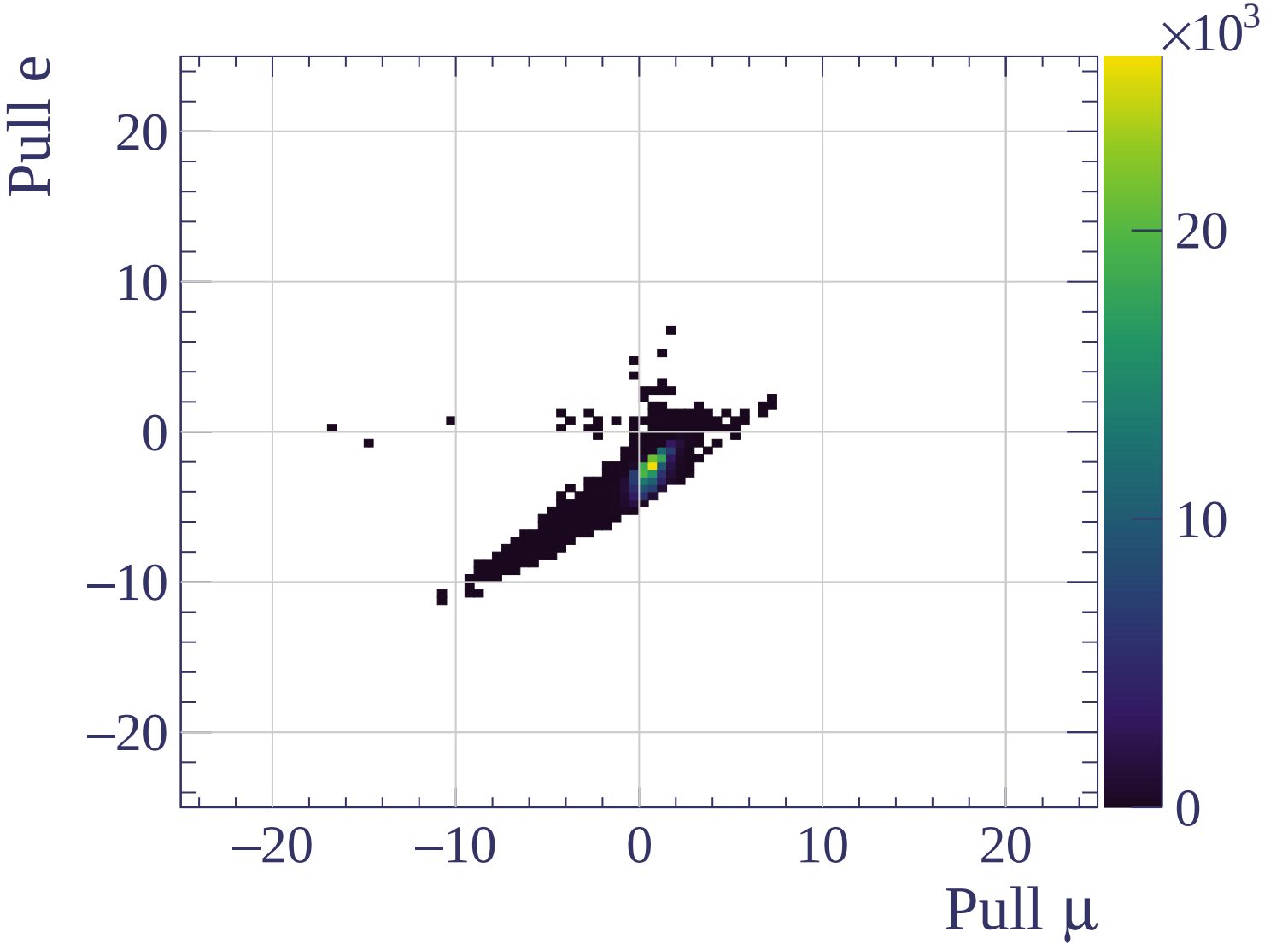


Pulls standard deviation

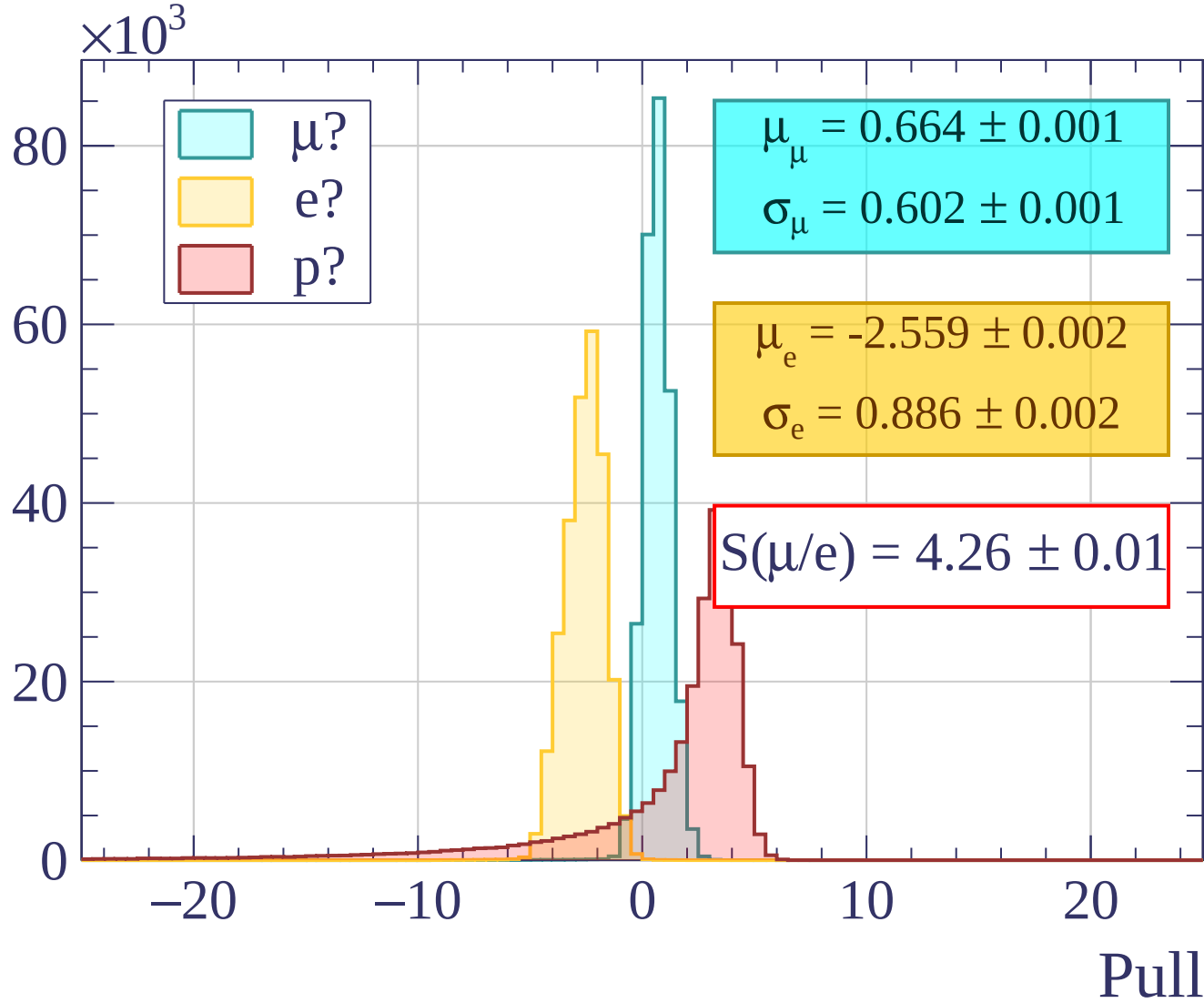


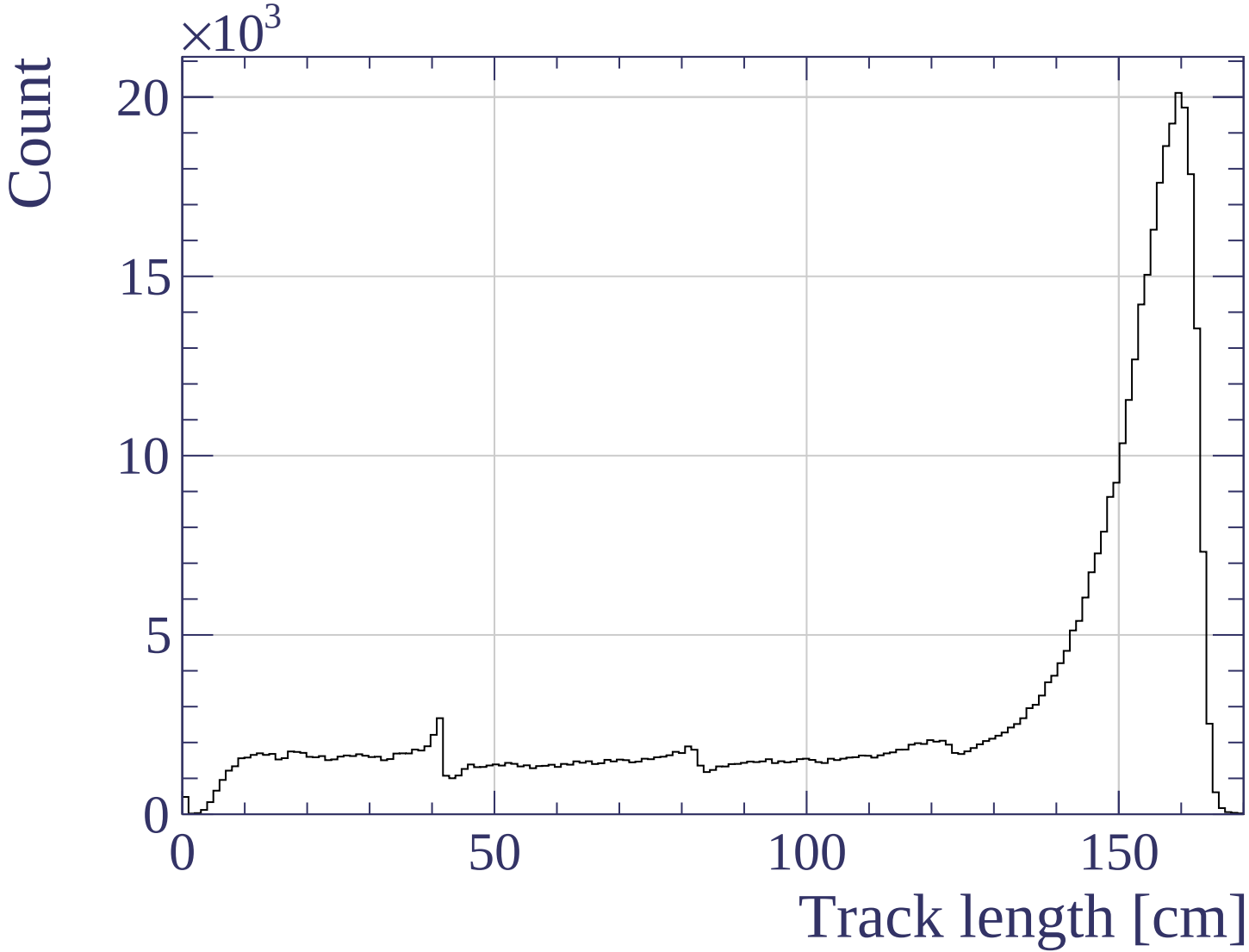
Pulls mean

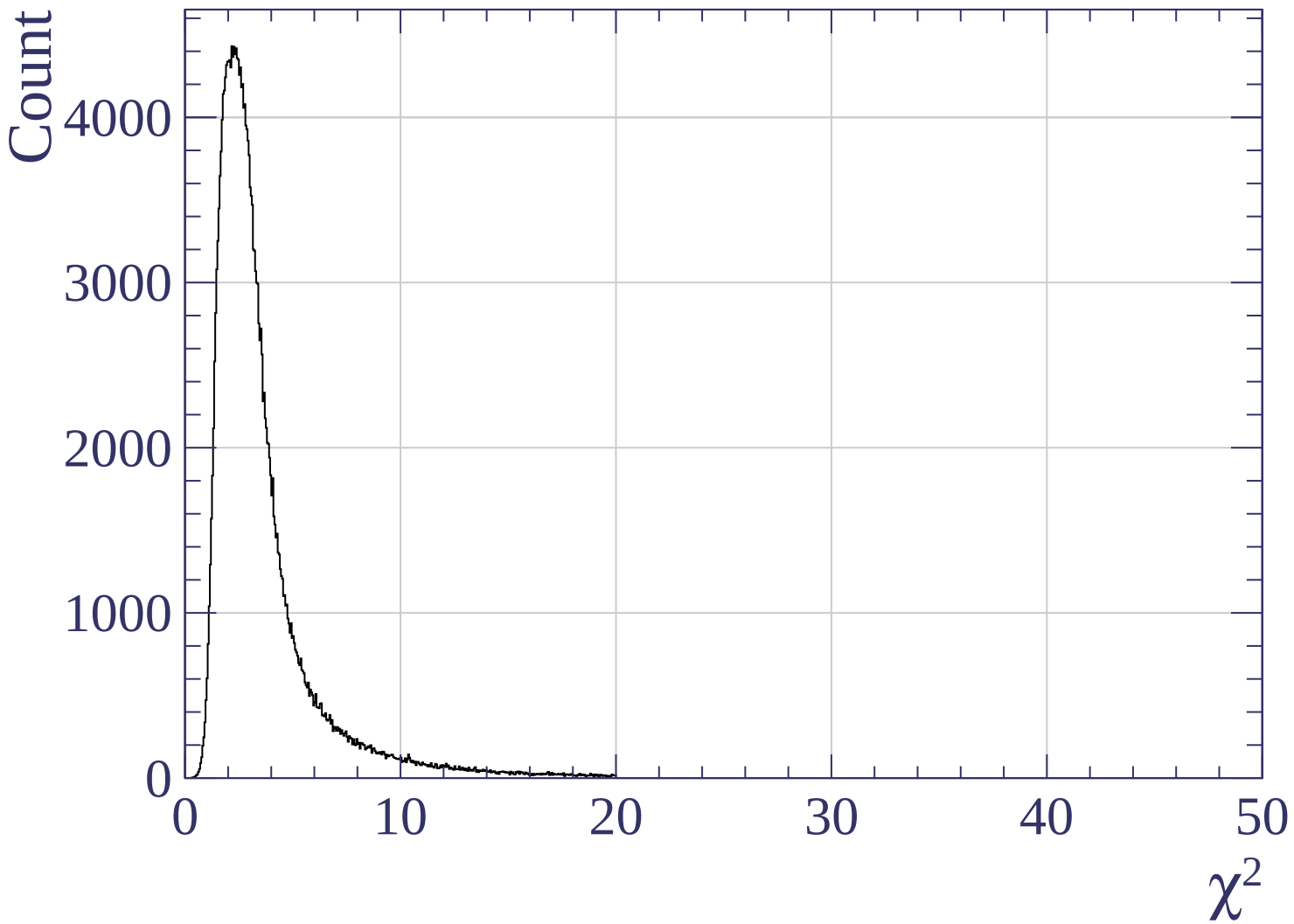


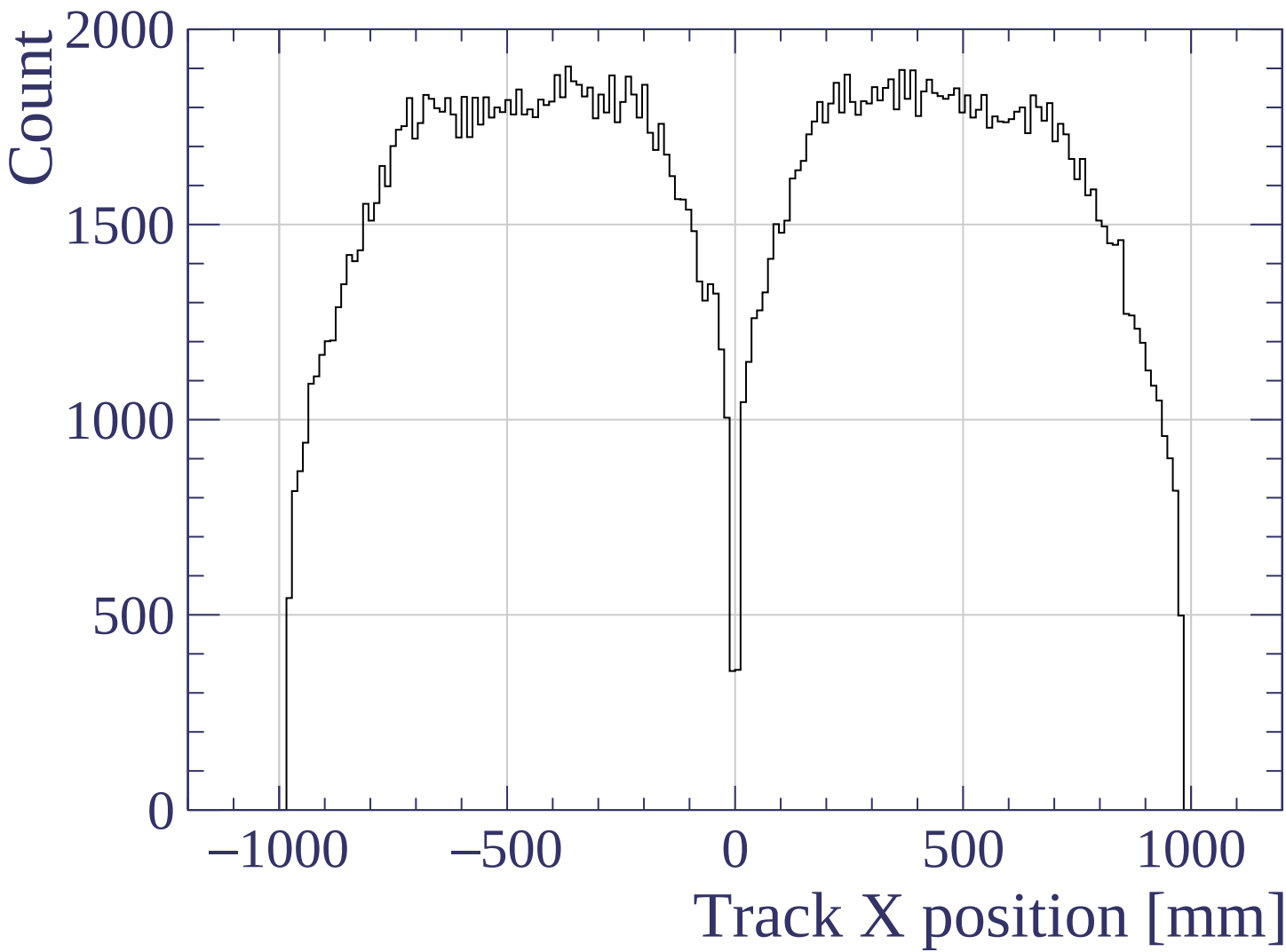


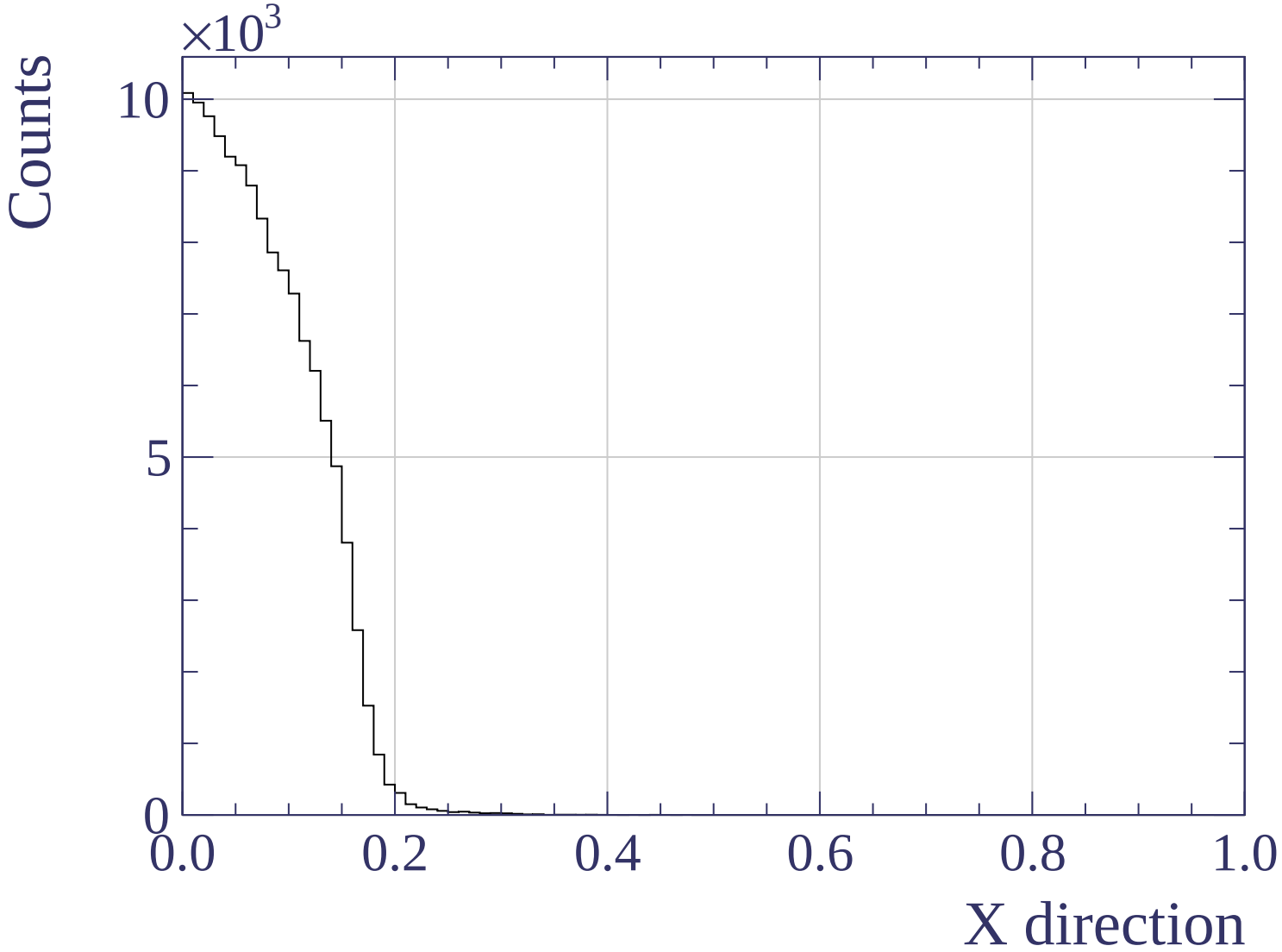
Count

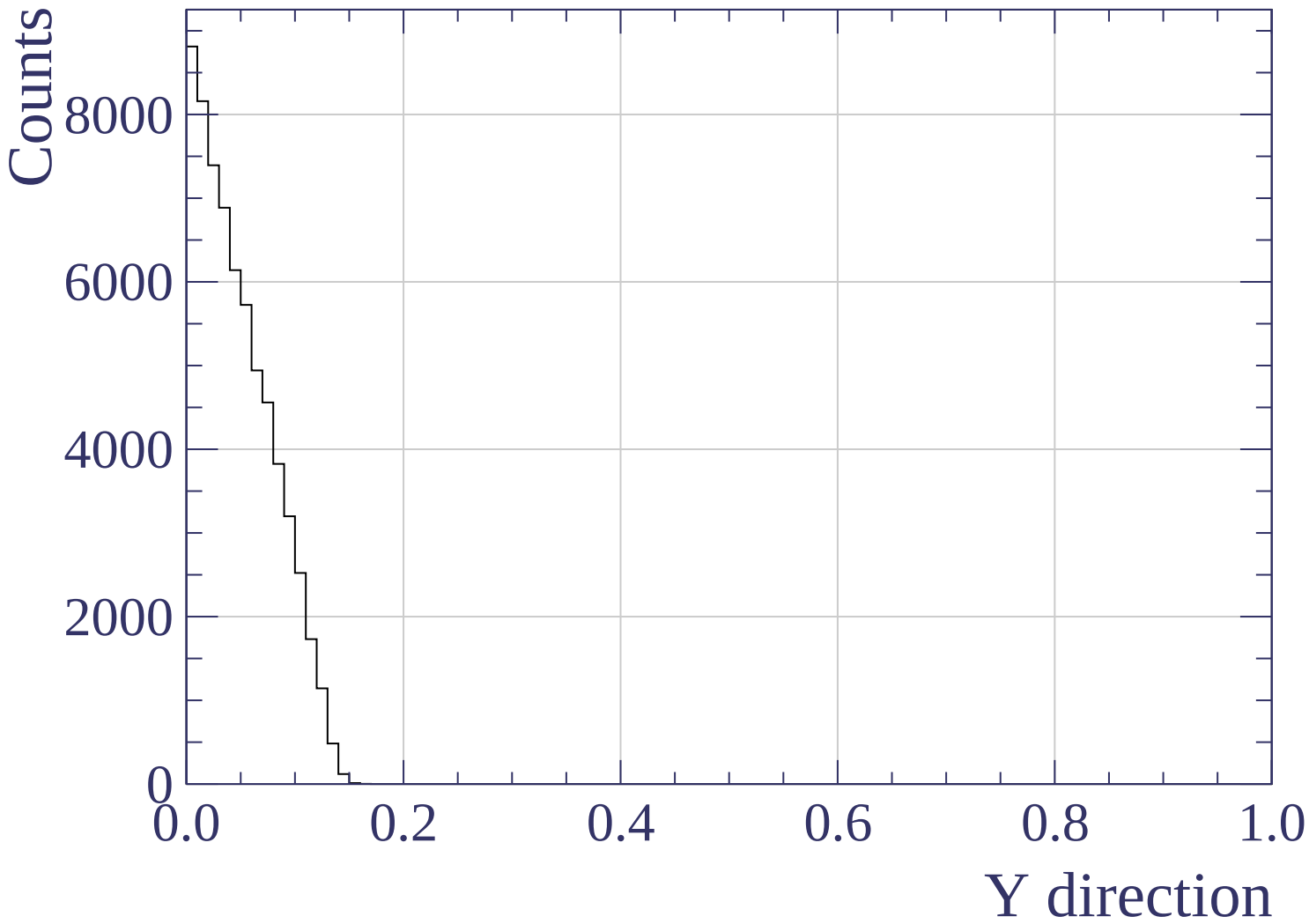


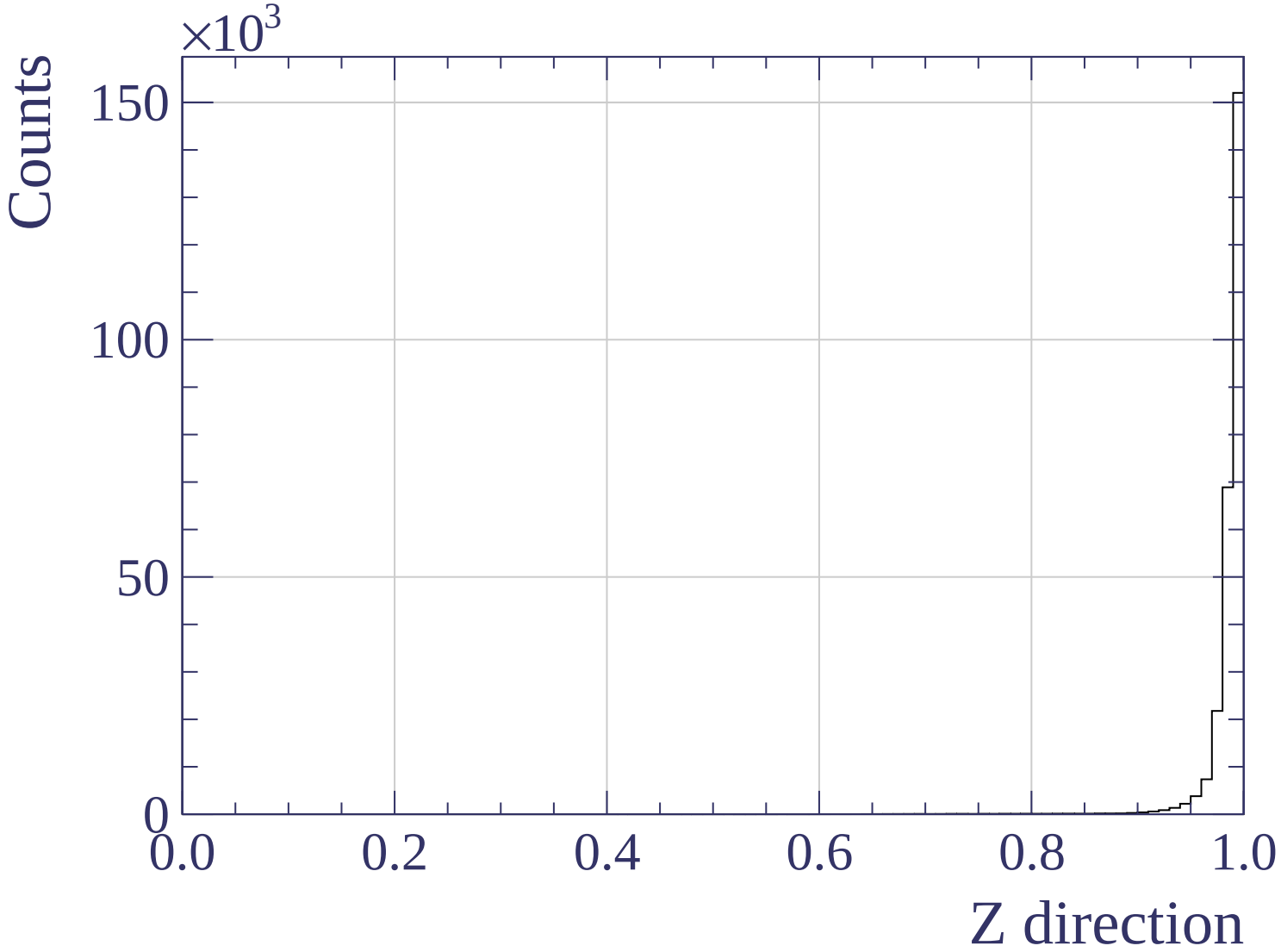


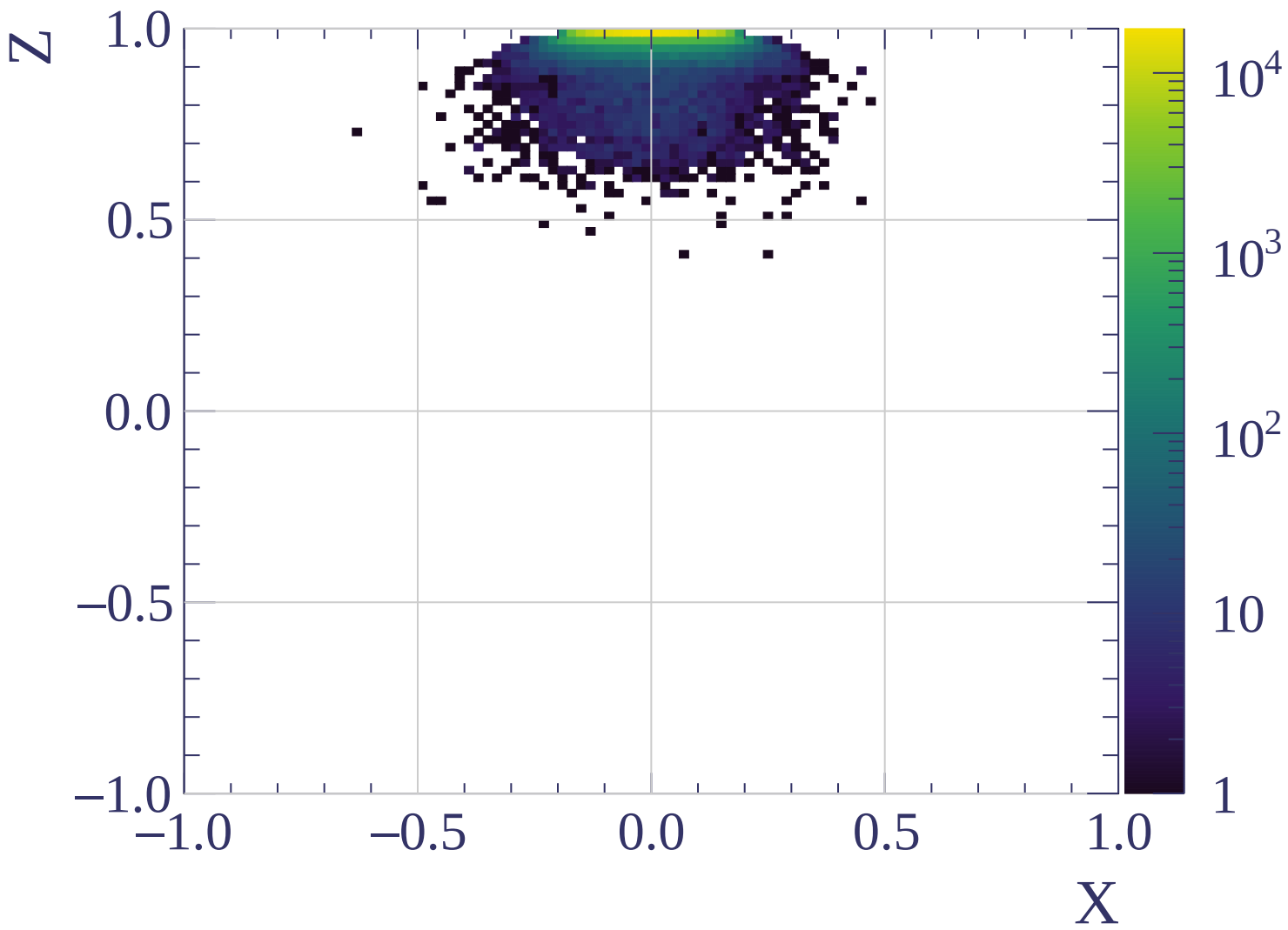


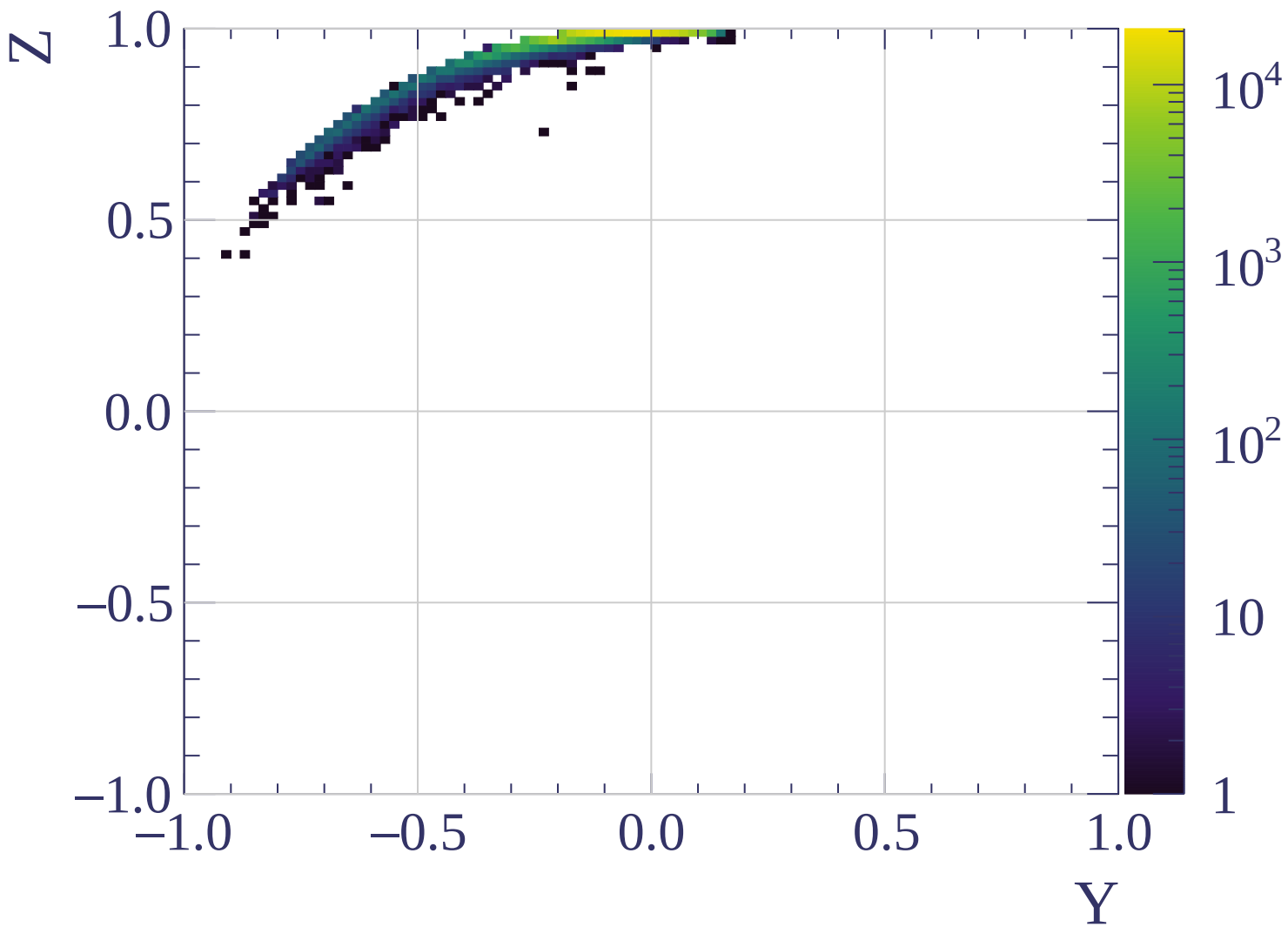


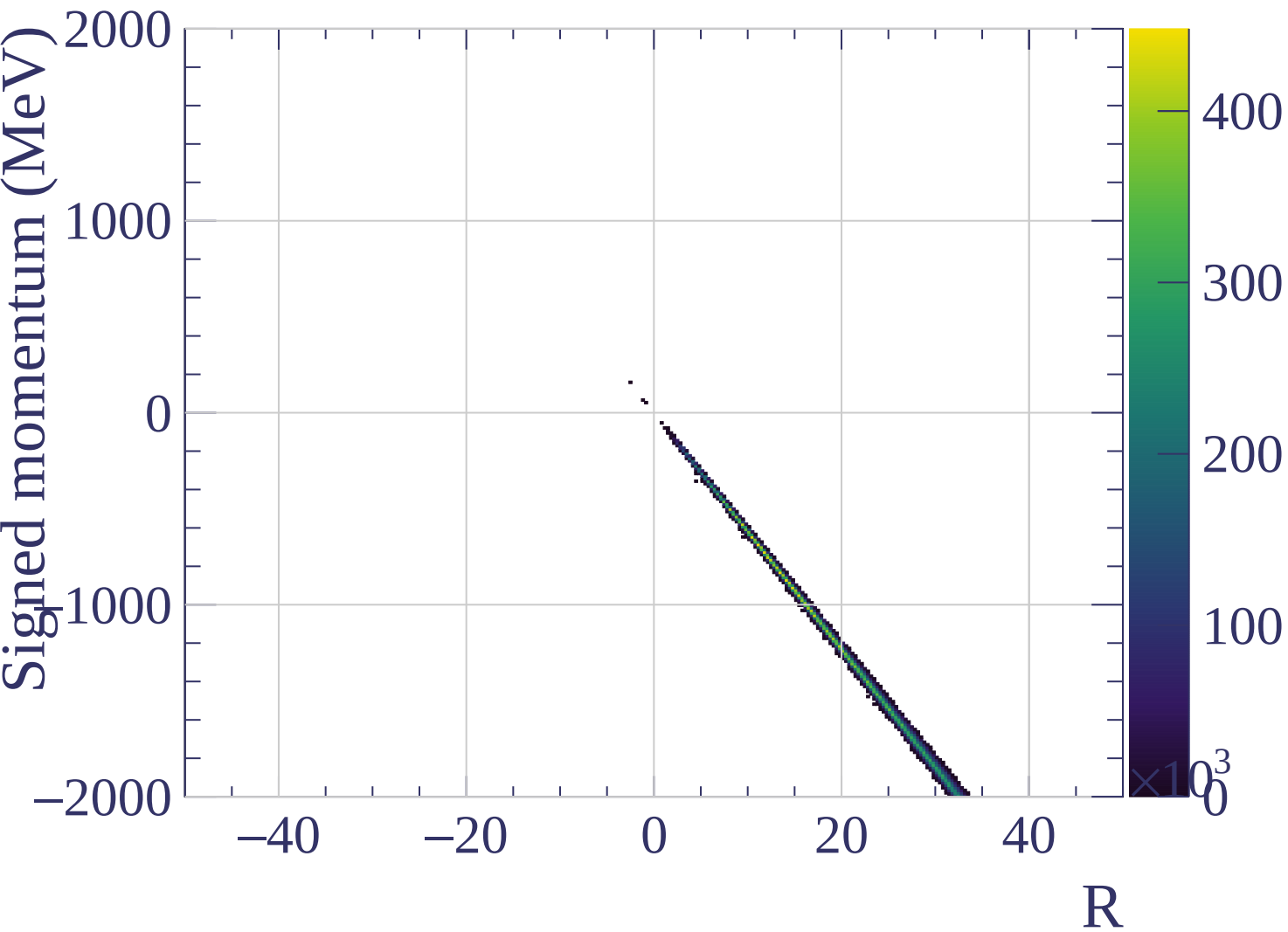


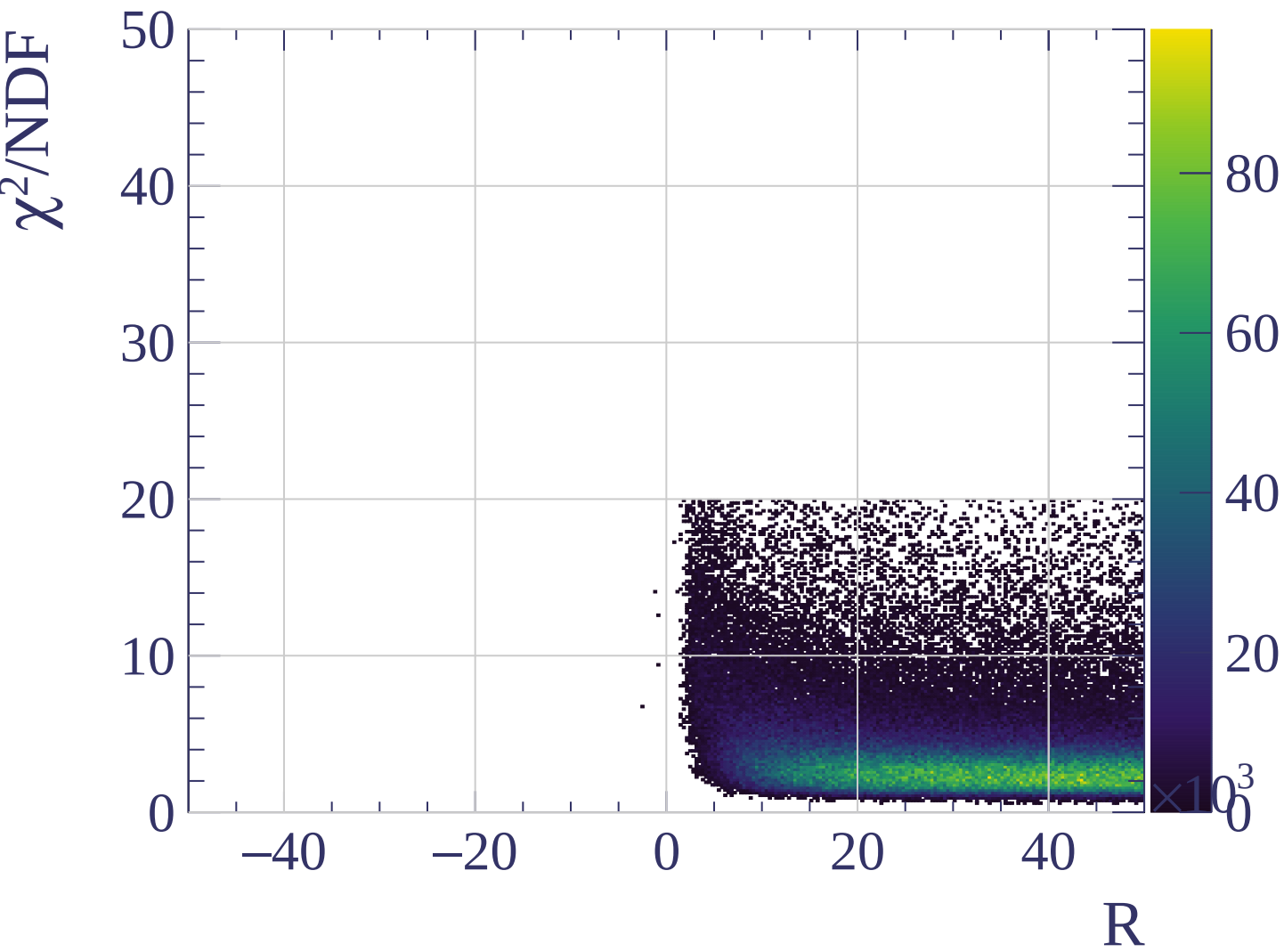






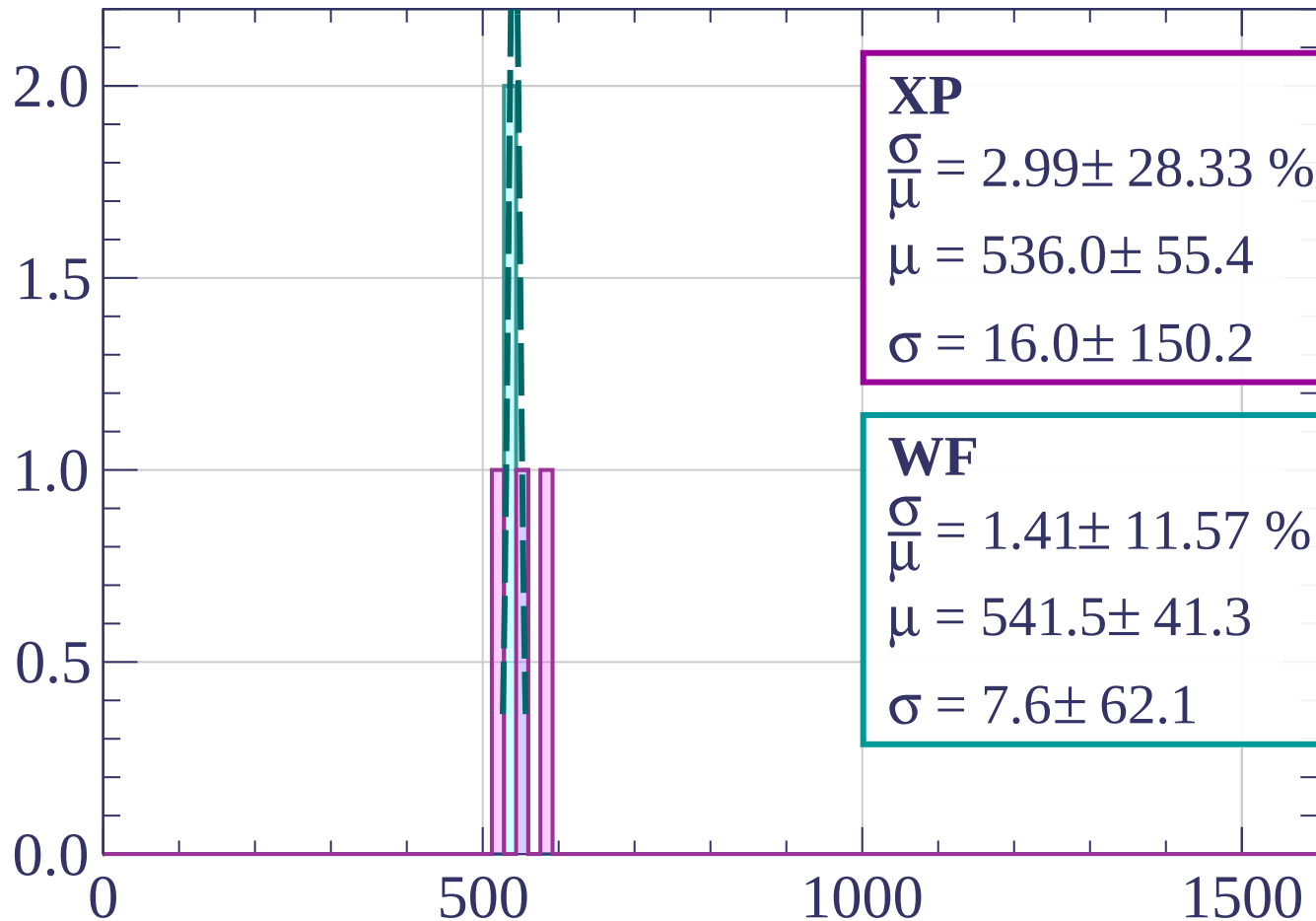






Count

Energy loss | $0 < p < 80$



XP

$$\frac{\sigma}{\mu} = 2.99 \pm 28.33 \%$$

$$\mu = 536.0 \pm 55.4$$

$$\sigma = 16.0 \pm 150.2$$

WF

$$\frac{\sigma}{\mu} = 1.41 \pm 11.57 \%$$

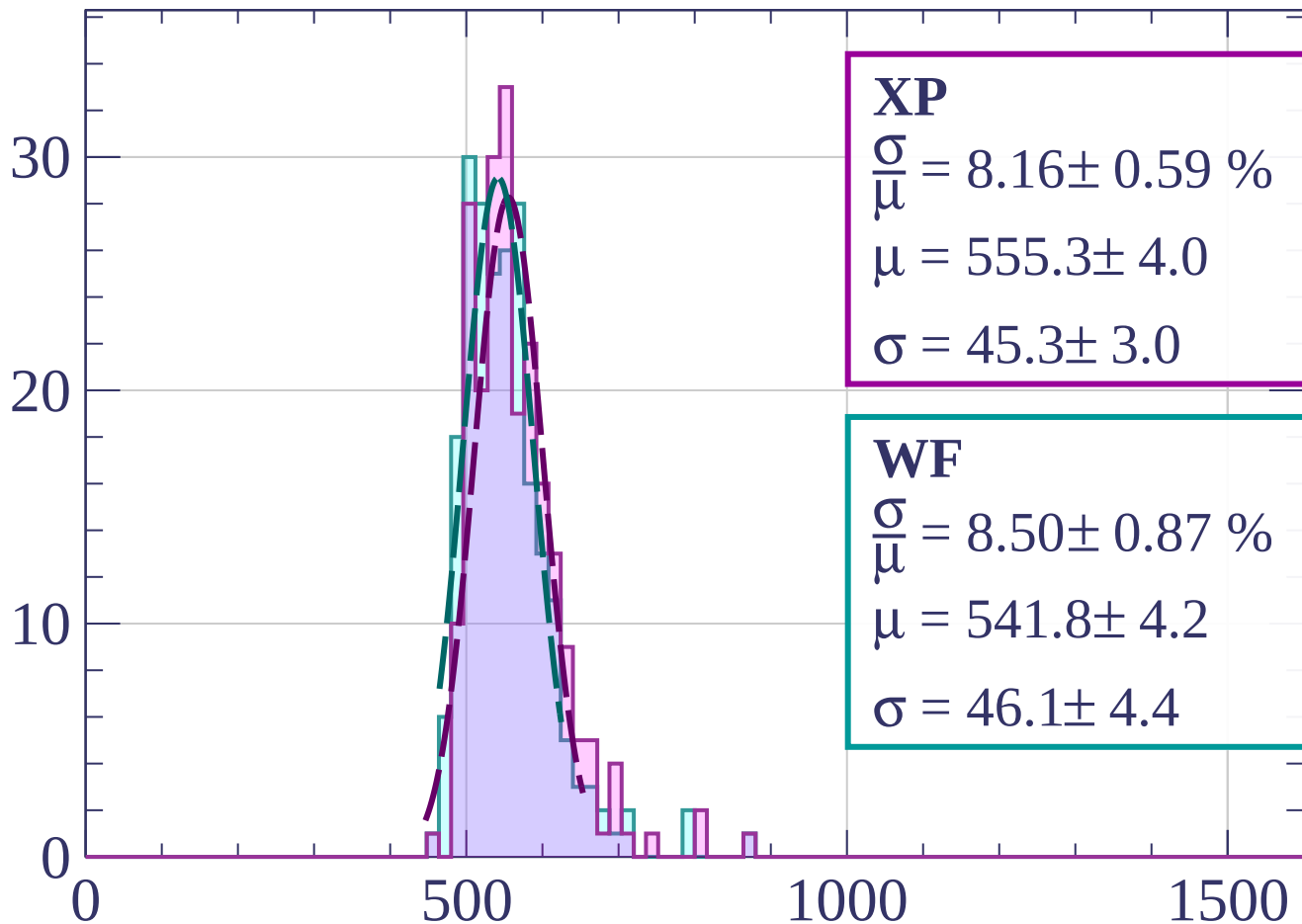
$$\mu = 541.5 \pm 41.3$$

$$\sigma = 7.6 \pm 62.1$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 8.16 \pm 0.59 \%$$

$$\mu = 555.3 \pm 4.0$$

$$\sigma = 45.3 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.87 \%$$

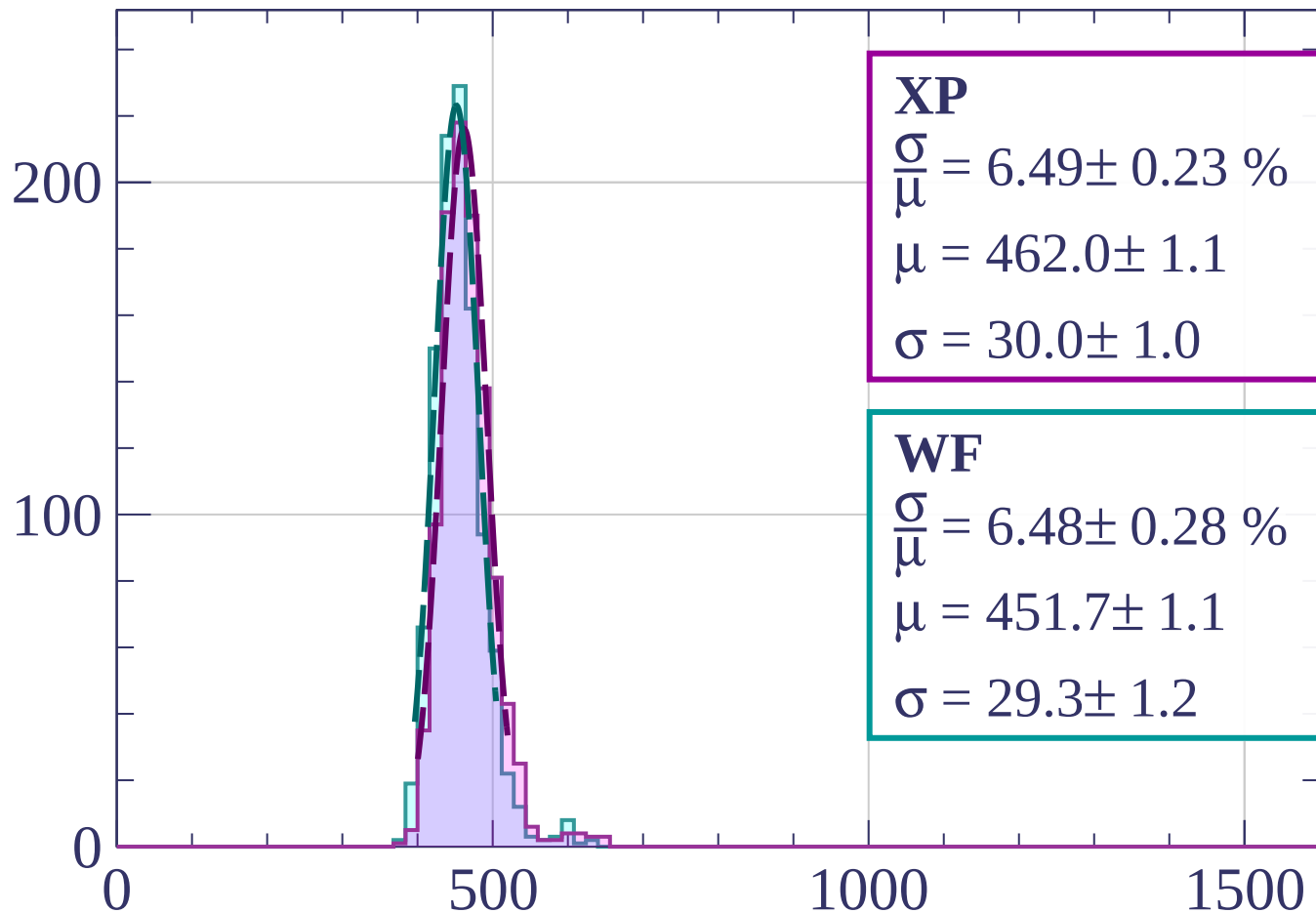
$$\mu = 541.8 \pm 4.2$$

$$\sigma = 46.1 \pm 4.4$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 6.49 \pm 0.23 \%$$

$$\mu = 462.0 \pm 1.1$$

$$\sigma = 30.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.28 \%$$

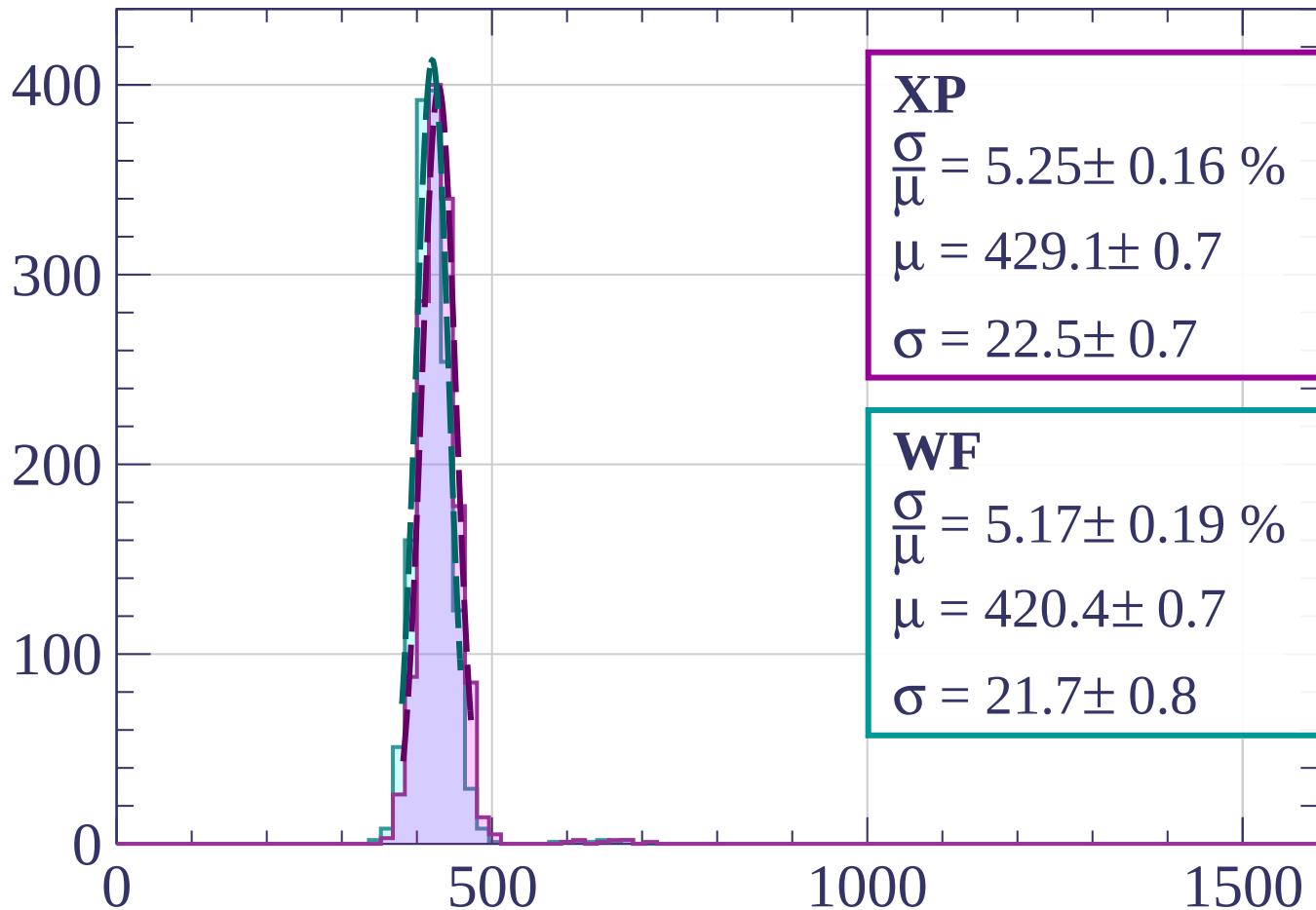
$$\mu = 451.7 \pm 1.1$$

$$\sigma = 29.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 5.25 \pm 0.16 \%$$

$$\mu = 429.1 \pm 0.7$$

$$\sigma = 22.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.17 \pm 0.19 \%$$

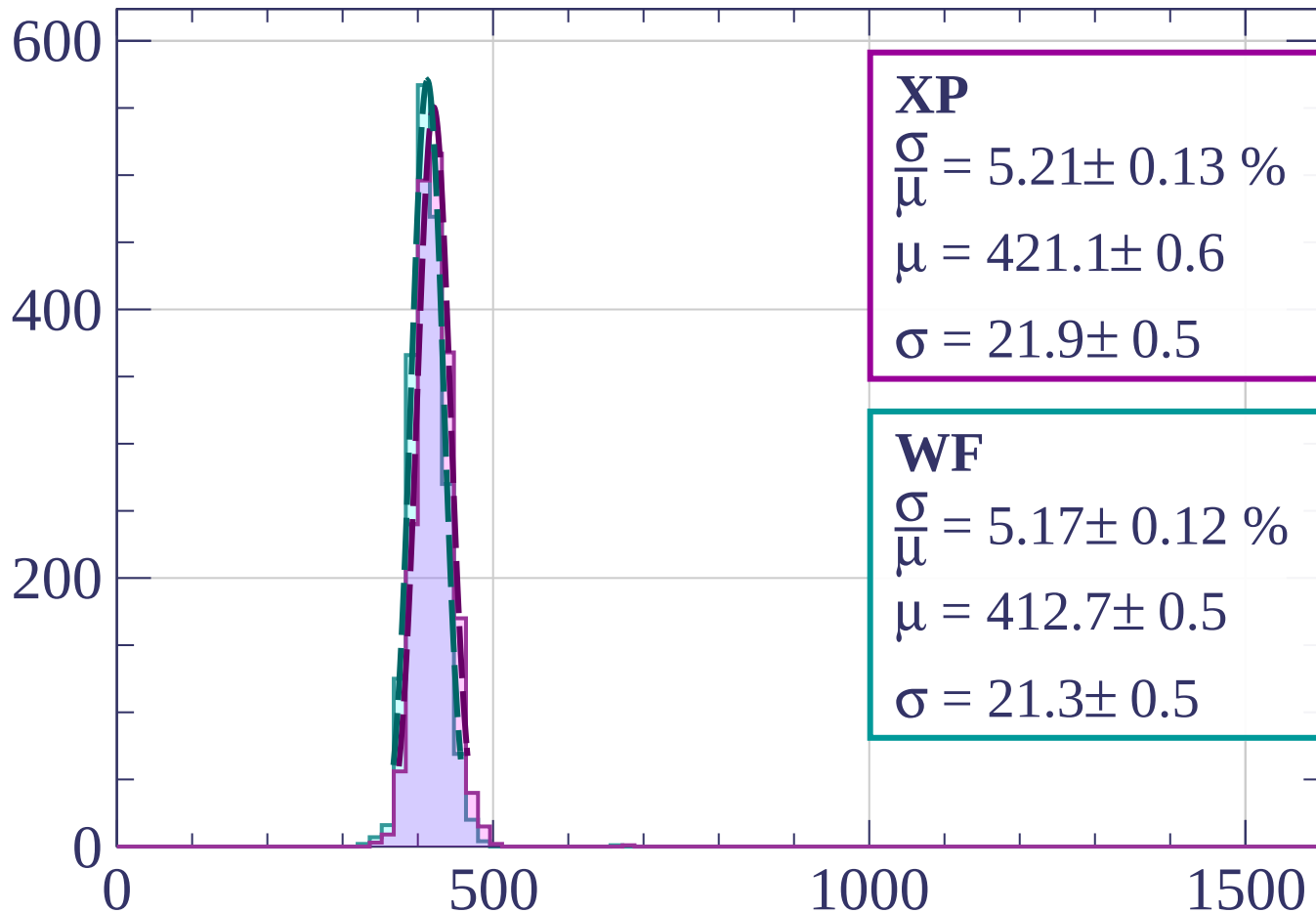
$$\mu = 420.4 \pm 0.7$$

$$\sigma = 21.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

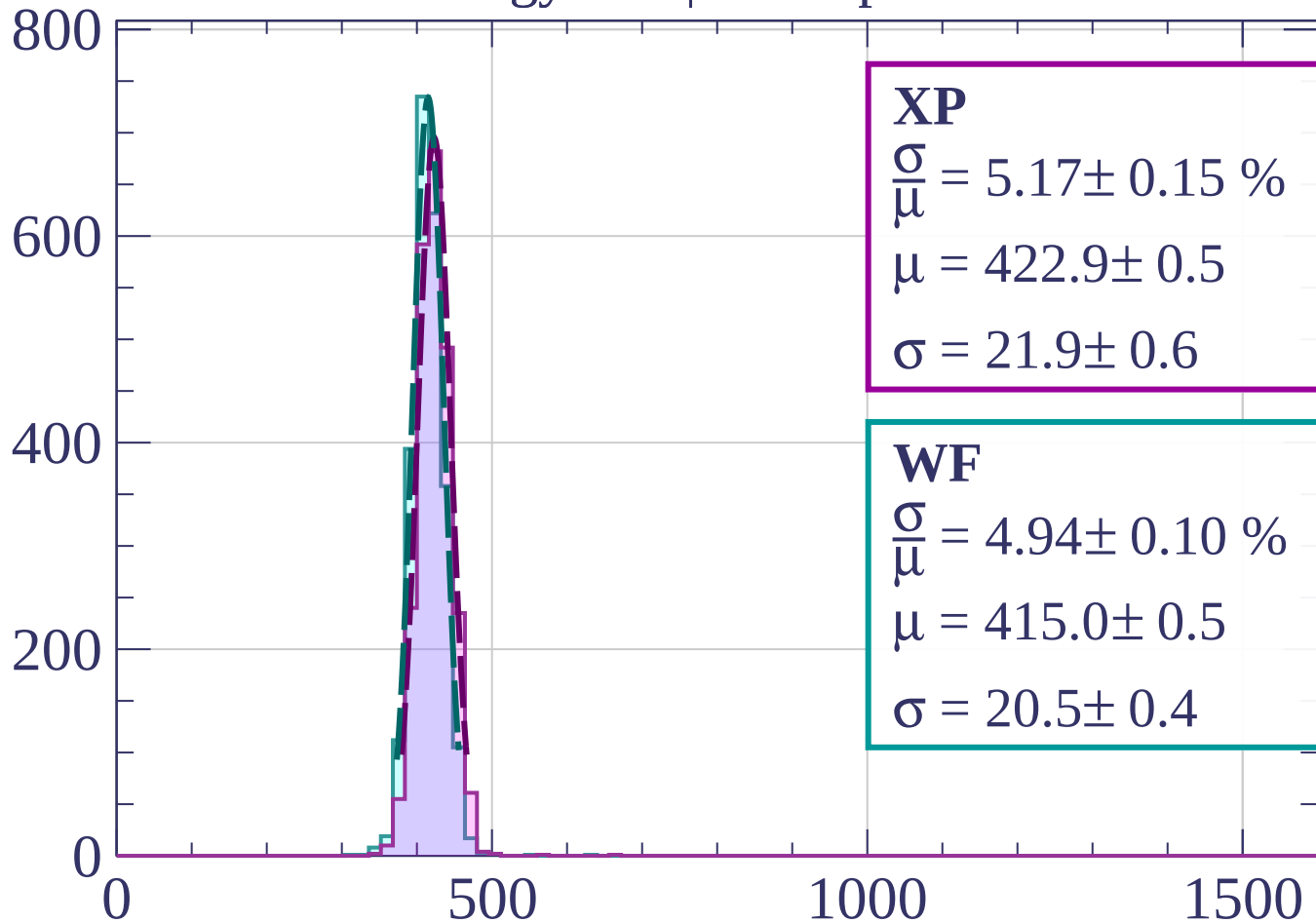
Count



dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 5.17 \pm 0.15 \%$$

$$\mu = 422.9 \pm 0.5$$

$$\sigma = 21.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 4.94 \pm 0.10 \%$$

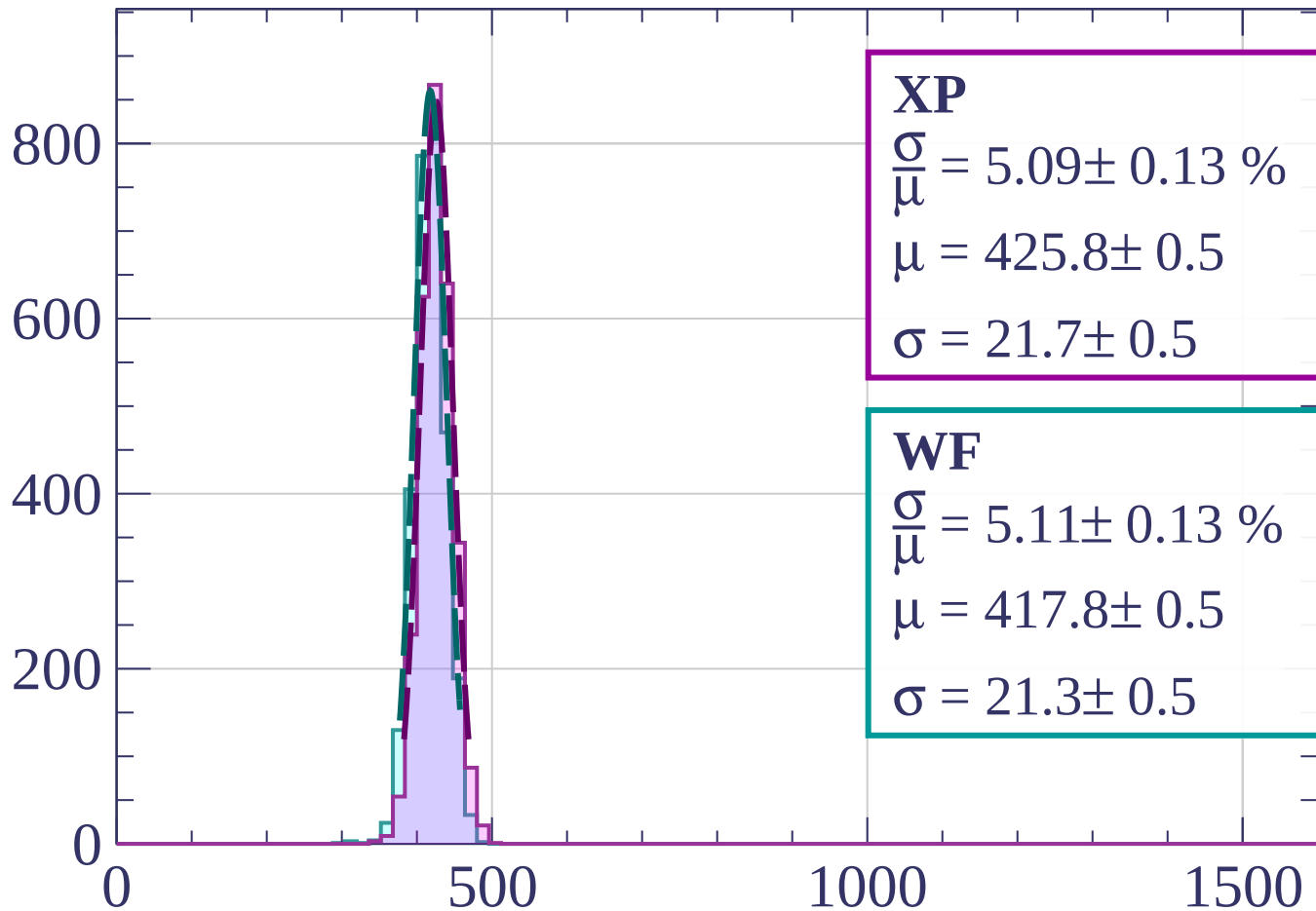
$$\mu = 415.0 \pm 0.5$$

$$\sigma = 20.5 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 5.09 \pm 0.13 \%$$

$$\mu = 425.8 \pm 0.5$$

$$\sigma = 21.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.11 \pm 0.13 \%$$

$$\mu = 417.8 \pm 0.5$$

$$\sigma = 21.3 \pm 0.5$$

dE/dx [ADC counts/cm]

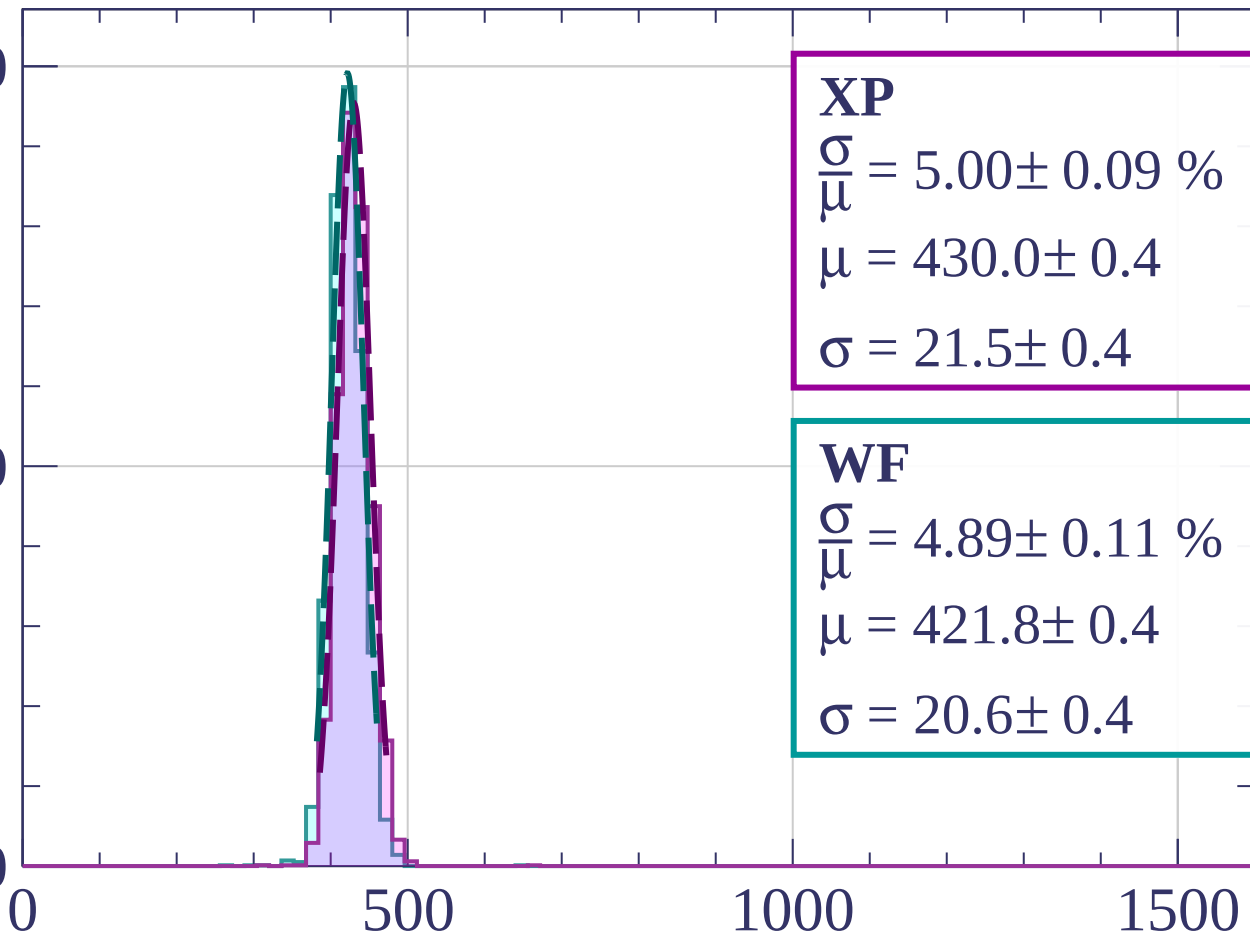
Energy loss | $560 < p < 640$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 5.00 \pm 0.09 \%$$

$$\mu = 430.0 \pm 0.4$$

$$\sigma = 21.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.89 \pm 0.11 \%$$

$$\mu = 421.8 \pm 0.4$$

$$\sigma = 20.6 \pm 0.4$$

dE/dx [ADC counts/cm]

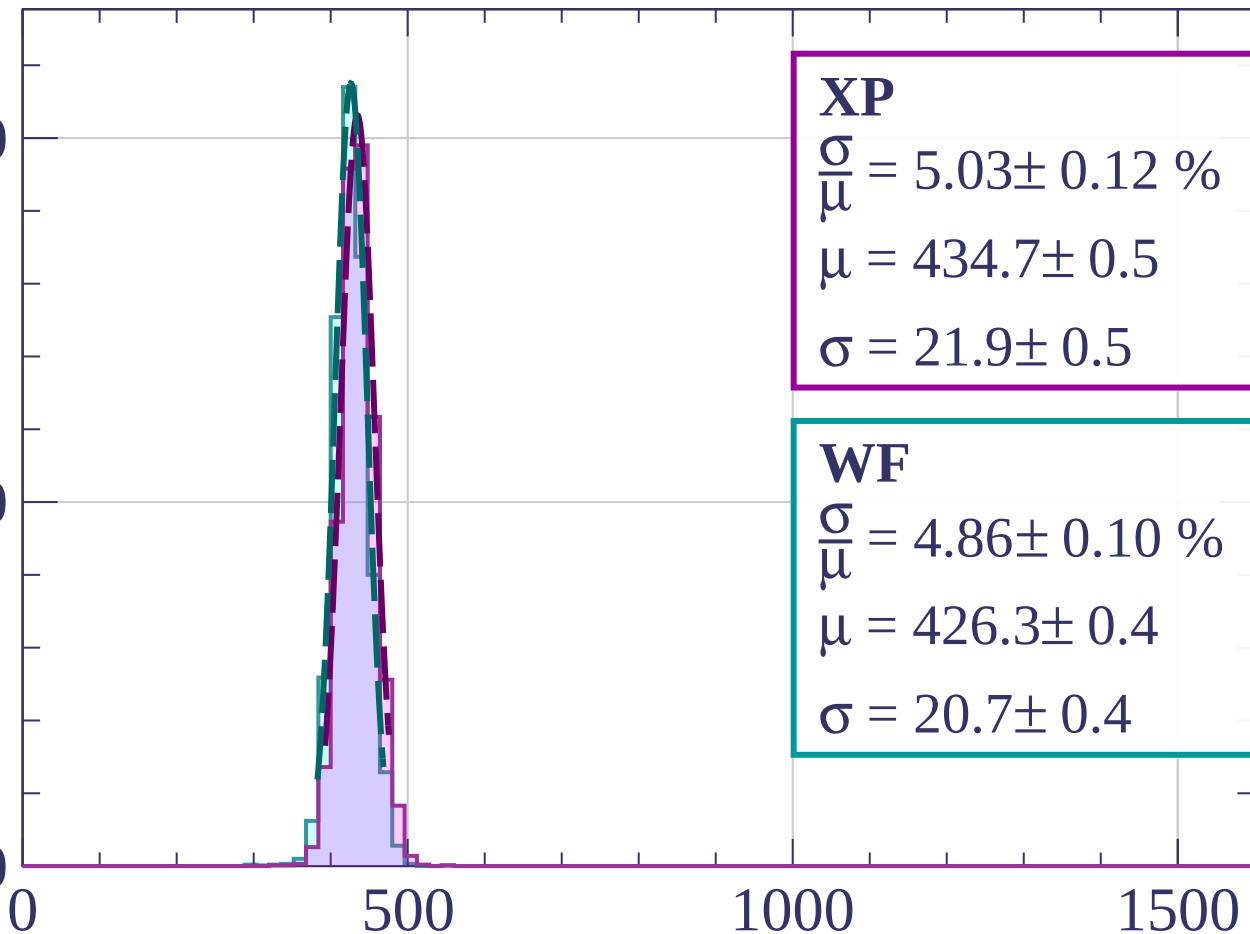
Energy loss | $640 < p < 720$

Count

1000

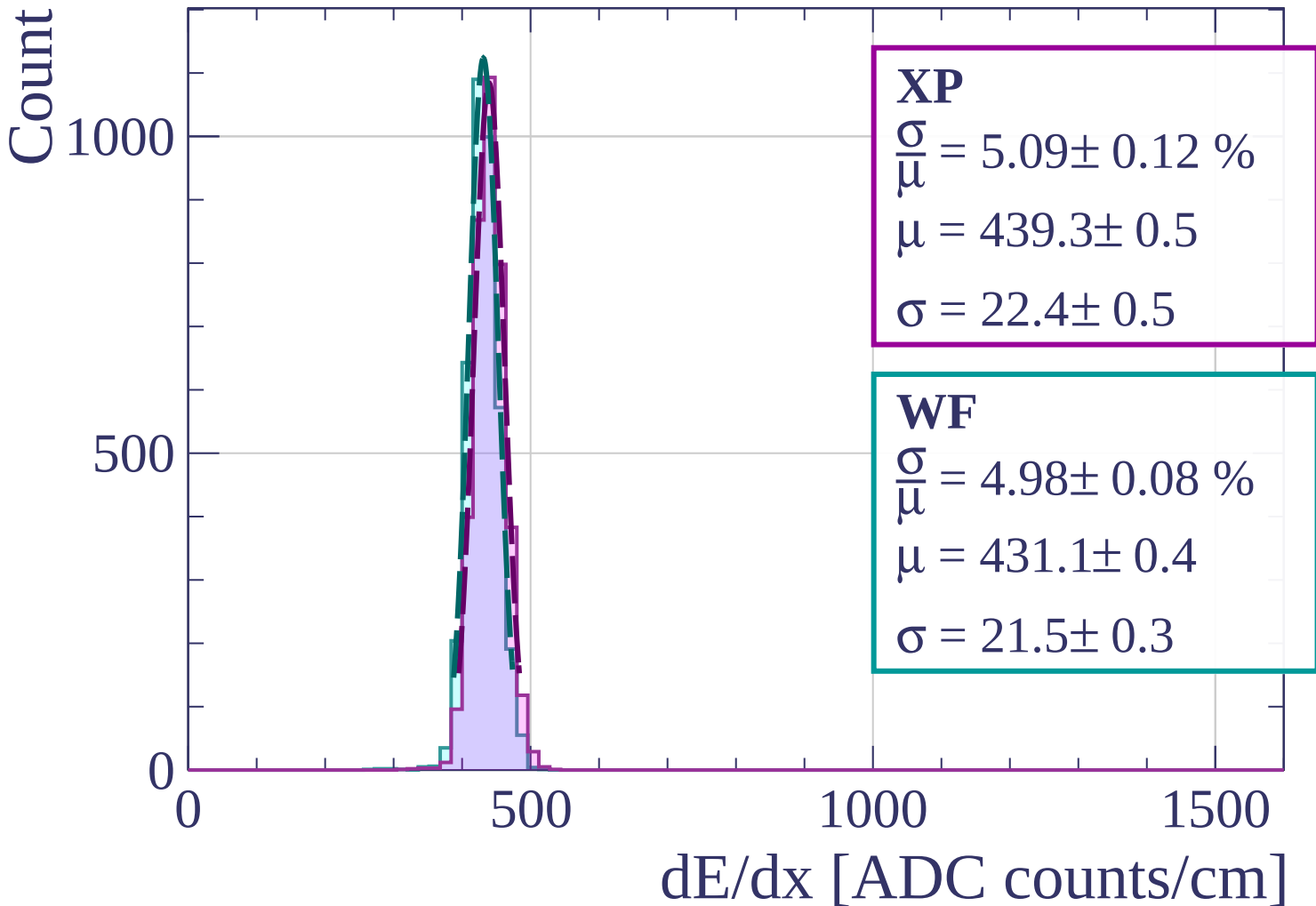
500

0



dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$



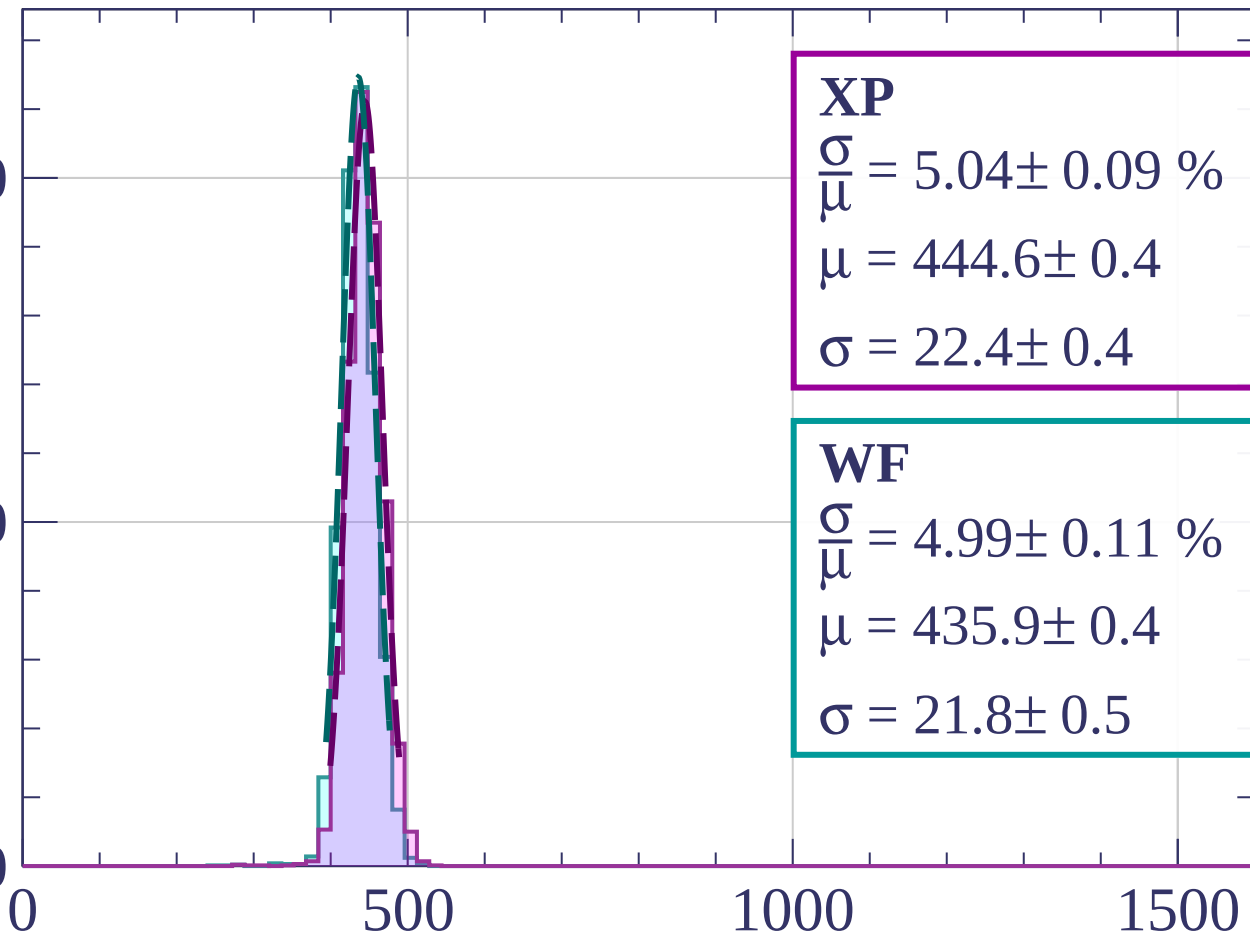
Energy loss | $800 < p < 880$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.09 \%$$

$$\mu = 444.6 \pm 0.4$$

$$\sigma = 22.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.99 \pm 0.11 \%$$

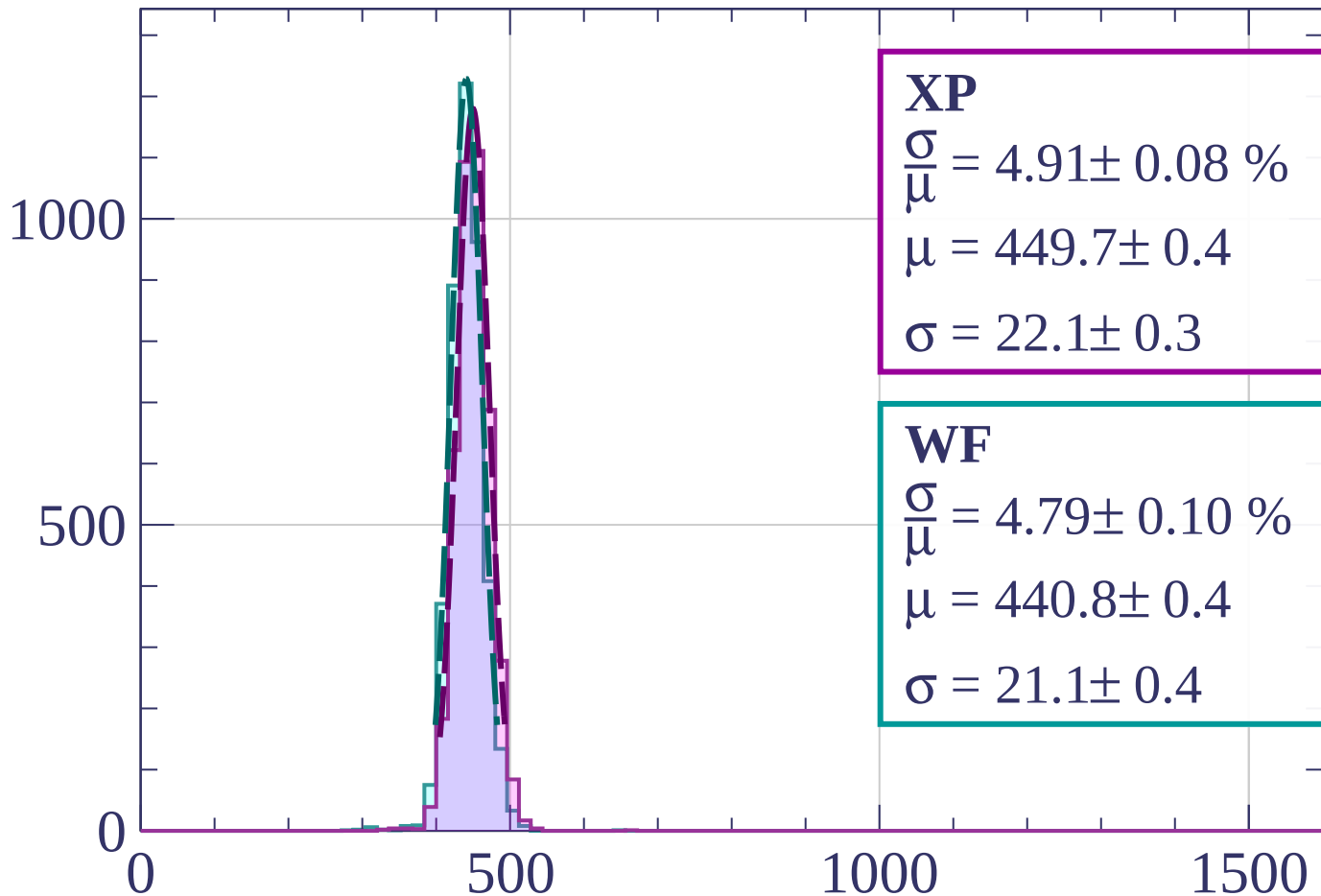
$$\mu = 435.9 \pm 0.4$$

$$\sigma = 21.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 4.91 \pm 0.08 \%$$

$$\mu = 449.7 \pm 0.4$$

$$\sigma = 22.1 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 4.79 \pm 0.10 \%$$

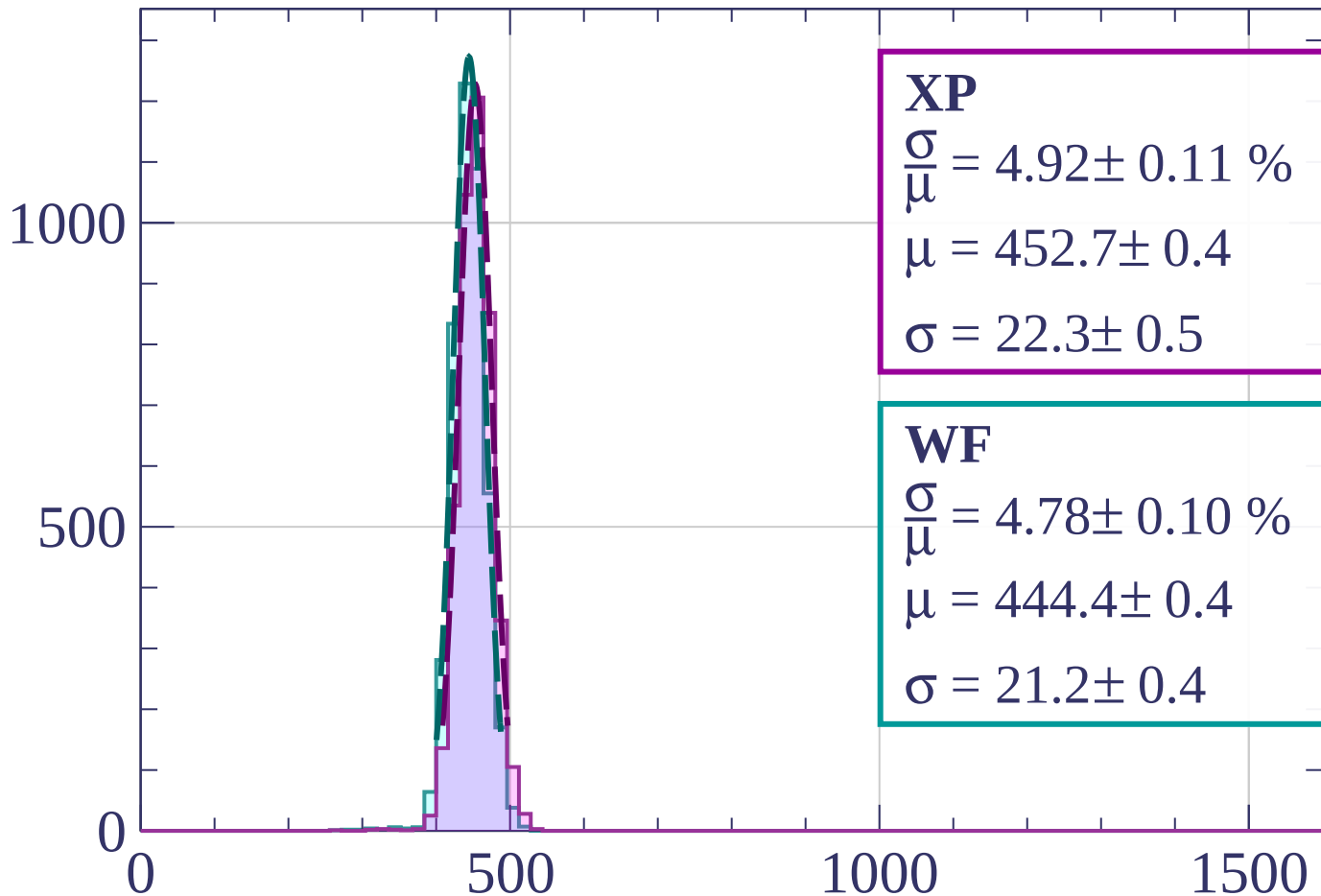
$$\mu = 440.8 \pm 0.4$$

$$\sigma = 21.1 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

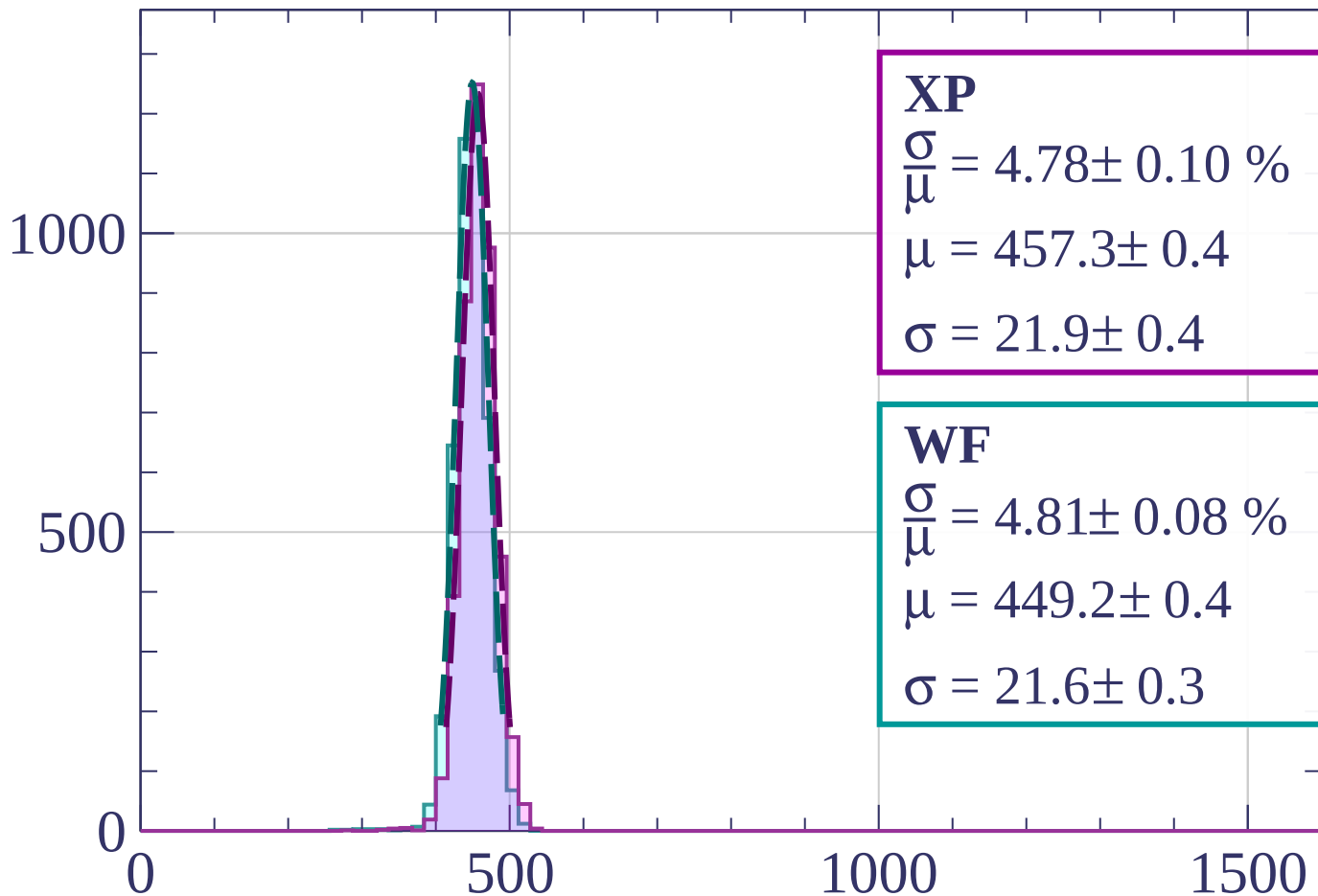
Count



dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.10 \%$$

$$\mu = 457.3 \pm 0.4$$

$$\sigma = 21.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.08 \%$$

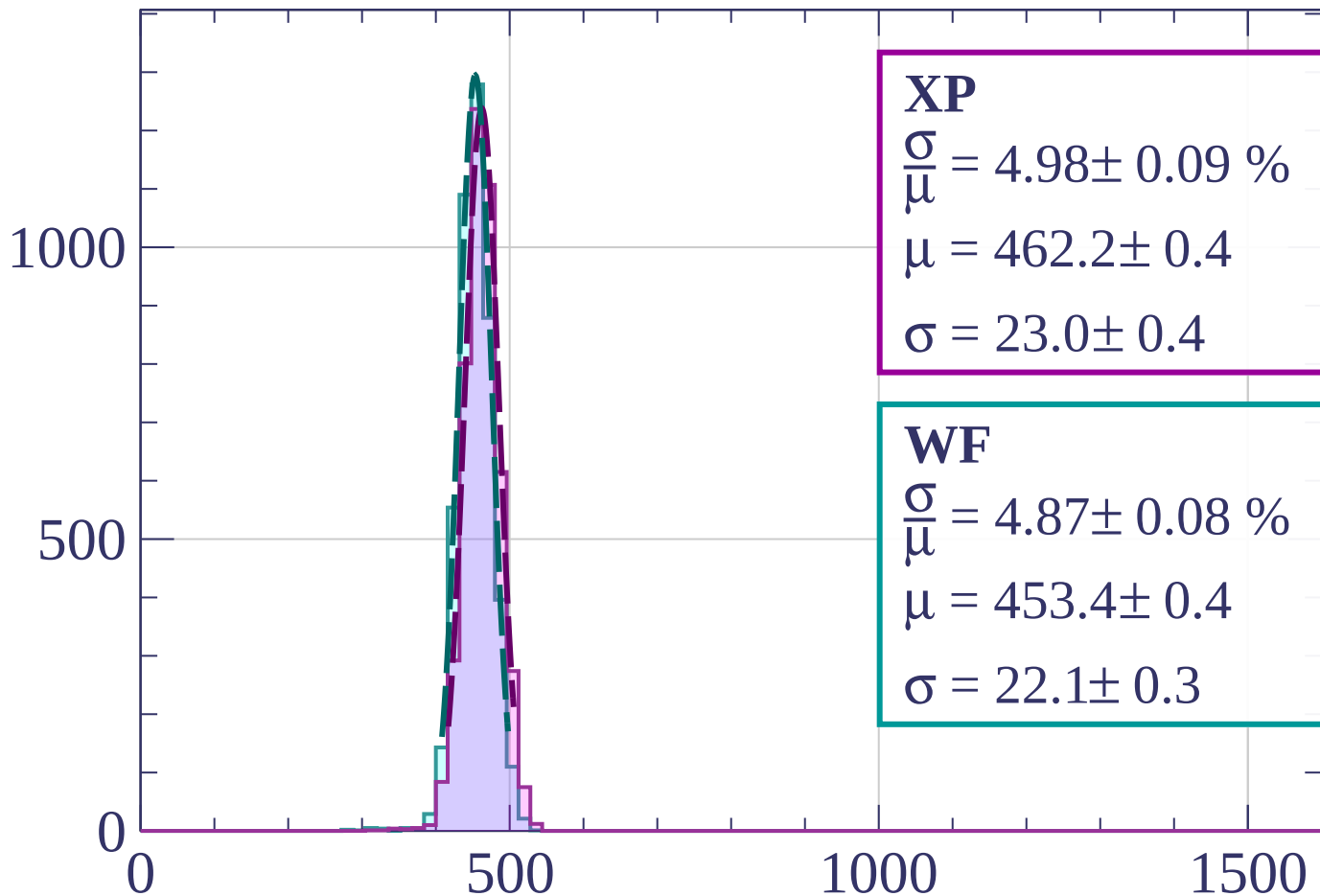
$$\mu = 449.2 \pm 0.4$$

$$\sigma = 21.6 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 4.98 \pm 0.09 \%$$

$$\mu = 462.2 \pm 0.4$$

$$\sigma = 23.0 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.87 \pm 0.08 \%$$

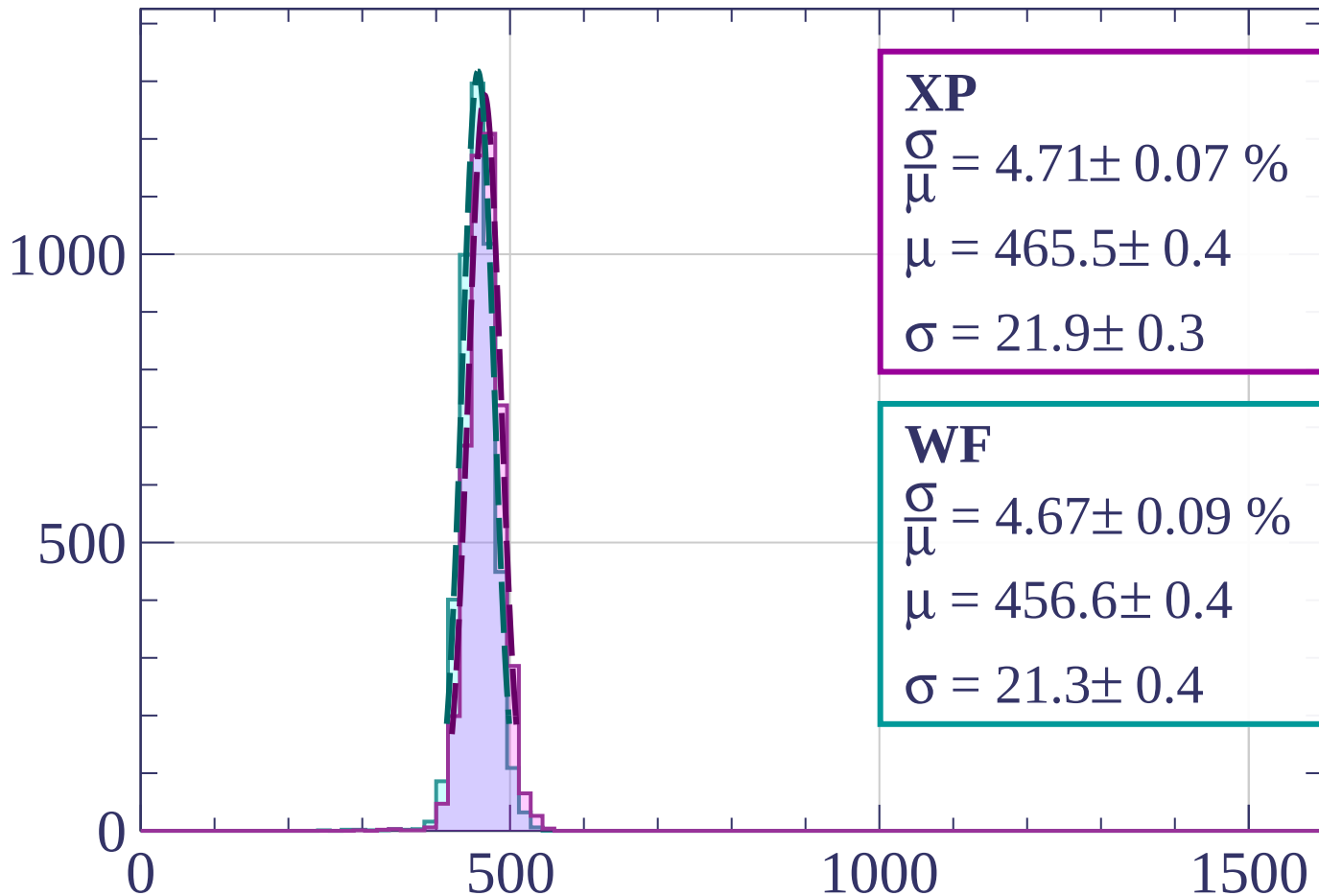
$$\mu = 453.4 \pm 0.4$$

$$\sigma = 22.1 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 4.71 \pm 0.07 \%$$

$$\mu = 465.5 \pm 0.4$$

$$\sigma = 21.9 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 4.67 \pm 0.09 \%$$

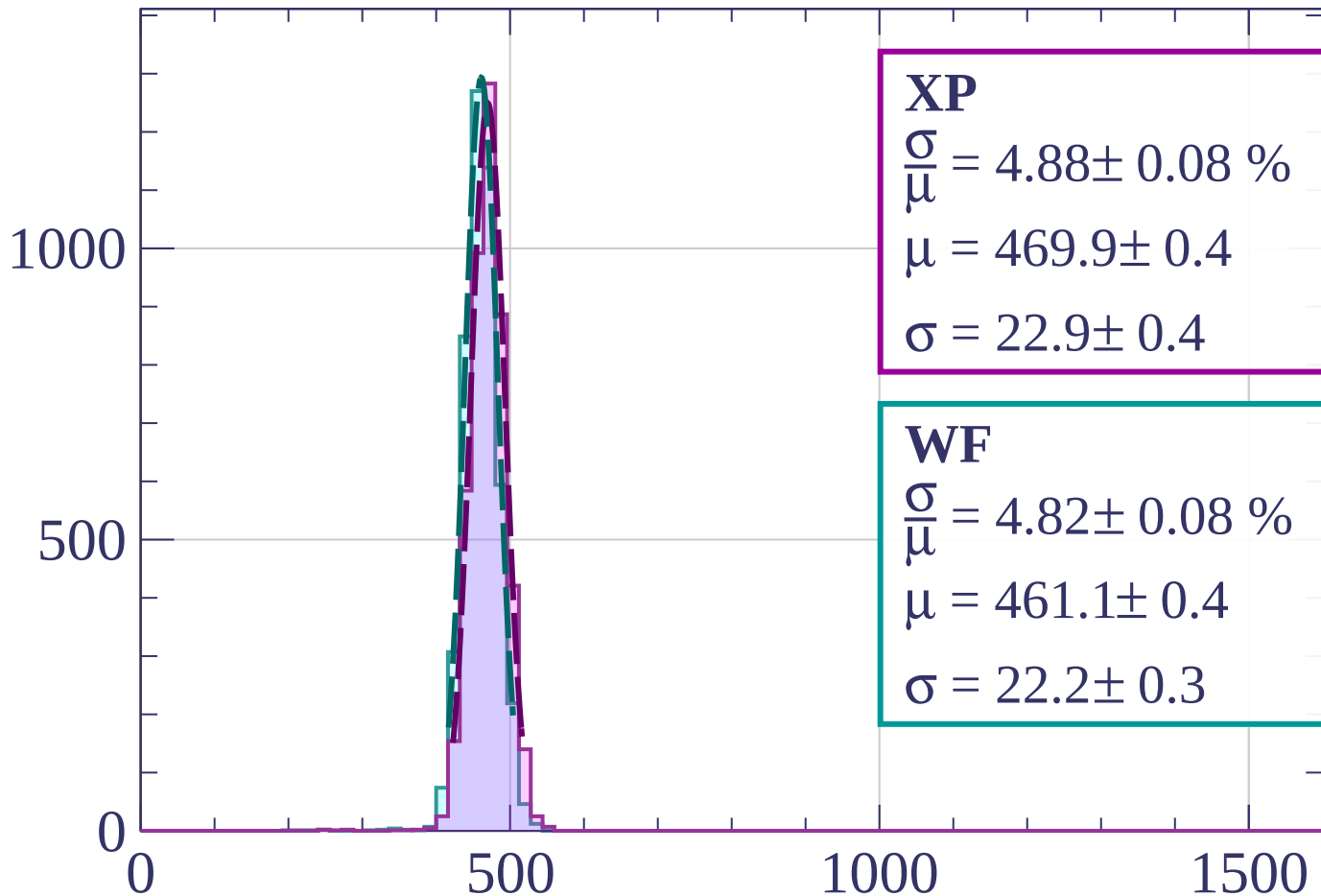
$$\mu = 456.6 \pm 0.4$$

$$\sigma = 21.3 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 4.88 \pm 0.08 \%$$

$$\mu = 469.9 \pm 0.4$$

$$\sigma = 22.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.08 \%$$

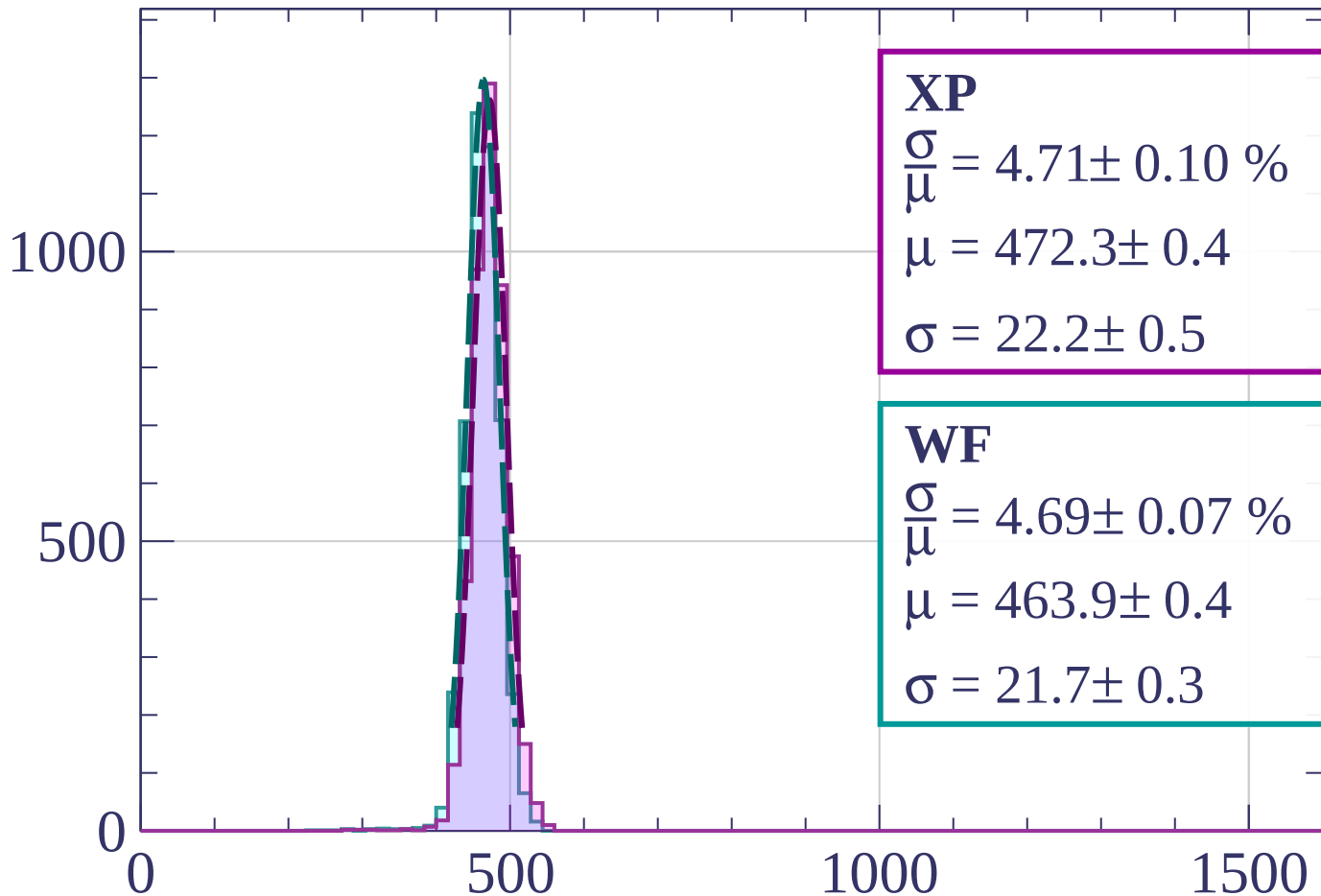
$$\mu = 461.1 \pm 0.4$$

$$\sigma = 22.2 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 4.71 \pm 0.10 \%$$

$$\mu = 472.3 \pm 0.4$$

$$\sigma = 22.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.69 \pm 0.07 \%$$

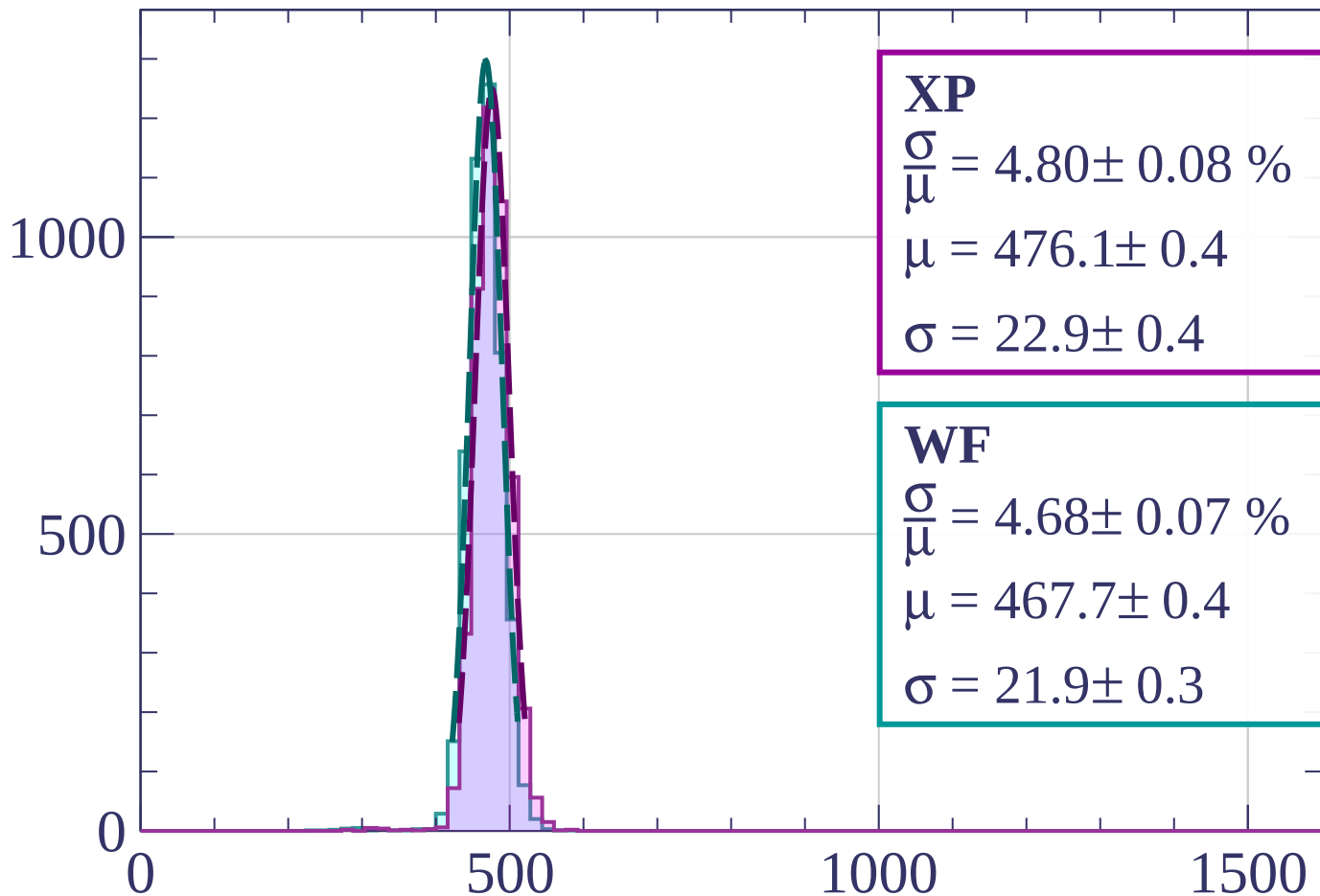
$$\mu = 463.9 \pm 0.4$$

$$\sigma = 21.7 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 4.80 \pm 0.08 \%$$

$$\mu = 476.1 \pm 0.4$$

$$\sigma = 22.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.68 \pm 0.07 \%$$

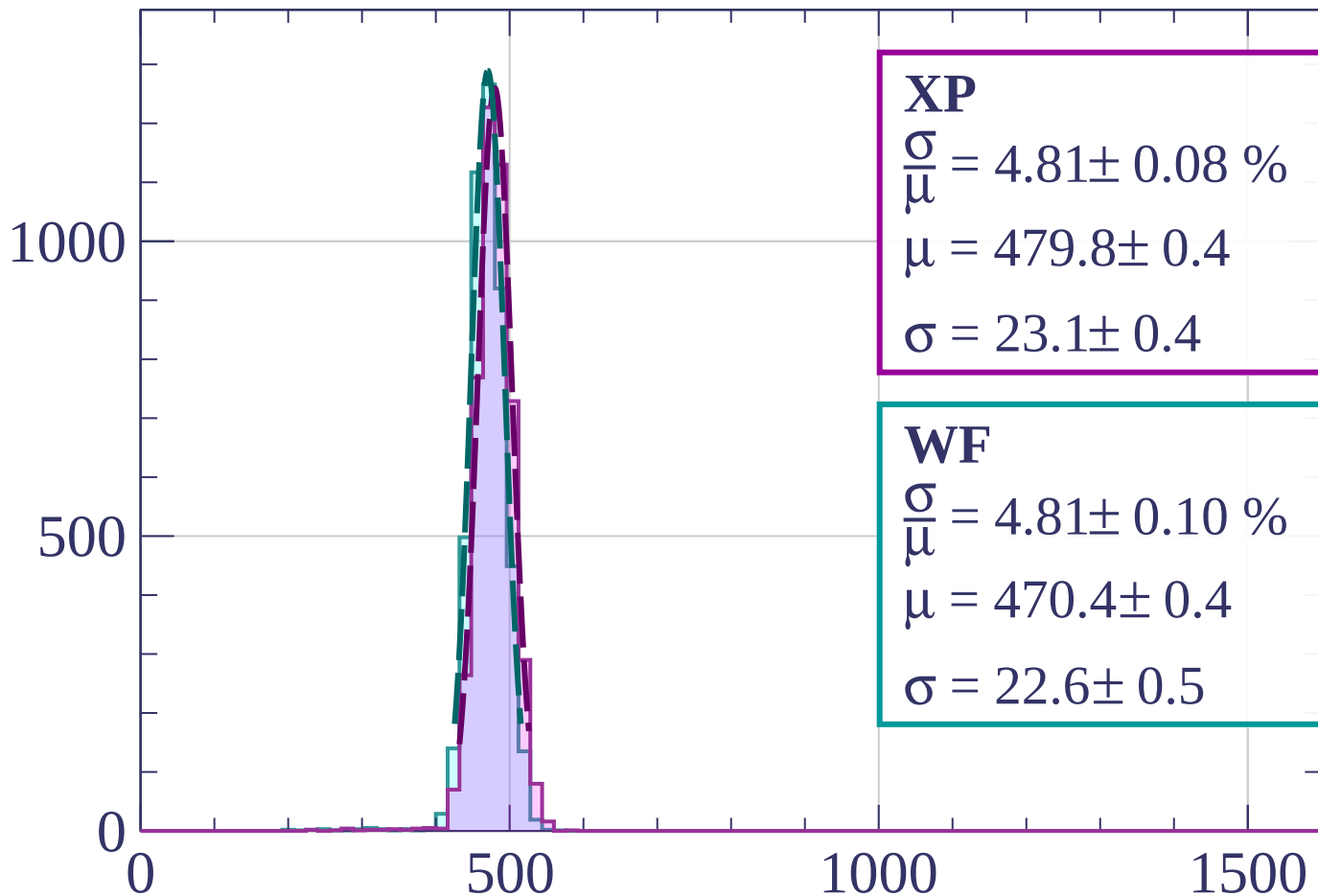
$$\mu = 467.7 \pm 0.4$$

$$\sigma = 21.9 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.08 \%$$

$$\mu = 479.8 \pm 0.4$$

$$\sigma = 23.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.10 \%$$

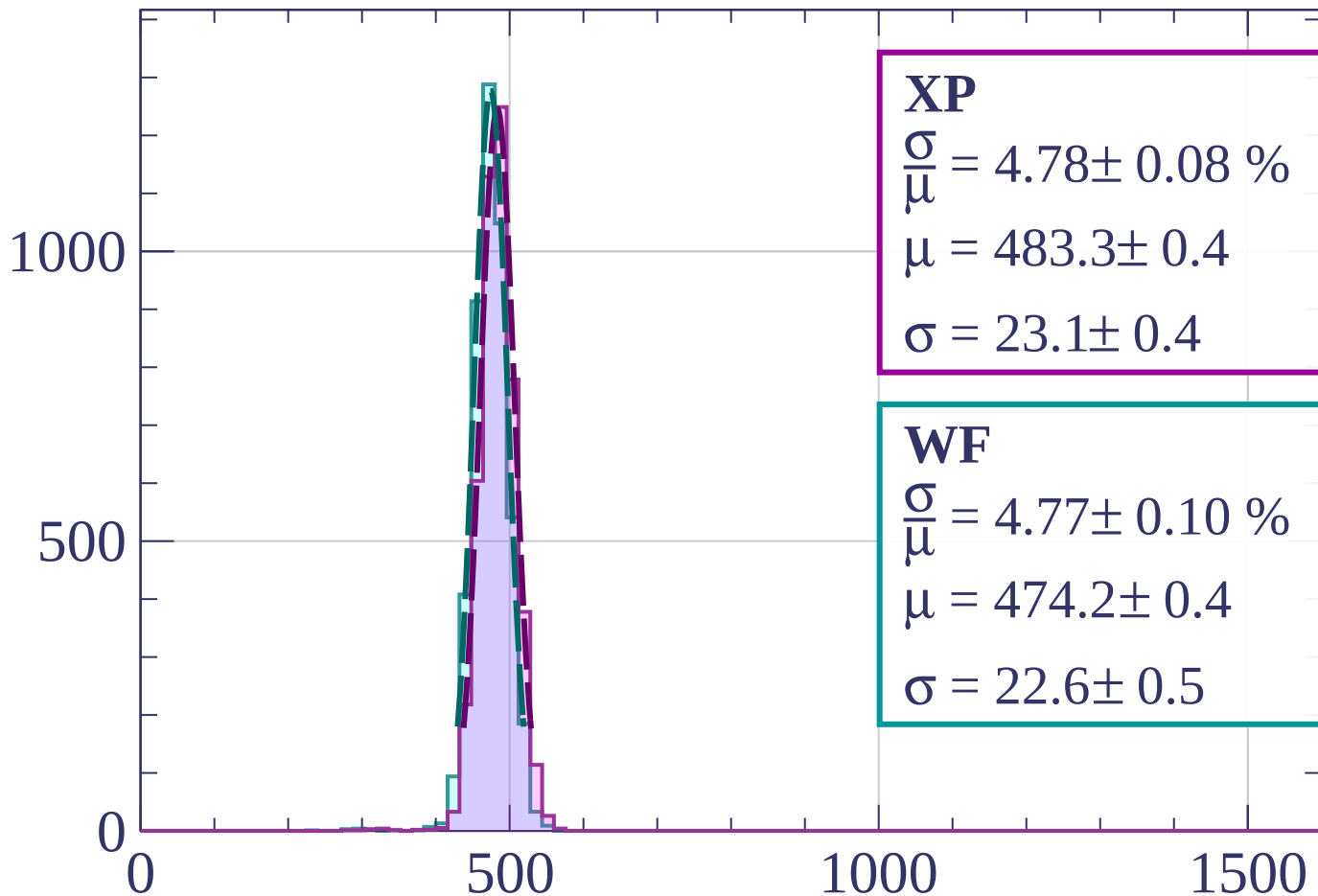
$$\mu = 470.4 \pm 0.4$$

$$\sigma = 22.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.08 \%$$

$$\mu = 483.3 \pm 0.4$$

$$\sigma = 23.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.10 \%$$

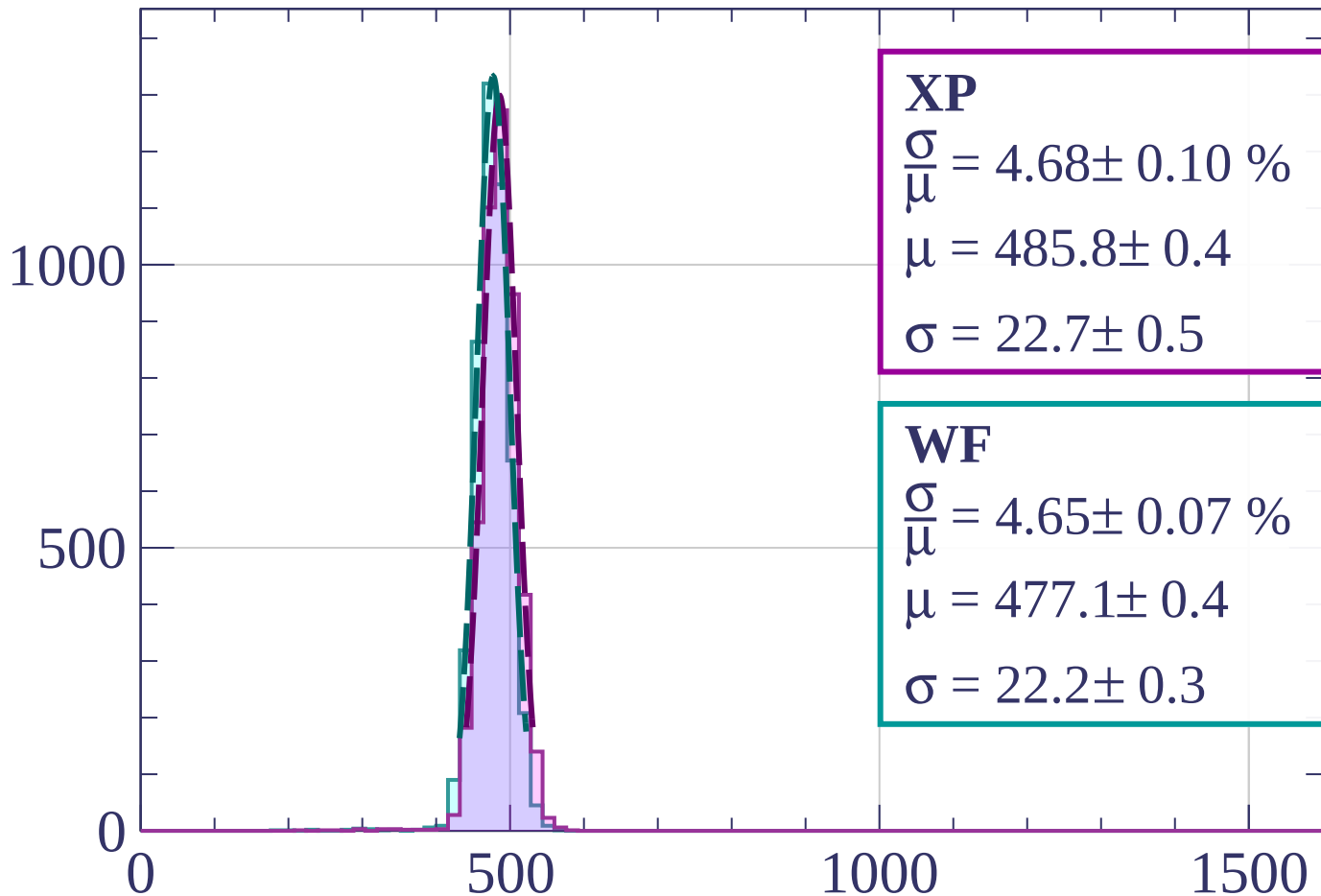
$$\mu = 474.2 \pm 0.4$$

$$\sigma = 22.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 4.68 \pm 0.10 \%$$

$$\mu = 485.8 \pm 0.4$$

$$\sigma = 22.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.65 \pm 0.07 \%$$

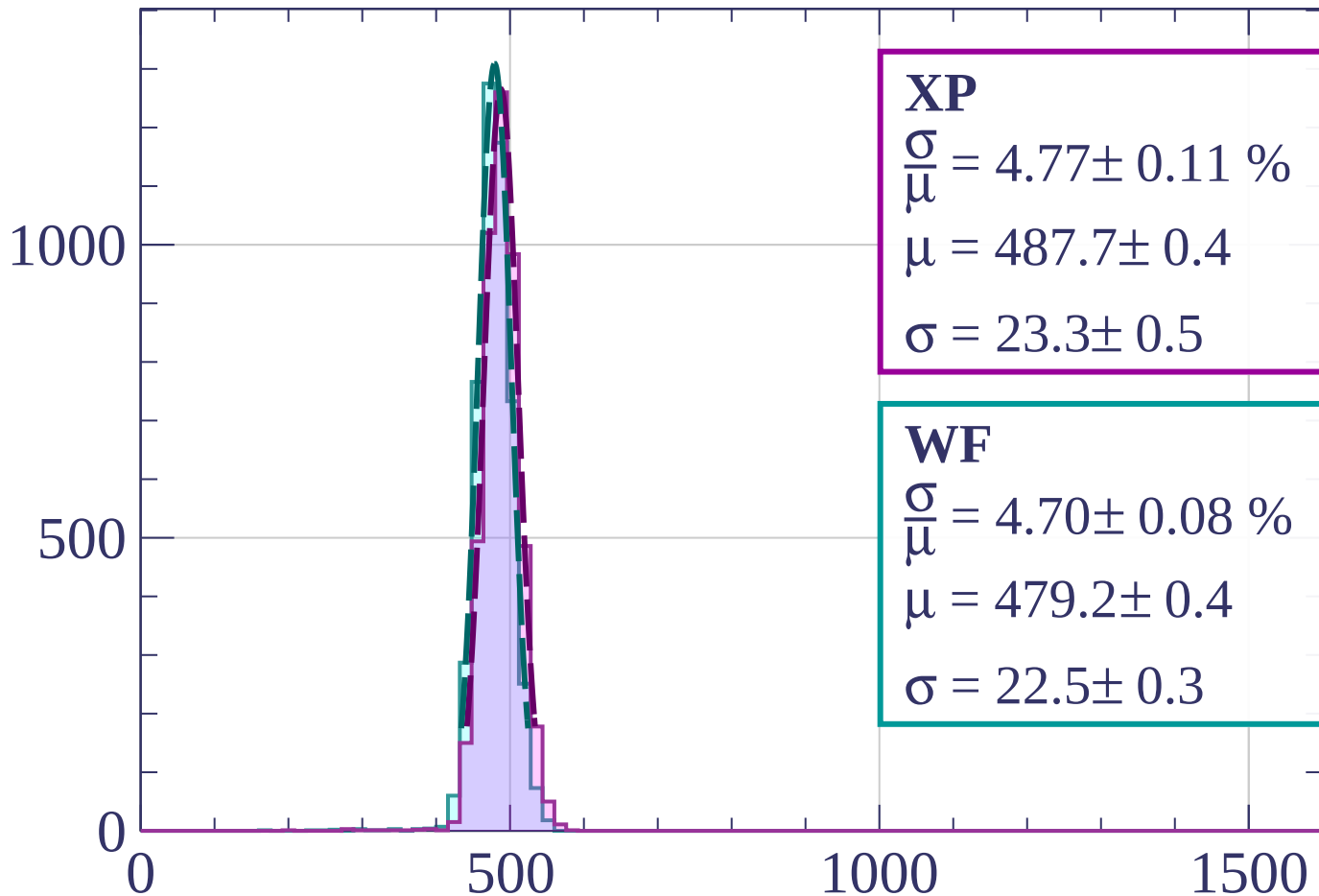
$$\mu = 477.1 \pm 0.4$$

$$\sigma = 22.2 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 4.77 \pm 0.11 \%$$

$$\mu = 487.7 \pm 0.4$$

$$\sigma = 23.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.70 \pm 0.08 \%$$

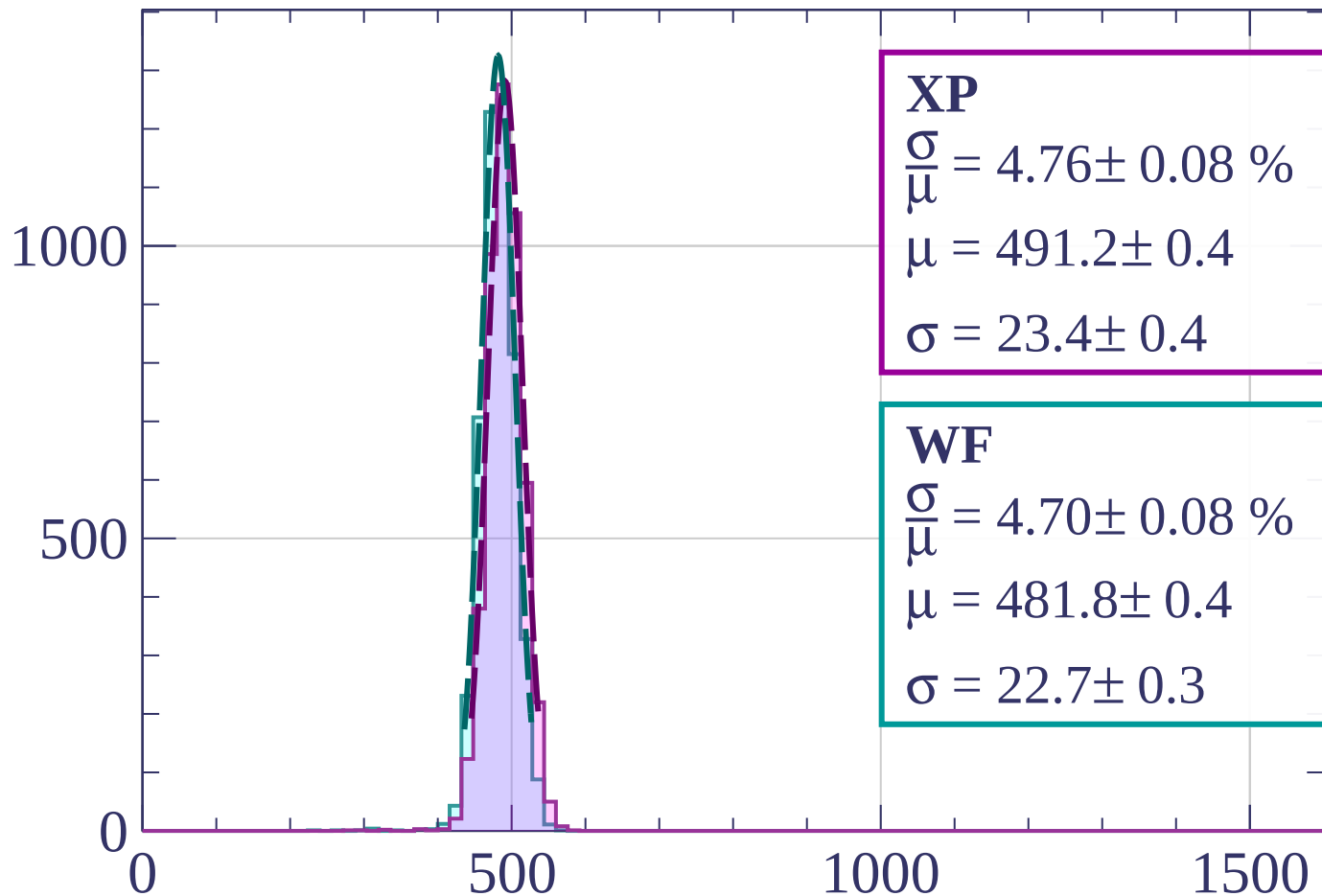
$$\mu = 479.2 \pm 0.4$$

$$\sigma = 22.5 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 4.76 \pm 0.08 \%$$

$$\mu = 491.2 \pm 0.4$$

$$\sigma = 23.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.70 \pm 0.08 \%$$

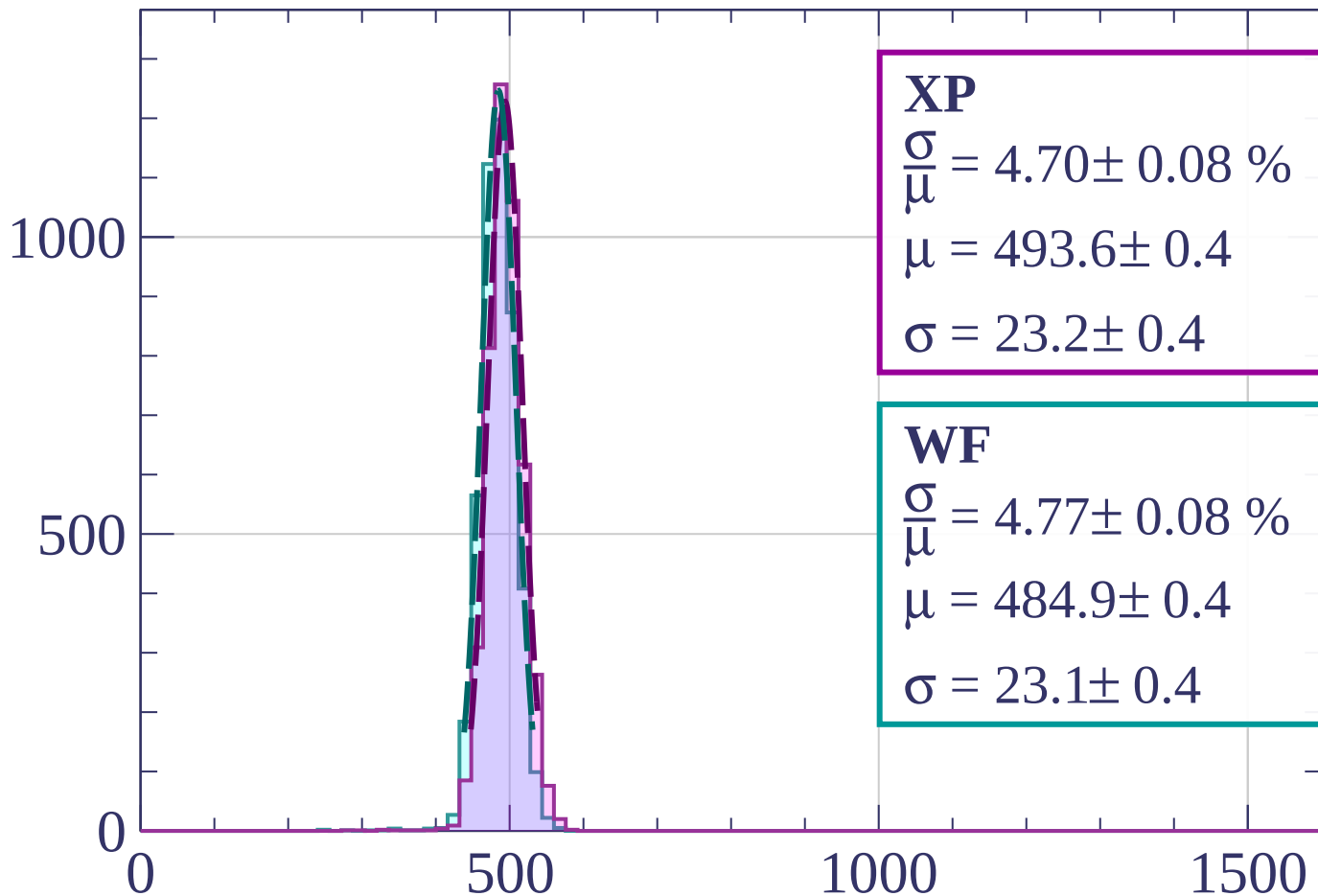
$$\mu = 481.8 \pm 0.4$$

$$\sigma = 22.7 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 4.70 \pm 0.08 \%$$

$$\mu = 493.6 \pm 0.4$$

$$\sigma = 23.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.08 \%$$

$$\mu = 484.9 \pm 0.4$$

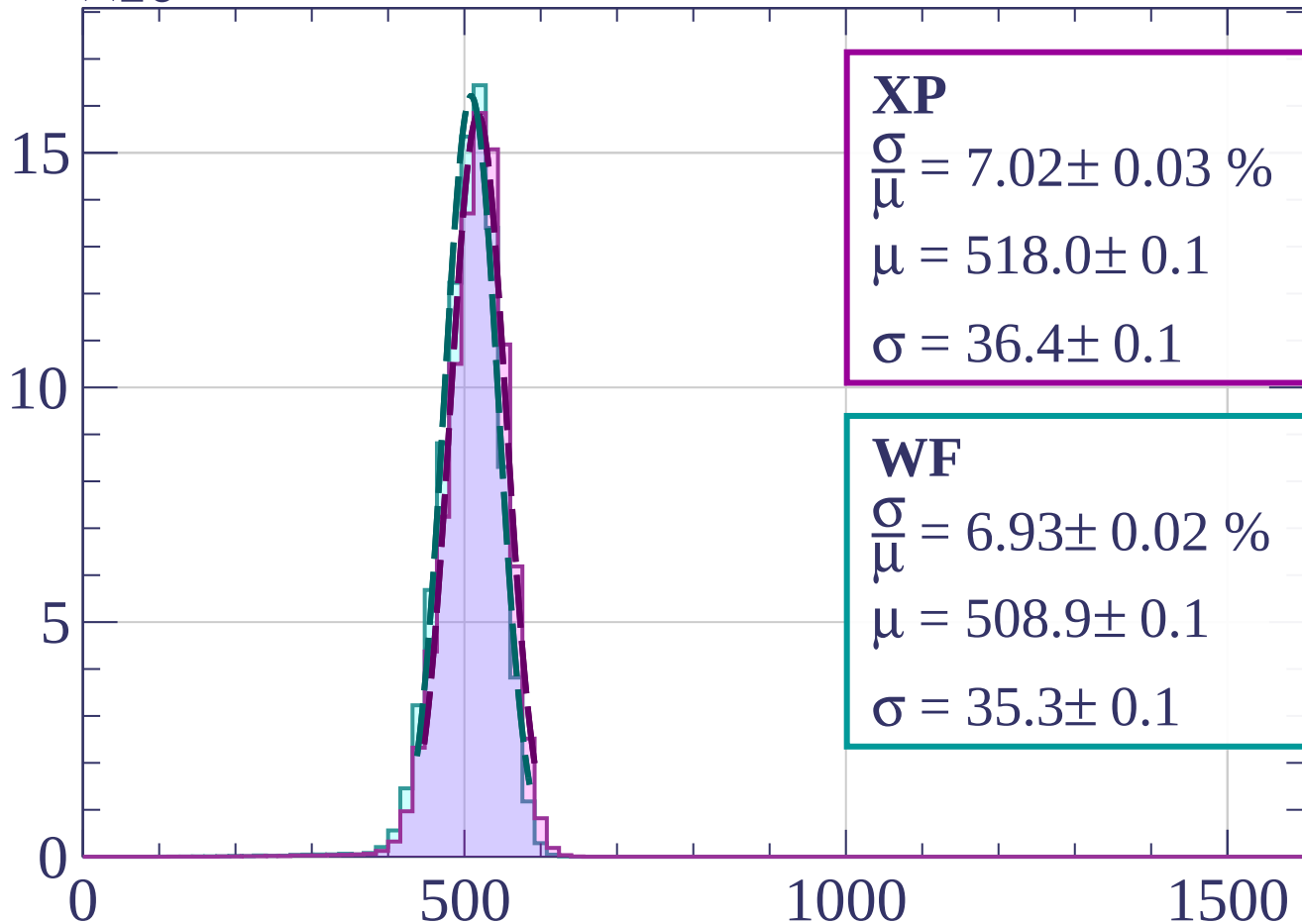
$$\sigma = 23.1 \pm 0.4$$

dE/dx [ADC counts/cm]

Count

$\times 10^3$

Energy loss | $0 < \phi < 3$



XP

$$\frac{\sigma}{\mu} = 7.02 \pm 0.03 \%$$

$$\mu = 518.0 \pm 0.1$$

$$\sigma = 36.4 \pm 0.1$$

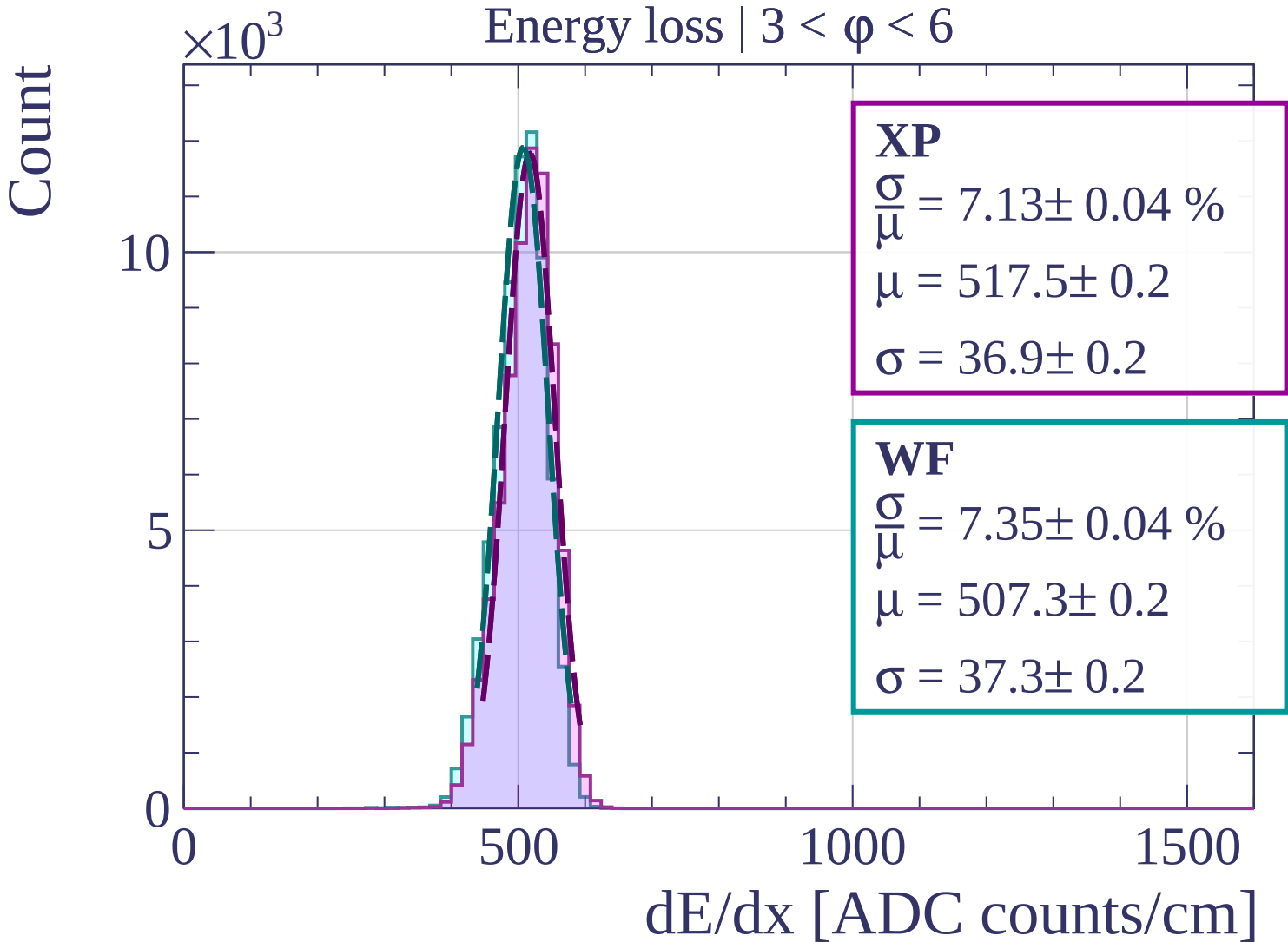
WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.02 \%$$

$$\mu = 508.9 \pm 0.1$$

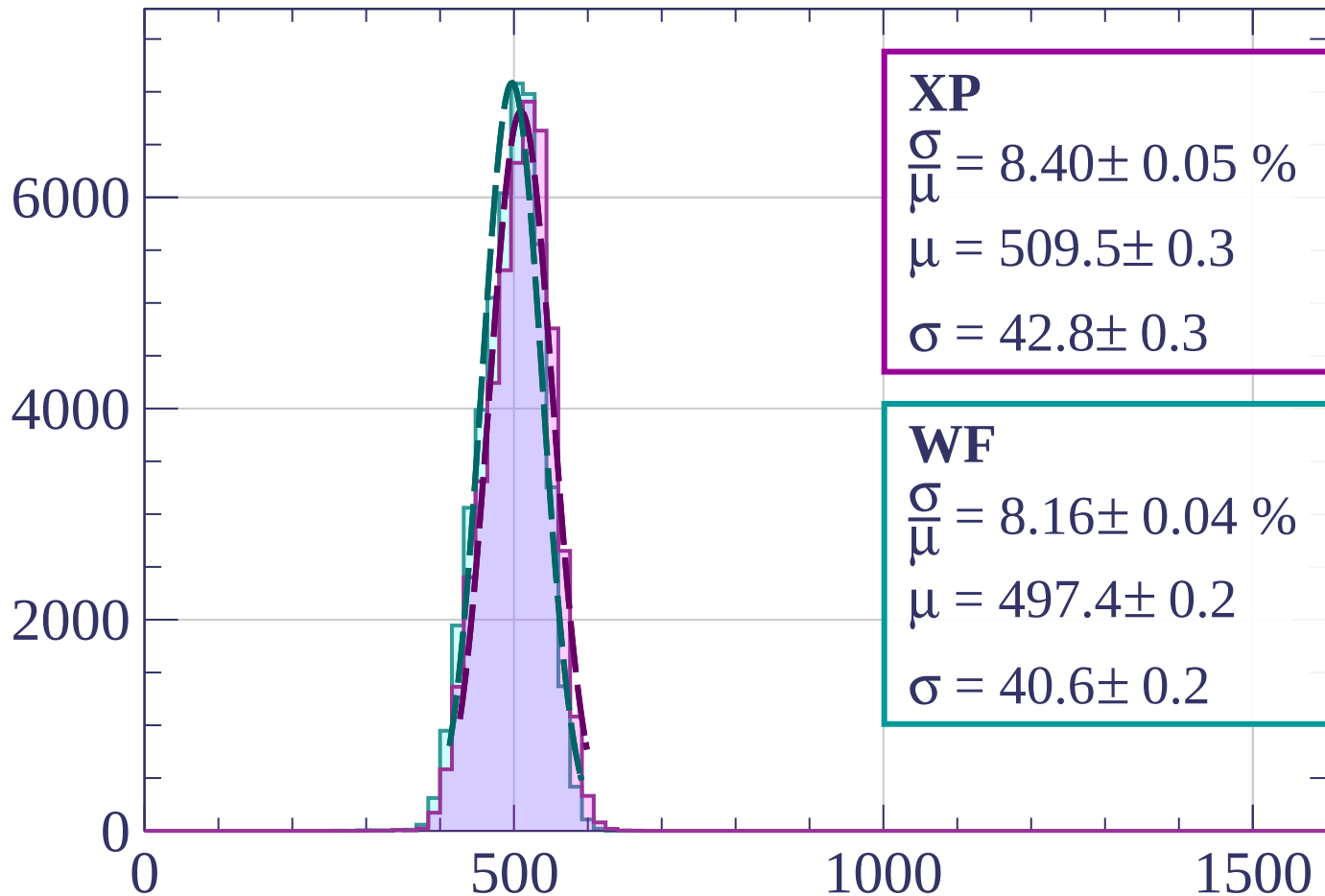
$$\sigma = 35.3 \pm 0.1$$

dE/dx [ADC counts/cm]



Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.05 \%$$

$$\mu = 509.5 \pm 0.3$$

$$\sigma = 42.8 \pm 0.3$$

WF

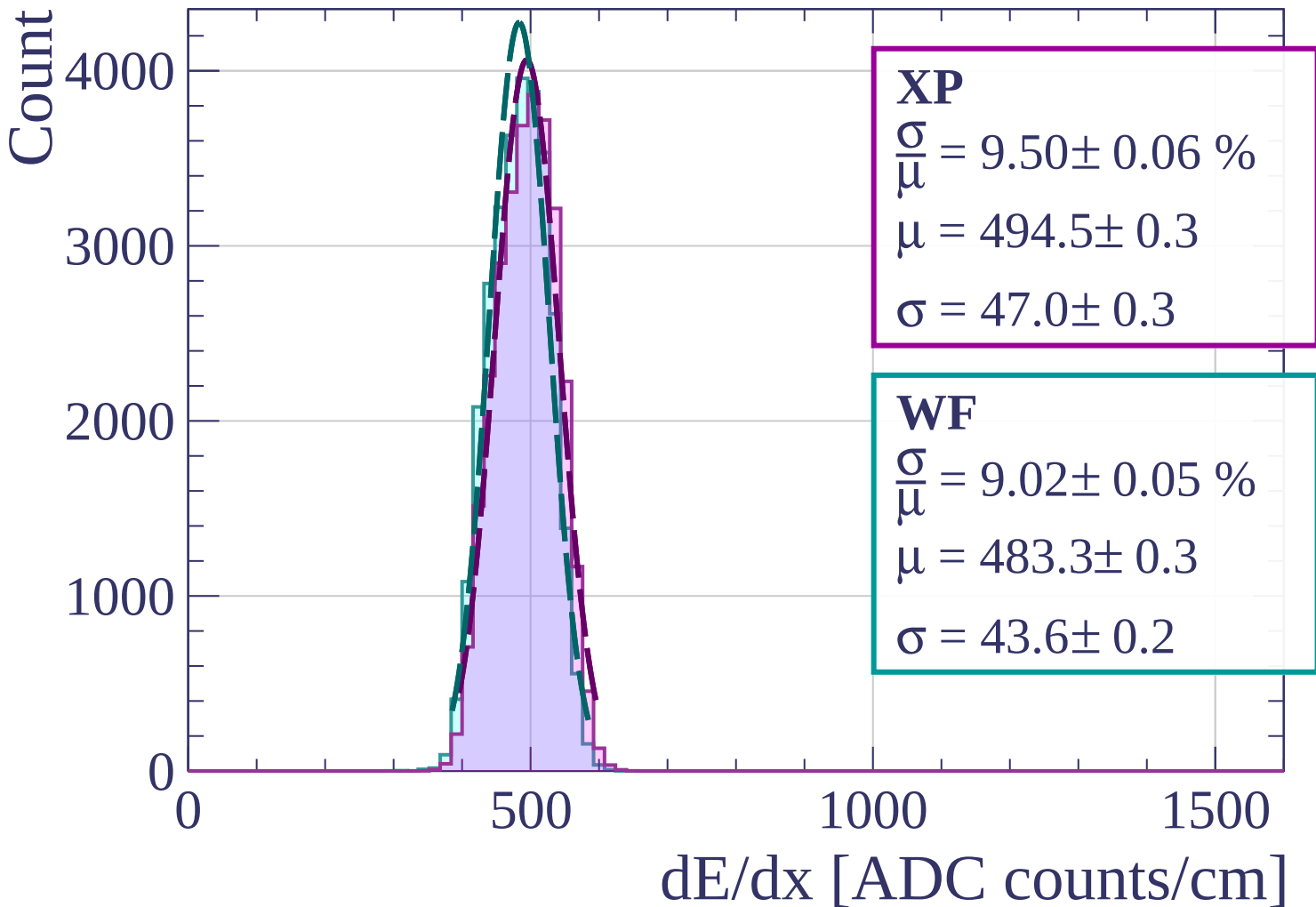
$$\frac{\sigma}{\mu} = 8.16 \pm 0.04 \%$$

$$\mu = 497.4 \pm 0.2$$

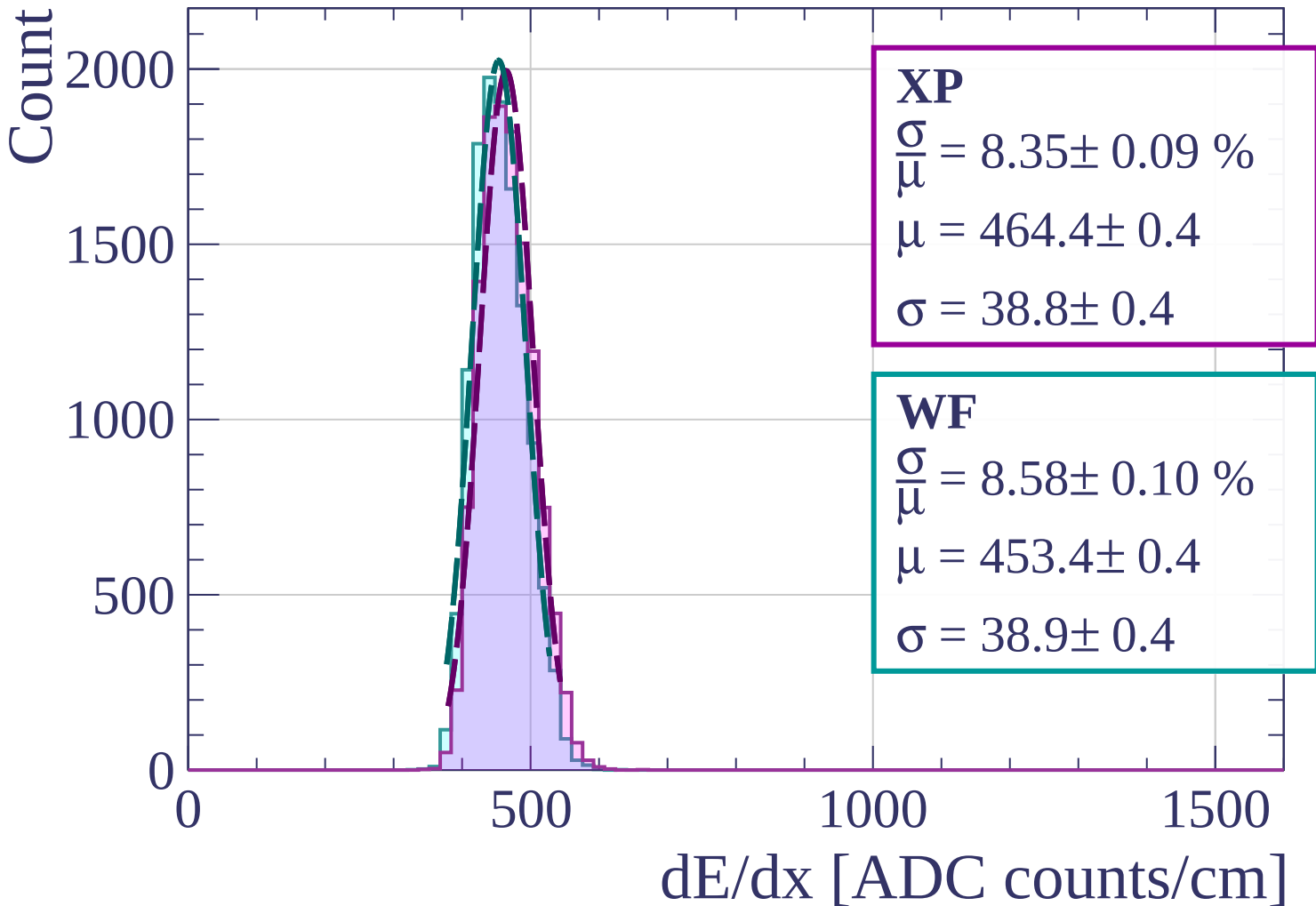
$$\sigma = 40.6 \pm 0.2$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

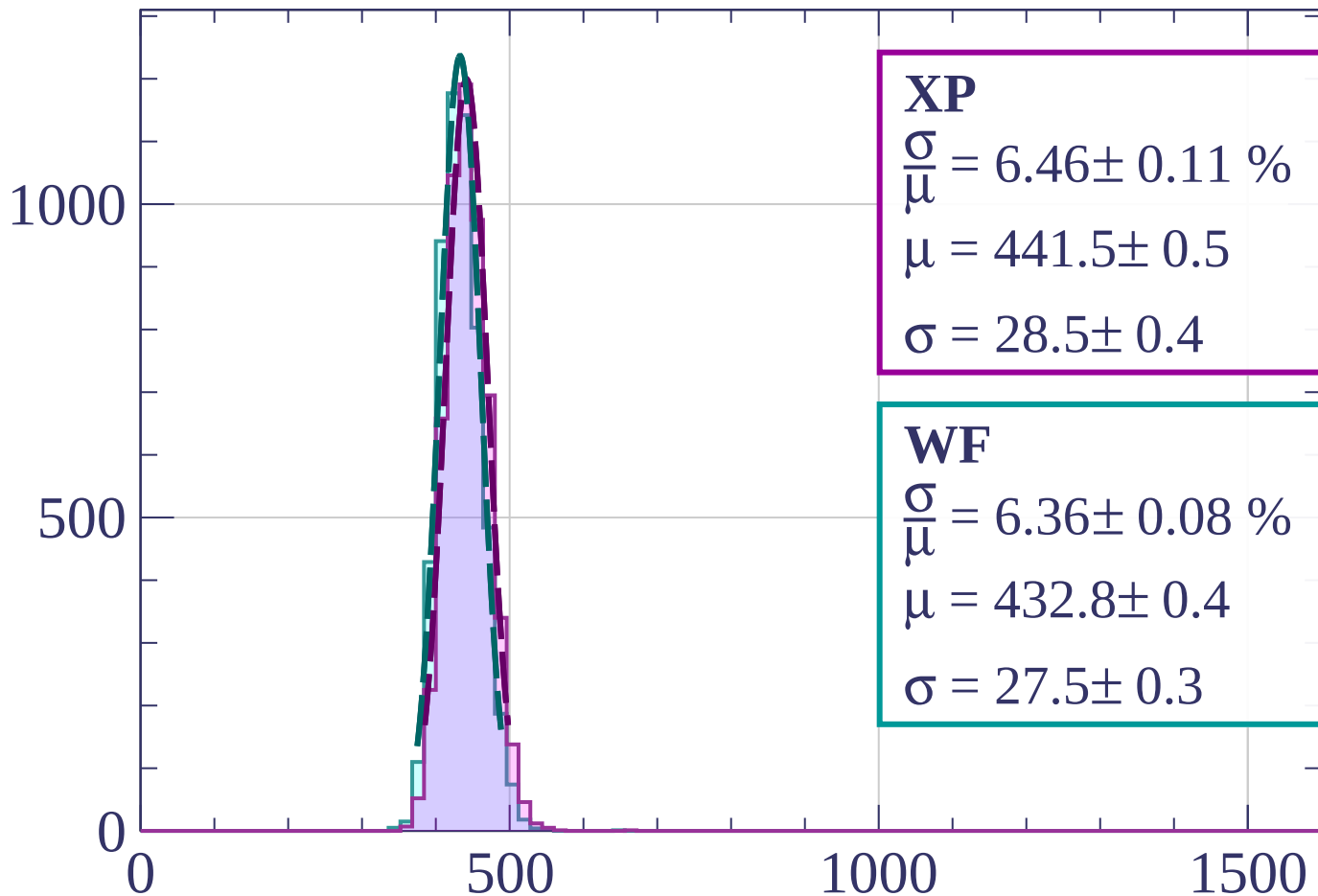


Energy loss | $12 < \phi < 15$



Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.46 \pm 0.11 \%$$

$$\mu = 441.5 \pm 0.5$$

$$\sigma = 28.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 6.36 \pm 0.08 \%$$

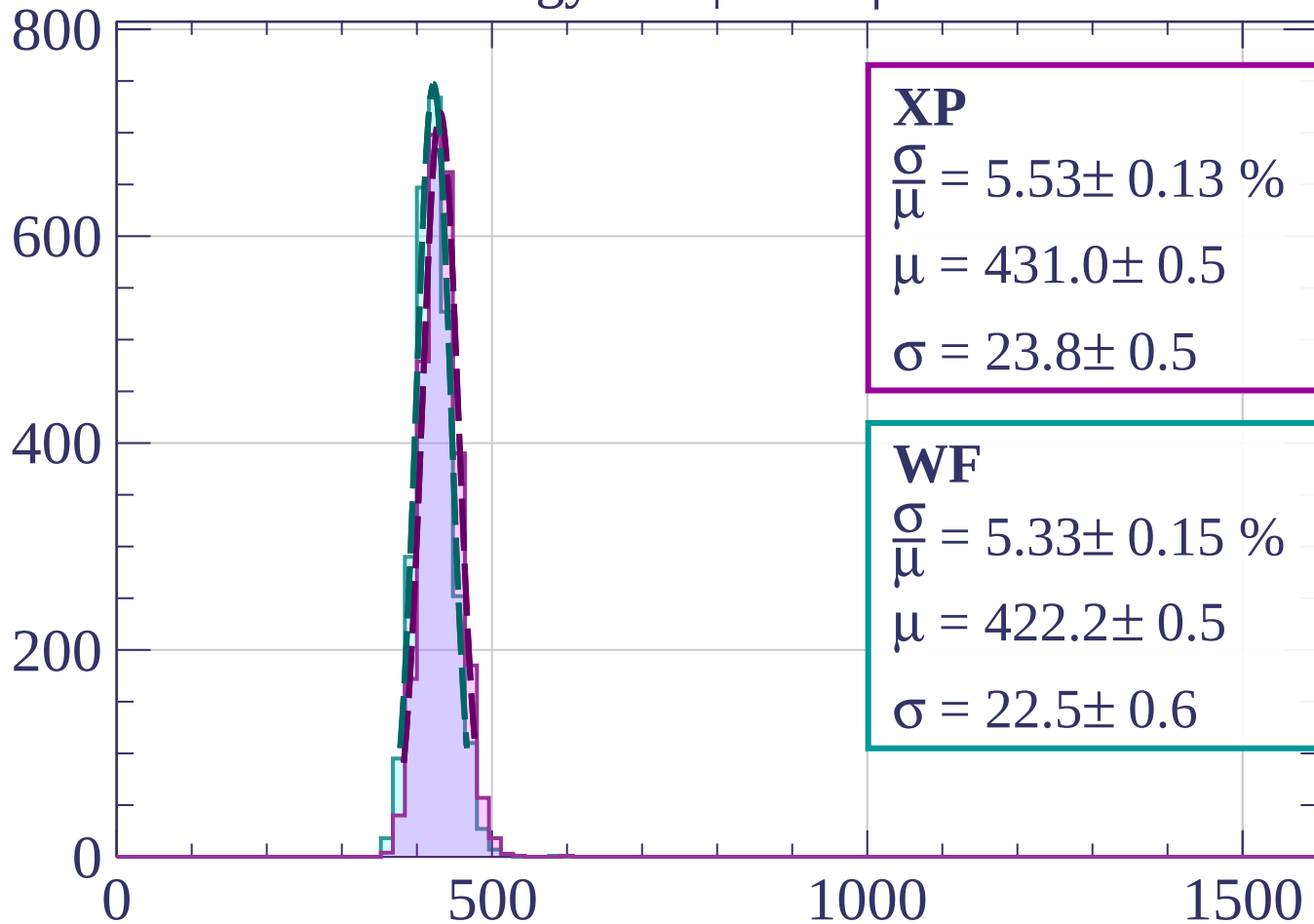
$$\mu = 432.8 \pm 0.4$$

$$\sigma = 27.5 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.53 \pm 0.13 \%$$

$$\mu = 431.0 \pm 0.5$$

$$\sigma = 23.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.33 \pm 0.15 \%$$

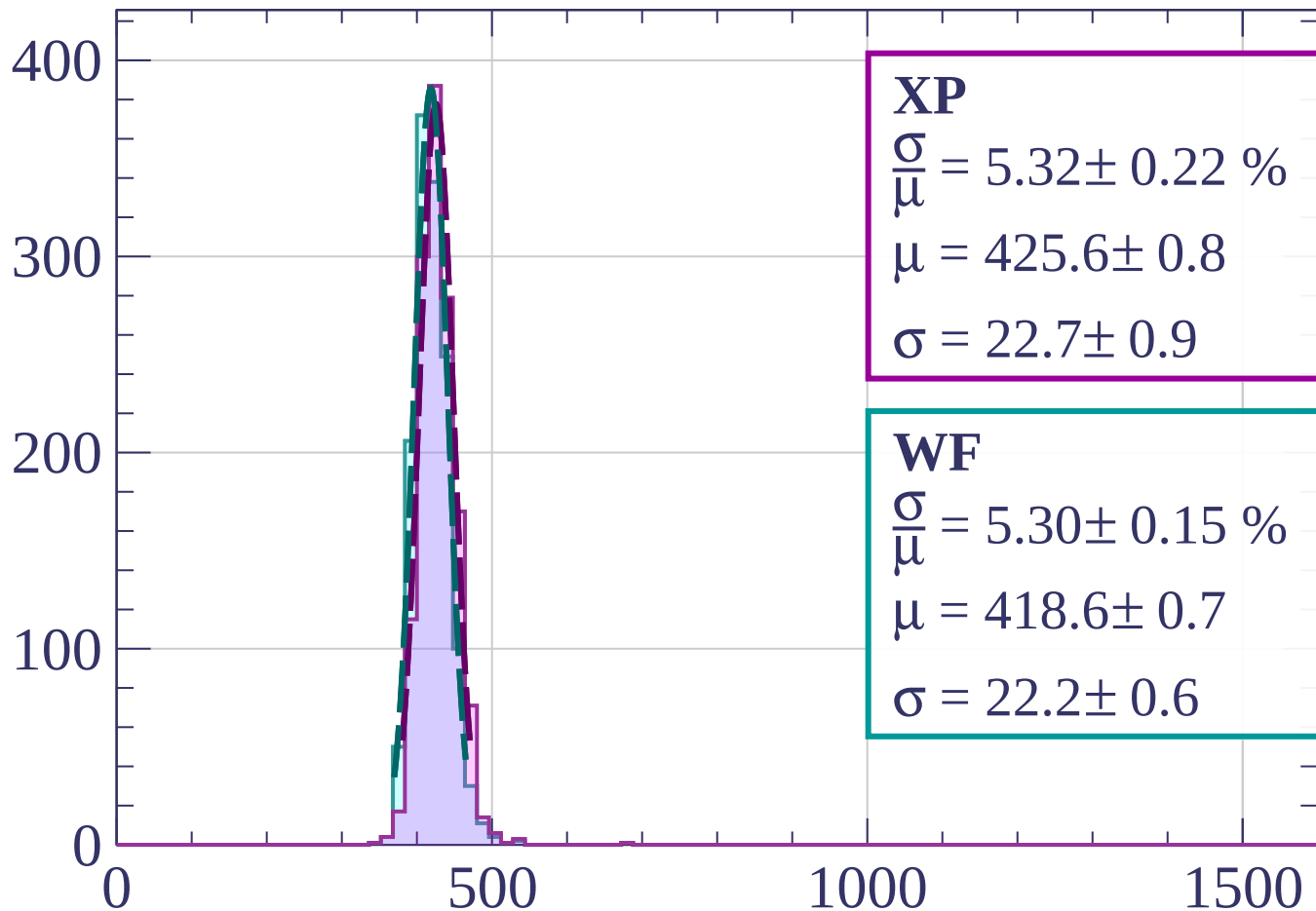
$$\mu = 422.2 \pm 0.5$$

$$\sigma = 22.5 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

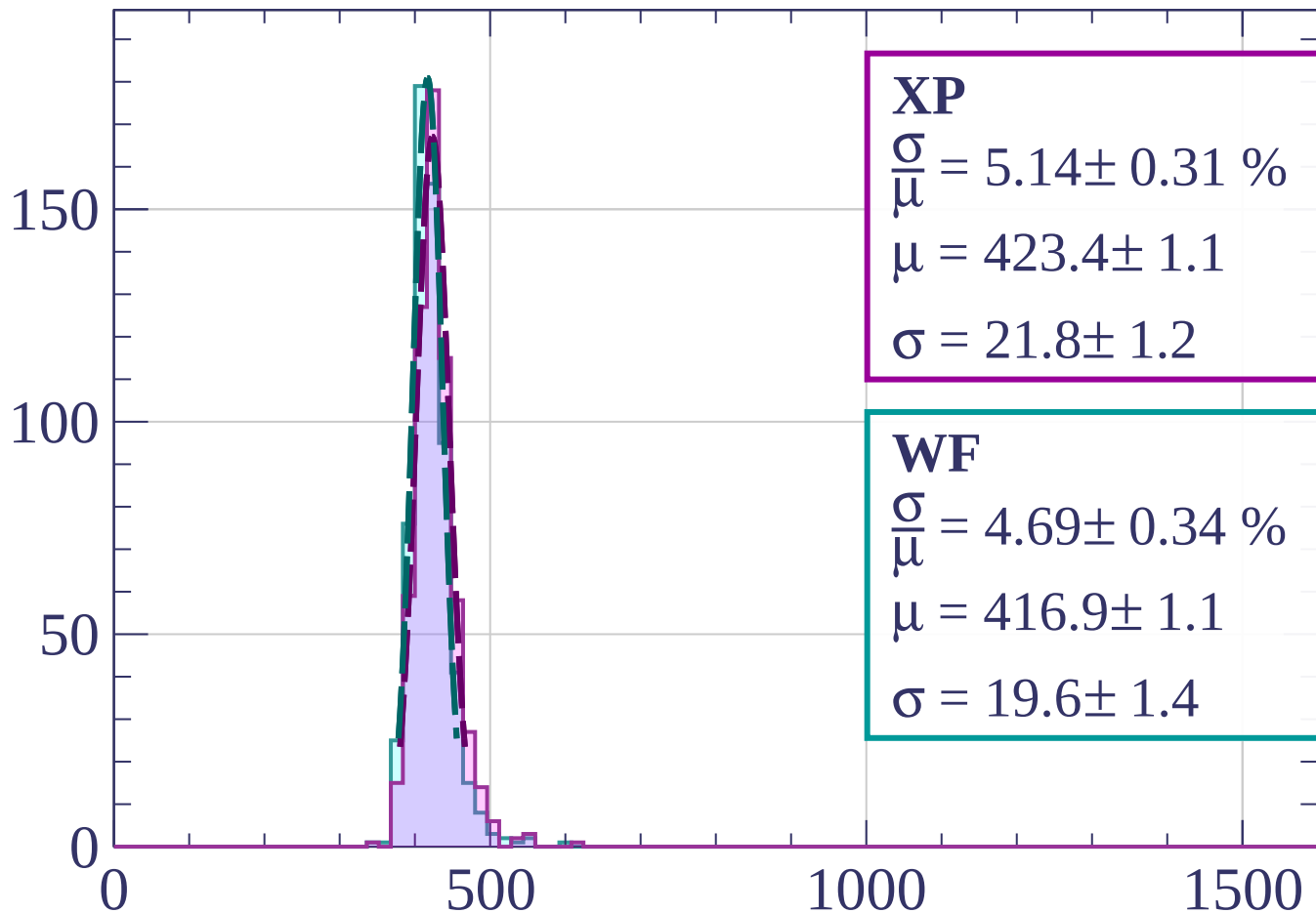
Count



dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.31 \%$$

$$\mu = 423.4 \pm 1.1$$

$$\sigma = 21.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 4.69 \pm 0.34 \%$$

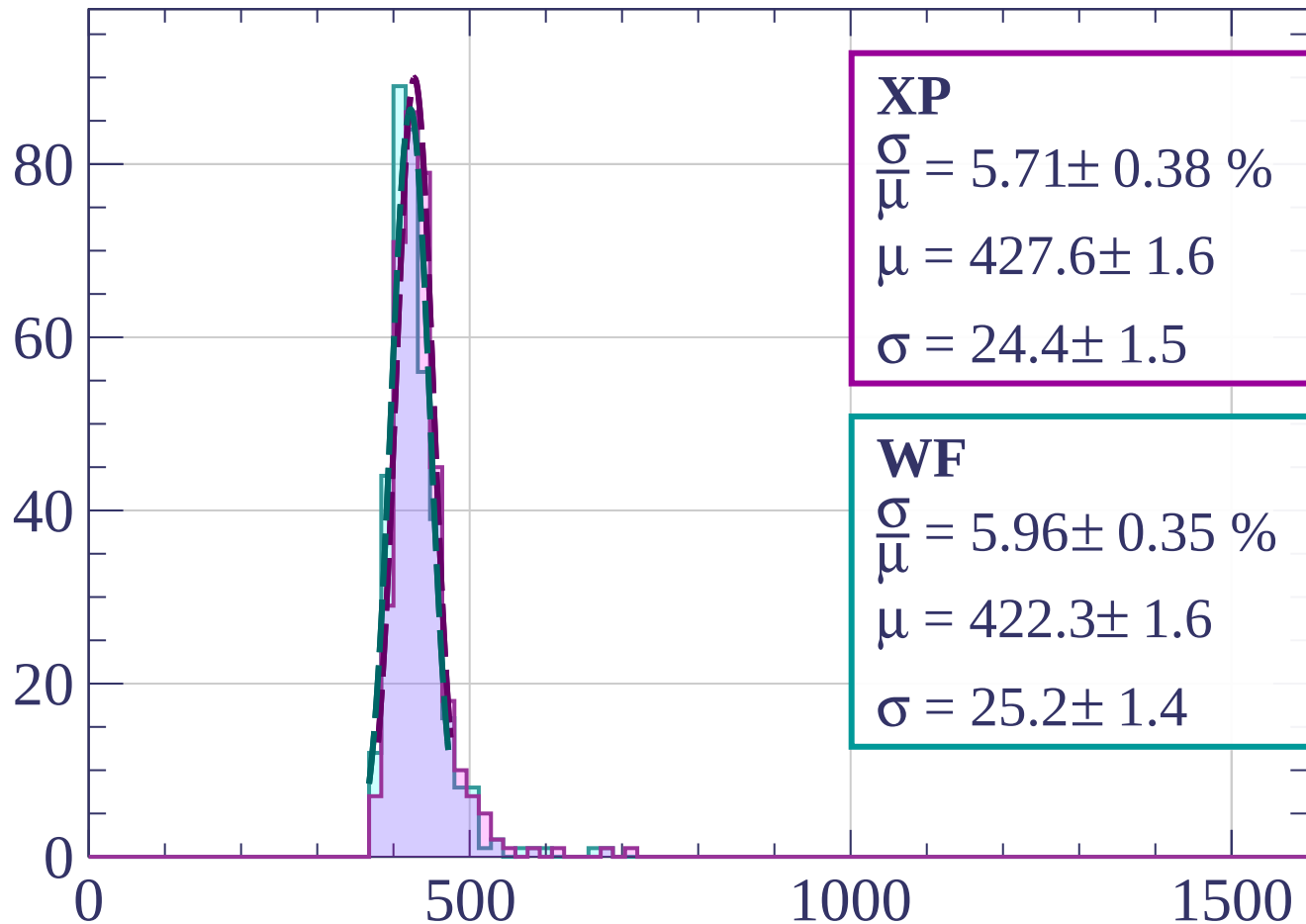
$$\mu = 416.9 \pm 1.1$$

$$\sigma = 19.6 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 5.71 \pm 0.38 \%$$

$$\mu = 427.6 \pm 1.6$$

$$\sigma = 24.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 5.96 \pm 0.35 \%$$

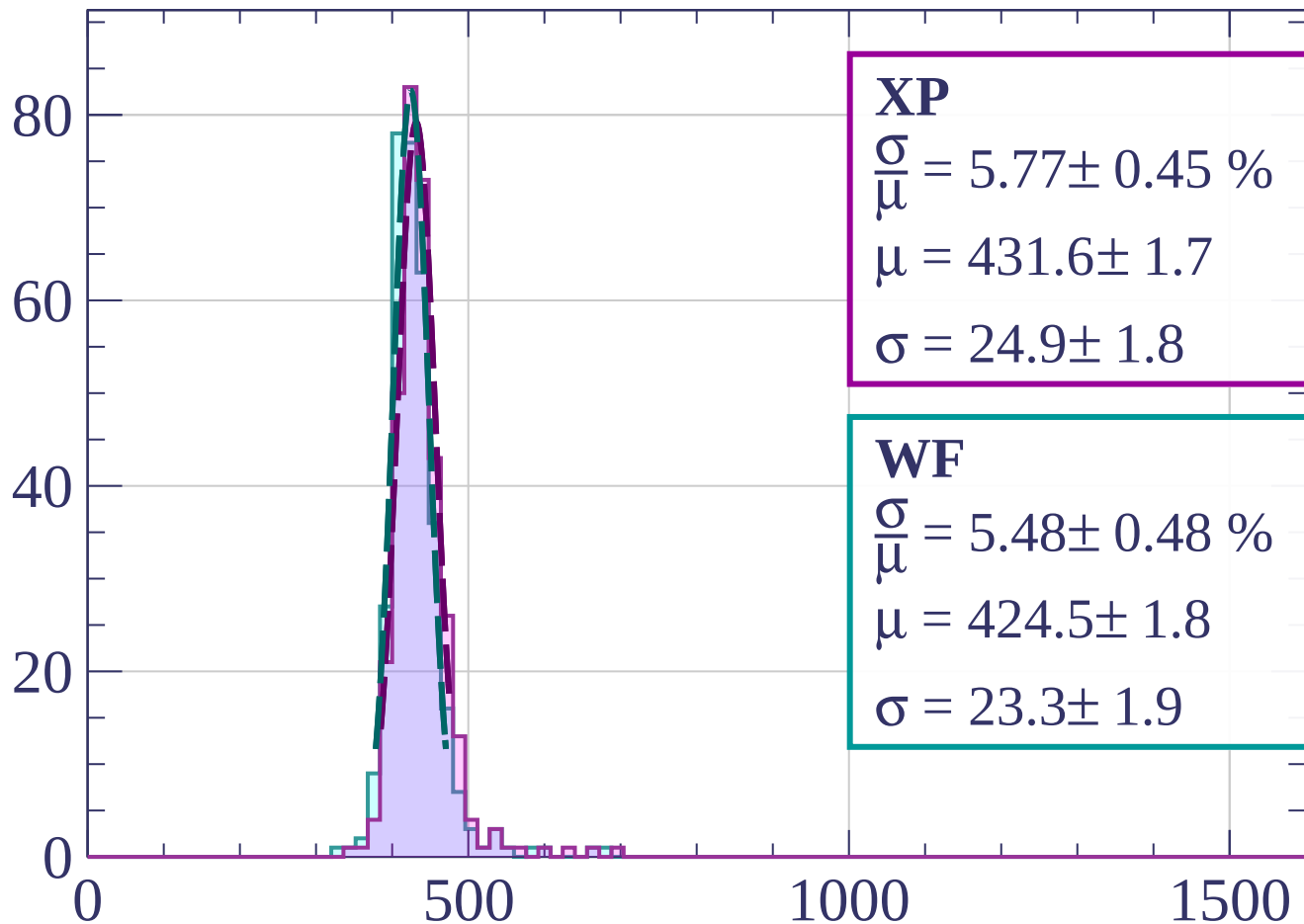
$$\mu = 422.3 \pm 1.6$$

$$\sigma = 25.2 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 5.77 \pm 0.45 \%$$

$$\mu = 431.6 \pm 1.7$$

$$\sigma = 24.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 5.48 \pm 0.48 \%$$

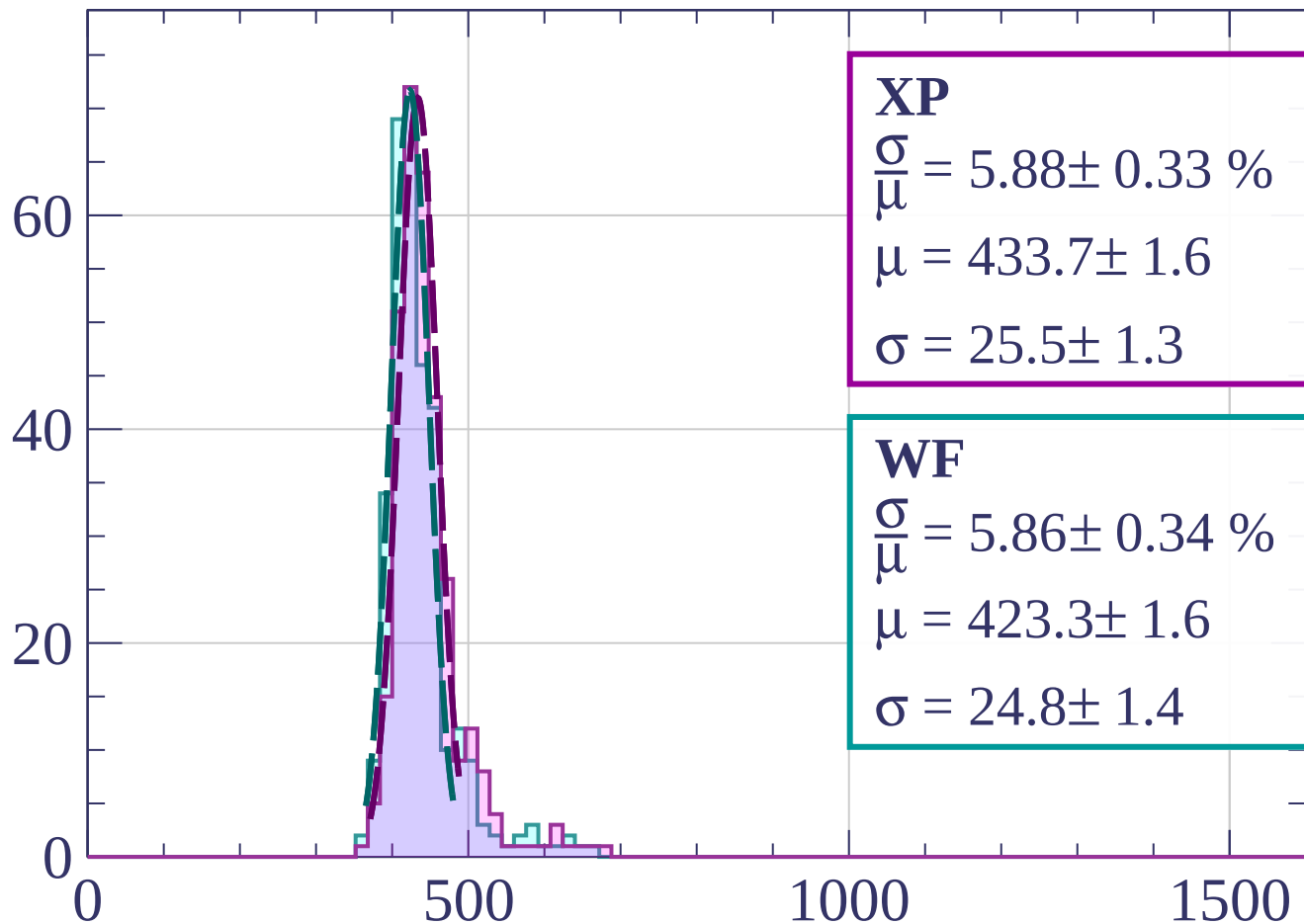
$$\mu = 424.5 \pm 1.8$$

$$\sigma = 23.3 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 5.88 \pm 0.33 \%$$

$$\mu = 433.7 \pm 1.6$$

$$\sigma = 25.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 5.86 \pm 0.34 \%$$

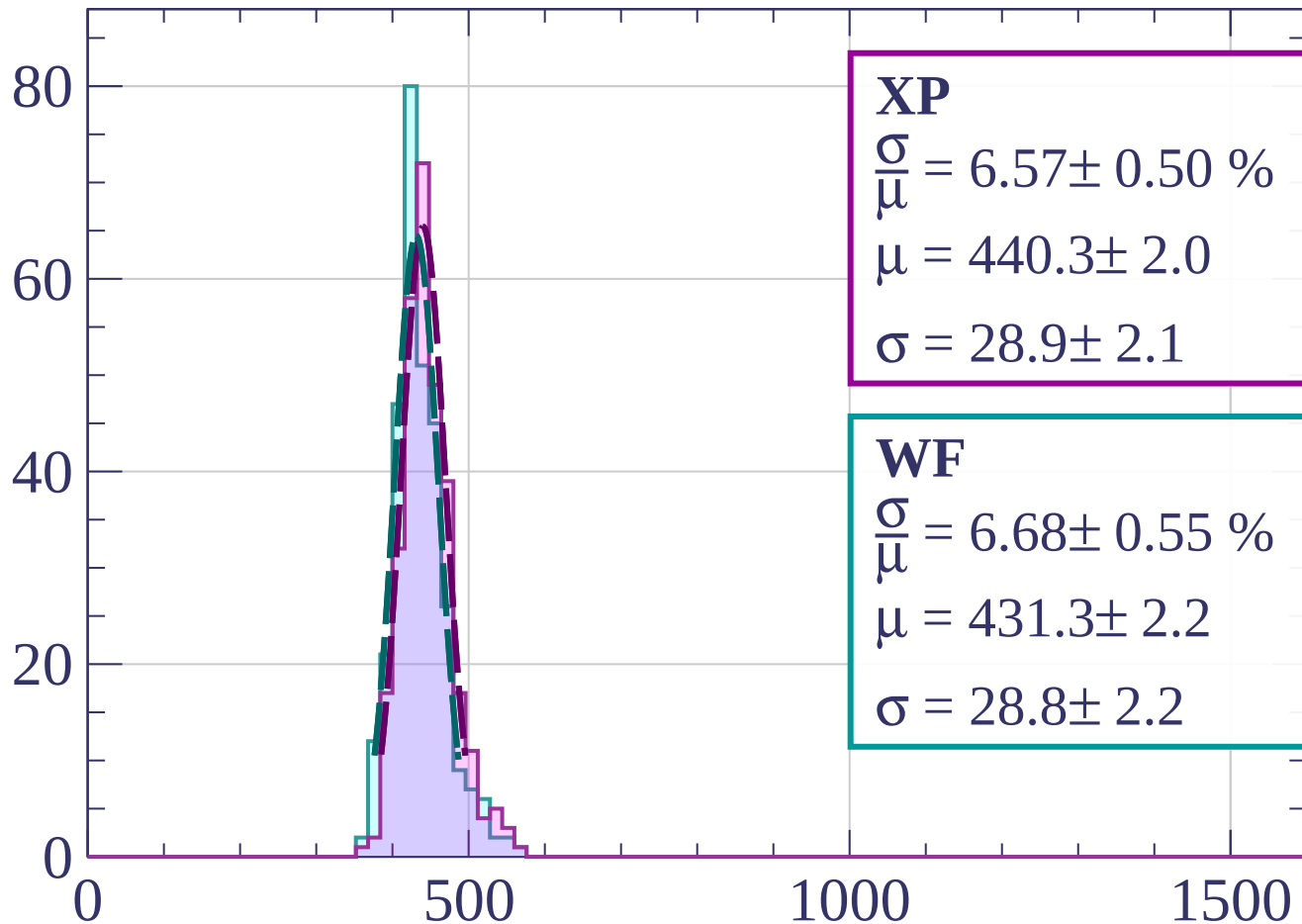
$$\mu = 423.3 \pm 1.6$$

$$\sigma = 24.8 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 6.57 \pm 0.50 \%$$

$$\mu = 440.3 \pm 2.0$$

$$\sigma = 28.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.55 \%$$

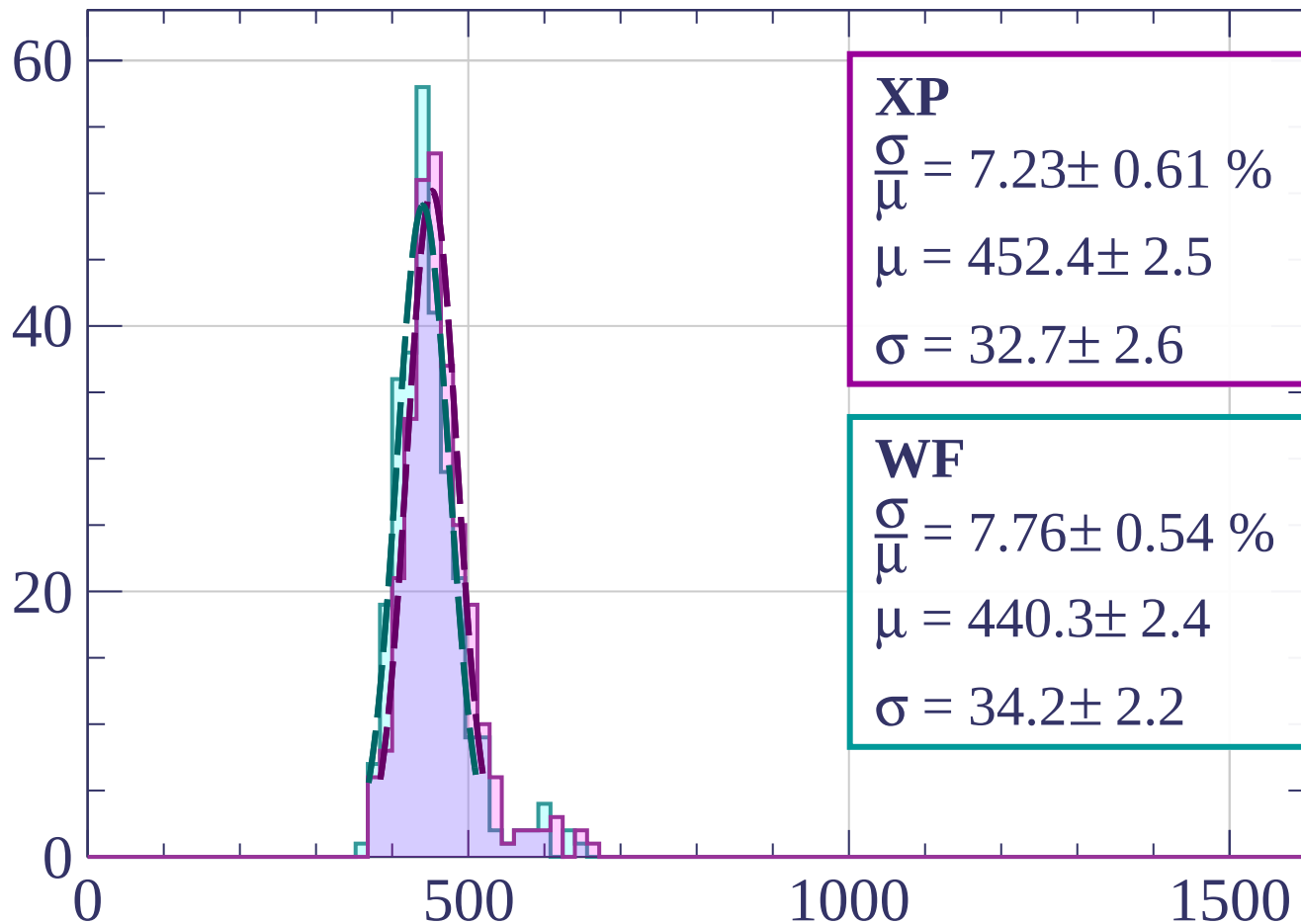
$$\mu = 431.3 \pm 2.2$$

$$\sigma = 28.8 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.23 \pm 0.61 \%$$

$$\mu = 452.4 \pm 2.5$$

$$\sigma = 32.7 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.76 \pm 0.54 \%$$

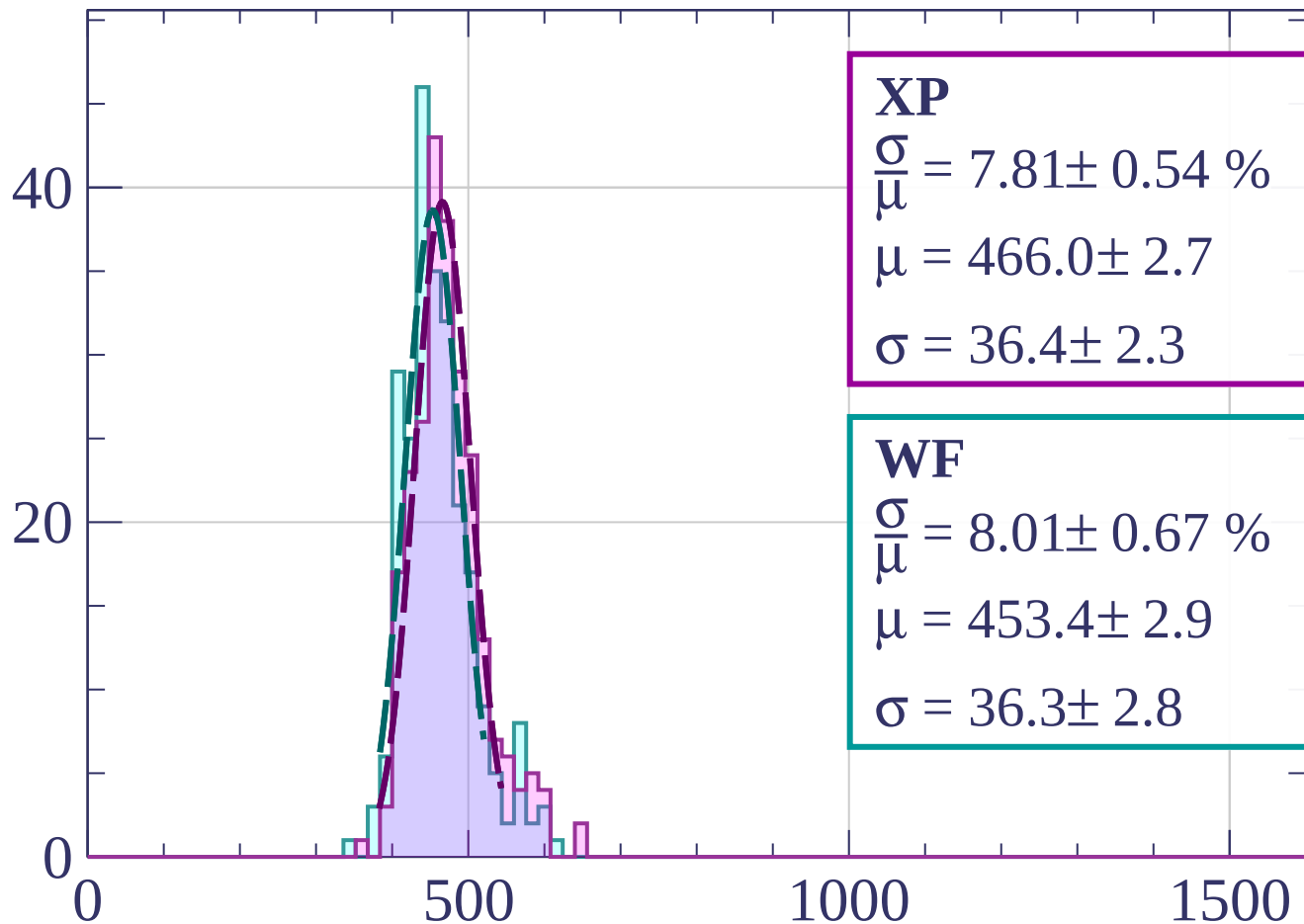
$$\mu = 440.3 \pm 2.4$$

$$\sigma = 34.2 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.81 \pm 0.54 \%$$

$$\mu = 466.0 \pm 2.7$$

$$\sigma = 36.4 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 8.01 \pm 0.67 \%$$

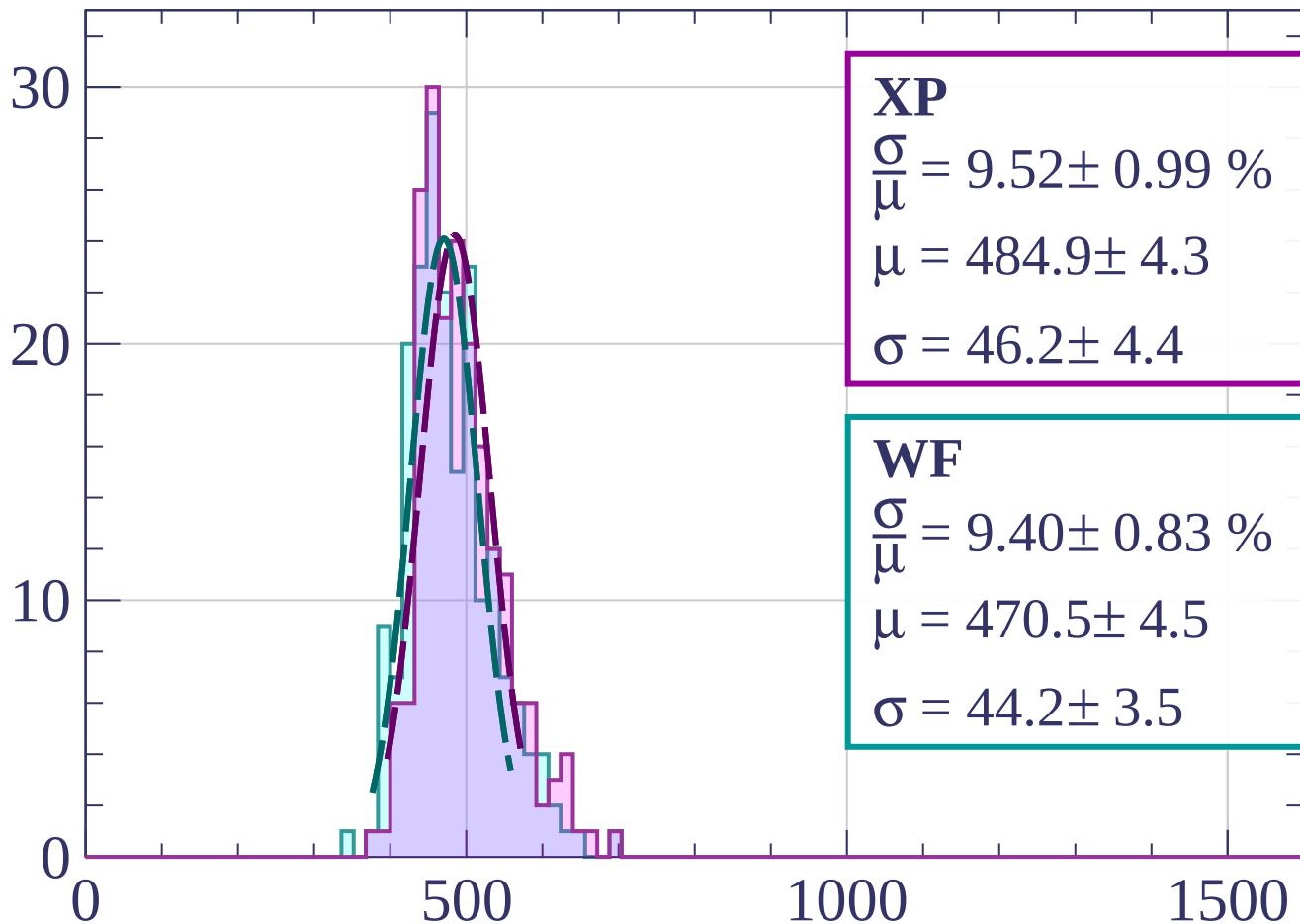
$$\mu = 453.4 \pm 2.9$$

$$\sigma = 36.3 \pm 2.8$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 9.52 \pm 0.99 \%$$

$$\mu = 484.9 \pm 4.3$$

$$\sigma = 46.2 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.83 \%$$

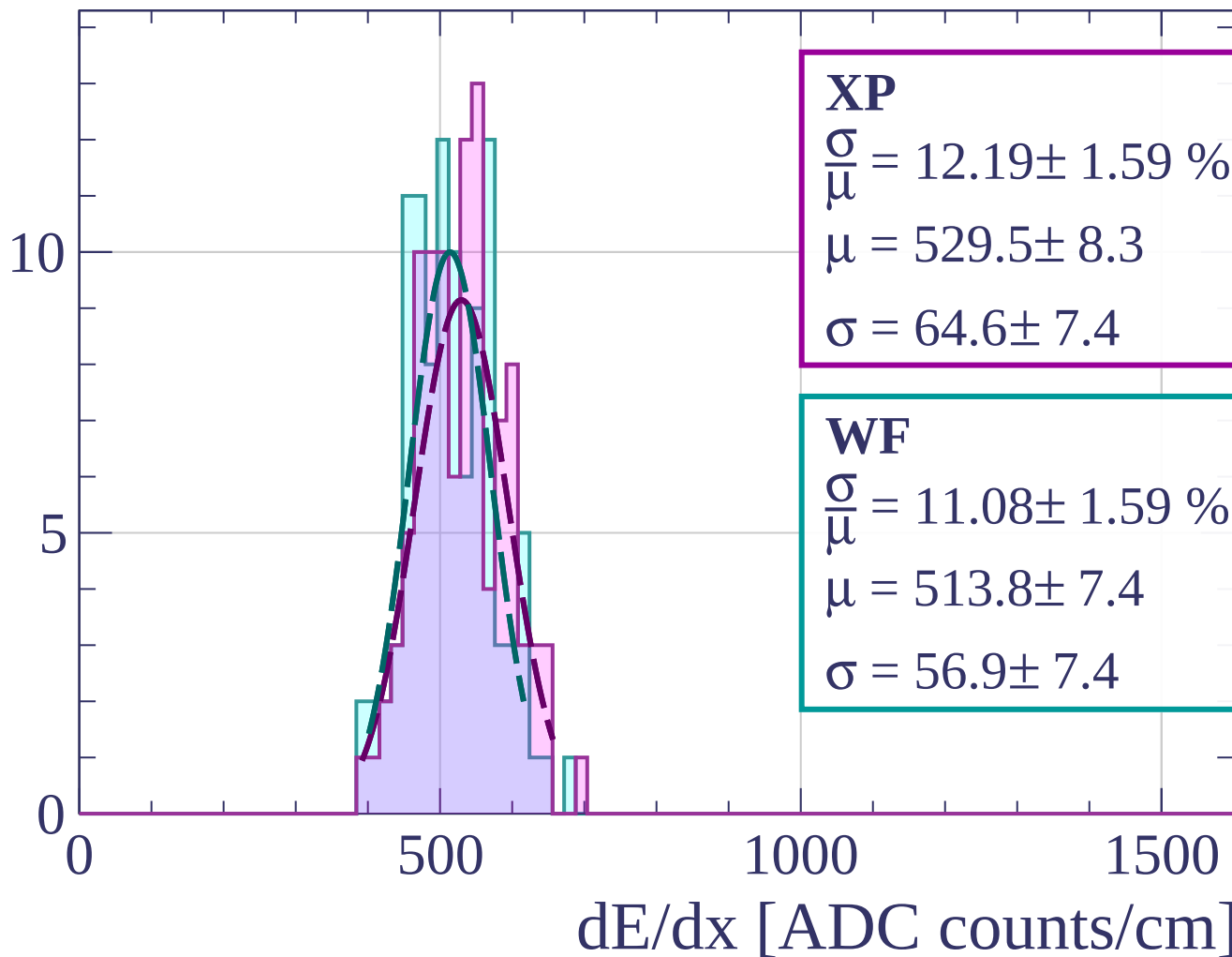
$$\mu = 470.5 \pm 4.5$$

$$\sigma = 44.2 \pm 3.5$$

dE/dx [ADC counts/cm]

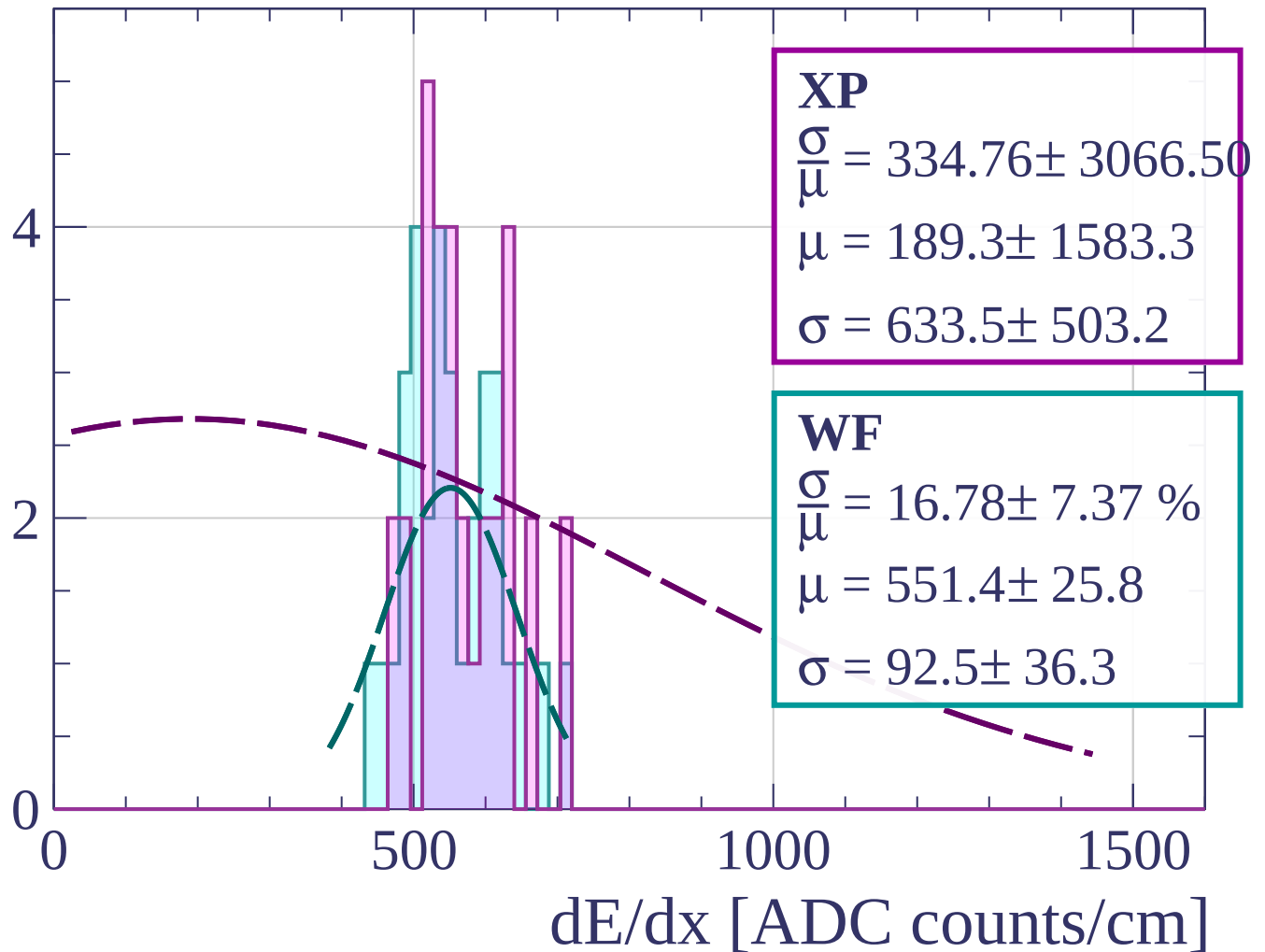
Energy loss | $48 < \phi < 51$

Count



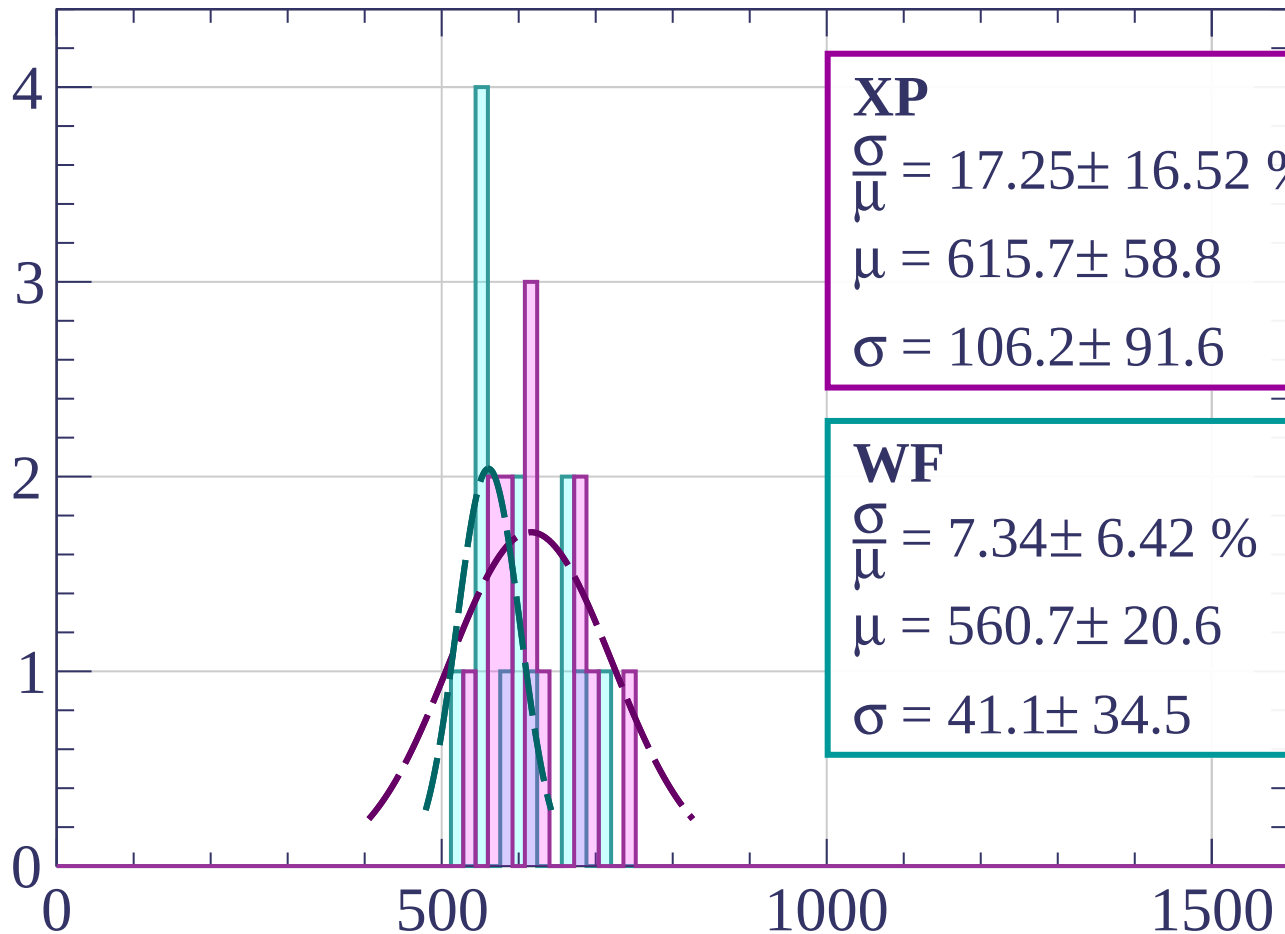
Energy loss | $51 < \phi < 54$

Count



Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 17.25 \pm 16.52 \%$$

$$\mu = 615.7 \pm 58.8$$

$$\sigma = 106.2 \pm 91.6$$

WF

$$\frac{\sigma}{\mu} = 7.34 \pm 6.42 \%$$

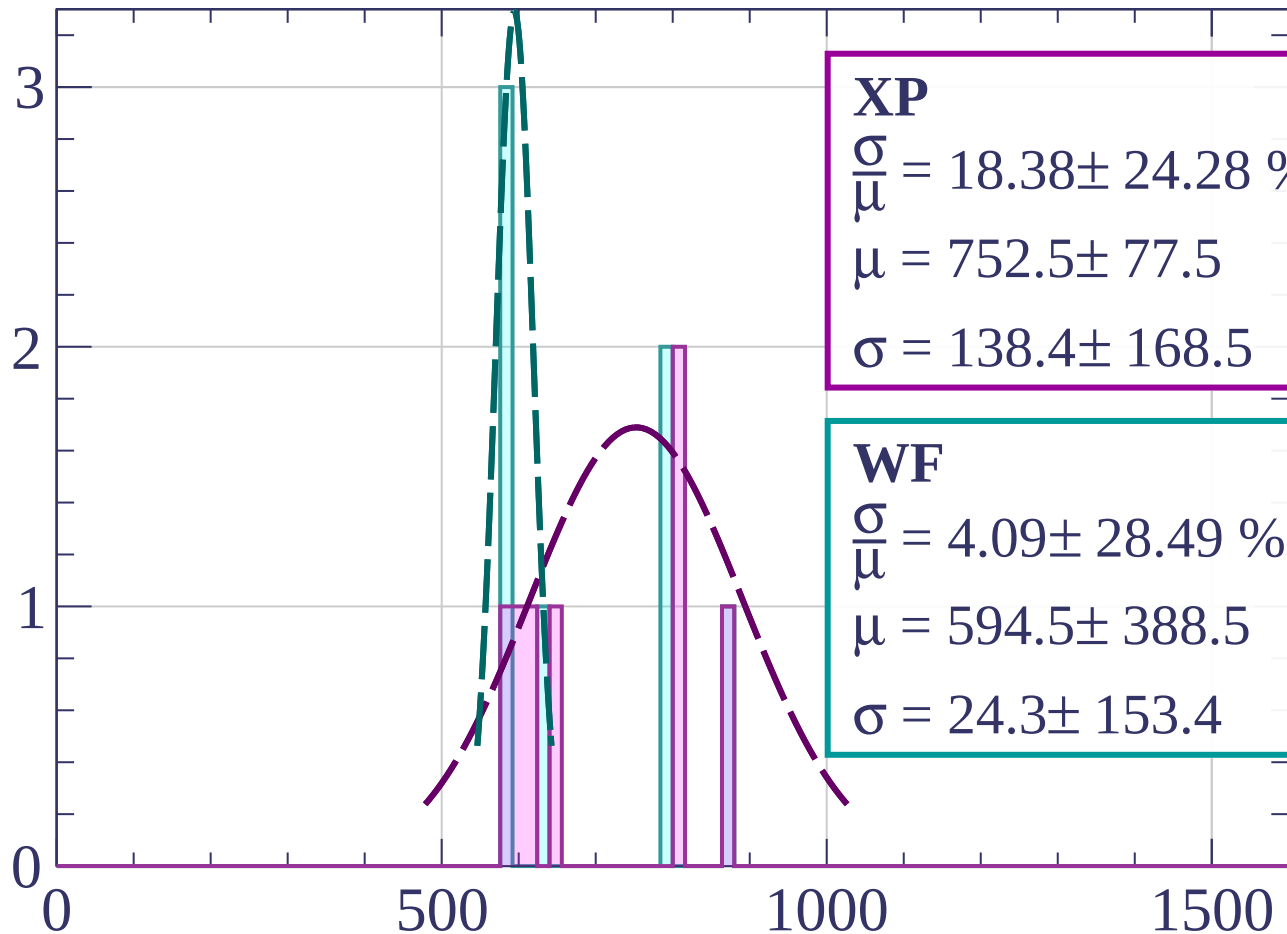
$$\mu = 560.7 \pm 20.6$$

$$\sigma = 41.1 \pm 34.5$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 18.38 \pm 24.28 \%$$

$$\mu = 752.5 \pm 77.5$$

$$\sigma = 138.4 \pm 168.5$$

WF

$$\frac{\sigma}{\mu} = 4.09 \pm 28.49 \%$$

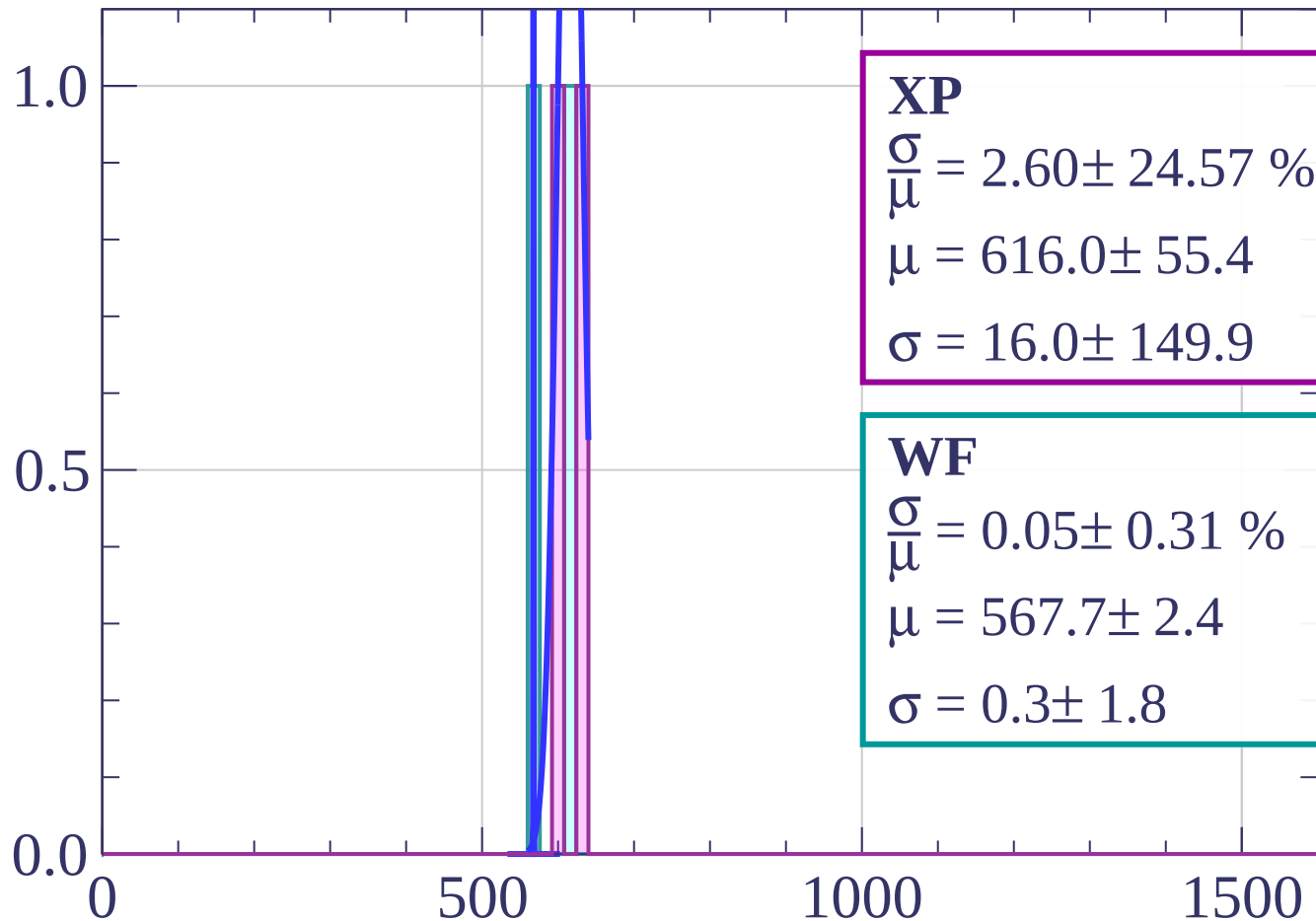
$$\mu = 594.5 \pm 388.5$$

$$\sigma = 24.3 \pm 153.4$$

dE/dx [ADC counts/cm]

Count

Energy loss | $60 < \phi < 63$



XP

$$\frac{\sigma}{\mu} = 2.60 \pm 24.57 \%$$

$$\mu = 616.0 \pm 55.4$$

$$\sigma = 16.0 \pm 149.9$$

WF

$$\frac{\sigma}{\mu} = 0.05 \pm 0.31 \%$$

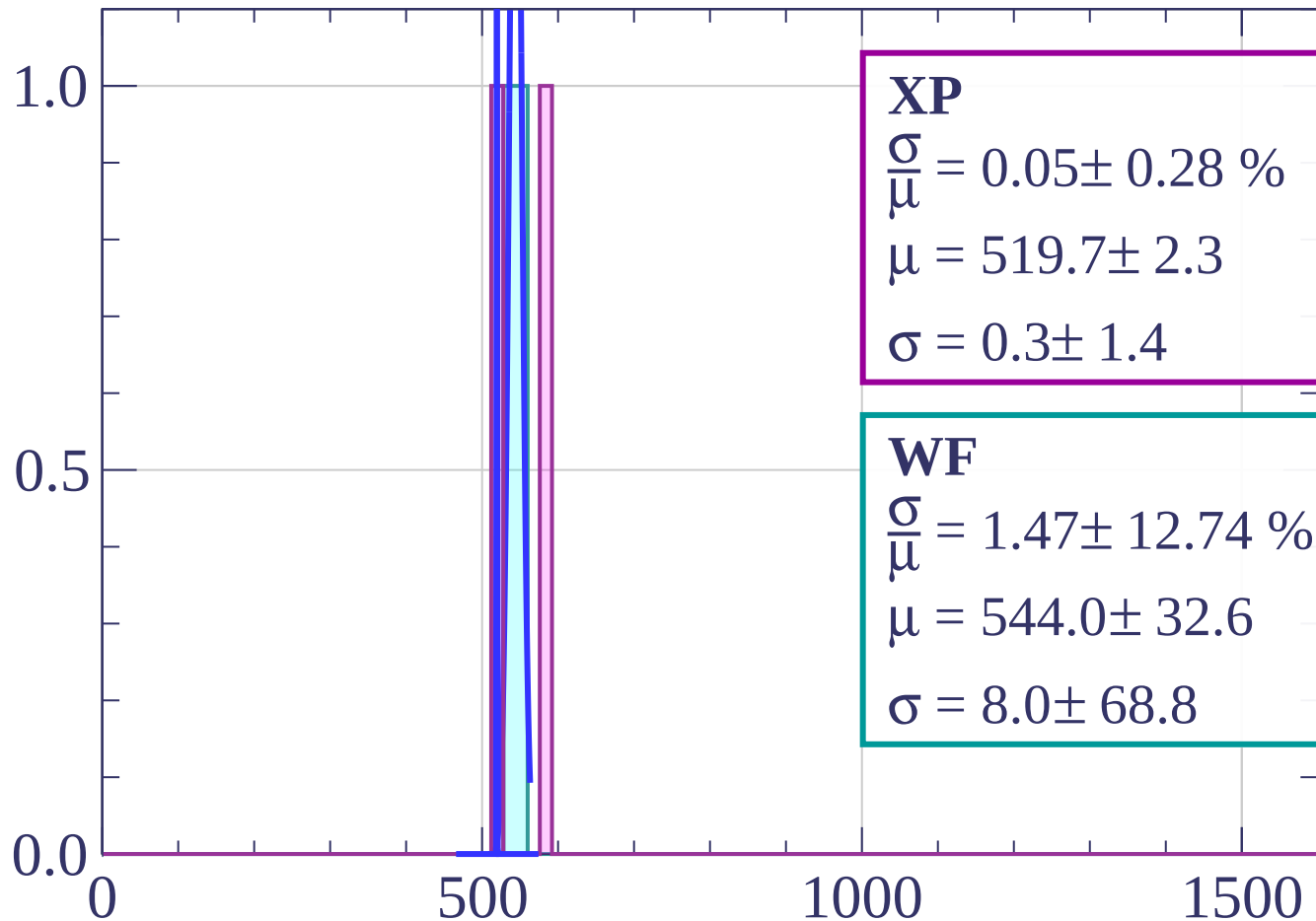
$$\mu = 567.7 \pm 2.4$$

$$\sigma = 0.3 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 0.05 \pm 0.28 \%$$

$$\mu = 519.7 \pm 2.3$$

$$\sigma = 0.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 1.47 \pm 12.74 \%$$

$$\mu = 544.0 \pm 32.6$$

$$\sigma = 8.0 \pm 68.8$$

dE/dx [ADC counts/cm]

Count

Energy loss | $66 < \phi < 69$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $69 < \phi < 72$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $72 < \phi < 75$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $75 < \phi < 78$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

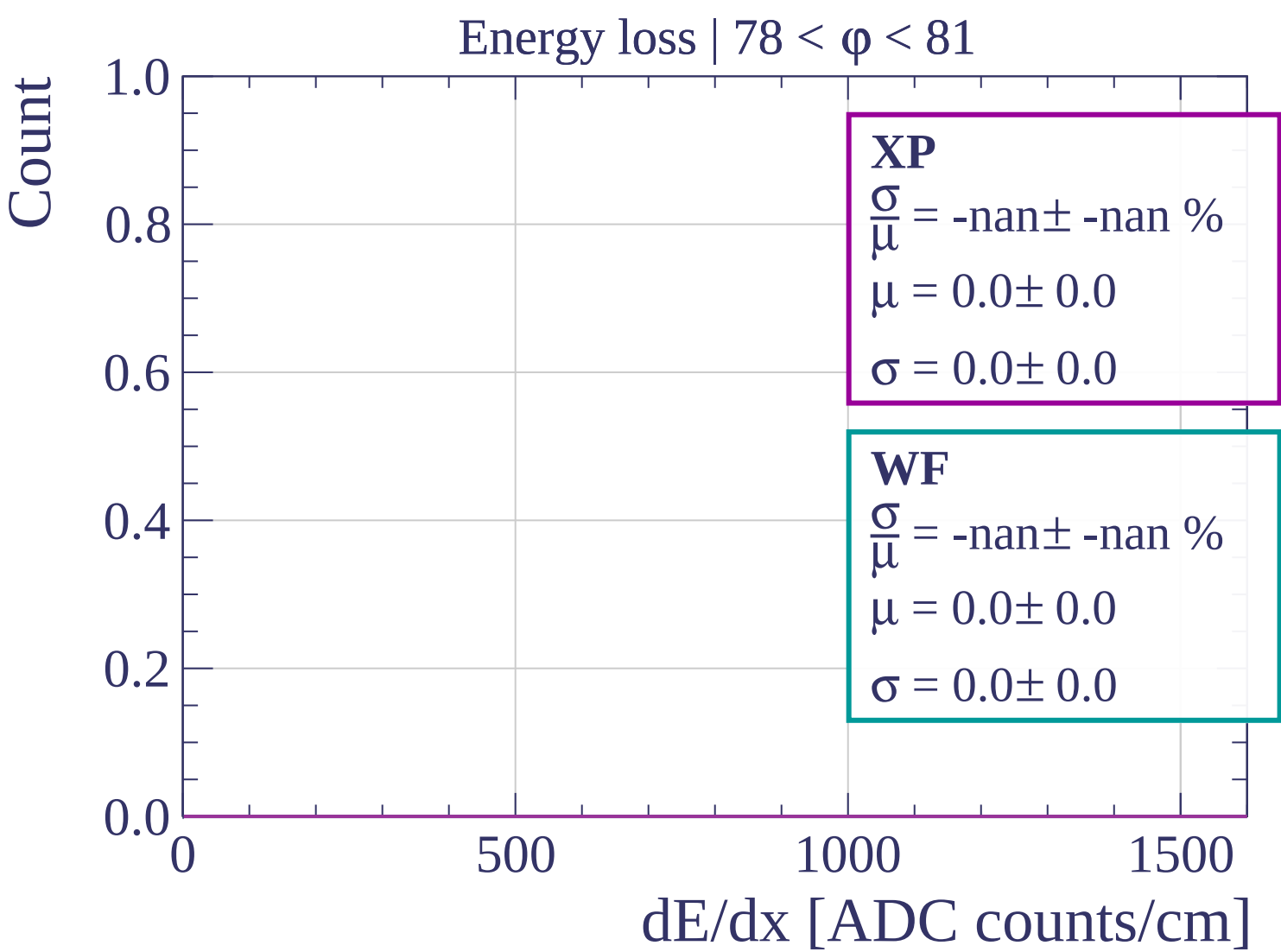
0

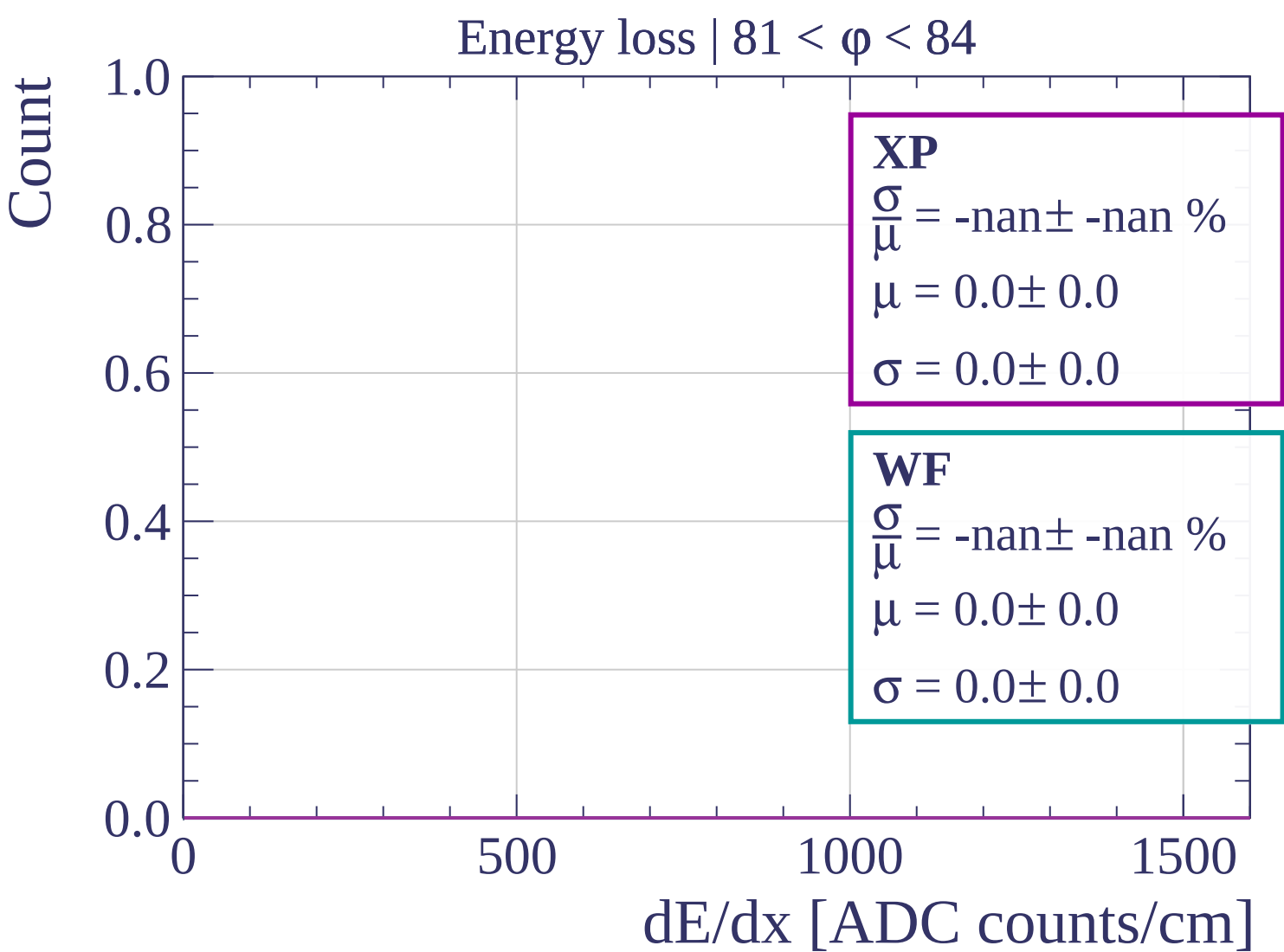
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $84 < \phi < 87$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $87 < \phi < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

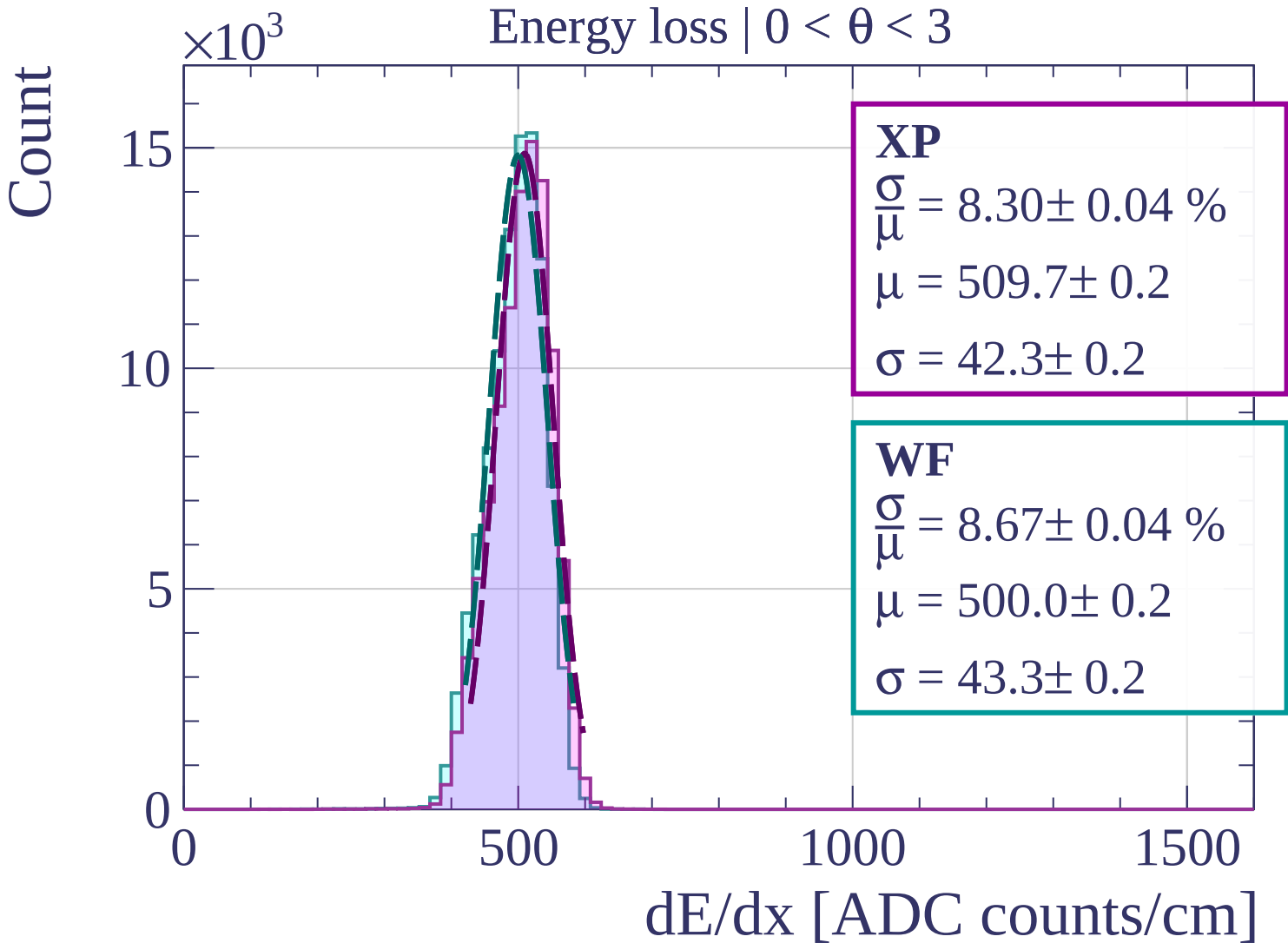
0

500

1000

1500

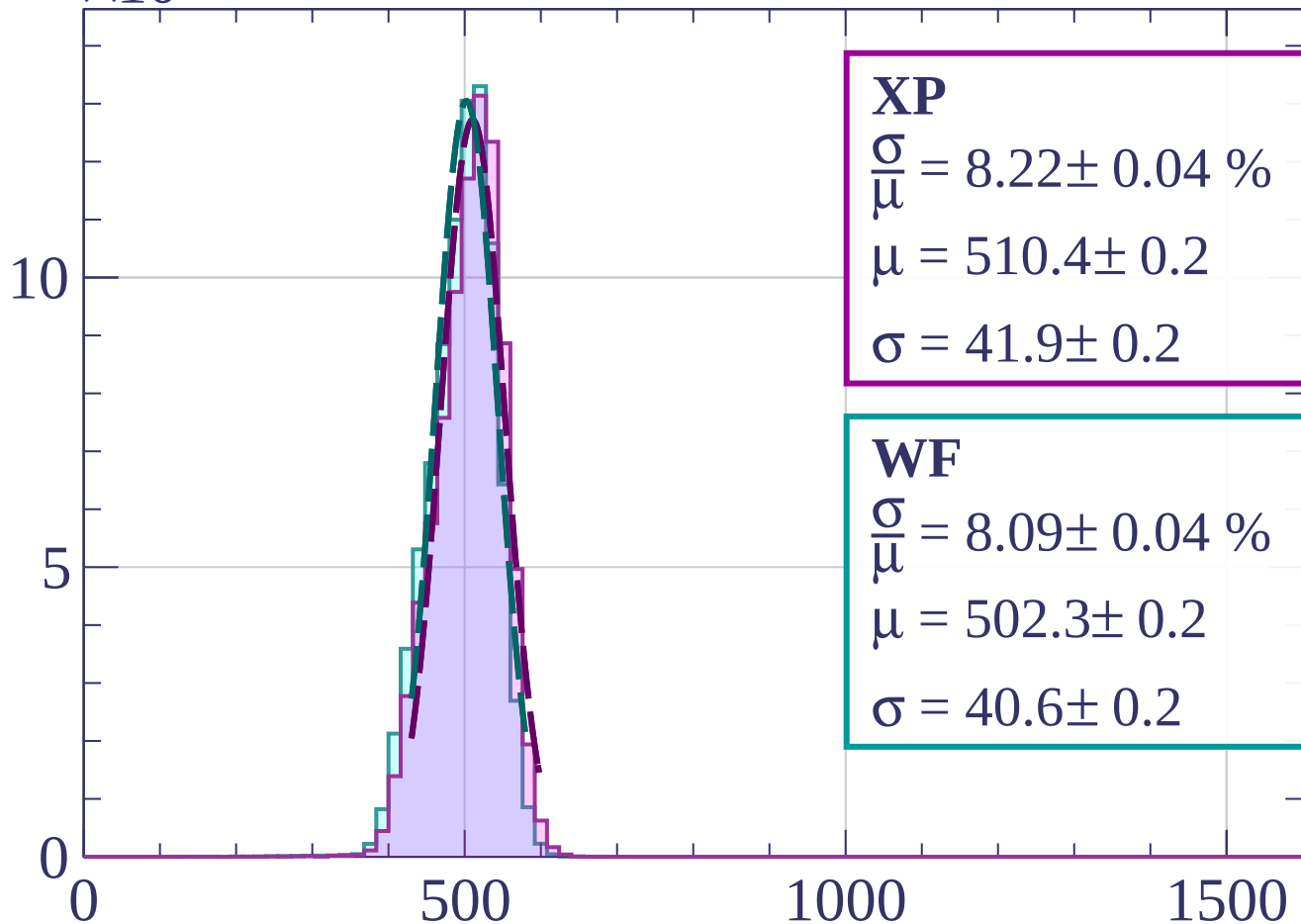
dE/dx [ADC counts/cm]



Count

$\times 10^3$

Energy loss | $3 < \theta < 6$



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.04 \%$$

$$\mu = 510.4 \pm 0.2$$

$$\sigma = 41.9 \pm 0.2$$

WF

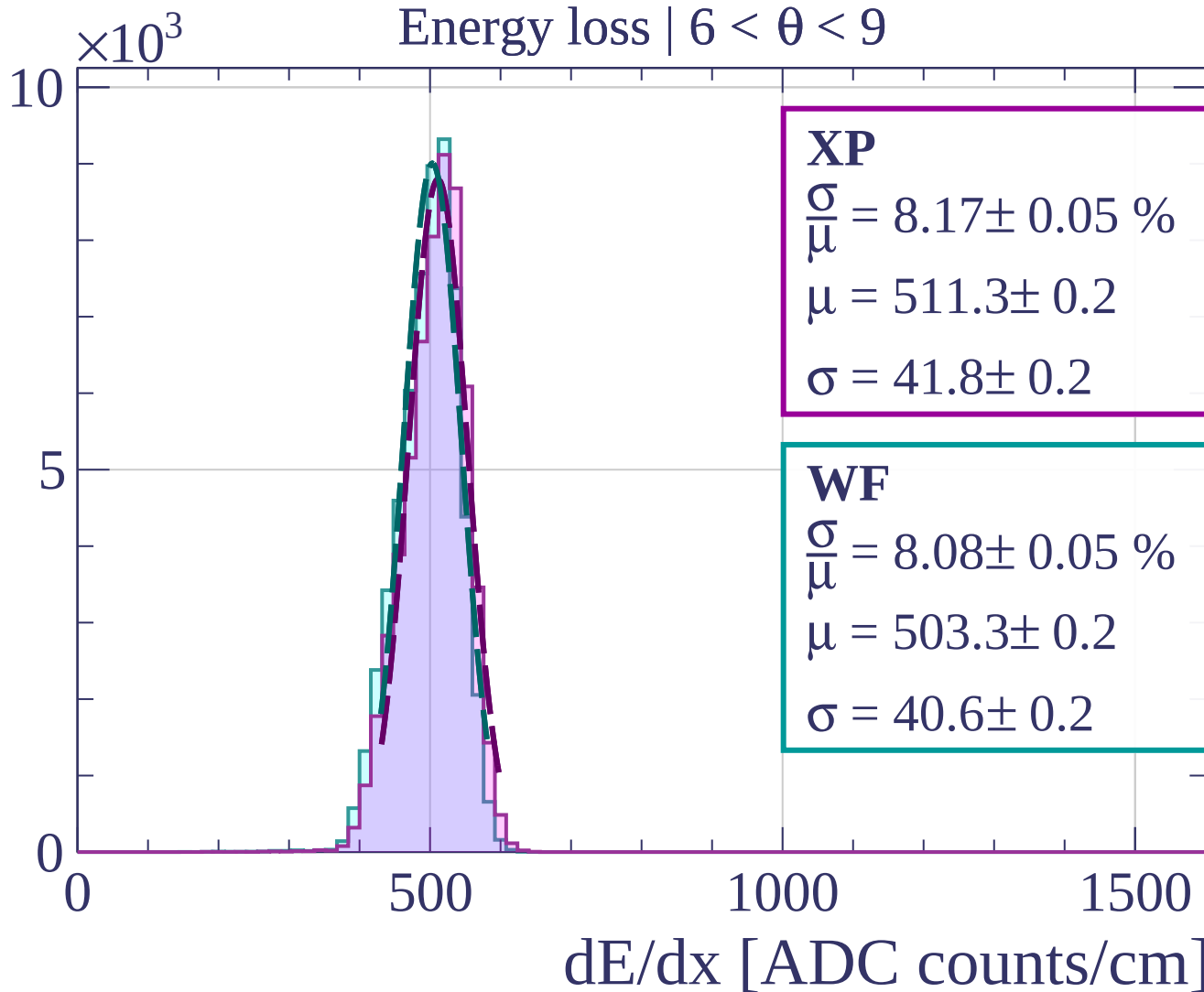
$$\frac{\sigma}{\mu} = 8.09 \pm 0.04 \%$$

$$\mu = 502.3 \pm 0.2$$

$$\sigma = 40.6 \pm 0.2$$

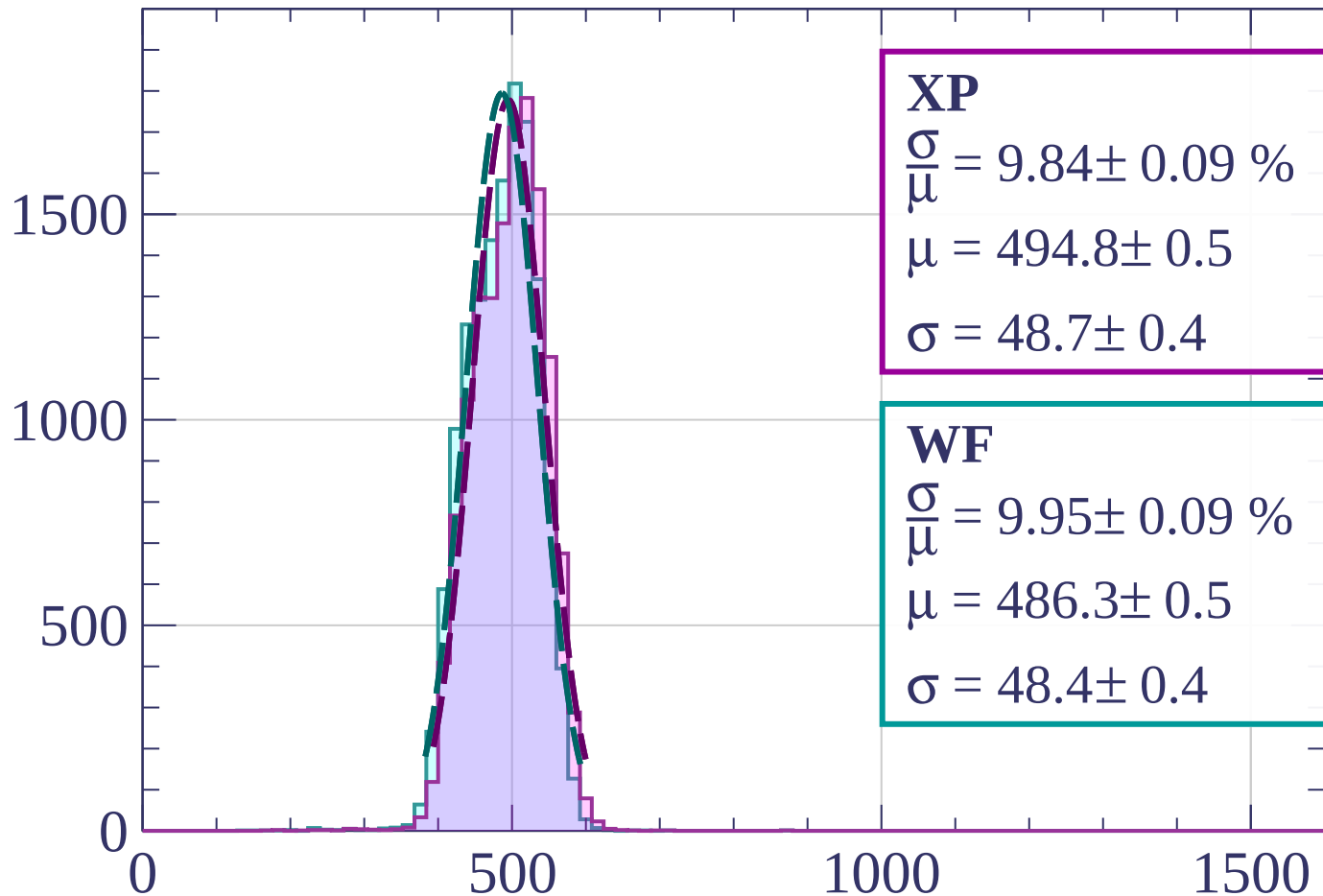
dE/dx [ADC counts/cm]

Count



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 9.84 \pm 0.09 \%$$

$$\mu = 494.8 \pm 0.5$$

$$\sigma = 48.7 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 0.09 \%$$

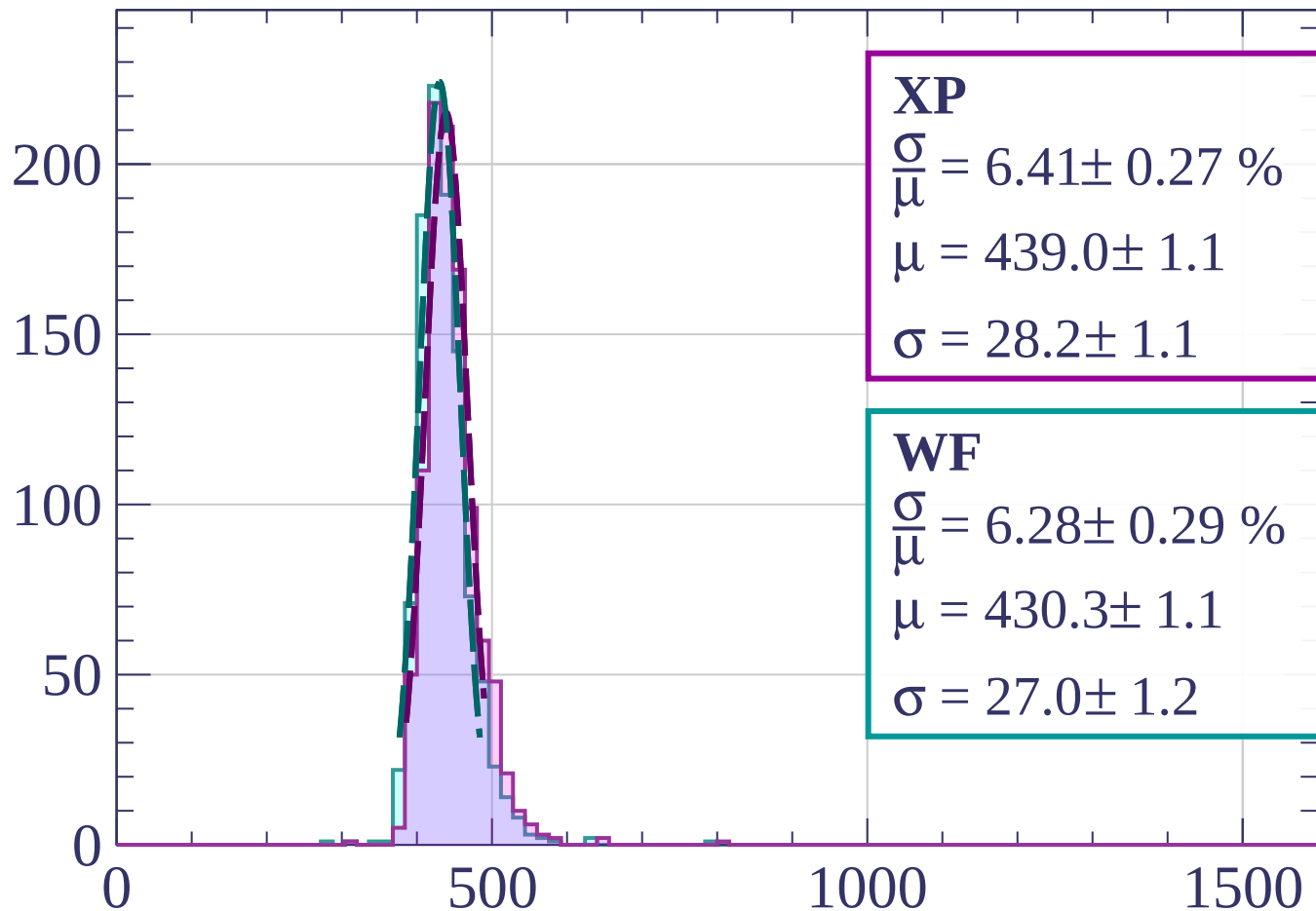
$$\mu = 486.3 \pm 0.5$$

$$\sigma = 48.4 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.41 \pm 0.27 \%$$

$$\mu = 439.0 \pm 1.1$$

$$\sigma = 28.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.28 \pm 0.29 \%$$

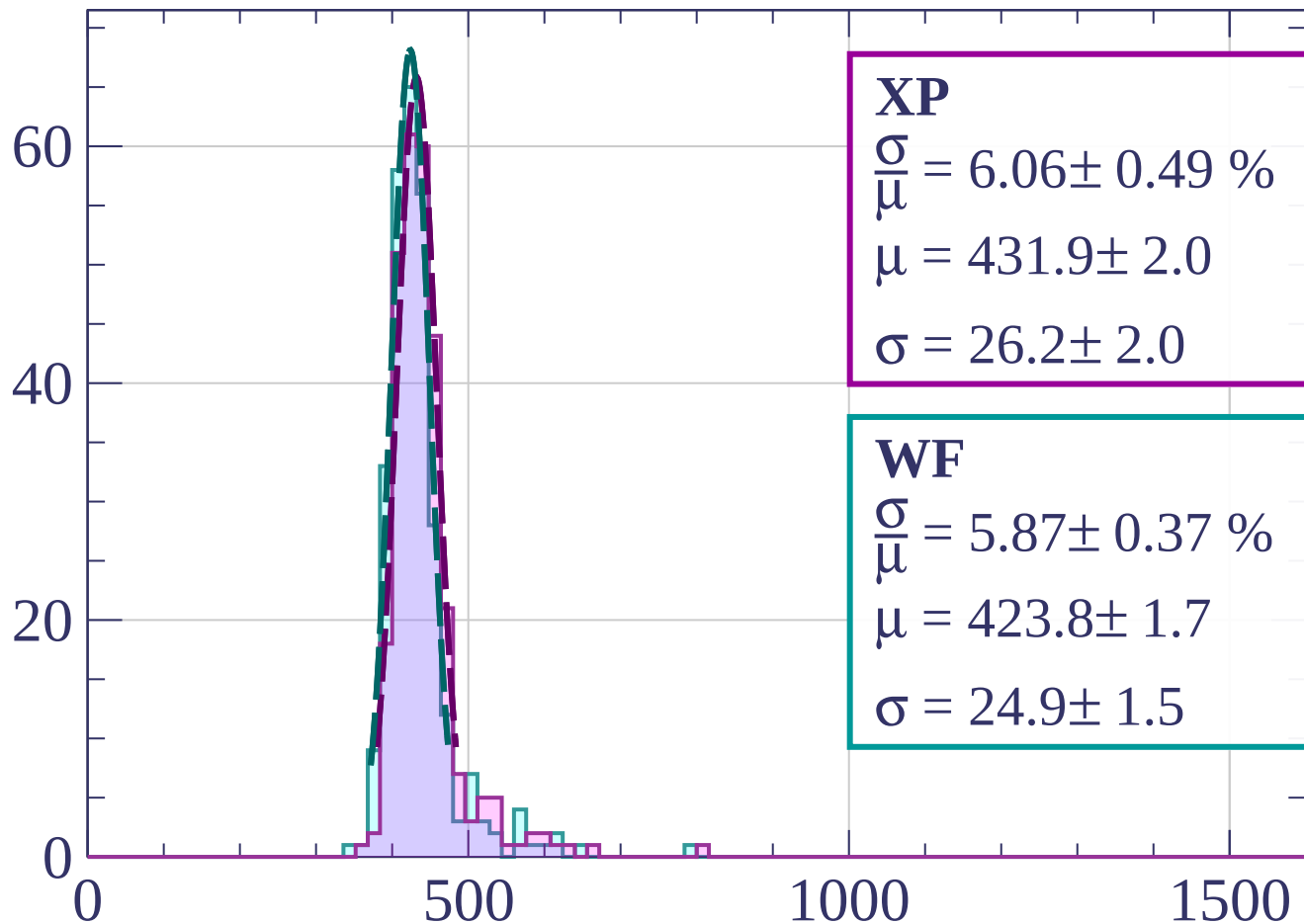
$$\mu = 430.3 \pm 1.1$$

$$\sigma = 27.0 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.49 \%$$

$$\mu = 431.9 \pm 2.0$$

$$\sigma = 26.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 5.87 \pm 0.37 \%$$

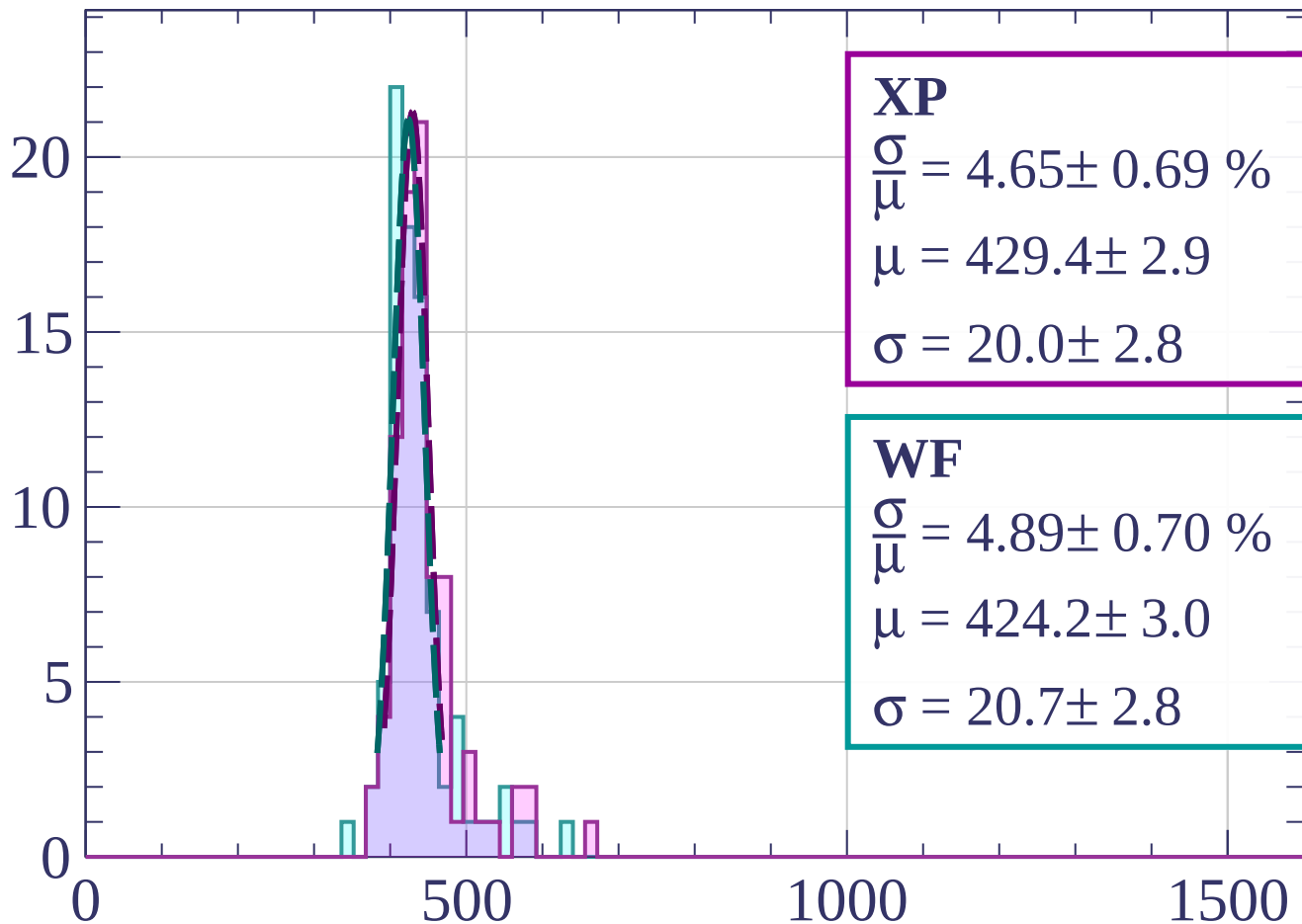
$$\mu = 423.8 \pm 1.7$$

$$\sigma = 24.9 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 4.65 \pm 0.69 \%$$

$$\mu = 429.4 \pm 2.9$$

$$\sigma = 20.0 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 4.89 \pm 0.70 \%$$

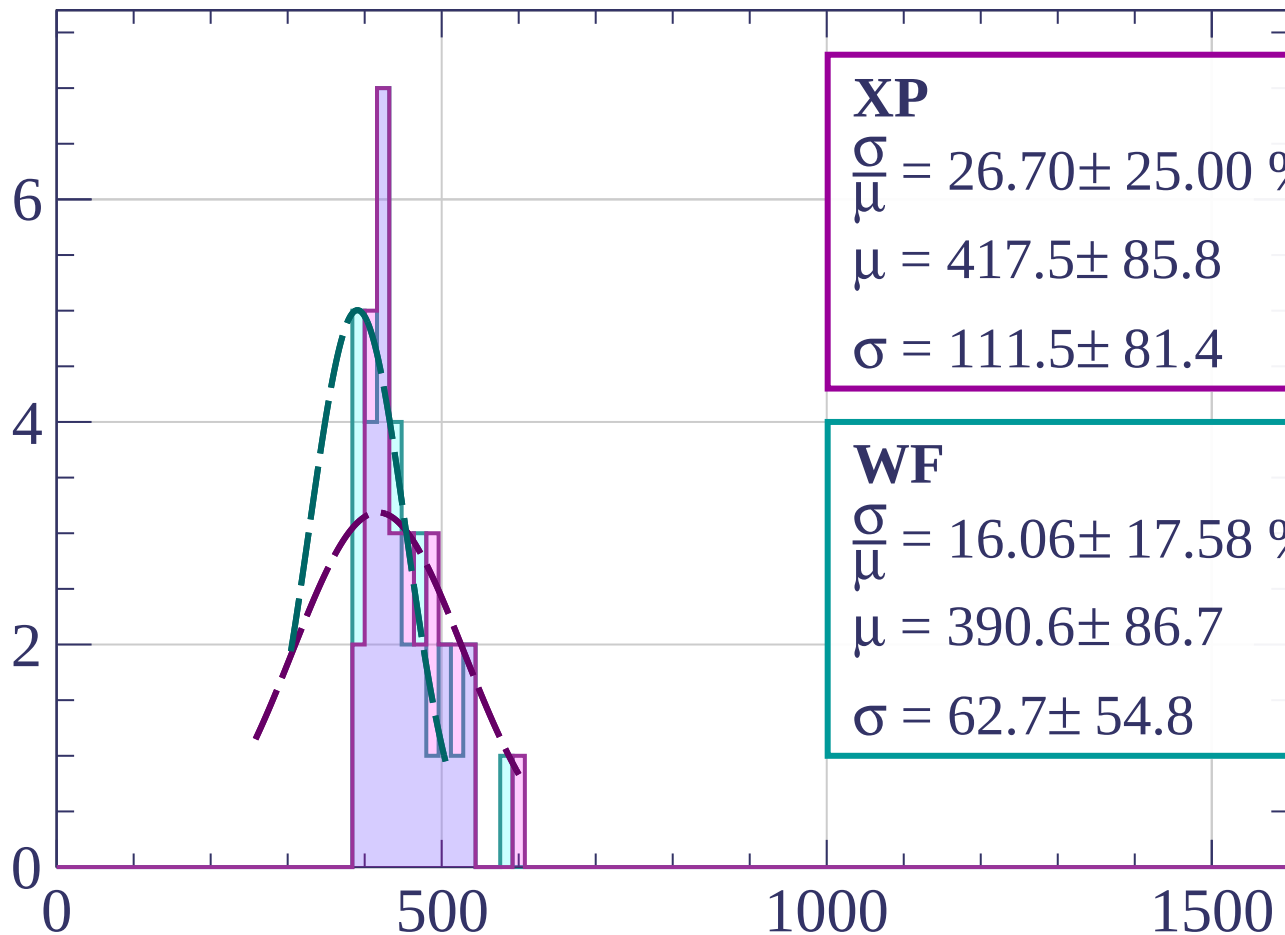
$$\mu = 424.2 \pm 3.0$$

$$\sigma = 20.7 \pm 2.8$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 26.70 \pm 25.00 \%$$

$$\mu = 417.5 \pm 85.8$$

$$\sigma = 111.5 \pm 81.4$$

WF

$$\frac{\sigma}{\mu} = 16.06 \pm 17.58 \%$$

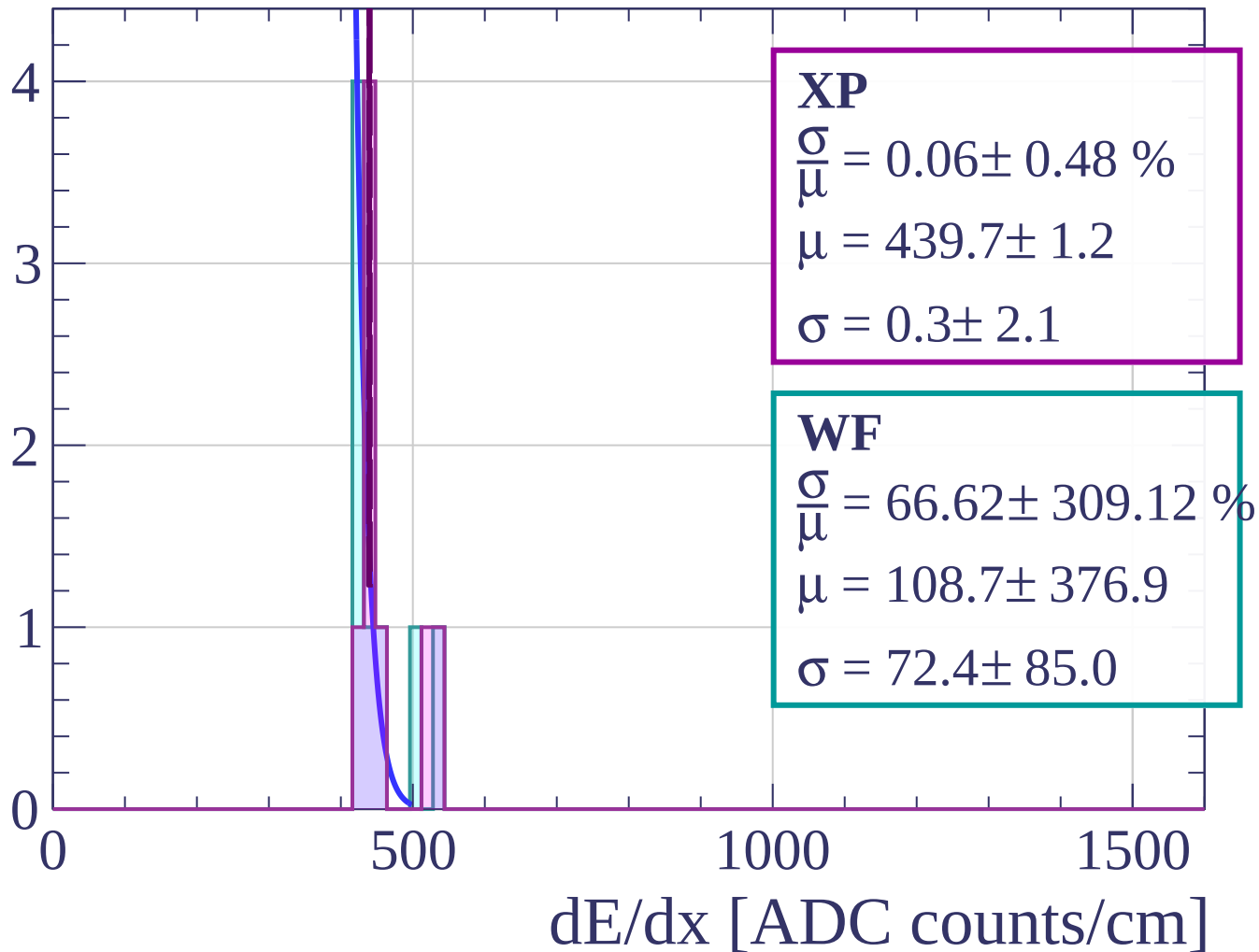
$$\mu = 390.6 \pm 86.7$$

$$\sigma = 62.7 \pm 54.8$$

dE/dx [ADC counts/cm]

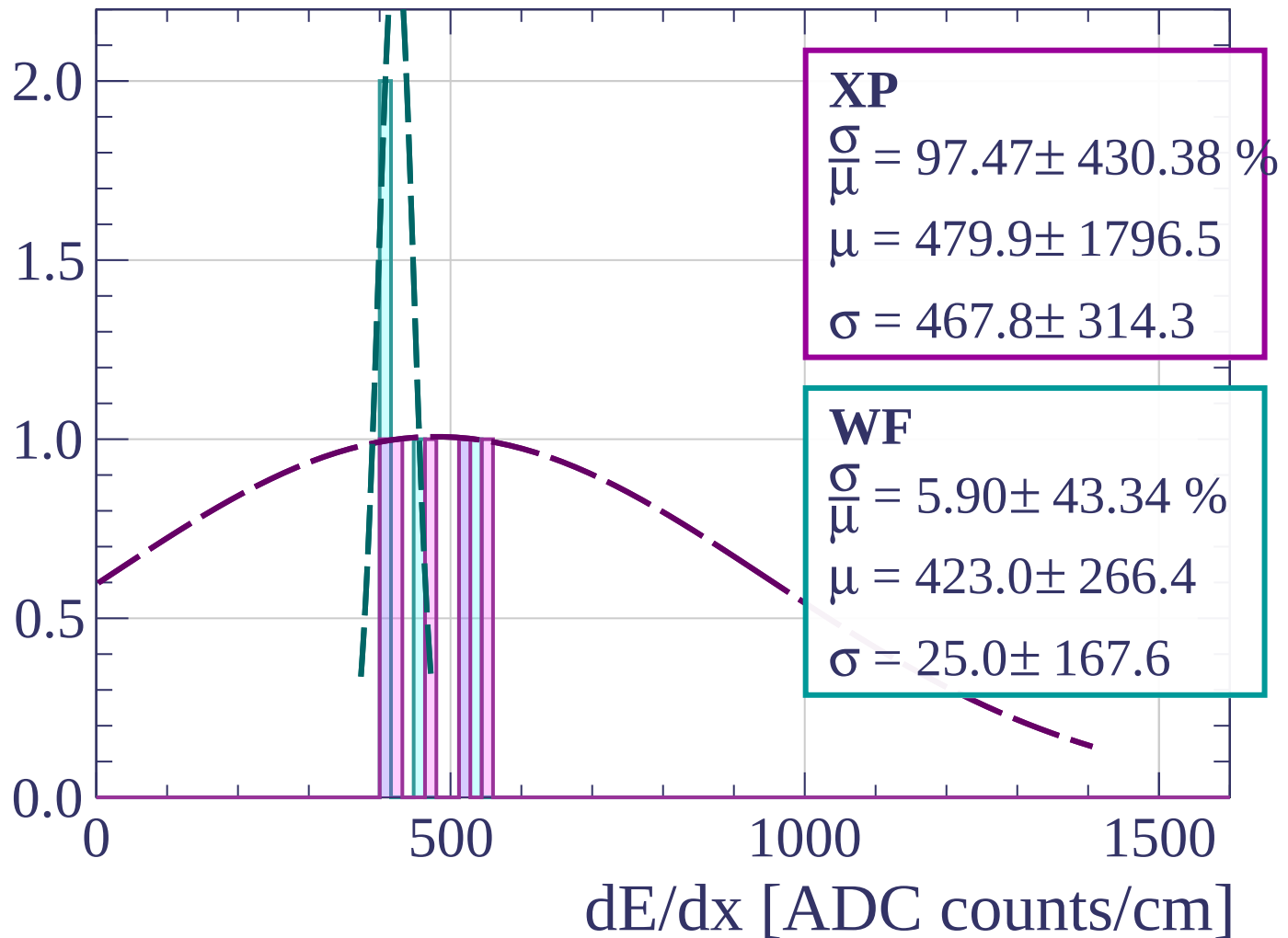
Energy loss | $24 < \theta < 27$

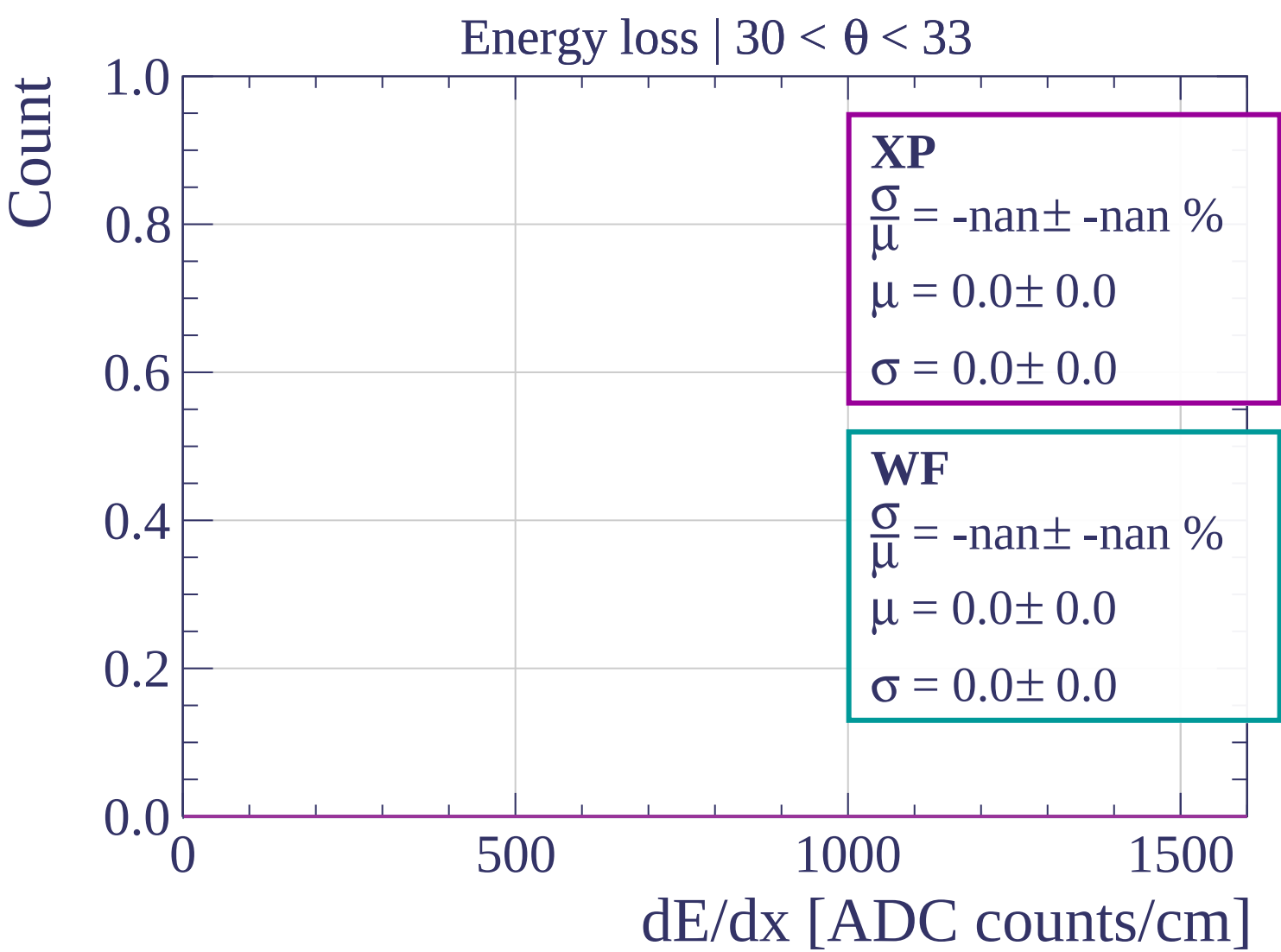
Count

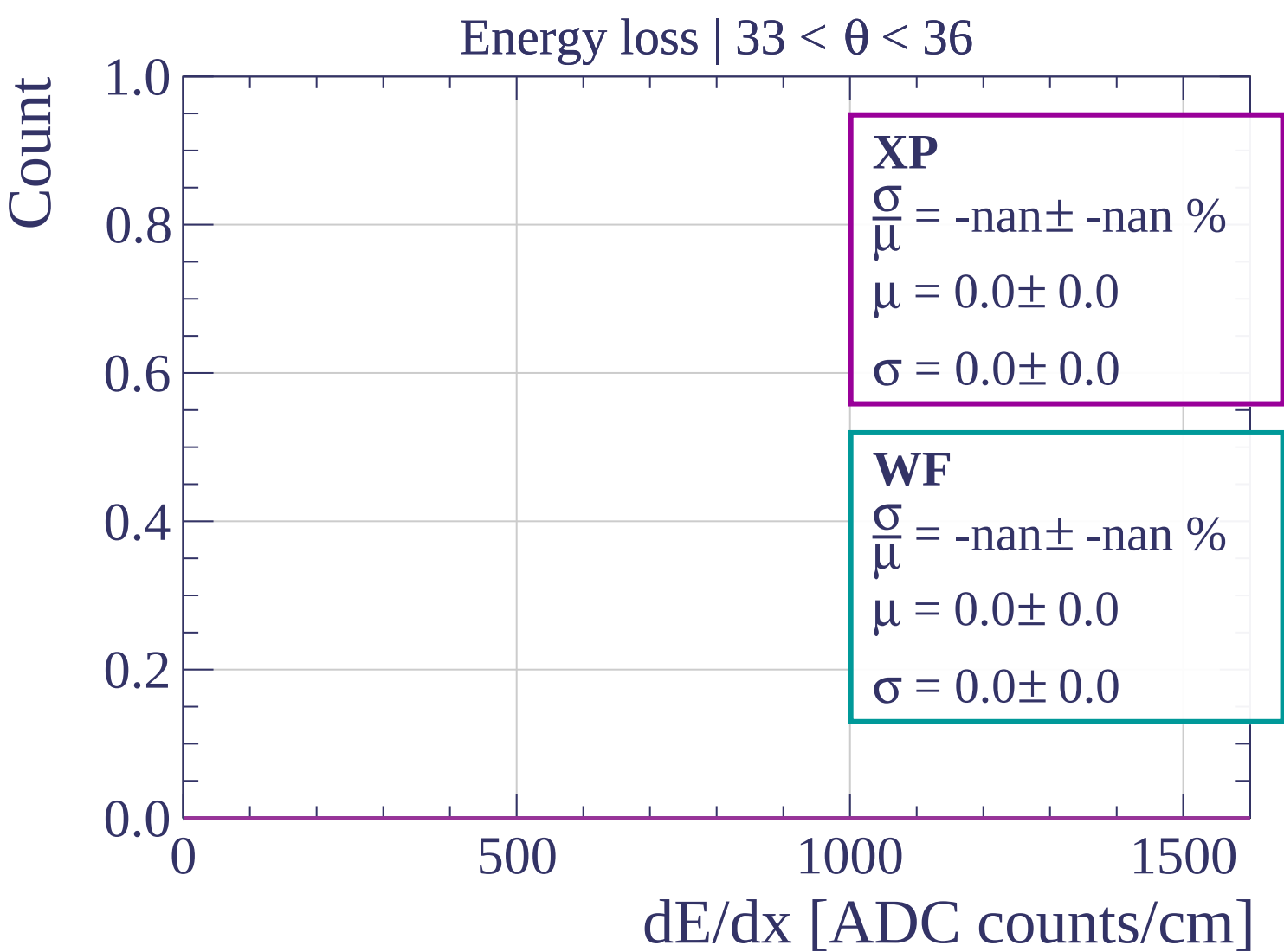


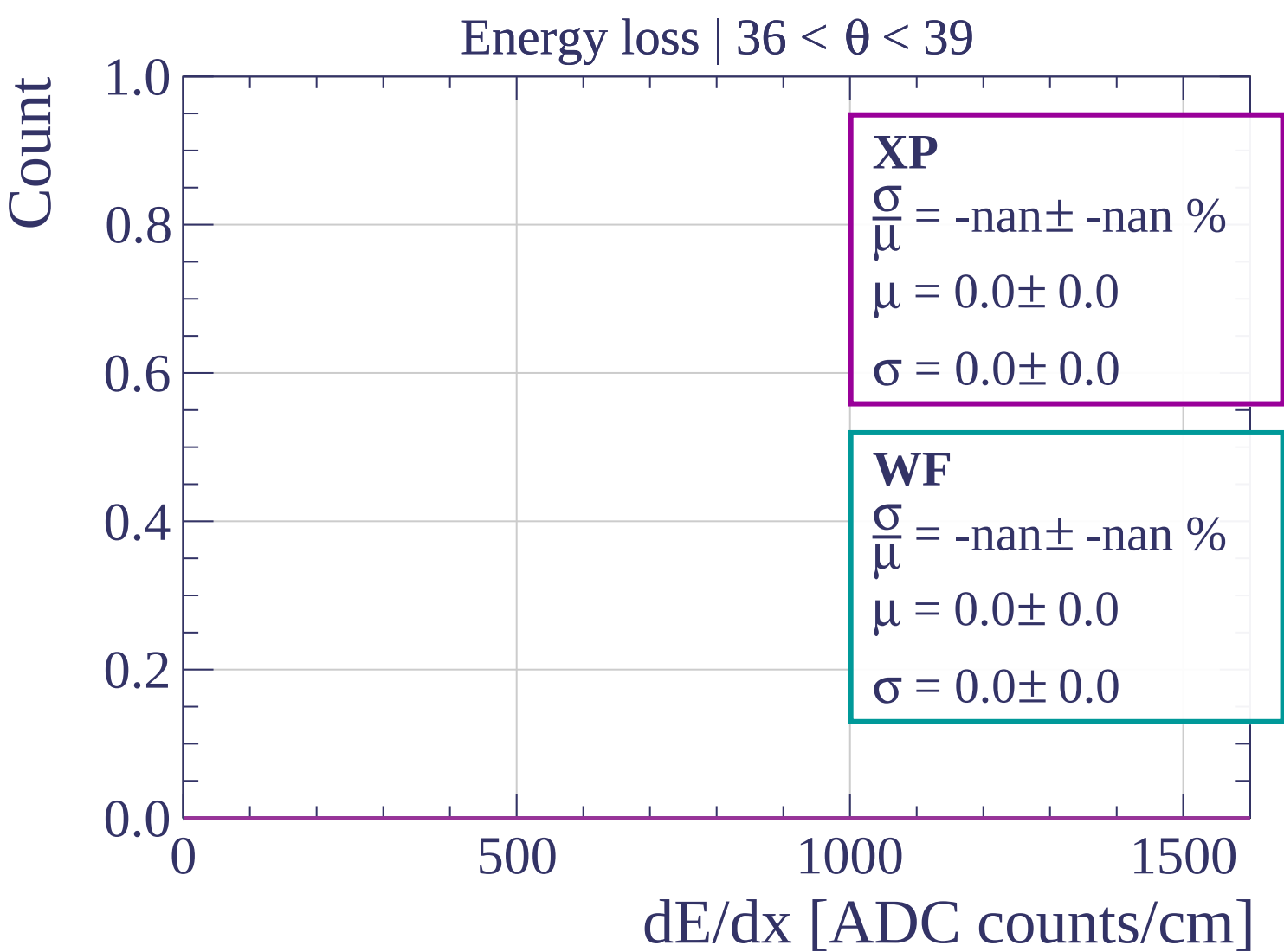
Energy loss | $27 < \theta < 30$

Count



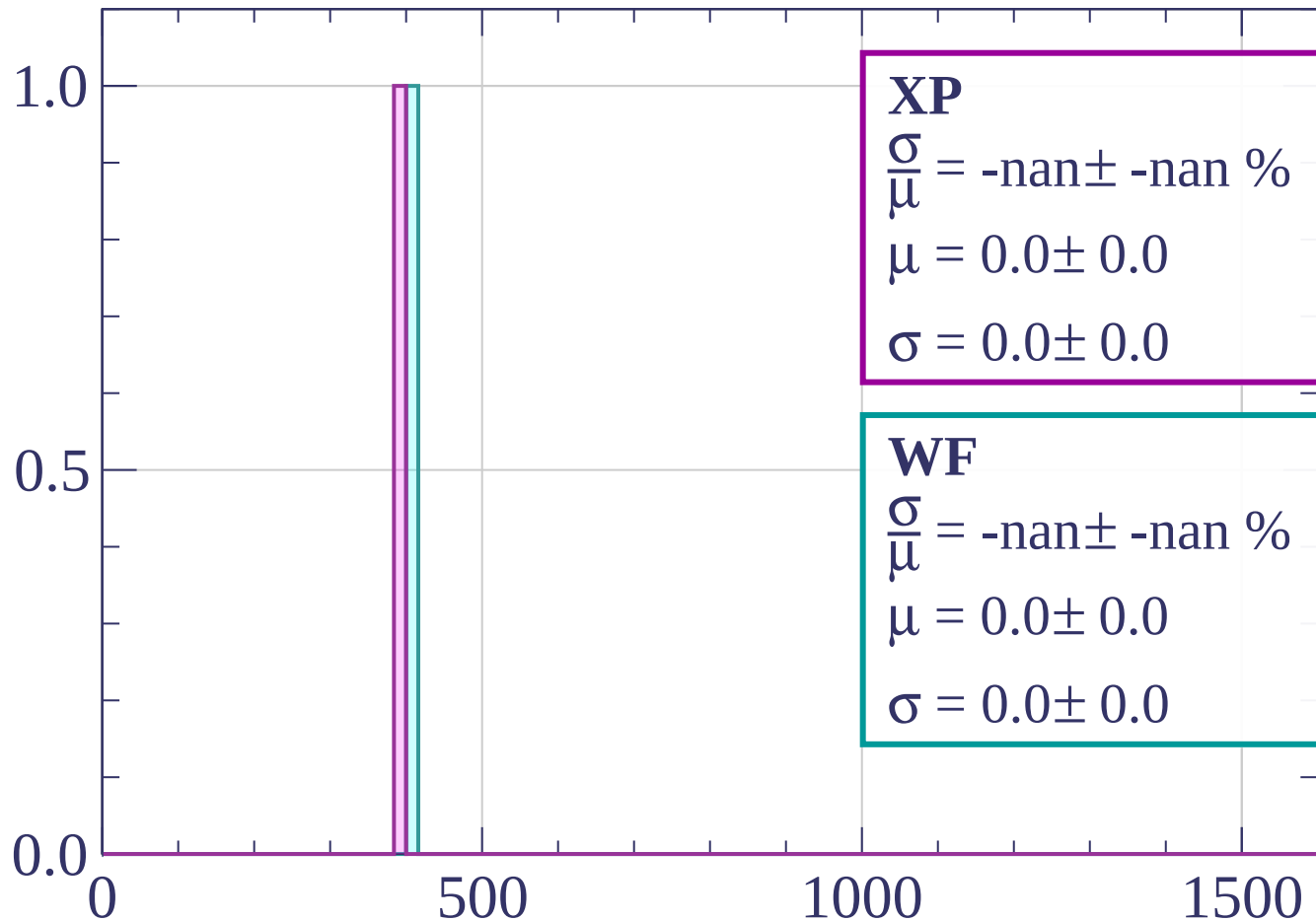






Energy loss | $39 < \theta < 42$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]

Count

Energy loss | $42 < \theta < 45$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

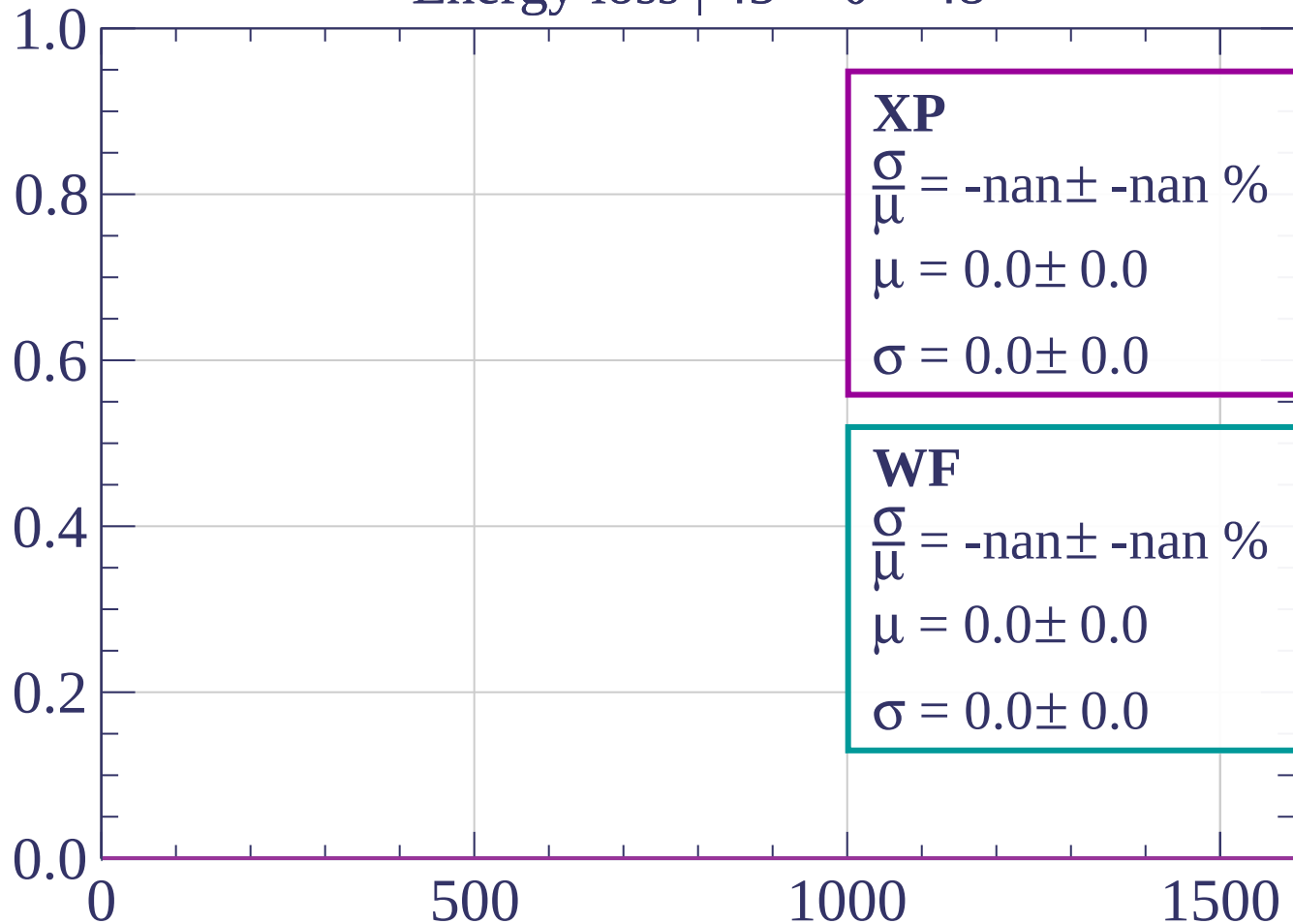
1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $45 < \theta < 48$



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

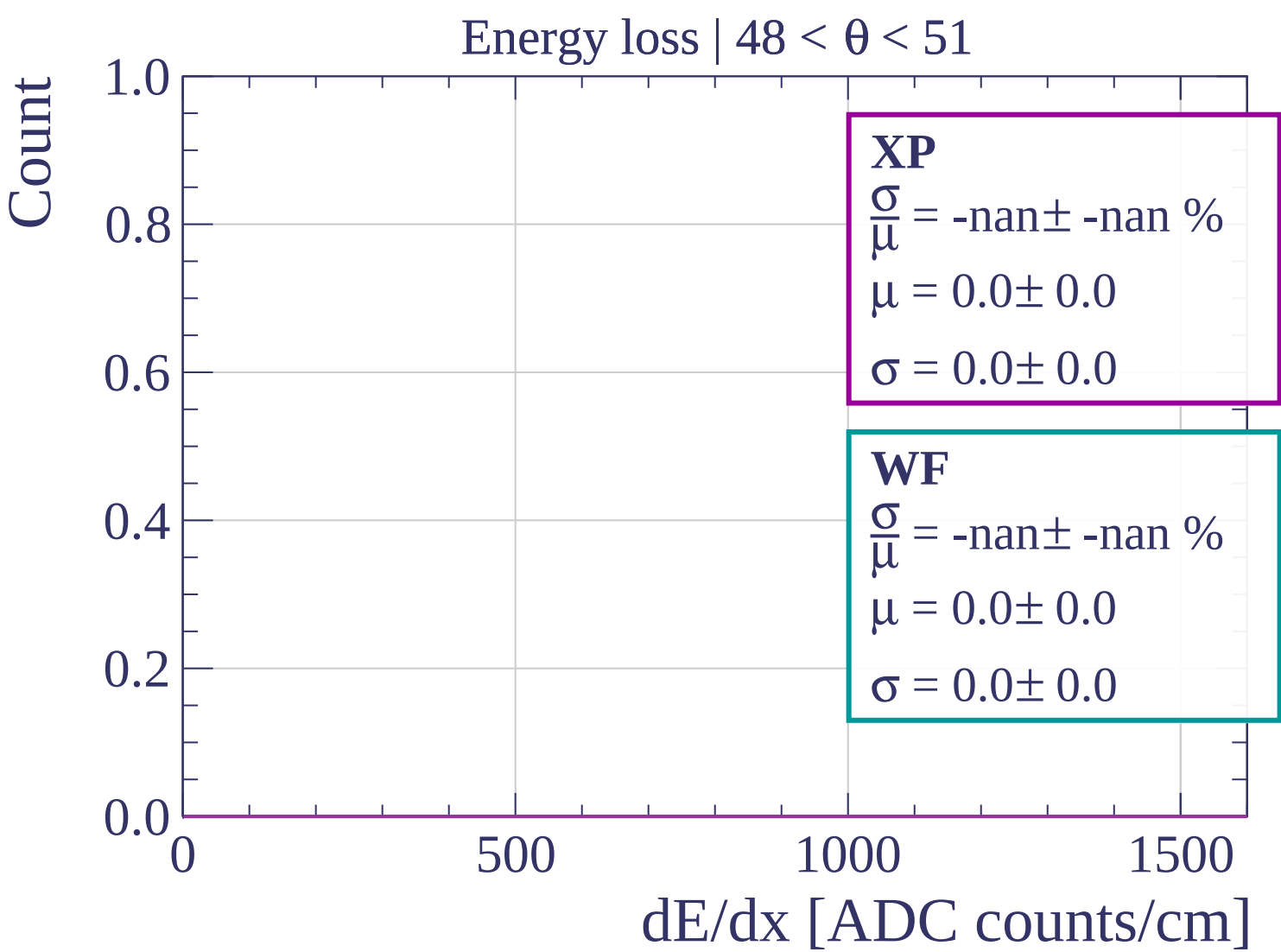
WF

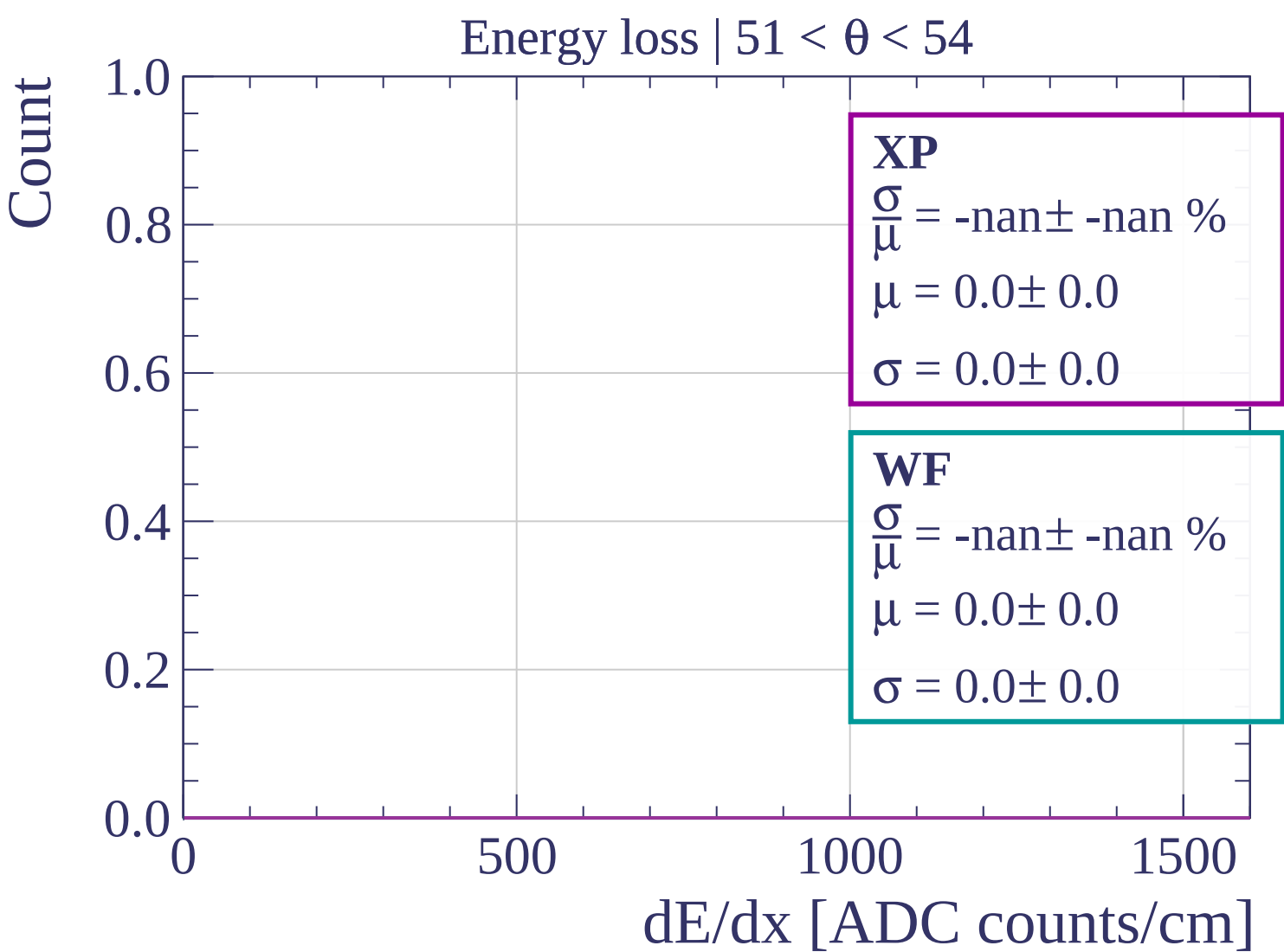
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

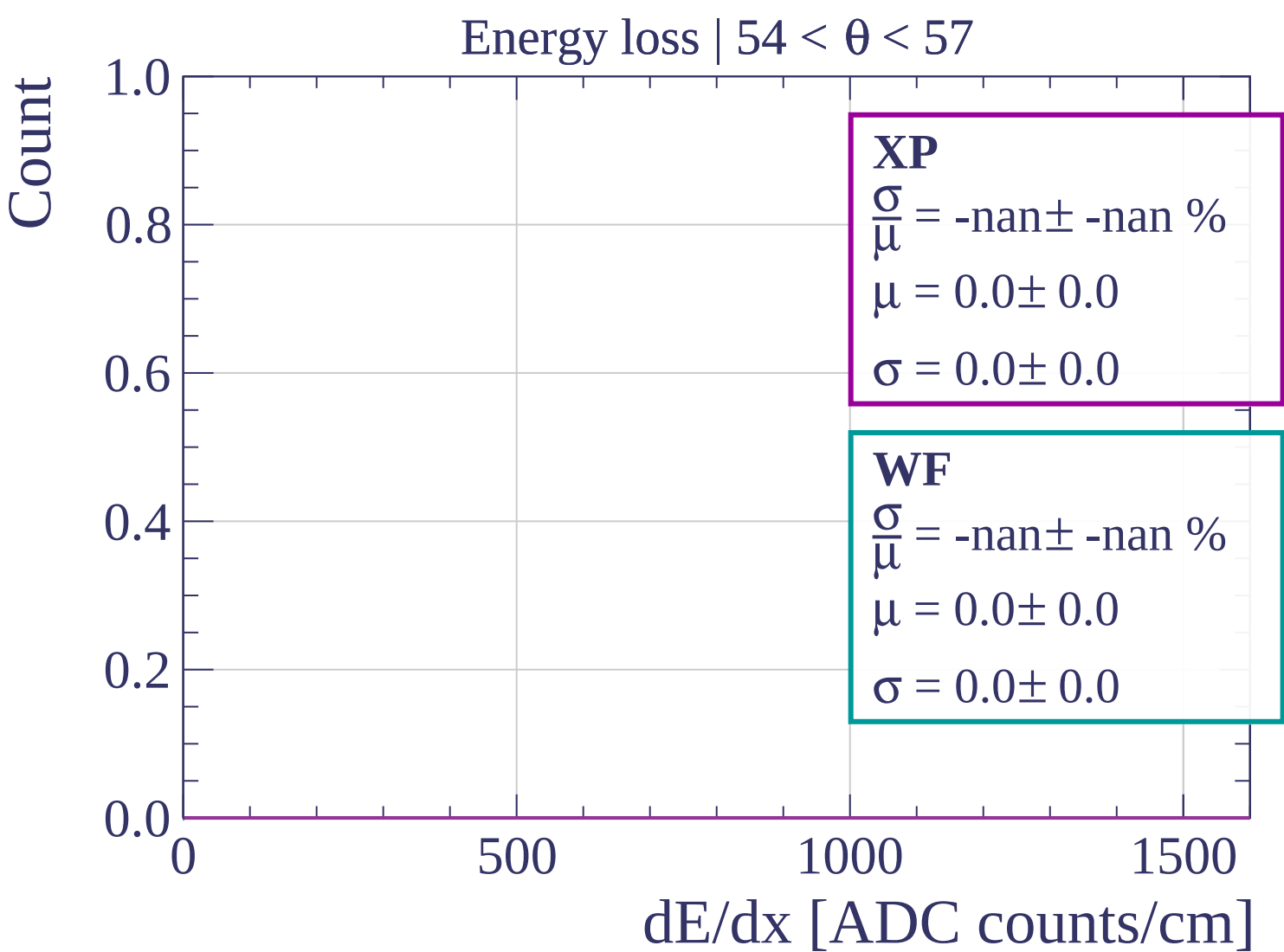
$\mu = 0.0 \pm 0.0$

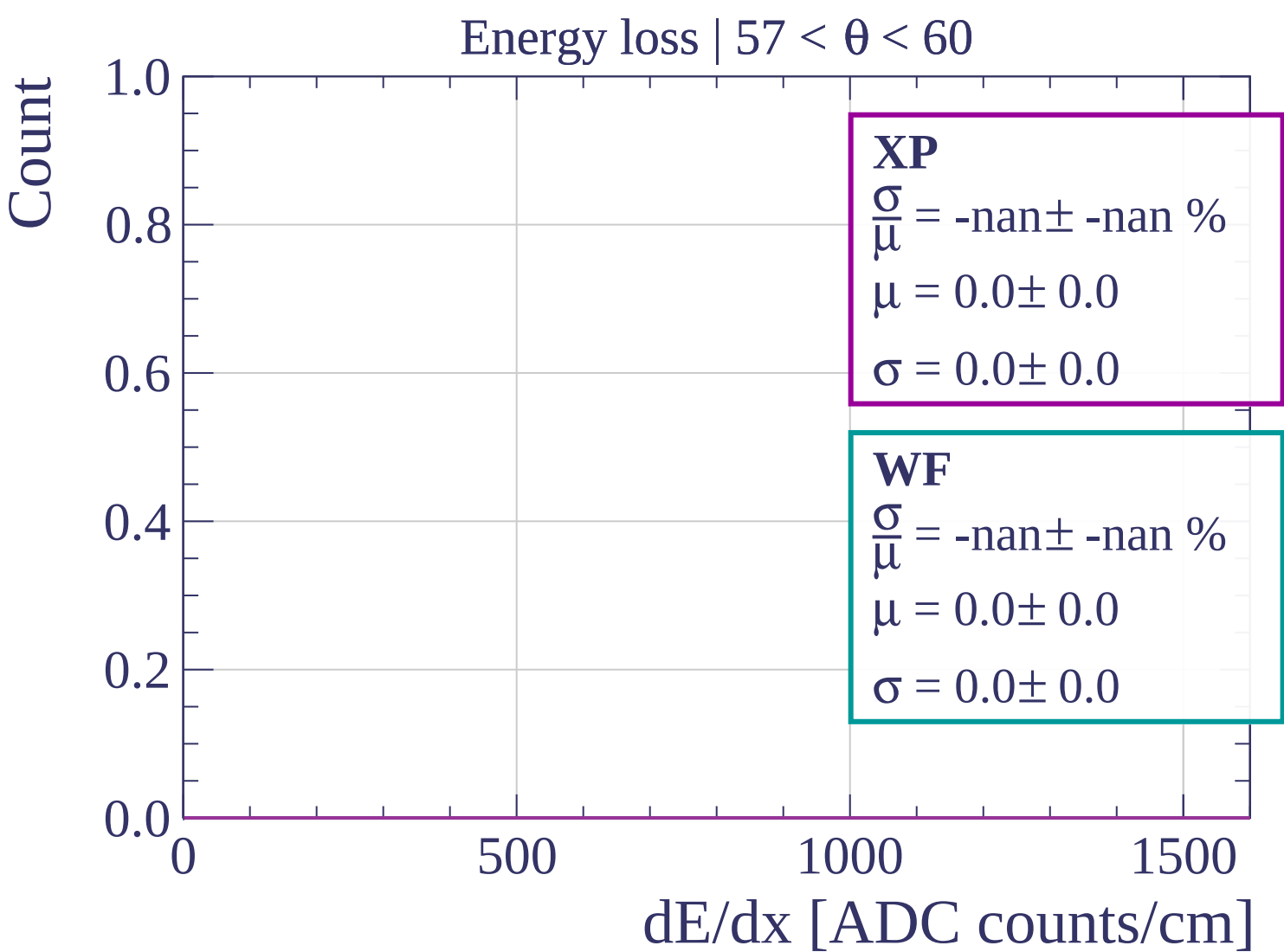
$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]









Count

Energy loss | $60 < \theta < 63$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

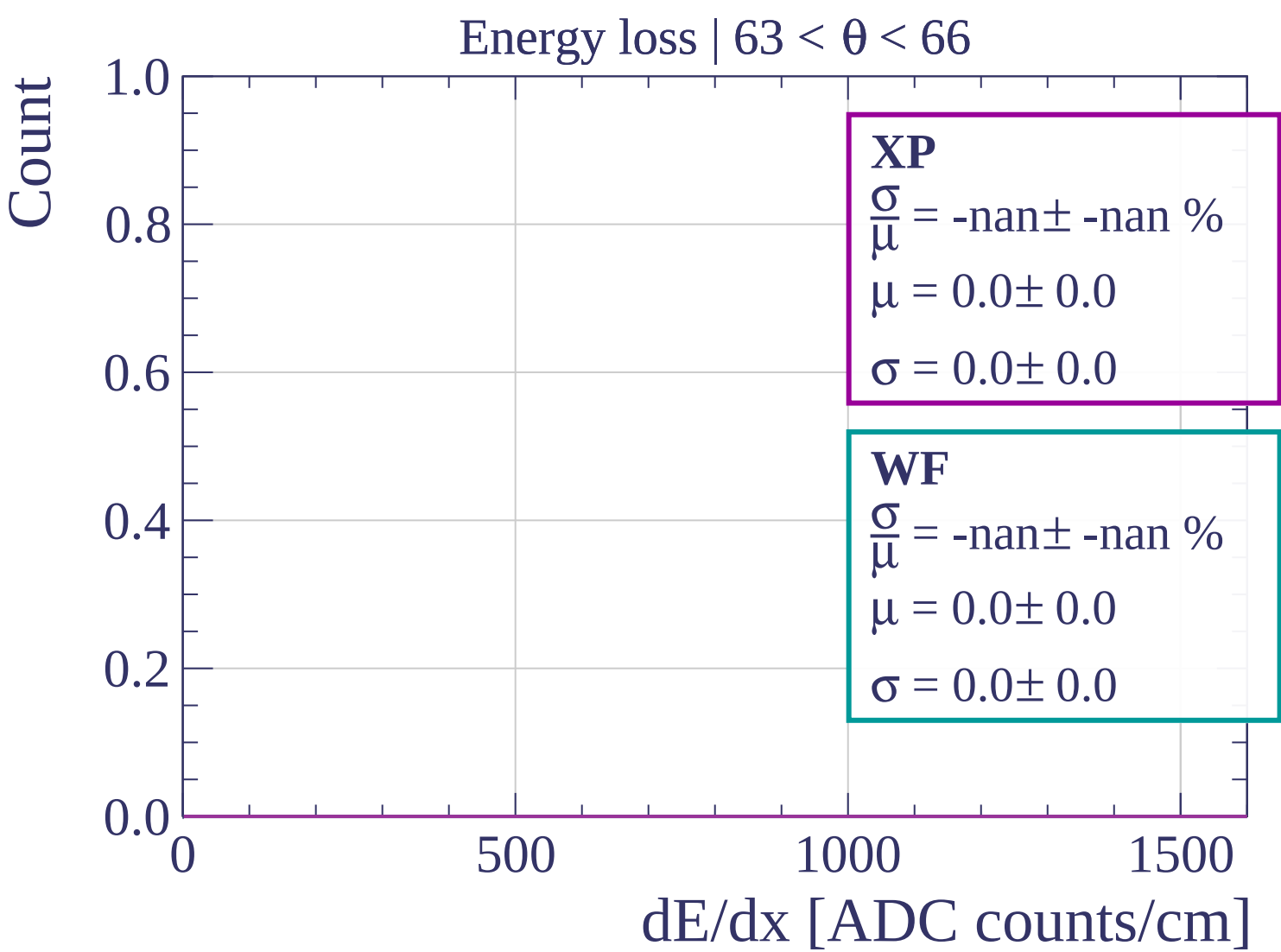
0

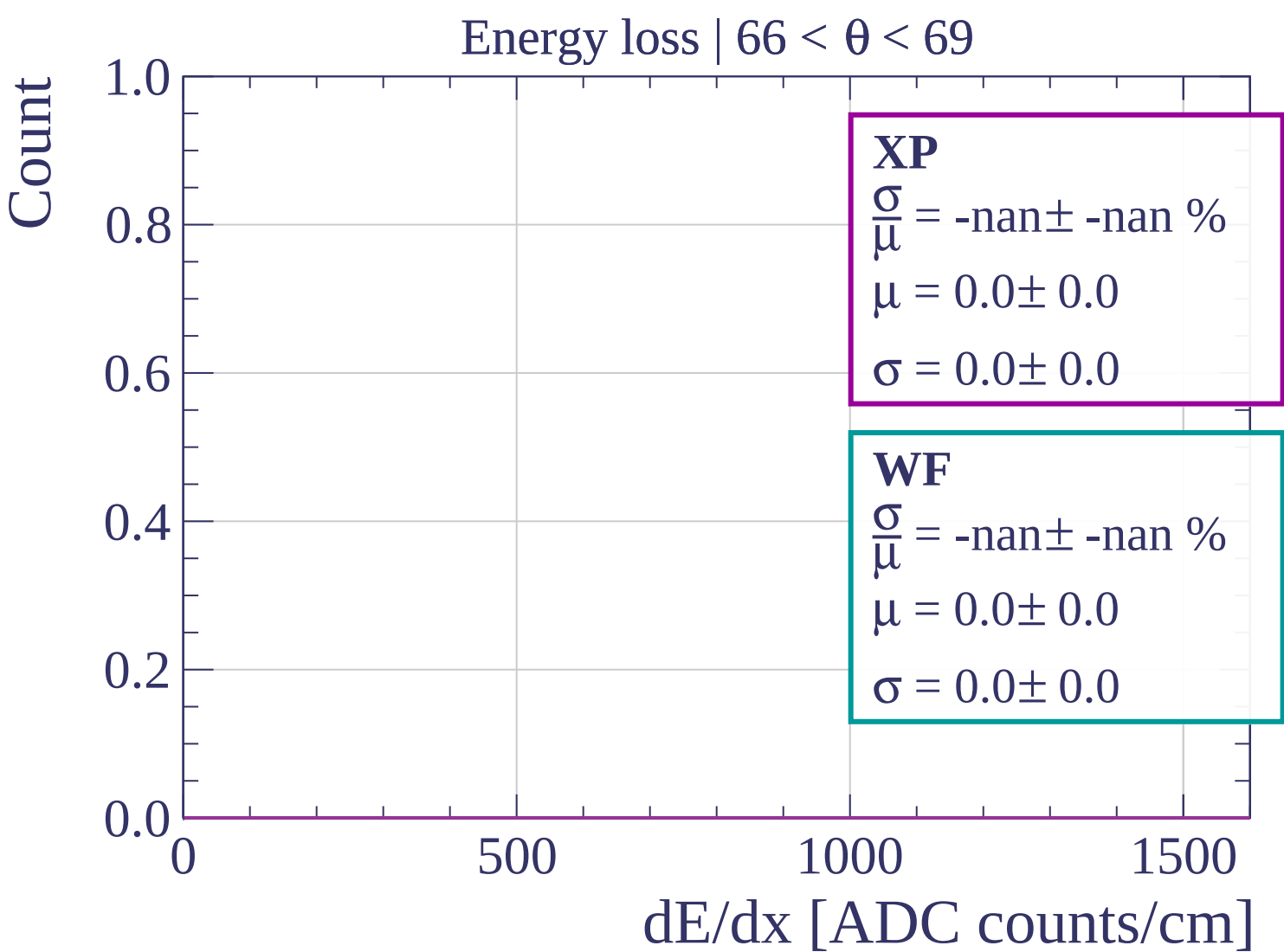
500

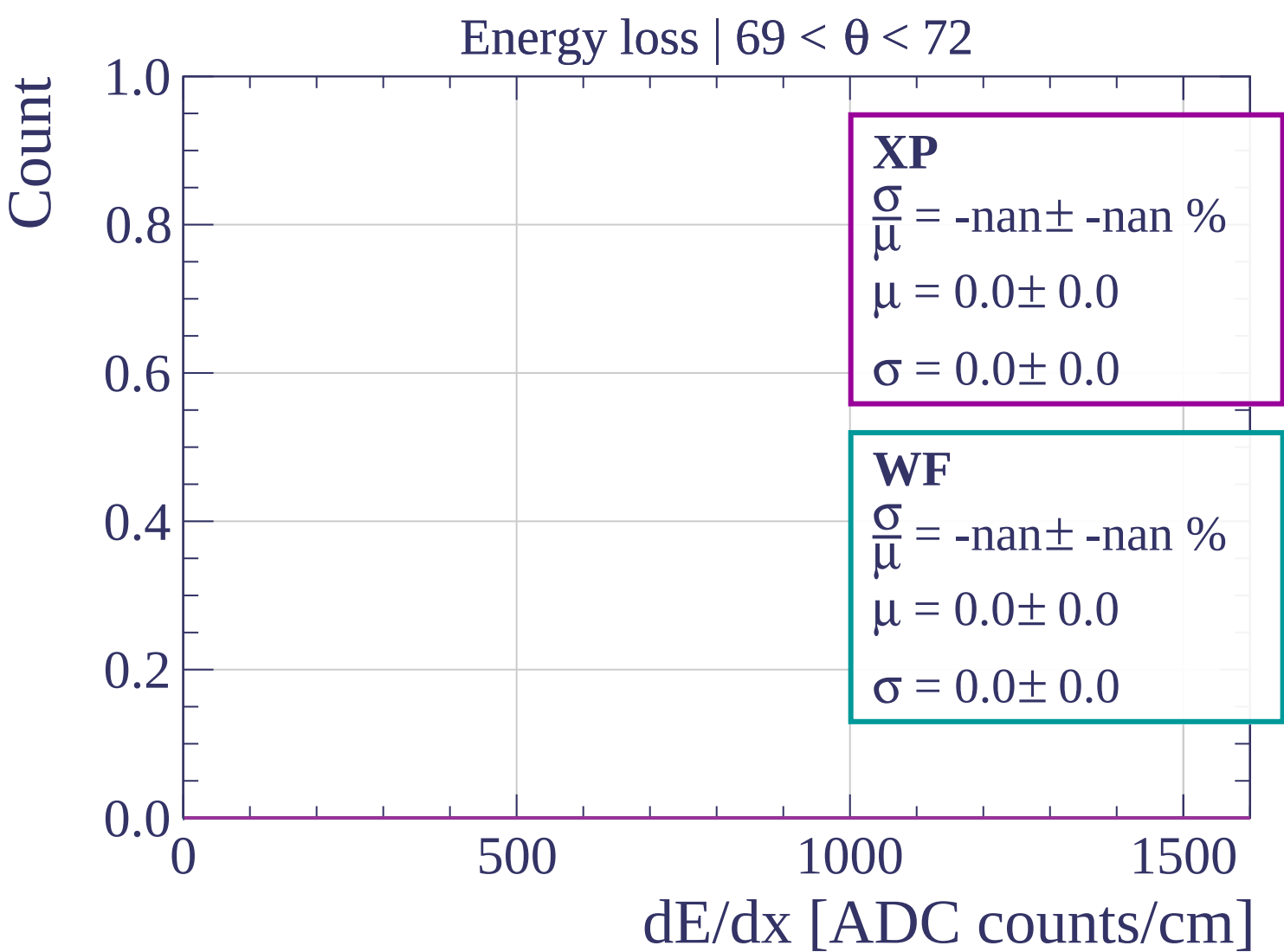
1000

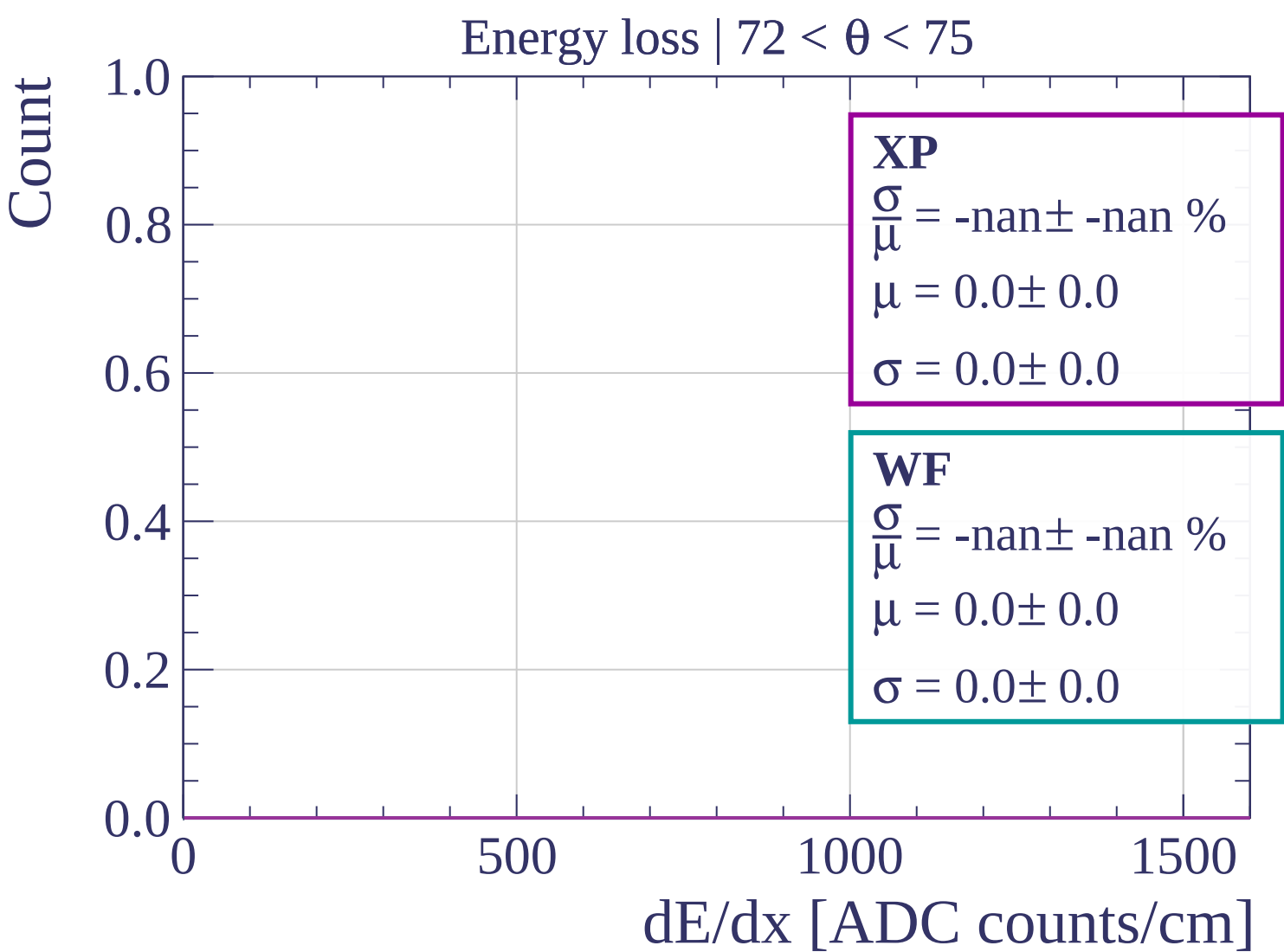
1500

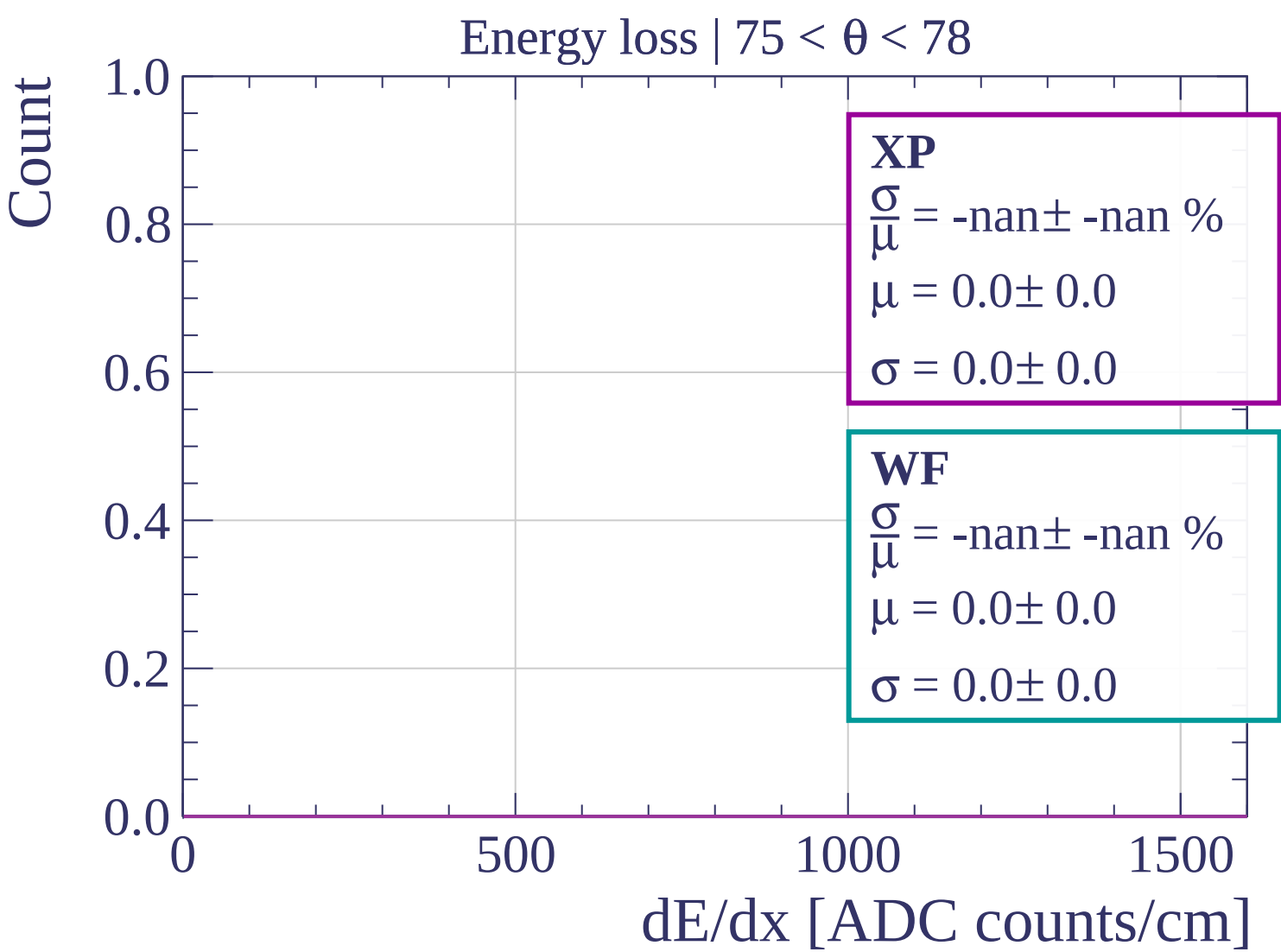
dE/dx [ADC counts/cm]

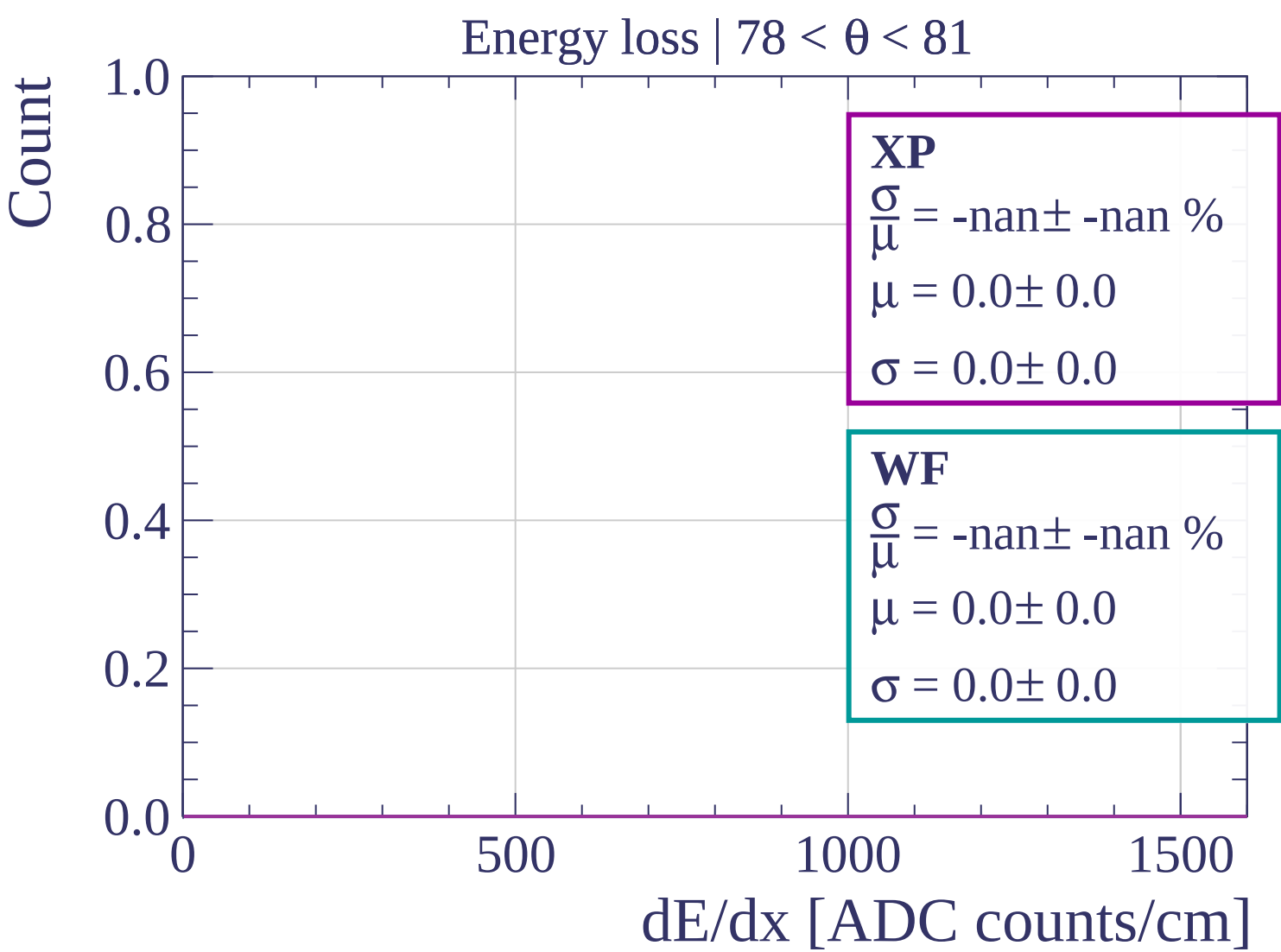


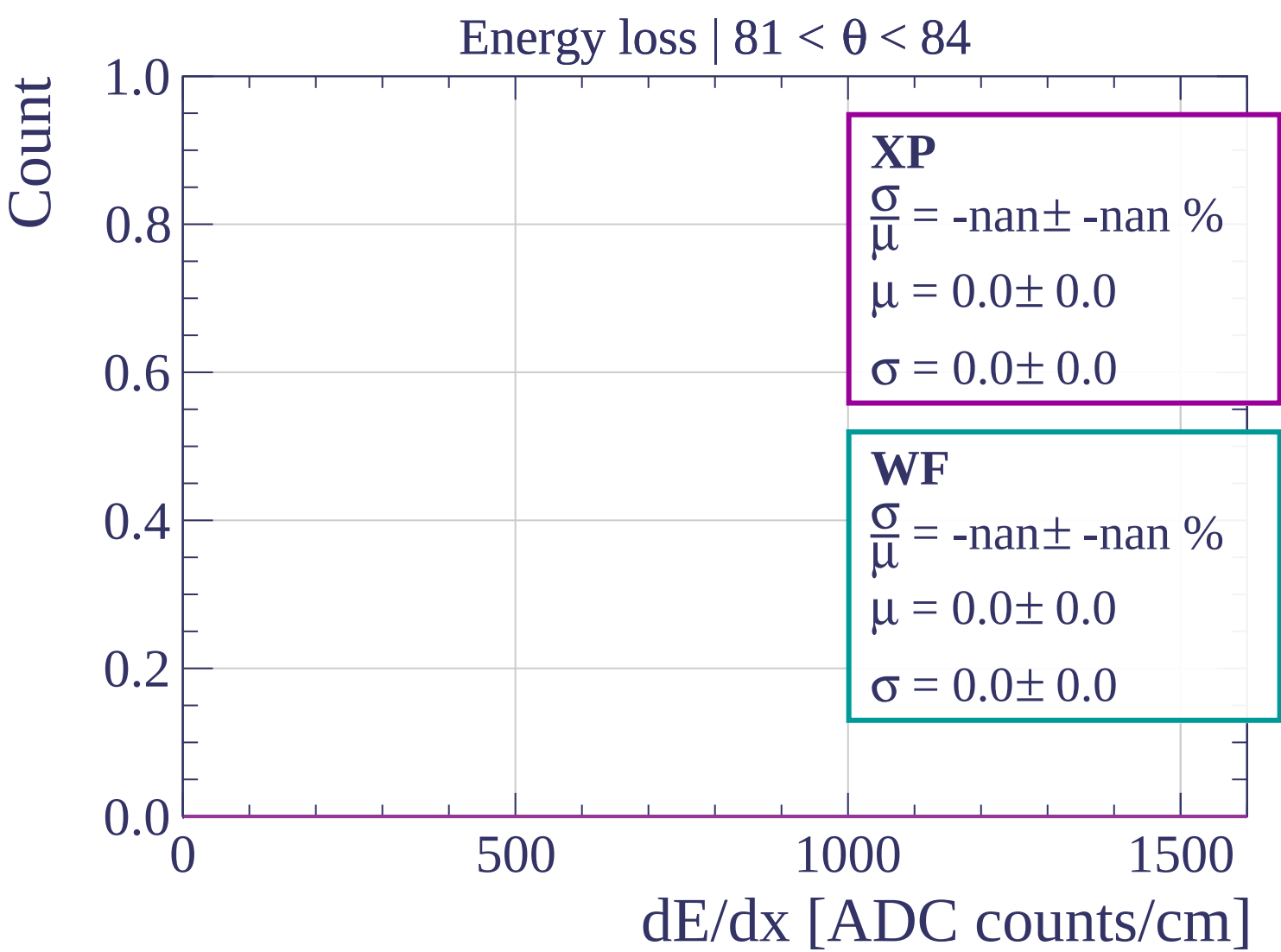


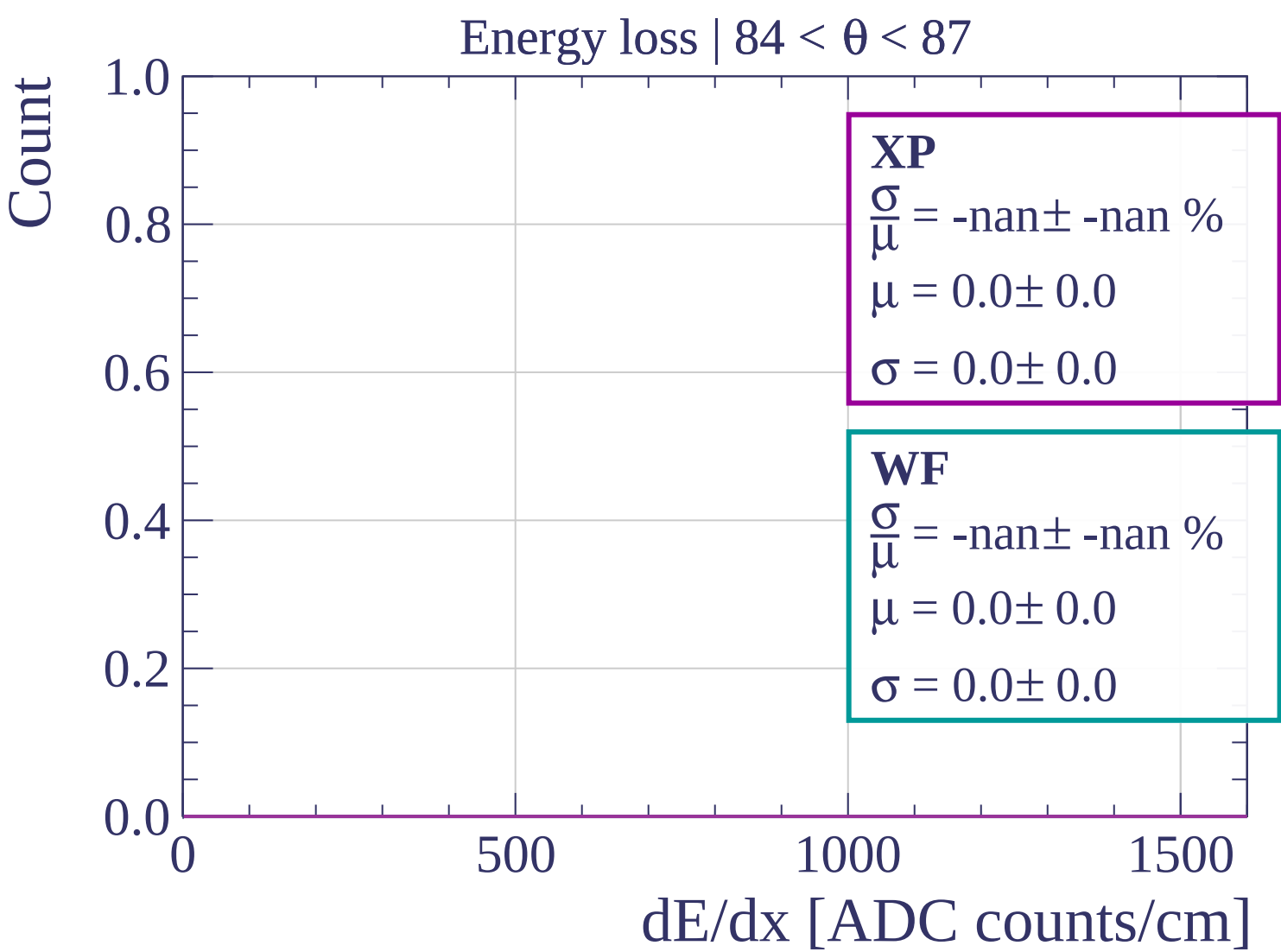


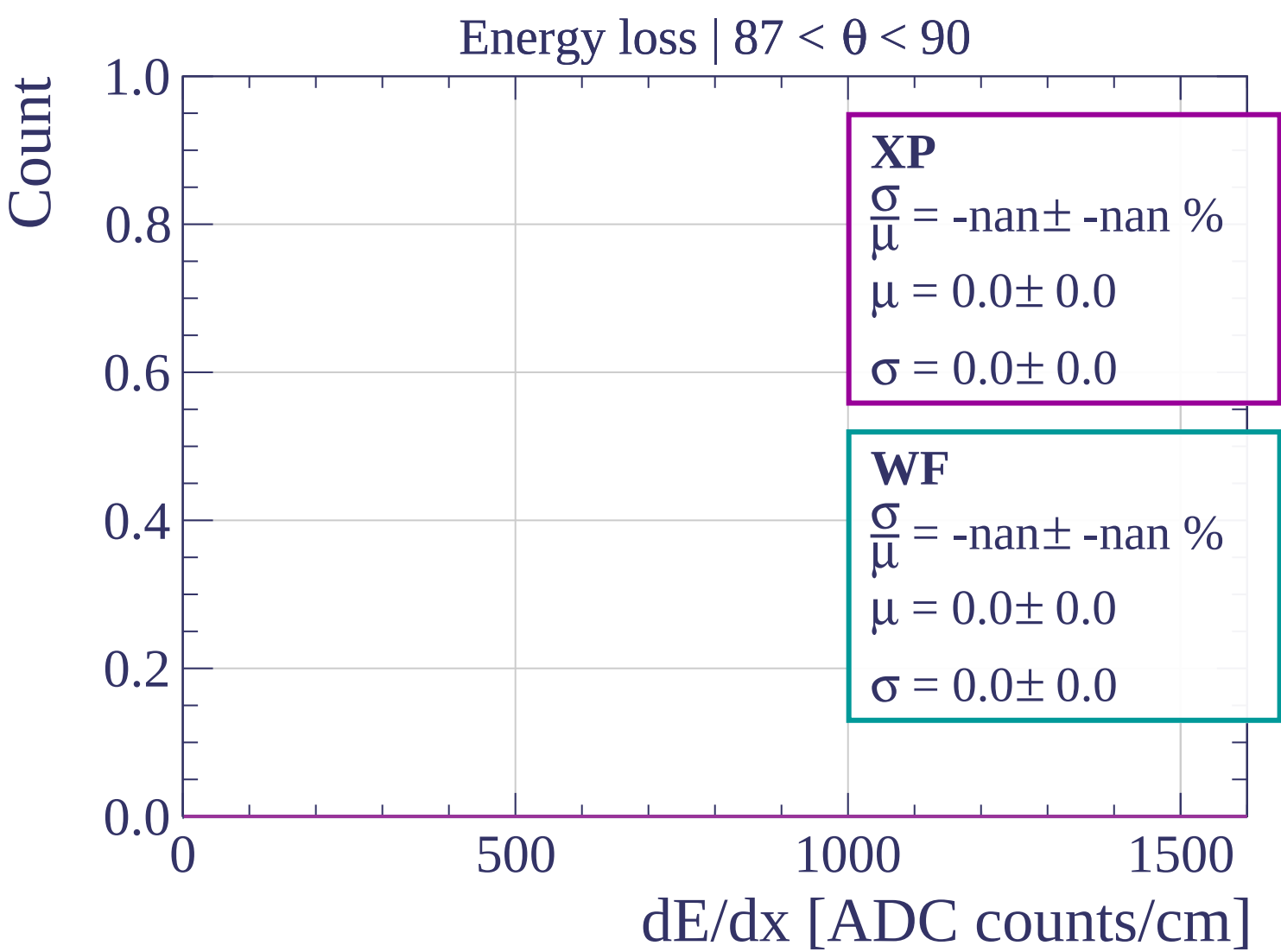




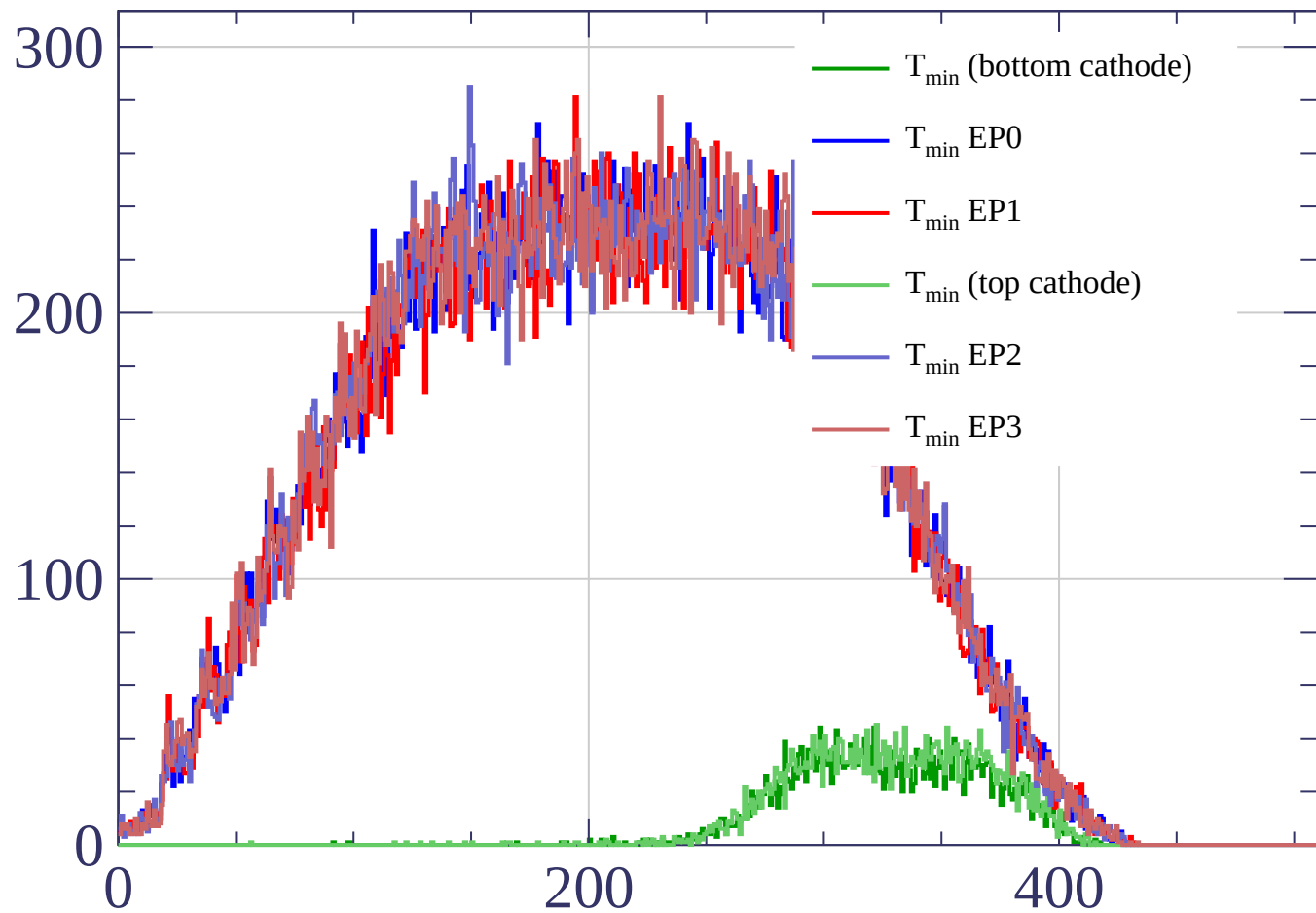








Count



time bin

